

Beta-Sorted Portfolios

Supplemental Appendix: Theory

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September 22, 2026

Abstract

This supplemental appendix gives the theoretical details for beta-sorted portfolio estimation and inference. Part [I](#) develops general high-level consistency and stochastic linearization results for beta-sorted estimators based on generated regressors and data-dependent portfolios. Part [II](#) specializes these results to the classical beta-sorted portfolio construction based on rolling-window first-step regressions and marginal-quantile portfolios, including convergence rates, distributional approximations, feasible variance estimation, uniform-in- β inference, and factor-cancelling contrasts. The companion Supplemental Appendix: Applied provides implementation guidance, simulation evidence, and empirical applications.

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Part I: General Beta-Sorted Portfolios

This part of the supplemental appendix considers a general portfolio scheme and establishes two main results under high-level conditions for the beta-sorted portfolio estimator: (i) uniform consistency, and (ii) stochastic linearization. Part II specializes this setup to the classical beta-sorted portfolio estimator, where portfolios are constructed using first-step local rolling-window regressions and second-step quantile-spaced partitions. Part II also provides primitive conditions for the high-level conditions imposed here.

SA-I.1 Setup

We consider the following asset return model throughout:

$$R_{it} = \alpha_{it} + \beta_{it}^\top f_t + \varepsilon_{it}, \quad \mathbb{E}[\varepsilon_{it} | \mathcal{F}_{t-1}, f_t] = 0, \quad i = 1, \dots, n_t, \quad t = 1, \dots, T,$$

where α_{it} and β_{it} are $\mathcal{F}_{t-1}$ -measurable, $f_t \in \mathbb{R}^d$ is a column vector of observable factors, and β_{it} takes values in a common compact support $\mathcal{B} \subset \mathbb{R}^d$ containing 0 as an interior point. For each finite sample, let n be a cross-sectional index bound such that $[n_t] \subseteq [n]$ for all $t \leq T$, where for any positive integer m , $[m] := \{1, \dots, m\}$. We define

$$\begin{aligned} \mathcal{G}_t &= \sigma(\{f_s\}_{s=1}^t, \{\aleph_{2s}\}_{s=1}^t), \\ \mathcal{F}_t &= \sigma(\{f_s\}_{s=1}^t, \{\{\varepsilon_{is}\}_{i=1}^{n_s}\}_{s=1}^t, \{\aleph_{1i}\}_{i=1}^n, \{\aleph_{2s}\}_{s=1}^t, \{\{\aleph_{3is}\}_{i=1}^{n_s}\}_{s=1}^t). \end{aligned}$$

Here $\aleph_{1i}$, $\aleph_{2t}$, and $\aleph_{3it}$ are auxiliary random vectors that may enter the data-generating process. The predictable coefficients α_{it} and β_{it} are $\mathcal{F}_{t-1}$ -measurable. Part II gives a primitive innovation-free representation in terms of factor, scale, and auxiliary histories dated at most $t-1$. Thus $\{\mathcal{F}_t\}$ and $\{\mathcal{G}_t\}$ are filtrations with $\mathcal{G}_t \subseteq \mathcal{F}_t$ and $\sigma(f_s : 1 \leq s \leq t) \subseteq \mathcal{G}_t$. We interpret $\mathcal{F}_t$ as the full information set and $\mathcal{G}_t$ as the factor-state information set. We assume that there is a fixed constant $c > 0$ such that $cn \leq n_t \leq n$ for all $t = 1, \dots, T$, and asymptotic approximations are taken as both n and T diverge to infinity.

We assume that there exists a nonrandom function $\mu_0(\cdot; \cdot)$ such that

$$\mathbb{E}[R_{it} | \mathcal{F}_{t-1}] = \mu_0(\beta_{it}; \mathcal{G}_{t-1}) =: \mu_t(\beta_{it}),$$

and the structural form is

$$R_{it} = \mu_t(\beta_{it}) + \beta_{it}^\top (f_t - \mathbb{E}[f_t | \mathcal{F}_{t-1}]) + \varepsilon_{it}.$$

We impose $\mathbb{E}[f_t | \mathcal{F}_{t-1}] = \mathbb{E}[f_t | \mathcal{G}_{t-1}]$ and therefore write

$$R_{it} = \mu_t(\beta_{it}) + \beta_{it}^\top \tilde{f}_t + \varepsilon_{it}, \quad \tilde{f}_t = f_t - \mathbb{E}[f_t | \mathcal{G}_{t-1}].$$

It follows that

$$\mu_t(\cdot) = \mu_0(\cdot; \mathcal{G}_{t-1}), \quad \alpha_t(b) := \mu_t(b) - b^\top \mathbb{E}[f_t | \mathcal{G}_{t-1}], \quad b \in \mathcal{B}, \quad \alpha_{it} = \alpha_t(\beta_{it}).$$

Let $H = H_{nT}$ be an integer satisfying $1 \leq H < T$. In this general part, H records the first-step

initialization period, so second-step estimation uses dates $t = H + 1, \dots, T$. Part II specializes H to the rolling-window length and imposes its asymptotic rate conditions.

At a fixed date, a cross-sectional regression of returns on risk exposures recovers the realized systematic return function

$$M_t(\beta) = \mu_t(\beta) + \beta^\top \tilde{f}_t,$$

rather than $\mu_t(\beta)$ itself. The time average of the second term, $(T - H)^{-1} \sum_{t=H+1}^T \beta^\top \tilde{f}_t$, has martingale-difference summands with respect to $\{\mathcal{G}_t\}$ because $\mathbb{E}[\beta^\top \tilde{f}_t | \mathcal{G}_{t-1}] = 0$. Hence this average is asymptotically negligible for consistency under the conditions below, although it contributes to the first-order distribution when $\beta \neq 0$.

The goal is to estimate and conduct inference on the *sample average of conditional expected returns (SACER)*:

$$\bar{\mu}_T(\beta; H) = \frac{1}{T - H} \sum_{t=H+1}^T \mu_t(\beta).$$

SA-I.1.1 Estimator

The beta-sorted portfolio procedure targets $\bar{\mu}_T(\cdot)$ by averaging over time cross-sectional piecewise polynomial regression fits constructed using estimated partitions (portfolios). More precisely, the estimator is constructed in two steps:

- (i) obtain an estimator $\hat{\beta}_{it}$ of β_{it} for each $i = 1, \dots, n_t$ and $t = H + 1, \dots, T$; and
- (ii) for each $t = H + 1, \dots, T$, construct a collection of $\bar{J}_t$ disjoint cells $\hat{\Delta}_t = \{\hat{P}_{jt} : j \in [\bar{J}_t]\}$, and then define the following estimator of $\bar{\mu}_T(\beta; H)$:

$$\hat{\mu}(\beta) = \frac{1}{T - H} \sum_{t=H+1}^T \hat{M}_t(\beta), \quad \hat{M}_t(\beta) = \hat{p}_t(\beta)^\top \left(\sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) \hat{p}_t(\hat{\beta}_{it})^\top \right)^{-1} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) R_{it},$$

where $\hat{p}_t(b)$ denotes a piecewise polynomial expansion based on the partition $\hat{\Delta}_t$.

Without loss of generality, we write this piecewise polynomial expansion in cell-normalized coordinates. For each cell $\hat{P}_{jt}$, choose a center $\hat{c}_{jt}$ and a nonsingular scaling/rotation matrix $\hat{D}_{jt}$ such that $\hat{D}_{jt}^{-1}(b - \hat{c}_{jt})$ is uniformly bounded over $b \in \hat{P}_{jt}$. Let $\hat{j}_t(b)$ denote the index of the cell containing b . For a nonnegative integer $\mathbf{q}$, let $\check{p}_{\mathbf{q}}(z) = (z^s : |s| \leq \mathbf{q})^\top$ denote the column vector collecting, in a fixed order, all d -variate monomials in z with total degree no greater than $\mathbf{q}$, using the multi-index notation collected in Section SA-I.1.2. For any cell P equipped with a center c_P and nonsingular scaling/rotation matrix D_P , define the localized degree- $\mathbf{q}$ block basis by

$$p_{\mathbf{q}}(b, P) = \mathbb{1}(b \in P) \check{p}_{\mathbf{q}}(D_P^{-1}(b - c_P)).$$

Cell indicators are understood with the fixed boundary tie-breaking convention for the corresponding partition, so a point contributes to at most one block. For any admissible partition $\Delta = \{P_j : j \in [\bar{J}]\}$ equipped

with its cell normalizations, define the associated stacked basis by

$$p(b; \Delta) = (p_{\mathbf{q}}(b, P_1)^\top, \dots, p_{\mathbf{q}}(b, P_{\bar{J}})^\top)^\top.$$

We then take $\widehat{p}_t(b) = p(b; \widehat{\Delta}_t)$. This stacked-block form emphasizes that $\widehat{p}_t(b)$ is the product of a cell-indicator and the polynomial basis used within the selected cell. When $\mathbf{q} = 0$, $\check{p}_{\mathbf{q}}$ is the scalar 1, so $p_{\mathbf{q}}(b, P) = \mathbf{1}(b \in P)$. If $K_{\mathbf{q}}$ denotes the number of monomials, then $\widehat{p}_t(\beta) \in \mathbb{R}^{\bar{J}_t K_{\mathbf{q}}}$.

At this level of generality, we allow for generic choices of the first-step estimator $\widehat{\beta}_{it}$ of β_{it} and the partitioning scheme $\{\widehat{\Delta}_t\}_{t=H+1}^T$, while in the second step, for each $t = H+1, \dots, T$, we allow for a piecewise polynomial regression of total degree $\mathbf{q}$ to obtain $\widehat{M}_t(\cdot)$.

Although β_{it} is assumed to have bounded support $\mathcal{B}$, the first-step estimator $\widehat{\beta}_{it}$ may lie outside $\mathcal{B}$. Thus, we fix $\delta_{\mathcal{B}} > 0$ and introduce a compact convex set $\mathcal{B}^+$ satisfying

$$\left\{ b \in \mathbb{R}^d : \inf_{a \in \mathcal{B}} \|b - a\| \leq \delta_{\mathcal{B}} \right\} \subseteq \mathcal{B}^+.$$

When $\widehat{\beta}_{it}$ is uniformly consistent for β_{it} in the sense that

$$\max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \|\widehat{\beta}_{it} - \beta_{it}\| = o_{\mathbb{P}}(1),$$

the estimated $\widehat{\beta}_{it}$ will belong to $\mathcal{B}^+$ for all $i \in [n_t]$ and $t = H+1, \dots, T$, with probability approaching one. In the theoretical analysis below, μ_t is understood as a bounded continuous extension from $\mathcal{B}$ to $\mathcal{B}^+$ whenever it is evaluated at $\widehat{\beta}_{it}$ outside $\mathcal{B}$. Thus, $\mathcal{B}^+$ is only an extension device for evaluating functions at first-step estimates and along Taylor paths; it does not change the support $\mathcal{B}$.

For the interior results below, fix an evaluation region $\mathcal{B}^\circ \subset \text{int}(\mathcal{B})$. For each t , define

$$\mathcal{J}_t = \{j \in [\bar{J}_t] : \widehat{P}_{jt} \cap \mathcal{B}^\circ \neq \emptyset\}.$$

Thus, $\mathcal{J}_t$ collects precisely the estimated cells that may contain evaluation points in $\mathcal{B}^\circ$.

SA-I.1.2 Notation

For a positive integer m , let $[m] = \{1, \dots, m\}$. All vectors are column vectors unless explicitly transposed. For a generic vector $v \in \mathbb{R}^d$, v_j denotes its j -th element and $\|v\|$ denotes its Euclidean norm. For a matrix $A \in \mathbb{R}^{m \times n}$, $\|A\|$ denotes its operator norm; for a square symmetric matrix A , $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ denote its minimum and maximum eigenvalues. For a scalar- or vector-valued random variable Y and $r \geq 1$, write $\|Y\|_r = (\mathbb{E}\|Y\|^r)^{1/r}$, where the inner norm is absolute value for scalars and Euclidean norm for vectors. For a set $S \subseteq \mathbb{R}^d$, define its diameter as $\text{diam}(S) = \sup\{\|s - s'\| : s, s' \in S\}$. We write $\otimes$ for the Kronecker product.

For $\ell, u \in \mathbb{R}^d$ such that $\ell \leq u$ elementwise, define the rectangle

$$[\ell, u] = \{b = (b_1, \dots, b_d)^\top : b_l \in [\ell_l, u_l], l \in [d]\},$$

where ℓ_l and u_l denote the l -th elements of ℓ and u . Thus, $\text{diam}([\ell, u]) = (\sum_{l \in [d]} (u_l - \ell_l)^2)^{1/2}$.

For a multi-index $m \in \mathbb{Z}_+^d$, let $|m| = \sum_{j=1}^d m_j$, and for $z \in \mathbb{R}^d$, write $z^m = z_1^{m_1} \dots z_d^{m_d}$. For a nonnegative integer $\mathbf{q}$, $(z^m : |m| \leq \mathbf{q})^\top$ denotes the column vector collecting, in a fixed order, all d -variate monomials in z .

with total degree no greater than q . For a multivariate function $g : \mathbb{R}^d \rightarrow \mathbb{R}$, $\nabla g = \nabla^1 g$ denotes its gradient, $\nabla^2 g$ its Hessian, and

$$g^{(m)}(z) = \frac{\partial^{|m|}}{\partial z_1^{m_1} \dots \partial z_d^{m_d}} g(z).$$

For a univariate function g , we write $g \in L^1(\mathbb{R})$ if $\int_{\mathbb{R}} |g(z)| dz < \infty$. For a function $g : \mathcal{X} \rightarrow \mathbb{R}$ on a compact set $\mathcal{X} \subseteq \mathbb{R}^d$ and some $s > 0$, we say g is Hölder- s on $\mathcal{X}$ if all partial derivatives $g^{(m)}$ with $|m| \leq \underline{s}$ exist and are continuous on $\mathcal{X}$, and all partial derivatives with $|m| = \underline{s}$ are Hölder continuous with exponent $s - \underline{s}$ uniformly on $\mathcal{X}$, where $\underline{s} = \max\{k \in \mathbb{Z}_+ : k < s\}$. Whenever a function enters the analysis through its Hessian, we additionally require it to be twice continuously differentiable with a uniformly bounded Hessian on its domain. This requirement is additional to Hölder- s smoothness when $s = 2$ under the preceding convention.

For sequences, $a_n \lesssim b_n$ denotes that $\limsup_n |a_n/b_n|$ is finite; $a_n = O_{\mathbb{P}}(b_n)$ or $a_n \lesssim_{\mathbb{P}} b_n$ means that $\lim_{\epsilon \rightarrow \infty} \limsup_n \mathbb{P}[|a_n/b_n| \geq \epsilon] = 0$; $a_n = o(b_n)$ or $a_n \ll b_n$ denotes $a_n/b_n \rightarrow 0$; and $a_n = o_{\mathbb{P}}(b_n)$ denotes that a_n/b_n converges to zero in probability. We write $a_n \asymp b_n$ when $a_n \lesssim b_n$ and $b_n \lesssim a_n$.

SA-I.2 Convergence Rate

For the remaining results in this part, define

$$\hat{n}_{jt} = \sum_{i=1}^{n_t} \mathbb{1}(\hat{\beta}_{it} \in \hat{P}_{jt}), \quad \hat{n}_{\max} = \max_{H+1 \leq t \leq T} \max_{j \in [J_t]} \hat{n}_{jt}, \quad \hat{n}_{\min} = \min_{H+1 \leq t \leq T} \min_{j \in [J_t]} \hat{n}_{jt}.$$

For each cell, define the estimated Gram block

$$\hat{Q}_{jt} = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{1}(\hat{\beta}_{it} \in \hat{P}_{jt}) \check{p}_q(\hat{D}_{jt}^{-1}(\hat{\beta}_{it} - \hat{c}_{jt})) \check{p}_q(\hat{D}_{jt}^{-1}(\hat{\beta}_{it} - \hat{c}_{jt}))^{\top}.$$

When $q = 0$, $\check{p}_q$ is the scalar 1, and $\hat{Q}_{jt} = \hat{n}_{jt}/n_t$ on every nonempty cell.

SA-I.2.1 Result

Theorem SA-I.2.1 (Convergence Rate). *Consider the second-step estimator with polynomial degree $q \geq 0$. Suppose that the following conditions hold:*

- (i) **DGP.** $\{\beta_{it} : i \in [n_t], t \in [T]\}$ take values in compact support $\mathcal{B}$; μ_t admits a bounded extension to the neighborhood $\mathcal{B}^+$ of $\mathcal{B}$ and this extension is uniformly Hölder- s_{μ} on $\mathcal{B}^+$ for some $s_{\mu} \geq 1$; $\mathbb{E}[f_t | \mathcal{F}_{t-1}] = \mathbb{E}[f_t | \mathcal{G}_{t-1}]$, $\mathbb{E}[\varepsilon_{it} | \mathcal{F}_{t-1}, f_t] = 0$, and ε_{it} is independent over i for each t conditional on $\mathcal{F}_{t-1}$. For some $q > 2$, $\max_{t \in [T]} \max_{i \in [n_t]} \mathbb{E}[|\varepsilon_{it}|^q | \mathcal{F}_{t-1}] \lesssim 1$ and $\max_{t \in [T]} \mathbb{E}[\|\tilde{f}_t\|^q | \mathcal{F}_{t-1}] \lesssim 1$.
- (ii) **First-Step.** $\hat{\beta}_{it}$ is $\mathcal{F}_{t-1}$ -measurable, and $H \leq cT$ for a constant $c \in (0, 1)$. With probability approaching one, $\hat{\beta}_{it} \in \mathcal{B}^+$ for all $i \in [n_t]$ and $t = H+1, \dots, T$. Define the realized first-step error

$$\hat{R}_{\text{FS}} = \max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \|\hat{\beta}_{it} - \beta_{it}\|.$$

(iii) **Partition and Design.** The estimated partition $\widehat{\Delta}_t$ and its associated centers and scaling matrices are $\mathcal{F}_{t-1}$ -measurable for each t . With probability approaching one, $\mathcal{B} \cup \{\widehat{\beta}_{it} : i \in [n_t]\}$ is contained in $\cup_{j \in [\bar{J}_t]} \widehat{P}_{jt}$ for each $t = H+1, \dots, T$, $\widehat{n}_{\min} > 0$, and, for all $t = H+1, \dots, T$ and $j \in [\bar{J}_t]$,

$$1 \lesssim \frac{n_t}{\widehat{n}_{jt}} \lambda_{\min}(\widehat{Q}_{jt}) \leq \frac{n_t}{\widehat{n}_{jt}} \lambda_{\max}(\widehat{Q}_{jt}) \lesssim 1.$$

Define the realized partition diameter

$$\widehat{R}_D = \max_{H+1 \leq t \leq T} \max_{j \in [\bar{J}_t]} \text{diam}(\widehat{P}_{jt}).$$

If $\frac{(nT)^{2/q} \log(nT)}{\widehat{n}_{\min}} \lesssim_{\mathbb{P}} 1$, then for any $\ell_n \rightarrow \infty$, with probability approaching one, the following bound holds for all $\beta \in \mathcal{B}$:

$$|\widehat{\mu}(\beta) - \bar{\mu}_T(\beta; H)| \lesssim \ell_n \left[\widehat{R}_{\text{FS}} + \widehat{R}_D^{\min\{s_\mu, q+1\}} + \sqrt{\frac{\log(nT)}{\widehat{n}_{\min}}} + \frac{\|\beta\|}{\sqrt{T}} \right].$$

All implicit constants in this theorem are independent of the evaluation point β and of n, T , but may depend on the Hölder and design constants.

SA-I.2.2 Discussion

- (i) The sequence $\ell_n \rightarrow \infty$ is an arbitrary high-probability slack sequence. Equivalently, the same display can be read as the corresponding $\lesssim_{\mathbb{P}}$ rate, while retaining the explicit dependence of the factor term on the evaluation point β .
- (ii) When $q = 0$, the within-cell polynomial basis p_q is the scalar 1 and $\widehat{Q}_{jt} = \widehat{n}_{jt}/n_t$ on every nonempty cell. Thus, the design condition in (iii) reduces to the nonempty-cell condition $\widehat{n}_{\min} > 0$, and since $s_\mu \geq 1$, the second-step approximation rate is governed by the cell-diameter term $\widehat{R}_D$.
- (iii) If the partition is quasi-uniform and each cell roughly contains the same number of observations, then $\widehat{R}_D \lesssim_{\mathbb{P}} J^{-1}$ and $n/J^d \lesssim_{\mathbb{P}} \widehat{n}_{\min} \leq \widehat{n}_{\max} \lesssim_{\mathbb{P}} n/J^d$. In that case, the theorem gives the rate

$$\ell_n \left[\widehat{R}_{\text{FS}} + J^{-\min\{s_\mu, q+1\}} + \sqrt{\frac{J^d \log(nT)}{n}} + \frac{\|\beta\|}{\sqrt{T}} \right].$$

- (iv) With a boundary-specific modification of the assumptions and proof, the same argument can allow boundary cells that are large outside the support but have a small intersection with the feasible support $\mathcal{B}$. For example, if $\beta = \underline{\beta}$ is on the lower boundary and the first cell intersects $\mathcal{B}$ only on a shrinking neighborhood of $\underline{\beta}$, then any $\widehat{\beta}_{it}$ in that cell is still close to β with high probability.

SA-I.2.3 Proof of Theorem SA-I.2.1

Work on the event where the first-step estimates belong to $\mathcal{B}^+$, the estimated partition covers $\mathcal{B}$ and the first-step estimates, and the design conditions in the theorem hold. This event has probability approaching one.

For each $\beta \in \mathcal{B}$, let $\hat{j}_t(\beta)$ denote the index of the cell containing β . The fitted value can be written as a weighted average,

$$\widehat{M}_t(\beta) = \sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) R_{it},$$

where

$$\widehat{w}_{it}(\beta) = \frac{\mathbb{1}(\widehat{\beta}_{it} \in \widehat{P}_{\hat{j}_t(\beta)t})}{n_t} \check{p}_{\mathbf{q}}(\widehat{D}_{\hat{j}_t(\beta)t}^{-1}(\beta - \widehat{c}_{\hat{j}_t(\beta)t}))^\top \widehat{Q}_{\hat{j}_t(\beta)t}^{-1} \check{p}_{\mathbf{q}}(\widehat{D}_{\hat{j}_t(\beta)t}^{-1}(\widehat{\beta}_{it} - \widehat{c}_{\hat{j}_t(\beta)t})).$$

The boundedness of the normalized polynomial basis and the scaled eigenvalue condition imply that with probability approaching one, uniformly over t and β ,

$$\sum_{i=1}^{n_t} |\widehat{w}_{it}(\beta)| \lesssim 1, \quad \sum_{i=1}^{n_t} \widehat{w}_{it}(\beta)^2 \lesssim \frac{1}{\widehat{n}_{\hat{j}_t(\beta)t}}.$$

Applying the same algebra to the constant polynomial gives $\sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) = 1$. Moreover, the weights reproduce all polynomials of total degree no greater than $\mathbf{q}$ on the selected cell:

$$\sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) \text{poly}(\widehat{\beta}_{it}) = \text{poly}(\beta)$$

for every such polynomial poly . This includes constant functions when $\mathbf{q} = 0$ and coordinate functions when $\mathbf{q} \geq 1$.

Using $R_{it} = \mu_t(\beta_{it}) + \beta_{it}^\top \widetilde{f}_t + \varepsilon_{it}$,

$$\begin{aligned} \widehat{\mu}(\beta) - \bar{\mu}_T(\beta; H) &= \frac{1}{T-H} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) \varepsilon_{it} \\ &\quad + \frac{1}{T-H} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) (\mu_t(\beta_{it}) - \mu_t(\beta)) \\ &\quad + \frac{1}{T-H} \sum_{t=H+1}^T \left(\sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) \beta_{it} \right)^\top \widetilde{f}_t. \end{aligned}$$

We first bound the approximation term. Since $s_\mu \geq 1$, the uniform Hölder condition implies a uniform Lipschitz bound on $\mathcal{B}^+$. Therefore,

$$\max_{H+1 \leq t \leq T} \max_{i \in [n_t]} |\mu_t(\beta_{it}) - \mu_t(\widehat{\beta}_{it})| \lesssim \widehat{R}_{\text{FS}}.$$

Together with the absolute-weight-sum bound, this controls the contribution from replacing β_{it} by $\widehat{\beta}_{it}$. For the remaining within-cell term, local polynomial approximation gives, uniformly over t and cells, a polynomial poly_{j_t} of degree no greater than $\mathbf{q}$ such that

$$\sup_{b \in \widehat{P}_{j_t} \cap \mathcal{B}^+} |\mu_t(b) - \text{poly}_{j_t}(b)| \lesssim \text{diam}(\widehat{P}_{j_t})^{\min\{s_\mu, \mathbf{q}+1\}}.$$

Using polynomial reproduction,

$$\left| \sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) (\mu_t(\widehat{\beta}_{it}) - \mu_t(\beta)) \right| \lesssim \widehat{\mathbf{R}}_{\mathbf{D}}^{\min\{s_\mu, q+1\}}.$$

Thus, uniformly over $\beta \in \mathcal{B}$,

$$\left| \frac{1}{T-H} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) (\mu_t(\beta_{it}) - \mu_t(\beta)) \right| \lesssim \widehat{\mathbf{R}}_{\text{FS}} + \widehat{\mathbf{R}}_{\mathbf{D}}^{\min\{s_\mu, q+1\}}.$$

Next, consider the idiosyncratic-error term. For each component $k \in [K_q]$ of $\check{p}_q(\widehat{D}_{jt}^{-1}(\widehat{\beta}_{it} - \widehat{c}_{jt}))$,

$$\frac{1}{\widehat{n}_{jt}} \sum_{i=1}^{n_t} \mathbb{1}(\widehat{\beta}_{it} \in \widehat{P}_{jt}) \left[\check{p}_q(\widehat{D}_{jt}^{-1}(\widehat{\beta}_{it} - \widehat{c}_{jt})) \right]_k \varepsilon_{it}$$

is a conditionally mean-zero average, independent over i given $\mathcal{F}_{t-1}$, with conditional variance bounded by a constant multiple of $1/\widehat{n}_{jt}$. On the event $\widehat{n}_{\min} > 0$, the number of cell-component averages is of order $\sum_{t=H+1}^T \bar{J}_t \leq \sum_{t=H+1}^T n_t \lesssim nT$. Since the normalized regressors are bounded, a standard truncation and union-bound argument for conditionally independent arrays yields

$$\max_{H+1 \leq t \leq T} \max_{j \in [\bar{J}_t]} \max_{k \in [K_q]} \left| \frac{1}{\widehat{n}_{jt}} \sum_{i=1}^{n_t} \mathbb{1}(\widehat{\beta}_{it} \in \widehat{P}_{jt}) \left[\check{p}_q(\widehat{D}_{jt}^{-1}(\widehat{\beta}_{it} - \widehat{c}_{jt})) \right]_k \varepsilon_{it} \right| \lesssim \ell_n \sqrt{\frac{\log(nT)}{\widehat{n}_{\min}}}$$

with probability approaching one; the rate restriction involving $\widehat{n}_{\min}$ ensures that the truncation contribution is absorbed into the displayed bound. For any evaluation point β , the weighted error is a bounded linear combination of these component averages in the selected cell. The boundedness of the normalized polynomial basis and the scaled eigenvalue bound for $\widehat{Q}_{jt}$ then gives, uniformly over $\beta \in \mathcal{B}$,

$$\left| \frac{1}{T-H} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) \varepsilon_{it} \right| \lesssim \ell_n \sqrt{\frac{\log(nT)}{\widehat{n}_{\min}}}.$$

Finally, decompose the factor term as

$$\begin{aligned} & \frac{1}{T-H} \sum_{t=H+1}^T \left(\sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) \beta_{it} \right)^\top \widetilde{f}_t \\ &= \frac{1}{T-H} \sum_{t=H+1}^T \beta^\top \widetilde{f}_t \\ & \quad + \frac{1}{T-H} \sum_{t=H+1}^T \left(\sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) (\beta_{it} - \widehat{\beta}_{it}) \right)^\top \widetilde{f}_t \\ & \quad + \frac{1}{T-H} \sum_{t=H+1}^T \left(\sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) \widehat{\beta}_{it} - \beta \right)^\top \widetilde{f}_t. \end{aligned}$$

By the martingale-difference property of $\tilde{f}_t$ with respect to $\mathcal{G}_t$ and the conditional moment bound,

$$\left\| \frac{1}{T-H} \sum_{t=H+1}^T \tilde{f}_t \right\| \lesssim_{\mathbb{P}} T^{-1/2},$$

so the first term is $\lesssim_{\mathbb{P}} \|\beta\| T^{-1/2}$ uniformly over β . Also,

$$\frac{1}{T-H} \sum_{t=H+1}^T \|\tilde{f}_t\| \lesssim_{\mathbb{P}} 1$$

by Markov's inequality. The second term is $\lesssim_{\mathbb{P}} \widehat{\mathbf{R}}_{\text{FS}}$ because the weights have uniformly bounded absolute sum. For the third term, polynomial reproduction gives

$$\sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) \widehat{\beta}_{it} = \beta$$

when $\mathbf{q} \geq 1$. When $\mathbf{q} = 0$, the same expression is the ordinary cell average of the $\widehat{\beta}_{it}$'s in the selected cell, and hence its distance from β is bounded by the diameter of that cell. Therefore, uniformly over $\beta \in \mathcal{B}$,

$$\left| \frac{1}{T-H} \sum_{t=H+1}^T \left(\sum_{i=1}^{n_t} \widehat{w}_{it}(\beta) \beta_{it} \right)^{\top} \tilde{f}_t \right| \lesssim_{\mathbb{P}} \frac{\|\beta\|}{\sqrt{T}} + \widehat{\mathbf{R}}_{\text{FS}} + \widehat{\mathbf{R}}_{\text{D}}^{\min\{s_{\mu}, \mathbf{q}+1\}}.$$

Combining the three bounds and absorbing the stochastic constants into the arbitrary high-probability slack sequence ℓ_n gives the claimed result, uniformly over $\beta \in \mathcal{B}$. $\square$

SA-I.3 Stochastic Linearization

This section derives a high-level stochastic linearization for second-step polynomial degree $\mathbf{q} \geq 0$. The result is stated in terms of generic first-step estimators and generic partitions, so it does not rely on the specific local-polynomial first step or marginal-quantile-spaced partition introduced later. Let J denote a generic partition-resolution sequence; the conditions below encode cell mass of order J^{-d} and cell diameter of order J^{-1} .

Let $\{\mathcal{D}_t\}$ denote an auxiliary design filtration with $\mathcal{D}_{t-1} \subseteq \mathcal{F}_{t-1}$. In this high-level section, $\mathcal{D}_{t-1}$ records the information used to construct the conditionally nonrandom reference partition; Section [SA-II.1.2](#) gives its concrete definition for the classical beta-sorted estimator. To state the linearization, introduce a reference partition,

$$\Delta_t = \{P_{jt} : j \in [\bar{J}_t]\}$$

that is $\mathcal{D}_{t-1}$ -measurable and has the same number of cells as the estimated partition $\widehat{\Delta}_t$ from Section [SA-I.1](#). For each reference cell paired with an estimated cell intersecting $\mathcal{B}^\circ$, choose a $\mathcal{D}_{t-1}$ -measurable center c_{jt} and nonsingular scaling/rotation matrix D_{jt} so that the normalized polynomial basis is uniformly bounded on the cell. Tail reference cells may instead be assigned any $\mathcal{D}_{t-1}$ -measurable nonsingular normalization; no uniform boundedness or design condition is imposed on them because they are never selected by an interior evaluation point. Thus, $\mathcal{D}_{t-1}$ governs the reference partition and reference objects, whereas the realized

estimated partition $\hat{\Delta}_t$ may depend on first-step estimates and is only required below to be $\mathcal{F}_{t-1}$ -measurable. The reference cells and estimated cells are paired by their common index j .

Define the reference basis maps by

$$p_t(b) = p(b; \Delta_t), \quad p_{jt}(b) = p_{\mathbf{q}}(b, P_{jt}).$$

Both basis maps are evaluated at the raw argument b ; in particular, the classical construction does not restrict the generated regressor to reside in $\mathcal{B}$. The estimated marginal partition in Section SA-II.3 uses the coordinatewise sample extrema as its outer endpoints and therefore contains every realized $\hat{\beta}_{it}$. Reference cells are paired with estimated cells by their split-quantile ranks. Only pairs whose estimated cell intersects $\mathcal{B}^\circ$ are normalized and used below; no reference design condition is imposed on the cells outside $\mathcal{B}^\circ$. Write $\hat{p}_{jt}(b)$ for the j -th $K_{\mathbf{q}}$ -dimensional block of the estimated basis $\hat{p}_t(b)$ defined in Section SA-I.1. Thus $p_t(\hat{\beta}_{it})$ denotes the reference basis evaluated at the first-step estimate, while $p_t(\beta_{it})$ denotes the reference basis evaluated at the true exposure. The reference Gram matrix below remains centered at the generated-regressor distribution, but the leading idiosyncratic term uses $p_t(\beta_{it})$ so that its coefficients are fixed by the innovation-free conditioning field in the classical verification. The estimated and reference Gram matrices are

$$\hat{Q}_t = \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) \hat{p}_t(\hat{\beta}_{it})^\top \quad \text{and} \quad Q_t = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{E}[p_t(\hat{\beta}_{it}) p_t(\hat{\beta}_{it})^\top \mid \mathcal{D}_{t-1}].$$

Both matrices are block diagonal. Write $\hat{Q}_{jt}$ and Q_{jt} for their j -th $K_{\mathbf{q}} \times K_{\mathbf{q}}$ diagonal blocks. Thus Q_t is the conditional reference Gram matrix for the generated regressor evaluated on the conditionally nonrandom reference partition. When $\mathbf{q} = 0$, $Q_{jt} = n_t^{-1} \sum_{i=1}^{n_t} \mathbb{P}(\hat{\beta}_{it} \in P_{jt} \mid \mathcal{D}_{t-1})$. In every expression involving Q_t^{-1} or $\hat{Q}_t^{-1}$ and an evaluation point in $\mathcal{B}^\circ$, the inverse is blockwise shorthand for the inverse of the selected block. Thus the algebra does not require invertibility or a polynomial normalization for tail blocks outside $\mathcal{J}_t$.

SA-I.3.1 Decomposition

Using the block Gram matrices defined above, the second-step estimator admits the decomposition

$$\begin{aligned} \hat{\mu}(\beta) - \bar{\mu}_T(\beta; H) &= \frac{1}{T-H} \sum_{t=H+1}^T (\hat{M}_t(\beta) - \mu_t(\beta)) \\ &= \mathcal{G}(\beta) + \mathcal{E}(\beta) - \mathcal{L}(\beta) + \mathcal{B}(\beta) \\ &\quad + \mathcal{R}_1(\beta) + \mathcal{R}_2(\beta) + \mathcal{R}_3(\beta) + \mathcal{R}_4(\beta) - \mathcal{R}_5(\beta), \end{aligned}$$

where the potentially leading terms are

$$\begin{aligned} \mathcal{G}(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \beta^\top \tilde{f}_t, \\ \mathcal{E}(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^\top Q_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} p_t(\beta_{it}) \varepsilon_{it}, \\ \mathcal{L}(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^\top \hat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) [\nabla \mu_t(\beta_{it})]^\top (\hat{\beta}_{it} - \beta_{it}), \end{aligned}$$

$$\mathcal{B}(\beta) = \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^\top \hat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) (\mu_t(\hat{\beta}_{it}) - \mu_t(\beta)),$$

and the higher-order remainder terms are

$$\begin{aligned} \mathcal{R}_1(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^\top \left(\hat{Q}_t^{-1} - Q_t^{-1} \right) \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) \varepsilon_{it}, \\ \mathcal{R}_2(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^\top Q_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} (\hat{p}_t(\hat{\beta}_{it}) - p_t(\beta_{it})) \varepsilon_{it}, \\ \mathcal{R}_3(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \left[\hat{p}_t(\beta)^\top \hat{Q}_t^{-1} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) \hat{\beta}_{it}^\top \right) - \beta^\top \right] \tilde{f}_t, \\ \mathcal{R}_4(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^\top \hat{Q}_t^{-1} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) (\beta_{it} - \hat{\beta}_{it})^\top \right) \tilde{f}_t, \\ \mathcal{R}_5(\beta) &= \frac{1}{2(T-H)} \sum_{t=H+1}^T \hat{p}_t(\beta)^\top \hat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) (\hat{\beta}_{it} - \beta_{it})^\top \nabla^2 \mu_t(\tilde{\beta}_{it}) (\hat{\beta}_{it} - \beta_{it}), \end{aligned}$$

with $\tilde{\beta}_{it} = a_{it} \hat{\beta}_{it} + (1 - a_{it}) \beta_{it}$ for some $a_{it} \in [0, 1]$.

The expanded set $\mathcal{B}^+$ from Section SA-I.1 handles first-step estimates and Taylor intermediate points that may lie outside the support $\mathcal{B}$. Recall that $\mathcal{B}^\circ$ is the interior evaluation region and $\mathcal{J}_t$ is the set of estimated cells intersecting it. For each $j \in \mathcal{J}_t$, the reference block Q_{jt} below is formed from the reference cell paired with the estimated cell $\hat{P}_{jt}$; the partition-design condition requires these paired reference cells to satisfy the same interior design bounds. Recall the previously defined quantity $\hat{R}_{\text{FS}}$, and also define

$$\begin{aligned} \hat{R}_D^\circ &= \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \text{diam}(\hat{P}_{jt}), \\ \hat{R}_Q &= \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \|\hat{Q}_{jt} - Q_{jt}\|, \\ \hat{R}_P &= \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \frac{1}{n_t} \sum_{i=1}^{n_t} \|\hat{p}_{jt}(\hat{\beta}_{it}) - p_{jt}(\beta_{it})\|^2. \end{aligned}$$

SA-I.3.2 Result

The following lemma establishes a general stochastic linearization for generic first-step estimators and partitioning schemes. Part II then verifies these high-level quantities for the classical beta-sorted portfolio estimator.

Lemma SA-I.3.1 (Stochastic Linearization). *Maintain the setup and notation of Section SA-I.1 and the definitions above. Suppose that conditions (i) and (ii) in Theorem SA-I.2.1 hold, and that μ_t is uniformly Hölder- s_μ on $\mathcal{B}^+$ and twice continuously differentiable on $\mathcal{B}^+$ with uniformly bounded first and second derivatives. Assume further that the following conditions hold:*

- (i) (Partition and Design) *the estimated partition $\hat{\Delta}_t$ and its associated centers and scaling matrices are $\mathcal{F}_{t-1}$ -measurable for each t , $\mathcal{B}^\circ \cup \{\hat{\beta}_{it} : i \in [n_t]\}$ is contained in $\cup_{j \in [\bar{J}_t]} \hat{P}_{jt}$ for each $t = H+1, \dots, T$ w.p.a.1, $\hat{R}_D^\circ \lesssim_{\mathbb{P}} J^{-1}$, and $J^{-d} \lesssim_{\mathbb{P}} \lambda_{\min}(Q_{jt}) \leq \lambda_{\max}(Q_{jt}) \lesssim_{\mathbb{P}} J^{-d}$ uniformly over $t = H+1, \dots, T$ and*

$j \in \mathcal{J}_t$. Also, uniformly over the same indices,

$$\lambda_{\max} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} p_{jt}(\beta_{it}) p_{jt}(\beta_{it})^\top \right) \lesssim_{\mathbb{P}} J^{-d}.$$

(ii) (Gram Convergence) $\widehat{\mathbf{R}}_{\mathbf{q}} = o_{\mathbb{P}}(J^{-d})$.

Then, for each fixed $\beta \in \mathcal{B}^\circ$,

$$\begin{aligned} |\mathcal{G}(\beta)| &\lesssim_{\mathbb{P}} \frac{\|\beta\|}{\sqrt{T}}, & |\mathcal{E}(\beta)| &\lesssim_{\mathbb{P}} \sqrt{\frac{J^d}{nT}}, \\ |\mathcal{R}_1(\beta)| &\lesssim_{\mathbb{P}} \sqrt{\frac{J^{3d}(\widehat{\mathbf{R}}_{\mathbf{q}})^2}{nT}}, & |\mathcal{R}_2(\beta)| &\lesssim_{\mathbb{P}} \sqrt{\frac{J^{2d}\widehat{\mathbf{R}}_{\mathbf{p}}}{nT}}, \\ |\mathcal{R}_3(\beta)| &\lesssim_{\mathbb{P}} \mathbf{1}(\mathbf{q} = 0) \frac{\widehat{\mathbf{R}}_{\mathbf{d}}^\circ}{\sqrt{T}}, & |\mathcal{R}_4(\beta)| &\lesssim_{\mathbb{P}} \frac{\widehat{\mathbf{R}}_{\mathbf{FS}}}{\sqrt{T}}. \end{aligned}$$

Moreover,

$$\begin{aligned} \sup_{\beta \in \mathcal{B}^\circ} |\mathcal{L}(\beta)| &\lesssim_{\mathbb{P}} \widehat{\mathbf{R}}_{\mathbf{FS}}, & \sup_{\beta \in \mathcal{B}^\circ} |\mathcal{R}_5(\beta)| &\lesssim_{\mathbb{P}} (\widehat{\mathbf{R}}_{\mathbf{FS}})^2, \\ \sup_{\beta \in \mathcal{B}^\circ} |\mathcal{B}(\beta)| &\lesssim_{\mathbb{P}} (\widehat{\mathbf{R}}_{\mathbf{d}}^\circ)^{\min\{s_\mu, \mathbf{q}+1\}}, & \sup_{\beta \in \mathcal{B}^\circ} |\mathcal{R}_3(\beta)| &\lesssim_{\mathbb{P}} \mathbf{1}(\mathbf{q} = 0) \widehat{\mathbf{R}}_{\mathbf{d}}^\circ. \end{aligned}$$

The constants implicit in $\lesssim_{\mathbb{P}}$ do not depend on the evaluation point $\beta \in \mathcal{B}^\circ$.

SA-I.3.3 Discussion

- (i) The condition $\widehat{\mathbf{R}}_{\mathbf{q}} = o_{\mathbb{P}}(J^{-d})$ is a relative Gram-convergence condition, because each reference block Q_{jt} has eigenvalues of order J^{-d} . Together with the reference design condition, it implies $\widehat{Q}_{jt} \asymp_{\mathbb{P}} J^{-d}$ uniformly over $j \in \mathcal{J}_t$ and controls the inverse perturbation $\widehat{Q}_{jt}^{-1} - Q_{jt}^{-1}$ used in the bound for $\mathcal{R}_1(\beta)$. The realized size of $\widehat{\mathbf{R}}_{\mathbf{p}}$ controls the generated-regressor basis-switching term $\mathcal{R}_2(\beta)$.
- (ii) Since the variance bounds used in the proof do not depend on β , the pointwise stochastic bounds for $\mathcal{E}(\beta)$ and $\mathcal{R}_j(\beta)$ can be read uniformly in the following sense: if $G(\beta)$ denotes one of these terms and r_n is the displayed rate, then for any deterministic sequence $\ell_n \rightarrow \infty$,

$$\limsup_{n, T \rightarrow \infty} \sup_{\beta \in \mathcal{B}^\circ} \mathbb{P}(|G(\beta)| > \ell_n r_n) = 0.$$

- (iii) The interior partition and design conditions may not hold for all bins. The index set $\mathcal{J}_t$ restricts attention to cells that can contain evaluation points in $\mathcal{B}^\circ$. Given the results in Section SA-II.3, this can be guaranteed if we focus on interior points β that are separated from the boundary of $\mathcal{B}$ by more than the cell width and the first-step smoothing scale.
- (iv) When $\mathbf{q} = 0$, the block basis is the scalar 1. In that special case, $\widehat{\mathbf{R}}_{\mathbf{q}}$ is the cell-average discrepancy and $\widehat{\mathbf{R}}_{\mathbf{p}}$ is the cell-switching discrepancy between the generated-regressor basis $\widehat{p}_t(\widehat{\beta}_{it})$ and the true-exposure reference basis $p_t(\beta_{it})$. The latter enters only through $\mathcal{R}_2(\beta)$.

- (v) For the classical marginal-quantile partition, Part II verifies the high-level quantities appearing in this lemma under primitive conditions. In particular, Section SA-II.3 establishes the partition, Gram, and basis-switching bounds, while Section SA-II.4 uses the structure of the rolling-window first step to control the generated-regressor terms. The rate substitutions are therefore postponed until the classical setup and its notation have been introduced.

SA-I.3.4 Proof of Lemma SA-I.3.1

Work on an event whose probability approaches one and on which the conditions in the lemma hold. By Weyl's inequality, for every $t = H + 1, \dots, T$ and $j \in \mathcal{J}_t$,

$$J^{-d} \lesssim \lambda_{\min}(\hat{Q}_{jt}) \leq \lambda_{\max}(\hat{Q}_{jt}) \lesssim J^{-d}.$$

Because $\mathbb{E}[f_t | \mathcal{F}_{t-1}] = \mathbb{E}[f_t | \mathcal{G}_{t-1}]$ and $\mathcal{G}_{t-1} \subseteq \mathcal{F}_{t-1}$, $\tilde{f}_t$ has mean zero conditional on $\mathcal{F}_{t-1}$. Thus any $\mathcal{F}_{t-1}$ -measurable coefficient multiplying $\tilde{f}_t$ generates a martingale-difference summand with respect to $\{\mathcal{F}_t\}$. The bound on $\mathcal{G}(\beta)$ is immediate from the martingale-difference property of $\tilde{f}_t$ and the conditional moment bound.

For $\mathcal{E}(\beta)$, let $\hat{j}_t(\beta)$ denote the cell containing β in $\hat{\Delta}_t$. Since $\beta \in \mathcal{B}^\circ$, $\hat{j}_t(\beta) \in \mathcal{J}_t$, and the paired indexing identifies the corresponding reference block. By the martingale structure and $\mathbb{E}[\varepsilon_{it}^2 | \mathcal{F}_{t-1}] \lesssim 1$, for each fixed $\beta \in \mathcal{B}^\circ$,

$$\begin{aligned} & \frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathbb{V} \text{ar} \left[\hat{p}_t(\beta)^\top Q_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} p_t(\beta_{it}) \varepsilon_{it} \middle| \mathcal{F}_{t-1} \right] \\ & \lesssim \frac{1}{T^2} \sum_{t=H+1}^T \frac{1}{n_t} \hat{p}_{\hat{j}_t(\beta)t}(\beta)^\top Q_{\hat{j}_t(\beta)t}^{-1} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} p_{\hat{j}_t(\beta)t}(\beta_{it}) p_{\hat{j}_t(\beta)t}(\beta_{it})^\top \right) Q_{\hat{j}_t(\beta)t}^{-1} \hat{p}_{\hat{j}_t(\beta)t}(\beta) \lesssim \frac{1}{T^2} \frac{J^d T}{n} = \frac{J^d}{nT}. \end{aligned}$$

The bound on $\mathcal{E}(\beta)$ follows by the Lengart–Rebolledo inequality.

For $\mathcal{L}(\beta)$ and $\mathcal{R}_5(\beta)$, bounded derivatives, bounded basis functions, and the eigenvalue bounds for $\hat{Q}_{jt}$ give

$$|\mathcal{R}_5(\beta)| \leq \frac{1}{2(T-H)} \sum_{t=H+1}^T \left[\frac{1}{n_t} \sum_{i=1}^{n_t} \left| \hat{p}_t(\beta)^\top \hat{Q}_t^{-1} \hat{p}_t(\hat{\beta}_{it}) \right| \|\nabla^2 \mu_t(\hat{\beta}_{it})\| \right] \hat{R}_{\text{FS}}^2 \lesssim \hat{R}_{\text{FS}}^2,$$

and

$$|\mathcal{L}(\beta)| \leq \frac{1}{T-H} \sum_{t=H+1}^T \left[\frac{1}{n_t} \sum_{i=1}^{n_t} \left| \hat{p}_t(\beta)^\top \hat{Q}_t^{-1} \hat{p}_t(\hat{\beta}_{it}) \right| \|\nabla \mu_t(\beta_{it})\| \right] \hat{R}_{\text{FS}} \lesssim \hat{R}_{\text{FS}}.$$

The same bounds hold uniformly over $\beta \in \mathcal{B}^\circ$.

Next, consider $\mathcal{B}(\beta)$. On each interior cell, let poly_{jt} denote a polynomial of degree at most $\mathbf{q}$ satisfying the usual approximation bound

$$\sup_{b \in \hat{P}_{jt} \cap \mathcal{B}^+} |\mu_t(b) - \text{poly}_{jt}(b)| \lesssim \text{diam}(\hat{P}_{jt})^{\min\{s_\mu, \mathbf{q}+1\}}.$$

Because the second-step basis reproduces every polynomial of degree at most $\mathbf{q}$ on the estimated cell con-

taining β ,

$$\widehat{p}_t(\beta)^\top \widehat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \widehat{p}_t(\widehat{\beta}_{it}) \text{poly}_{\widehat{J}_t(\beta)t}(\widehat{\beta}_{it}) = \text{poly}_{\widehat{J}_t(\beta)t}(\beta).$$

The boundedness of the absolute sum of the local polynomial weights then gives

$$\sup_{\beta \in \mathcal{B}^\circ} |\mathcal{R}(\beta)| \lesssim (\widehat{\mathbf{R}}_\mathbf{D}^\circ)^{\min\{s_\mu, \mathbf{q}+1\}}.$$

For $\mathcal{R}_3(\beta)$, if $\mathbf{q} \geq 1$, the global linear function $g(b) = b$ is reproduced exactly by the local polynomial basis, and therefore $\mathcal{R}_3(\beta) = 0$. If $\mathbf{q} = 0$,

$$\left\| \widehat{p}_t(\beta)^\top \widehat{Q}_t^{-1} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} \widehat{p}_t(\widehat{\beta}_{it}) \widehat{\beta}_{it}^\top \right) - \beta^\top \right\| \leq \widehat{\mathbf{R}}_\mathbf{D}^\circ,$$

because the left-hand side is the distance between β and the average of the estimated betas in the same cell as β . This gives

$$|\mathcal{R}_3(\beta)| \lesssim_{\mathbb{P}} \mathbf{1}(\mathbf{q} = 0) \frac{\widehat{\mathbf{R}}_\mathbf{D}^\circ}{\sqrt{T}} \quad \text{and} \quad \sup_{\beta \in \mathcal{B}^\circ} |\mathcal{R}_3(\beta)| \lesssim_{\mathbb{P}} \mathbf{1}(\mathbf{q} = 0) \widehat{\mathbf{R}}_\mathbf{D}^\circ$$

because the term is identically zero when $\mathbf{q} \geq 1$.

Finally, consider $\mathcal{R}_j(\beta)$, $j = 1, 2, 4$. The block-diagonal structure gives

$$\|\widehat{Q}_{jt}^{-1} - Q_{jt}^{-1}\| \lesssim J^{2d} \widehat{\mathbf{R}}_\mathbf{Q}$$

uniformly over $t = H+1, \dots, T$ and $j \in \mathcal{J}_t$. Boundedness of the estimated basis and the eigenvalue bounds for $\widehat{Q}_{jt}$ control the conditional second moment of the selected cell average. Therefore,

$$\begin{aligned} & \frac{1}{T} \max_{H+1 \leq t \leq T} \mathbb{V}\text{ar} \left[\widehat{p}_t(\beta)^\top \left(\widehat{Q}_t^{-1} - Q_t^{-1} \right) \frac{1}{n_t} \sum_{i=1}^{n_t} \widehat{p}_t(\widehat{\beta}_{it}) \varepsilon_{it} \middle| \mathcal{F}_{t-1} \right] \\ & \lesssim \frac{1}{nT J^d} (J^{2d} \widehat{\mathbf{R}}_\mathbf{Q})^2 = \frac{J^{3d} \widehat{\mathbf{R}}_\mathbf{Q}^2}{nT}. \end{aligned}$$

For $\mathcal{R}_2(\beta)$,

$$\begin{aligned} & \frac{1}{T} \max_{H+1 \leq t \leq T} \mathbb{V}\text{ar} \left[\widehat{p}_t(\beta)^\top Q_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} (\widehat{p}_t(\widehat{\beta}_{it}) - p_t(\beta_{it})) \varepsilon_{it} \middle| \mathcal{F}_{t-1} \right] \\ & \lesssim \frac{J^{2d}}{T} \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \frac{1}{n_t^2} \sum_{i=1}^{n_t} \|\widehat{p}_{jt}(\widehat{\beta}_{it}) - p_{jt}(\beta_{it})\|^2 \lesssim \frac{J^{2d} \widehat{\mathbf{R}}_\mathbf{P}}{nT}. \end{aligned}$$

For $\mathcal{R}_4(\beta)$,

$$\frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathbb{V}\text{ar} \left[\widehat{p}_t(\beta)^\top \widehat{Q}_t^{-1} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} \widehat{p}_t(\widehat{\beta}_{it}) (\beta_{it} - \widehat{\beta}_{it})^\top \right) \widetilde{f}_t \middle| \mathcal{F}_{t-1} \right] \lesssim \frac{\widehat{\mathbf{R}}_{\mathbf{FS}}^2}{T}.$$

The desired pointwise bounds follow again by the Lengart–Rebolledo inequality. $\square$

Part II: Classical Beta-Sorted Portfolios

This part specializes the general setup in Part I to the classical beta-sorted portfolio estimator, where the portfolios are constructed using local rolling-window least-squares estimates of β_{it} .

We keep the notation from Part I, including the estimator $\hat{\mu}(\beta)$, the estimated and reference partitions $\hat{\Delta}_t$ and Δ_t , and the basis maps $\hat{p}_t(\cdot)$ and $p_t(\cdot)$. The specialization has two ingredients: the first-step betas are estimated by rolling-window least-squares regressions, and the second-step portfolios are formed by constructing tensor products based on marginal empirical quantiles of the estimated betas. Throughout Part II, we take $\mathcal{B} = [\underline{\beta}, \bar{\beta}]$ to be a rectangle with $\underline{\beta} = (\underline{\beta}_1, \dots, \underline{\beta}_d)^\top$, $\bar{\beta} = (\bar{\beta}_1, \dots, \bar{\beta}_d)^\top$, and 0 in its interior.

SA-II.1 Assumptions

All notation from Part I remains in force.

We employ the following simple notion of time-series dependence. For $r \geq 1$, a sequence of random variables $\{Y_t\}_{t \in \mathbb{Z}}$ with finite r -th moments is L_r -near-epoch dependent on a sequence $\{\xi_t\}_{t \in \mathbb{Z}}$ with approximation constants a_m if, for $\mathcal{H}_{t,m} = \sigma(\xi_{t-m}, \dots, \xi_{t+m})$,

$$\|Y_t - \mathbb{E}[Y_t | \mathcal{H}_{t,m}]\|_r \leq a_m,$$

where $a_m \rightarrow 0$ as $m \rightarrow \infty$.

SA-II.1.1 Primitive Conditions

The following primitive conditions specialize the high-level conditions from Part I to the classical beta-sorted estimator. They govern the data-generating process, the predictability and cross-sectional distribution of the time-varying coefficients, and the dependence and moment conditions for the factors and idiosyncratic errors. The precise definitions of the filtrations and limit sigma-fields appearing in the assumption are collected in Section SA-II.1.2.

Assumption SA-DGP (DGP). *The following conditions hold:*

- (i) **Smoothness.** μ_t admits a bounded extension to the neighborhood $\mathcal{B}^+$ of $\mathcal{B}$, and this extension is uniformly Hölder- s_μ on $\mathcal{B}^+$ for some $s_\mu > 0$.
- (ii) **Idiosyncratic Error.** $\varepsilon_{it} = \varsigma_{it}\eta_{it}$, where (a) η_{it} is i.i.d. across i and over t with mean 0 and variance 1, is independent of $\mathcal{E}_\infty$, and has a Lebesgue density g_η with weak derivative $g_\eta^{(1)} \in L^1(\mathbb{R})$; moreover, there exists a moment exponent $q > 2(3+d)$ such that $\mathbb{E}[|\eta_{it}|^q] < \infty$; (b) ς_{it} is $\mathcal{E}_t$ -measurable, bounded, and bounded away from zero for all $i \in [n_t]$ and $t \in [T]$.
- (iii) **Varying Coefficients.**
 - (a) **Predictability and Boundedness.** The coefficients α_{it} and β_{it} are $\mathcal{E}_{t-1}$ -measurable, $\beta_{it} \in \mathcal{B}$, and $|\alpha_{it}| + \|\beta_{it}\|$ is bounded a.s. by a deterministic finite bound uniformly over i , t , and T .
 - (b) **Independence.** Conditional on $\mathcal{D}_{t-1}$, the unit-specific predictable histories

$$((\alpha_{i1}, \beta_{i1}^\top), \dots, (\alpha_{it}, \beta_{it}^\top))$$

are independent (but not necessarily identically distributed) over $i \in [n_t]$, and β_{it} has a density $f_{\beta_{it}}(b|\mathcal{D}_{t-1})$ conditional on $\mathcal{D}_{t-1}$, which is bounded, bounded away from zero, and Lipschitz continuous on $\mathcal{B}$.

(iv) **Factors.** $\mathbb{E}[f_t|\mathcal{F}_{t-1}] = \mathbb{E}[f_t|\mathcal{G}_{t-1}]$, $\lambda_{\min}(\mathbb{E}[\tilde{f}_t\tilde{f}_t^\top]) \asymp 1$ with $\tilde{f}_t = (1, f_t^\top)^\top$, for all $t \in [T]$, and $\max_{1 \leq t \leq T} \mathbb{E}[\|\tilde{f}_t\|^4|\mathcal{F}_{t-1}] \lesssim_{\mathbb{P}} 1$. There exists a deterministic positive-definite matrix $\Omega_f \in \mathbb{R}^{d \times d}$ with $0 < \lambda_{\min}(\Omega_f) \leq \lambda_{\max}(\Omega_f) < \infty$ such that

$$\left\| \frac{1}{T-H} \sum_{t=H+1}^T \text{Var}(f_t | \mathcal{G}_{t-1}) - \Omega_f \right\| = o_{\mathbb{P}}(1). \quad (\text{SA-II.1.1})$$

For the moment exponent q in part (ii), there exists a fixed $v > 0$ such that, for each coordinate $k \in [d]$, $\sup_{t \in \mathbb{Z}} \mathbb{E}[\|f_{t,k}\|^{q+v}] < \infty$ and the process $f_{t,k}$ is L_{q+v} -near-epoch dependent on a strictly stationary strongly mixing process $\{\xi_t\}$ with mixing coefficients $\alpha(m)$ and approximation constants $a_m \downarrow 0$. Assume there exist constants $C_a, C_\alpha > 0$ and $\rho_a, \rho_\alpha \in (0, 1)$ such that

$$a_m \leq C_a \rho_a^m, \quad \alpha(m) \leq C_\alpha \rho_\alpha^m, \quad m \geq 1.$$

Remark SA-II.1.1 (Primitive verification of factor variance stabilization). Condition (SA-II.1.1) holds, for example, if $\{\tilde{f}_t\}$ is strictly stationary, its coordinates are L_q -near-epoch dependent on the same strongly mixing process with the exponential rates in Assumption SA-DGP(iv), and $\Omega_f = \mathbb{E}[\tilde{f}_0\tilde{f}_0^\top]$ is positive definite. Indeed, because $q > 4$, the NED law of large numbers applies to $\tilde{f}_t\tilde{f}_t^\top$. Moreover, $\tilde{f}_t\tilde{f}_t^\top - \mathbb{E}[\tilde{f}_t\tilde{f}_t^\top | \mathcal{G}_{t-1}]$ is a matrix-valued martingale difference whose average is $O_{\mathbb{P}}((T-H)^{-1/2})$ by the fourth-moment bound and martingale orthogonality. Hence the average conditional variance converges to Ω_f whenever $T-H \rightarrow \infty$.

Assumption SA-SEP (Interior separation). Let $J = J_{nT} \rightarrow \infty$ denote the common order of the coordinatewise portfolio counts used to construct the second-step partition. Writing $b = (b_1, \dots, b_d)^\top$, the interior evaluation region $\mathcal{B}^\circ$ introduced in Section SA-I.1 satisfies

$$\inf_{b \in \mathcal{B}^\circ} \min_{l \in [d]} \min\{|b_l - \underline{\beta}_l|, |\bar{\beta}_l - b_l|\} \gg J^{-1} + \frac{\log(nT)}{\sqrt{H}}.$$

SA-II.1.2 Filtrations

The idiosyncratic-error condition in Assumption SA-DGP separates predictable model information from the i.i.d. innovations. In Part II we use four nested sigma-fields, ordered from common factor information to full information:

$$\begin{aligned} \mathcal{G}_t &= \sigma(\{f_s\}_{s=1}^t, \{\aleph_{2s}\}_{s=1}^t), \\ \mathcal{D}_t &= \sigma(\{f_s\}_{s=1}^t, \{\{\varsigma_{is}\}_{i=1}^{n_s}\}_{s=1}^t, \{\aleph_{2s}\}_{s=1}^t), \\ \mathcal{E}_t &= \sigma(\{f_s\}_{s=1}^t, \{\{\varsigma_{is}\}_{i=1}^{n_s}\}_{s=1}^t, \{\aleph_{1i}\}_{i=1}^n, \{\aleph_{2s}\}_{s=1}^t, \{\{\aleph_{3is}\}_{i=1}^{n_s}\}_{s=1}^t), \\ \mathcal{F}_t &= \sigma(\{f_s\}_{s=1}^t, \{\{\varsigma_{is}\}_{i=1}^{n_s}\}_{s=1}^t, \{\{\eta_{is}\}_{i=1}^{n_s}\}_{s=1}^t, \{\aleph_{1i}\}_{i=1}^n, \{\aleph_{2s}\}_{s=1}^t, \{\{\aleph_{3is}\}_{i=1}^{n_s}\}_{s=1}^t). \end{aligned}$$

Thus

$$\mathcal{G}_t \subseteq \mathcal{D}_t \subseteq \mathcal{E}_t \subseteq \mathcal{F}_t.$$

Here $\mathcal{G}_t$ is the common information used to form conditional factor means and variances. The design field $\mathcal{D}_t$ adds realized volatility scales and is the conditioning set for reference portfolios, conditional beta densities, and reference cell objects. It excludes the unit-specific auxiliary variables $\aleph_{1i}$ and $\aleph_{3is}$, so after conditioning on $\mathcal{D}_{t-1}$ the predictable unit histories in Assumption SA-DGP(iii)(b) remain independent across i . The environment field $\mathcal{E}_t$ adds those unit-specific auxiliary variables and therefore contains the realized predictable paths α_{is} and β_{is} , but still excludes the i.i.d. innovations. The full field $\mathcal{F}_t$ adds the realized innovations η_{is} ; under the bounded-away-from-zero condition on ς_{it} , this is equivalently the field obtained by augmenting $\mathcal{E}_t$ with the idiosyncratic errors ε_{is} .

The corresponding limit sigma-fields are

$$\mathcal{G}_\infty = \sigma(\cup_{t \geq 1} \mathcal{G}_t), \quad \mathcal{D}_\infty = \sigma(\cup_{t \geq 1} \mathcal{D}_t), \quad \mathcal{E}_\infty = \sigma(\cup_{t \geq 1} \mathcal{E}_t), \quad \mathcal{F}_\infty = \sigma(\cup_{t \geq 1} \mathcal{F}_t).$$

Remark SA-II.1.2 (Interpretation of the auxiliary variables). *The variables $\aleph_{1i}$, $\aleph_{2t}$, and $\aleph_{3it}$ are book-keeping devices for features of the data-generating process not otherwise specified. Their inclusion in the sigma-fields above does not require that they be observed by the econometrician or used to construct the estimator. The variable $\aleph_{1i}$ collects time-invariant unit-specific information, such as a fixed industry classification or a permanent firm type. The variable $\aleph_{2t}$ collects time-varying common information not already summarized by f_t , such as an aggregate volatility or liquidity state. Finally, $\aleph_{3it}$ collects time-varying unit-specific information, such as a predetermined exogenous firm characteristic (e.g., firm age) or an unobserved firm-specific state governing the evolution of its risk factor exposure. Thus, β_{it} may depend on $\aleph_{1i}$ and on the histories of $\aleph_{2s}$, $\aleph_{3is}$, and f_s through date $t-1$, with α_{it} then restricted to satisfy $\alpha_{it} = \alpha_t(\beta_{it})$.*

All of the examples remain subject to Assumption SA-DGP. In particular, Assumption SA-DGP(iii)(b) separately requires the resulting predictable coefficient histories to be independent, although not necessarily identically distributed, across assets conditional on $\mathcal{D}_{t-1}$. For example, fixed industry membership may be included in $\aleph_{1i}$, whereas a time-varying state shared by firms in an industry should be included as a component of the common vector $\aleph_{2t}$, and hence in the conditioning information. Moreover, because the auxiliary variables belong to $\mathcal{E}_\infty$, Assumption SA-DGP(ii)(a) requires the primitive idiosyncratic innovations η_{it} to be independent of them.

SA-II.1.3 Discussion

The filtration notation in this part should be read as the primitive information structure for the classical beta-sorted estimator. Although the same letters $\mathcal{F}_t$ and $\mathcal{G}_t$ are used in Part I, the high-level fields in Part I and the primitive fields in this part are local to their respective arguments. Part I uses a compact two-field notation to state abstract measurability and mean-zero conditions. This part refines that structure by separating common factor information, design information, innovation-free environment information, and full information.

This distinction matters for verifying the high-level idiosyncratic-error arguments under Assumption SA-DGP. The contemporaneous scale ς_{it} is $\mathcal{E}_t$ -measurable and need not be measurable with respect to the past full field $\mathcal{F}_{t-1}$ as defined in this part. Thus, if the high-level Part I arguments were applied literally under

the primitive assumptions in this part, the conditioning field used for period- t idiosyncratic cross-sectional arrays would have to fix the current innovation-free scales and design objects while still excluding the current primitive innovations. Equivalently, one may introduce the intermediate field

$$\mathcal{C}_t = \mathcal{E}_t \vee \sigma(\{\{\eta_{is}\}_{i=1}^{n_s}\}_{s=1}^{t-1}),$$

which contains the past full information and the current innovation-free environment, but not the current shocks $\{\eta_{it} : i \in [n_t]\}$. Conditional on $\mathcal{C}_t$, the variables $\varepsilon_{it} = \varsigma_{it}\eta_{it}$ are independent over i and have conditional mean zero. The first-step estimates, partitions, and reference weights are measurable with respect to subfields of $\mathcal{C}_t$ in the arguments where they are held fixed, and martingale or mean-zero statements relative to coarser past information follow by iterated expectations. The specialized proofs below use this conditioning logic, sometimes through proof-specific stabilized design fields, without changing the rates or substantive conclusions.

SA-II.2 First-Step Beta Estimates

Our main inference results focus on a local-polynomial least-squares estimator of β_{it} , which is commonly used in practice. When the panel is unbalanced, the notation $i \in [n_t]$ in the rolling-window formulas below refers to units observed throughout the window ending at t , so that every lagged observation $R_{i(t-s)}$ with $s \in [H]$ is available. Specifically, let $\mathfrak{p} \geq 0$, let $K(\cdot)$ be a kernel function supported on $[0, 1]$, and define

$$r_{\mathfrak{p}}(u) = (1, u, \dots, u^{\mathfrak{p}})^{\top}, \quad X_s(t) = r_{\mathfrak{p}}(s/H) \otimes (1, f_{t-s}^{\top})^{\top} \in \mathbb{R}^{(d+1)(\mathfrak{p}+1)}.$$

Let $S = (e_2^{\top}, e_3^{\top}, \dots, e_{d+1}^{\top})^{\top}$, where each $e_j \in \mathbb{R}^{(\mathfrak{p}+1)(d+1)}$ is the j -th standard basis vector. Define

$$\hat{\beta}_{it} = S\hat{\theta}_{it}, \quad \hat{\theta}_{it} = \arg \min_{\theta \in \mathbb{R}^{(d+1)(\mathfrak{p}+1)}} \sum_{s=1}^H \left(R_{i(t-s)} - X_s(t)^{\top} \theta \right)^2 K\left(\frac{s}{H}\right). \quad (\text{SA-II.2.1})$$

In this case, we can write

$$\hat{\beta}_{it} = \frac{1}{H} \sum_{s=1}^H w_s(t) R_{i(t-s)}, \quad w_s(t) = S\hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right), \quad \hat{A}(t) = \frac{1}{H} \sum_{s=1}^H K\left(\frac{s}{H}\right) X_s(t) X_s(t)^{\top}.$$

Assumption SA-FS (First step: least squares). $\hat{\beta}_{it}$ is the least-squares estimator defined in (SA-II.2.1), where $H \rightarrow \infty$ and $h = H/T = o(1)$. The kernel $K(\cdot) : [0, 1] \rightarrow \mathbb{R}_+$ is bounded and Lipschitz continuous, and $\int_0^1 r_{\mathfrak{p}}(s) r_{\mathfrak{p}}(s)^{\top} K(s) \, ds$ is positive definite.

SA-II.2.1 Decomposition

The estimation error admits the decomposition

$$\hat{\beta}_{it} - \beta_{it} = (\check{\beta}_{it} - \beta_{it}) + \frac{1}{H} \sum_{s=1}^H w_s(t) \varepsilon_{i(t-s)}, \quad (\text{SA-II.2.2})$$

where

$$\check{\beta}_{it} = \frac{1}{H} \sum_{s=1}^H w_s(t) (\alpha_{i(t-s)} + \beta_{i(t-s)}^\top f_{t-s}).$$

The term $\check{\beta}_{it} - \beta_{it}$ is the first-step bias component, while the second term is the first-step variance component. To characterize the bias component, define

$$\widehat{\mathbf{B}}_{\text{FS}} = \max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \|\check{\beta}_{it} - \beta_{it}\|.$$

Assumption SA-FSB (First step: bias). *There exists a deterministic sequence $\mathbf{B}_{\text{FS}} > 0$ such that $\mathbf{B}_{\text{FS}} = o(H^{-1/2})$ and*

$$H^{(d-1)/2} \max_{H+1 \leq t \leq T} \mathbb{P}(\widehat{\mathbf{B}}_{\text{FS}} > \mathbf{B}_{\text{FS}} \mid \mathcal{D}_{t-1}) \lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}} + H^{-1}.$$

The uniform rate is summarized by

$$\mathbf{R}_{\text{FS}} = \sqrt{\frac{\log(nT)}{H}} + \frac{(nT)^{1/q} \log(nT)}{H} + \mathbf{B}_{\text{FS}}, \quad h = \frac{H}{T}.$$

Assumption [SA-FSB](#) implies

$$\widehat{\mathbf{B}}_{\text{FS}} \leq \mathbf{B}_{\text{FS}} \quad \text{w.p.a.1.}$$

Consequently, $\mathbf{R}_{\text{FS}}$ controls the uniform first-step error. If $(nT)^{2/q} \log(nT)/H \rightarrow 0$, the middle term is dominated by $\sqrt{\log(nT)/H}$. Lemma [SA-II.2.4](#) gives primitive bounds for $\mathbf{B}_{\text{FS}}$ under several smoothness models.

Remark SA-II.2.1 (Stronger and weaker first-step bias conditions). *A stronger sufficient condition for Assumption [SA-FSB](#) is $\widehat{\mathbf{B}}_{\text{FS}} \leq \mathbf{B}_{\text{FS}}$ a.s. for all sufficiently large n, T . In addition, inspection of our proofs shows that the weaker conditions $\widehat{\mathbf{B}}_{\text{FS}} \lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}}$ and $H^{(d-1)/2} \max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \mathbb{P}(\|\check{\beta}_{it} - \beta_{it}\| > \mathbf{B}_{\text{FS}} \mid \mathcal{D}_{t-1}) \lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}} + H^{-1}$ are sufficient. We impose the stronger condition in Assumption [SA-FSB](#) for clarity of presentation.*

SA-II.2.2 Results

The first result verifies the uniform concentration of the local-polynomial Gram matrix. This regularity condition is used below to control the inverse Gram matrix uniformly over the rolling windows.

Lemma SA-II.2.2 (First-Step Gram Matrix). *Suppose that Assumptions [SA-DGP\(iv\)](#) and [SA-FS](#) hold. Then,*

$$\max_{H+1 \leq t \leq T} \|\widehat{\mathbf{A}}(t) - \mathbb{E}[\widehat{\mathbf{A}}(t)]\| \lesssim_{\mathbb{P}} \frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}}.$$

Together with Assumptions [SA-DGP\(iv\)](#) and [SA-FS](#), this implies that, if $T^{2/q}/H + \sqrt{\log T/H} = o(1)$, then $\sup_{H+1 \leq t \leq T} \|\widehat{\mathbf{A}}(t)^{-1}\| \lesssim_{\mathbb{P}} 1$. The lower bound follows from the uniform positive definiteness of $\mathbb{E}[\bar{f}_t \bar{f}_t^\top]$ and the positive definiteness of the kernel-weighted polynomial Gram matrix in Assumption [SA-FS](#).

The next result gives the uniform first-step estimation error rate R_{FS} in the least-squares case, which provides the high-level bound $\hat{R}_{\text{FS}} \lesssim_{\mathbb{P}} R_{\text{FS}}$ for Theorem SA-I.2.1.

Lemma SA-II.2.3 (First-Step Convergence Rate). *Suppose Assumptions SA-DGP(ii) and (iv) and SA-FS and SA-FSB hold, and that $T^{2/q}/H + \sqrt{\log T/H} = o(1)$. Then,*

$$\hat{R}_{\text{FS}} = \max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \|\hat{\beta}_{it} - \beta_{it}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{H}} + \frac{(nT)^{1/q} \log(nT)}{H} + B_{\text{FS}} = R_{\text{FS}}.$$

Assumption SA-DGP does not impose temporal smoothness on the realized coefficient paths. The next result gives primitive bounds on the deterministic threshold B_{FS} under several smoothness models.

Lemma SA-II.2.4 (First-Step Bias Threshold).

Suppose Assumptions SA-DGP(iv) and SA-FS hold, and

$$\frac{T^{2/(q-d-1)}}{H} = o(1).$$

- (i) **Nonrandom $\beta_i(\cdot)$: Hölder- s .** *Suppose that $\alpha_{it} = \alpha_i(t/T)$ and $\beta_{it} = \beta_i(t/T)$ for all i and t , where $\alpha_i(\cdot)$ and $\beta_i(\cdot)$ are nonrandom, bounded, and Hölder- s on $[0, 1]$ uniformly in $i \in [n]$ for some $s > 0$. If*

$$\sqrt{H} h^{\min\{s, p+1\}} \rightarrow 0,$$

then Assumption SA-FSB holds with

$$B_{\text{FS}} \lesssim h^{\min\{s, p+1\}}.$$

- (ii) **Stochastic $\beta_i(\cdot)$: Brownian Motion.** *Suppose that $\beta_{it} = \beta_i^0 + \sigma_i W_i(t/T)$ for all i and t , where $\beta_i^0 \in \mathbb{R}^d$ is nonrandom, $W_i(\cdot)$ is a d -dimensional standard Brownian motion on $[0, 1]$, independent across i , and $0 < \underline{\sigma} \leq \sigma_i \leq \bar{\sigma} < \infty$ uniformly in i . Suppose also that the conditional intercept function satisfies*

$$\alpha_t(b) = \alpha(t/T, b), \quad b \in \mathcal{B},$$

where the possibly random map $\alpha(\cdot, \cdot)$ is bounded, Hölder- s in its first argument uniformly over its second argument for some $s > 1/2$, and Lipschitz in its second argument uniformly over its first argument, a.s. with deterministic uniform bounds. If

$$\sqrt{H h \log(nT)} \rightarrow 0,$$

then Assumption SA-FSB holds with

$$B_{\text{FS}} \lesssim h^{1/2} (\log(nT))^{1/2}.$$

- (iii) **Stochastic $\beta_i(\cdot)$: Nonrandom Trend Plus Brownian Motion.** *Suppose that the conditional intercept function satisfies*

$$\alpha_t(b) = \alpha(t/T, b), \quad b \in \mathcal{B},$$

where the possibly random map $\alpha(\cdot, \cdot)$ is bounded, Hölder- s in its first argument uniformly over its second argument, and Lipschitz in its second argument uniformly over its first argument, a.s. with deterministic uniform bounds. Suppose also that

$$\beta_{it} = \bar{\beta}_i(t/T) + \sigma_i W_i(t/T),$$

where $\bar{\beta}_i(\cdot)$ is nonrandom and Hölder- s as in case (i) with $s \in (0, \mathfrak{p} + 1]$, $W_i(\cdot)$ is a d -dimensional standard Brownian motion on $[0, 1]$, independent across i , and the scale condition in case (ii) holds. If

$$\sqrt{H} \max \left\{ h^s, h^{1/2} (\log(nT))^{1/2} \right\} \rightarrow 0,$$

then Assumption [SA-FSB](#) holds with

$$\mathbf{B}_{\text{FS}} \lesssim \max \left\{ h^s, h^{1/2} (\log(nT))^{1/2} \right\}.$$

(iv) **Stochastic $\beta_i(\cdot)$: Integrated Brownian Motion.** Suppose that

$$\beta_{it} = \beta_i^0 + \sigma_i \int_0^{t/T} W_i(v) \, dv,$$

where $\beta_i^0 \in \mathbb{R}^d$ is nonrandom, $W_i(\cdot)$ is a d -dimensional standard Brownian motion on $[0, 1]$, independent across i , and the scale condition in case (ii) holds. Suppose also that

$$\alpha_t(b) = \alpha(t/T, b), \quad b \in \mathcal{B},$$

where the possibly random map $\alpha(\cdot, \cdot)$ is twice continuously differentiable jointly in its arguments, a.s., and the function and all its derivatives through order two are uniformly bounded by deterministic constants. If $h \log(nT) = o(1)$ and

$$\sqrt{H} h^{\min\{\mathfrak{p}+1, 3/2\}} (\log(nT))^{1/2} \rightarrow 0,$$

then Assumption [SA-FSB](#) holds with

$$\mathbf{B}_{\text{FS}} \lesssim h^{\min\{\mathfrak{p}+1, 3/2\}} (\log(nT))^{1/2}.$$

SA-II.2.3 Discussion

(i) *First-step rate under smooth trends.* Lemma [SA-II.2.4](#) translates the path restrictions stated directly in that lemma into primitive bounds on the deterministic threshold $\mathbf{B}_{\text{FS}}$. If $(nT)^{2/q} \log(nT)/H \rightarrow 0$, then the term $(nT)^{1/q} \log(nT)/H$ in $\mathbf{R}_{\text{FS}}$ is dominated by $\sqrt{\log(nT)/H}$, and Lemma [SA-II.2.3](#) gives

$$\mathbf{R}_{\text{FS}} \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{H}} + \mathbf{B}_{\text{FS}}.$$

In the nonrandom Hölder- s case, writing $h = H/T$, Lemma [SA-II.2.4](#)(i) gives $\mathbf{B}_{\text{FS}} \lesssim h^{\min\{s, \mathfrak{p}+1\}}$. Thus the local-polynomial order reduces the first-step bias only up to the smoothness ceiling s .

In the integrated-Brownian case (iv), the analogous distinction is

$$\mathbf{B}_{\text{FS}} \lesssim \begin{cases} h(\log(nT))^{1/2}, & \mathbf{p} = 0, \\ h^{3/2}(\log(nT))^{1/2}, & \mathbf{p} \geq 1. \end{cases}$$

A degree-one fit absorbs the current random slope $\sigma_i W_i(t/T)$ of the integrated path. Higher polynomial orders do not further improve this component because that slope itself has Brownian local variation. In particular, the displayed primitive $\mathbf{B}_{\text{FS}}$ bound is $o(H^{-1/2})$ whenever

$$\sqrt{H} h^{\min\{\mathbf{p}+1, 3/2\}} (\log(nT))^{1/2} \rightarrow 0.$$

The restriction $T^{2/(q-d-1)}/H = o(1)$ in Lemma SA-II.2.4 is stronger than the concentration rate $T^{2/q}/H = o(1)$ used for the first-step Gram matrix because the lemma verifies the conditional-tail requirement in Assumption SA-FSB through the stronger $H^{(d+1)/2}$ -weighted global-tail bound. It is implied by $T^{(3+d)/q}/H = o(1)$, the rate used in the partition results, because $q > 2(3+d)$ implies $2/(q-d-1) < (3+d)/q$.

- (ii) *Conditional intercepts in the Brownian cases.* The conditions in cases (ii)–(iii) are imposed on the conditional intercept function $\alpha_t(b) = \mu_t(b) - b^\top \mathbb{E}[f_t \mid \mathcal{G}_{t-1}]$, rather than on a separate nonrandom path for α_{it} . Because $\alpha_{it} = \alpha_t(\beta_{it})$, temporal smoothness of $\alpha_t(b)$ at fixed b controls variation in the conditional intercept function, while Lipschitz continuity in b transfers the local modulus of β_{it} to α_{it} . In case (iv), the stronger joint twice-continuous differentiability condition permits a first-order expansion of $\tau \mapsto \alpha(\tau, \beta_i(\tau))$. The local polynomial can therefore absorb both its time derivative and the contribution of the current random beta slope, leaving a remainder of integrated-Brownian order.
- (iii) *Compact support in the Brownian cases.* Cases (ii)–(iii) use the Brownian component only through its local modulus of continuity. The proof requires

$$\max_{i \in [n]} \sup_{|u-v| \leq h} \|W_i(u) - W_i(v)\| \lesssim_{\mathbb{P}} h^{1/2} (\log(nT))^{1/2}$$

and does not otherwise use the global level of $W_i(\cdot)$. Since Assumption SA-DGP(iii)(a) imposes a fixed compact support $\mathcal{B}$ on β_{it} , the Brownian specifications should be read as a convenient shorthand for any compact-state process with this same local modulus, such as reflected Brownian motion on $\mathcal{B}$ or a diffusion on a compact state space. For such a compact-state process, the threshold bound $\mathbf{B}_{\text{FS}} \lesssim h^{1/2}(\log(nT))^{1/2}$ is unchanged, provided the remaining requirements of Assumption SA-DGP(iii) are maintained.

The same interpretation applies to case (iv). There the proof uses

$$\max_{i \in [n]} \sup_{|u-v| \leq h} \left\| \int_v^u (W_i(z) - W_i(v)) \, dz \right\| \lesssim_{\mathbb{P}} h^{3/2} (\log(nT))^{1/2}.$$

Thus the displayed integrated-Brownian specification may be replaced by a compact-state process with continuously differentiable paths whose derivative has the same Brownian local modulus; the rates in case (iv) are unchanged.

SA-II.2.4 Proof of Lemma SA-II.2.2

Recall $X_s(t) = r_p(s/H) \otimes (1, f_{t-s}^\top)^\top$ and $K_{\max} = \sup_{u \in [0,1]} |K(u)| < \infty$.

Define

$$V_{s,t} = \frac{1}{H} X_s(t) X_s(t)^\top K\left(\frac{s}{H}\right),$$

so that $\hat{A}(t) = \sum_{s=1}^H V_{s,t}$.

Put $r_* = (q+v)/2 > q/2 > 2$.

Step 1: NED property of $V_{s,t}$. Since $X_s(t) = r_p(s/H) \otimes (1, f_{t-s}^\top)^\top$ is a function of f_{t-s} , Assumption SA-DGP(iv) implies that $X_s(t) X_s(t)^\top$ is L_{r_*} -NED on $\{\xi_t\}$ with geometric approximation constants. To see this, let $\mathcal{H}_{t-s,k} = \sigma(\xi_{t-s-k}, \dots, \xi_{t-s+k})$, $V_{s,t}^{(k)} = \mathbb{E}[V_{s,t} \mid \mathcal{H}_{t-s,k}]$, and $X_s^{(k)}(t) = \mathbb{E}[X_s(t) \mid \mathcal{H}_{t-s,k}]$. The contraction property of conditional expectation and the product inequality give

$$\|V_{s,t} - V_{s,t}^{(k)}\|_{r_*} \leq \frac{2K_{\max}}{H} (\|X_s(t)\|_{q+v} + \|X_s^{(k)}(t)\|_{q+v}) \|X_s(t) - X_s^{(k)}(t)\|_{q+v} \lesssim H^{-1} \rho_a^k.$$

Step 2: Moment bound. Uniformly in s and t , $\mathbb{E}[\|V_{s,t}\|^{r_*}] \lesssim H^{-r_*} \mathbb{E}[\|X_s(t)\|^{q+v}] \lesssim H^{-r_*}$.

Step 3: Rolling-sum bound. Let Y_u be any centered scalar entry of $(1, f_u^\top)^\top (1, f_u^\top)$. It has a uniformly bounded r_* -th moment and is geometrically L_{r_*} -NED. Set $m = C_0 \log T$ and $Y_u^{(m)} = \mathbb{E}[Y_u \mid \mathcal{H}_{u,m}]$. Because Y_u is centered, so is $Y_u^{(m)}$, and conditional-expectation contraction gives $\|Y_u^{(m)}\|_{r_*} \leq \|Y_u\|_{r_*} \lesssim 1$. Moreover, $\{Y_u^{(m)}\}$ is strongly mixing: observations separated by more than $2m$ have mixing coefficient at most $\alpha(j-2m)$. Therefore, at probability level v , its inverse mixing coefficient is at most $2m + C \log(e/v)$, while Markov's inequality bounds the quantile of $|Y_u^{(m)}|$ by Cv^{-1/r_*} . For sufficiently large x , these two bounds give

$$\int_{\{0 < v < 1: C\{m + \log(e/v)\}v^{-1/r_*} > x\}} C v^{-1/r_*} dv \lesssim x^{-(r_*-1)} \{m + \log(e+x)\}^{r_*-1}.$$

This verifies the tail-integral condition in Rio (2017, Theorem 6.2).

For two entries Y_u and Y_{u+j} , approximate each one using $\mathcal{H}_{u, \lfloor j/3 \rfloor}$ and $\mathcal{H}_{u+j, \lfloor j/3 \rfloor}$. The covariance inequality in Rio (2017, Theorem 1.1(a)) and the geometric NED bound give, for some $\rho \in (0, 1)$,

$$|\text{Cov}(Y_u, Y_{u+j})| \lesssim \rho_a^{\lfloor j/3 \rfloor} + \alpha(\lfloor j/3 \rfloor)^{1-2/r_*} \lesssim \rho^j.$$

Thus the sum of the absolute covariances is uniformly bounded. Since $\|Y_u - Y_u^{(m)}\|_2 \lesssim \rho_a^m$, on every interval of length at most $2H$ the sum of the absolute covariances of $Y_u^{(m)}$ is at most $CH + CH^2 \rho_a^m \lesssim CH$ when C_0 is sufficiently large.

The variables $Y_u^{(m)}$ have now been verified to be centered and strongly mixing, with the moment, tail-integral, and covariance bounds required by Rio (2017, Theorem 6.2). Apply that theorem on consecutive blocks of length H . An interval crossing a block boundary splits into at most two partial sums, and there are $O(T/H)$ blocks. Also, for any fixed $A > 0$, C_0 can be chosen so that, uniformly for $\lambda \geq 1$,

$$\mathbb{P}\left(\sum_{u=1}^T |Y_u - Y_u^{(m)}| > \lambda\right) \leq \lambda^{-1} \sum_{u=1}^T \mathbb{E}|Y_u - Y_u^{(m)}| \leq \frac{CT\rho_a^m}{\lambda} = o(T^{-A}).$$

Choose C_0 large enough to absorb this approximation error. Consequently, for any $a \geq 1$ and $1 \leq \lambda \leq T$

with λ/a sufficiently large, we have the following bound, where the maximum is over time intervals I :

$$\mathbb{P}\left(\max_{|I|\leq H}\left|\sum_{u\in I}Y_u\right|>C\lambda\right)\lesssim\frac{T}{H}\left(1+\frac{\lambda^2}{aH}\right)^{-a/2}+Ta^{r_*-1}\lambda^{-r_*}(\log T)^{r_*-1}. \quad (\text{SA-II.2.3})$$

Indeed, the tail-integral term on one block is the preceding integral multiplied by H/λ and evaluated at $x = \lambda/a$.

For each matrix entry, the deterministic weights formed from $r_{\mathbf{p}}(s/H)$ and $K(s/H)$ are uniformly bounded and have bounded total change over $s = 1, \dots, H$. Expanding a weighted sum into partial sums and successive changes in these weights therefore bounds it by a constant times the largest unweighted partial sum in (SA-II.2.3). Take $a = M \log T$ and $\lambda = M\{T^{2/q} + \sqrt{H \log T}\}$. The first term in (SA-II.2.3) tends to zero for sufficiently large M . For the second term, the moment gap gives

$$T\lambda^{-r_*}(\log T)^{2(r_*-1)} \lesssim T^{1-2r_*/q}(\log T)^{2(r_*-1)} = T^{-v/q}(\log T)^{2(r_*-1)} = o(1).$$

A finite union over the matrix entries now yields

$$\max_{H+1\leq t\leq T}\left\|\widehat{A}(t)-\mathbb{E}[\widehat{A}(t)]\right\|\lesssim_{\mathbb{P}}\frac{T^{2/q}}{H}+\sqrt{\frac{\log T}{H}}.$$

The covariance bound above also gives $(T-H)^{-1}\sum_{t=H+1}^T\|\widehat{A}(t)-\mathbb{E}[\widehat{A}(t)]\|^2 = O_{\mathbb{P}}(H^{-1})$ by taking expectations and applying Markov's inequality. The same rolling-sum argument applies to the fixed powers of $\|f_t\|$ used below because their available moment is strictly above the working moment used in the corresponding rate. $\square$

SA-II.2.5 Proof of Lemma SA-II.2.3

Define

$$N_i(t) = \frac{1}{H}\sum_{s=1}^HX_s(t)R_{i(t-s)}K\left(\frac{s}{H}\right), \quad \text{and} \quad U_{it} = \frac{1}{H}\sum_{s=1}^HX_s(t)\varepsilon_{i(t-s)}K\left(\frac{s}{H}\right).$$

The normal equations give $\widehat{\theta}_{it} = \widehat{A}(t)^{-1}N_i(t)$. Since

$$N_i(t) = \frac{1}{H}\sum_{s=1}^HX_s(t)(\alpha_{i(t-s)} + \beta_{i(t-s)}^\top f_{t-s})K\left(\frac{s}{H}\right) + U_{it},$$

the definition of $\check{\beta}_{it}$ implies the exact identity

$$\widehat{\beta}_{it} - \check{\beta}_{it} = S\widehat{A}(t)^{-1}U_{it}.$$

Step 1: Bounding $\widehat{A}(t)^{-1}$. By Assumptions SA-DGP(iv) and SA-FS, $\lambda_{\min}(\mathbb{E}[\widehat{A}(t)]) \geq c > 0$ uniformly in $t = H+1, \dots, T$. Indeed, writing $\bar{f}_r = (1, f_r^\top)^\top$, any vector a conformable with $X_s(t)$ can be written as polynomial coefficients $a_0, \dots, a_{\mathbf{p}} \in \mathbb{R}^{d+1}$, so that $(X_s(t)^\top a)^2 = (\bar{f}_{t-s}^\top a(s/H))^2$, where $a(u) = \sum_{k=0}^{\mathbf{p}} a_k u^k$. The uniform lower bound on $\mathbb{E}[\bar{f}_r \bar{f}_r^\top]$ gives a constant multiple of $H^{-1}\sum_{s=1}^HK(s/H)\|a(s/H)\|^2$, and this discrete polynomial Gram matrix is bounded away from zero for all large H by the positive definite integral

in Assumption SA-FS. By Lemma SA-II.2.2, $\|\hat{A}(t) - \mathbb{E}[\hat{A}(t)]\| \lesssim_{\mathbb{P}} T^{2/q}/H + \sqrt{\log T/H} = o_{\mathbb{P}}(1)$ uniformly in t . Weyl's inequality then gives

$$\lambda_{\min}(\hat{A}(t)) \geq \lambda_{\min}(\mathbb{E}[\hat{A}(t)]) - \|\hat{A}(t) - \mathbb{E}[\hat{A}(t)]\| \geq c - o_{\mathbb{P}}(1) > 0$$

uniformly over $t \in [H+1, T]$ with probability approaching one. Hence

$$\sup_{H+1 \leq t \leq T} \|\hat{A}(t)^{-1}\| \lesssim_{\mathbb{P}} 1. \quad (\text{SA-II.2.4})$$

Step 2: Bounding U_{it} . Condition on $\mathcal{E}_{\infty}$. Then the factor history and volatility scales are fixed, and $\varepsilon_{i(t-s)} = \varsigma_{i(t-s)} \eta_{i(t-s)}$ with independent mean-zero innovations. Hence, for fixed (i, t_0) , the array $\{X_s(t_0)\varepsilon_{i(t_0-s)} : s \in [H]\}$ is a martingale difference array in the index s with respect to the filtration $\mathcal{H}_{s, it_0} = \mathcal{E}_{\infty} \vee \sigma(\eta_{i(t_0-r)} : 1 \leq r \leq s)$. The probability bounds below can therefore be read conditionally on $\mathcal{E}_{\infty}$ and then integrated. Abel's summation formula gives, for any $t_0 \in [H+1, T]$,

$$\begin{aligned} & \left\| \sum_{s=1}^H K\left(\frac{s}{H}\right) X_s(t_0) \varepsilon_{i(t_0-s)} \right\| \\ & \leq \max_{\ell \in [H]} \left\| \sum_{s=1}^{\ell} X_s(t_0) \varepsilon_{i(t_0-s)} \right\| \left\| \sum_{s=1}^{H-1} \left| K\left(\frac{s+1}{H}\right) - K\left(\frac{s}{H}\right) \right| \right\| + \left\| \sum_{s=1}^H X_s(t_0) \varepsilon_{i(t_0-s)} \right\| K(1). \end{aligned}$$

Since K is Lipschitz continuous, $\sum_{s=1}^{H-1} |K(\frac{s+1}{H}) - K(\frac{s}{H})| \leq L_K$ for a finite constant L_K , and $K(1) \leq \|K\|_{\infty} < \infty$. Therefore,

$$\max_{H+1 \leq t_0 \leq T} \max_{i \in [n_{t_0}]} \|U_{it_0}\| \lesssim \frac{1}{H} \max_{H+1 \leq t_0 \leq T} \max_{i \in [n_{t_0}]} \max_{\ell \in [H]} \left\| \sum_{s=1}^{\ell} X_s(t_0) \varepsilon_{i(t_0-s)} \right\|.$$

Fix t_0 and i , put $\tau = (nT)^{1/q}$, and write

$$Y_{s, it_0} = X_s(t_0) \varepsilon_{i(t_0-s)}, \quad s \in [H].$$

For a fixed constant $M > 0$, define the conditionally recentered truncated increment

$$Y_{s, it_0}^{(M)} = Y_{s, it_0} \mathbf{1}(\|Y_{s, it_0}\| \leq M\tau) - \mathbb{E}[Y_{s, it_0} \mathbf{1}(\|Y_{s, it_0}\| \leq M\tau) \mid \mathcal{H}_{s-1, it_0}].$$

By construction,

$$\mathbb{E}[Y_{s, it_0}^{(M)} \mid \mathcal{H}_{s-1, it_0}] = 0, \quad \|Y_{s, it_0}^{(M)}\| \leq 2M\tau.$$

Thus the truncated array remains a martingale-difference array with respect to the original filtration; no conditioning on a future-shock event is involved.

Its predictable quadratic variation satisfies, conditional on $\mathcal{E}_{\infty}$,

$$\sum_{s=1}^H \mathbb{E}[\|Y_{s, it_0}^{(M)}\|^2 \mid \mathcal{H}_{s-1, it_0}] \lesssim \sum_{s=1}^H \|X_s(t_0)\|^2,$$

because the volatility scales are bounded and $\mathbb{E}[\eta_{it}^2] = 1$. Moreover,

$$\max_{H+1 \leq t \leq T} \frac{1}{H} \sum_{s=1}^H \|X_s(t)\|^2 \lesssim_{\mathbb{P}} 1,$$

by applying the same maximal Fuk–Nagaev inequality used in Lemma SA-II.2.2 to the $L_{q/2}$ -NED sequence $\|\tilde{f}_t\|^2$. Hence, on an $\mathcal{E}_\infty$ -measurable event whose probability can be made arbitrarily close to one, the coordinatewise maximal martingale Bernstein inequality (Freedman, 1975) gives

$$\mathbb{P}\left(\max_{\ell \in [H]} \left\| \sum_{s=1}^{\ell} Y_{s,it_0}^{(M)} \right\| > x \mid \mathcal{E}_\infty\right) \lesssim \exp\left\{-\frac{cx^2}{H + M\tau x}\right\}. \quad (\text{SA-II.2.5})$$

The dimension of Y_{s,it_0} is fixed, so the coordinatewise inequality and a finite union bound imply the displayed norm bound.

It remains to transfer this inequality to the original increments. Let

$$\mathcal{A}_M = \left\{ \max_{H+1 \leq t_0 \leq T} \max_{i \in [n_{t_0}]} \max_{s \in [H]} \|Y_{s,it_0}\| \leq M\tau \right\}.$$

Since $r_p(\cdot)$ is bounded, the maximum over rolling-window summands is controlled by the maximum over the nT calendar shock pairs. Markov's inequality, the finite q -th moments of $(1 + \|f_r\|)\varepsilon_{ir}$, and a union bound give

$$\mathbb{P}(\mathcal{A}_M^c) \lesssim M^{-q}. \quad (\text{SA-II.2.6})$$

Because $\mathbb{E}[Y_{s,it_0} \mid \mathcal{H}_{s-1,it_0}] = 0$, on $\mathcal{A}_M$,

$$\sum_{s=1}^{\ell} Y_{s,it_0} = \sum_{s=1}^{\ell} Y_{s,it_0}^{(M)} - \sum_{s=1}^{\ell} \mathbb{E}[Y_{s,it_0} \mathbf{1}(\|Y_{s,it_0}\| > M\tau) \mid \mathcal{H}_{s-1,it_0}], \quad \ell \in [H].$$

The recentering term is negligible at the target scale. Indeed, uniformly over (i, t_0) , conditional Markov's inequality and the bounded volatility scales give

$$\begin{aligned} \sum_{s=1}^H \|\mathbb{E}[Y_{s,it_0} \mathbf{1}(\|Y_{s,it_0}\| > M\tau) \mid \mathcal{H}_{s-1,it_0}]\| &\leq (M\tau)^{1-q} \sum_{s=1}^H \mathbb{E}[\|Y_{s,it_0}\|^q \mid \mathcal{E}_\infty] \\ &\lesssim_{\mathbb{P}} (M\tau)^{1-q} \sum_{r=1}^T (1 + \|f_r\|)^q \lesssim_{\mathbb{P}} M^{1-q} \frac{\tau}{n} \lesssim_{\mathbb{P}} M^{1-q} \tau, \end{aligned}$$

where $\tau^q = nT$ and $T^{-1} \sum_{r=1}^T (1 + \|f_r\|)^q = O_{\mathbb{P}}(1)$ by the uniform q -moment bound in Assumption SA-DGP(iv).

For fixed M , taking x to be a sufficiently large multiple of $\sqrt{H \log(nT)} + \tau \log(nT)$ in (SA-II.2.5) and applying a union bound over at most nT pairs (i, t_0) gives a probability tending to zero. The recentering term is $o_{\mathbb{P}}(\sqrt{H \log(nT)} + \tau \log(nT))$, while (SA-II.2.6) can be made arbitrarily small by choosing M large.

Therefore,

$$\max_{H+1 \leq t_0 \leq T} \max_{i \in [n_{t_0}]} \max_{\ell \in [H]} \left\| \sum_{s=1}^{\ell} Y_{s, it_0} \right\| \lesssim_{\mathbb{P}} \sqrt{H \log(nT)} + (nT)^{1/q} \log(nT).$$

Dividing by H in the definition of U_{it} yields

$$\max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \|U_{it}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{H}} + \frac{(nT)^{1/q} \log(nT)}{H}.$$

Step 3: Combining with the realized bias. By the exact identity above, (SA-II.2.4), and the preceding rate bound,

$$\max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \|\hat{\beta}_{it} - \check{\beta}_{it}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{H}} + \frac{(nT)^{1/q} \log(nT)}{H}.$$

The triangle inequality and Assumption SA-FSB, which gives $\hat{\mathbf{B}}_{\text{FS}} \leq \mathbf{B}_{\text{FS}}$ w.p.a.1, yield

$$\max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \|\hat{\beta}_{it} - \beta_{it}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{H}} + \frac{(nT)^{1/q} \log(nT)}{H} + \mathbf{B}_{\text{FS}}.$$

□

SA-II.2.6 Proof of Lemma SA-II.2.4

We first give an argument common to all four cases. Apply (SA-II.2.3) with the fixed threshold $\lambda = cH$ and $a = q + 2$, both to the rolling Gram matrices and to the rolling averages of $(1 + \|f_t\|)^2$. Because $T^{2/(q-d-1)}/H = o(1)$, H grows at a positive power of T and hence $H^{-v/2}(\log T)^{(q+v)/2-1} = o(1)$. The two terms in (SA-II.2.3) are therefore bounded by $CTH^{-q/2}$. It follows that, for a sufficiently large C ,

$$\begin{aligned} & \mathbb{P} \left(\max_{H+1 \leq t \leq T} \|\hat{A}(t)^{-1}\| > C \right) \\ & + \mathbb{P} \left(\max_{H+1 \leq t \leq T} \frac{1}{H} \sum_{s=1}^H (1 + \|f_{t-s}\|)^2 > C \right) \\ & \lesssim TH^{-q/2} = o(H^{-(d+1)/2}). \end{aligned} \tag{SA-II.2.7}$$

Indeed,

$$H^{(d+1)/2} TH^{-q/2} = \left(\frac{T^{2/(q-d-1)}}{H} \right)^{(q-d-1)/2} \rightarrow 0.$$

The inverse bound follows from the uniform lower eigenvalue bound for $\mathbb{E}[\hat{A}(t)]$ and the same fixed-threshold inequality.

Now let $a_h > 0$ be deterministic. Suppose that, outside an event with probability $o(H^{-(d+1)/2})$, for every (i, t) there exists a coefficient vector $\gamma_{it} = (\gamma_{it,0}^\top, \dots, \gamma_{it,p}^\top)^\top$ satisfying $S\gamma_{it} = \beta_{it}$ and, uniformly over $s \in [H]$,

$$\alpha_{i(t-s)} + \beta_{i(t-s)}^\top f_{t-s} = X_s(t)^\top \gamma_{it} + e_{its}, \quad |e_{its}| \leq Ca_h(1 + \|f_{t-s}\|). \tag{SA-II.2.8}$$

Using the definition of $\check{\beta}_{it}$ and the normal equations,

$$\check{\beta}_{it} - \beta_{it} = S\hat{A}(t)^{-1} \frac{1}{H} \sum_{s=1}^H X_s(t) e_{its} K\left(\frac{s}{H}\right).$$

Consequently, (SA-II.2.7) and boundedness of the kernel imply, for a sufficiently large fixed constant C ,

$$\mathbb{P}\left(\hat{\mathbf{B}}_{\text{FS}} > Ca_h\right) = o(H^{-(d+1)/2}). \quad (\text{SA-II.2.9})$$

Set $\mathbf{B}_{\text{FS}} = Ca_h$. If $\sqrt{H} a_h \rightarrow 0$, then $\mathbf{B}_{\text{FS}} = o(H^{-1/2})$. Moreover, because

$$\mathbb{E}\left[\mathbf{1}\left(\hat{\mathbf{B}}_{\text{FS}} > \mathbf{B}_{\text{FS}}\right) \middle| \mathcal{D}_{t-1}\right], \quad t = H+1, \dots, T,$$

is a nonnegative martingale sampled along the nested design filtration, Doob's maximal inequality and (SA-II.2.9) give, for every $\epsilon > 0$,

$$\mathbb{P}\left(H^{(d+1)/2} \max_{H+1 \leq t \leq T} \mathbb{P}\left(\hat{\mathbf{B}}_{\text{FS}} > \mathbf{B}_{\text{FS}} \middle| \mathcal{D}_{t-1}\right) > \epsilon\right) \leq \frac{H^{(d+1)/2}}{\epsilon} \mathbb{P}\left(\hat{\mathbf{B}}_{\text{FS}} > \mathbf{B}_{\text{FS}}\right) \rightarrow 0.$$

Consequently,

$$\frac{H^{(d-1)/2}}{\mathbf{B}_{\text{FS}} + H^{-1}} \max_{H+1 \leq t \leq T} \mathbb{P}\left(\hat{\mathbf{B}}_{\text{FS}} > \mathbf{B}_{\text{FS}} \middle| \mathcal{D}_{t-1}\right) \leq H^{(d+1)/2} \max_{H+1 \leq t \leq T} \mathbb{P}\left(\hat{\mathbf{B}}_{\text{FS}} > \mathbf{B}_{\text{FS}} \middle| \mathcal{D}_{t-1}\right) = o_{\mathbb{P}}(1),$$

which verifies Assumption SA-FSB. Thus it remains only to verify (SA-II.2.8) with the stated a_h and tail probability.

Case (i). The standard Taylor–Hölder expansion of $\alpha_i(\cdot)$ and each coordinate of $\beta_i(\cdot)$ around t/T gives a local polynomial in s/H of degree at most $\mathfrak{p}$, with a deterministic remainder bounded by $a_h = h^{\min\{s, \mathfrak{p}+1\}}$ uniformly over i , t , and $s \in [H]$. The common argument therefore proves part (i).

Case (ii). For every fixed $M > 0$, the Gaussian maximal inequality and the Brownian modulus of continuity give a fixed constant C_M such that

$$\mathbb{P}\left(\max_{i \in [n]} \sup_{|u-v| \leq h} \|W_i(u) - W_i(v)\| > C_M h^{1/2} (\log(nT))^{1/2}\right) \lesssim (nT)^{-M}. \quad (\text{SA-II.2.10})$$

Take $M > (d+1)/2$. Put the current Brownian level in the constant beta coefficient, so that $S\gamma_{it} = \beta_{it}$. For the intercept component, use the degree- $\mathfrak{p}$ local Taylor–Hölder polynomial for $\tau \mapsto \alpha(\tau, \beta_{it})$ around t/T . Its temporal remainder is $O(h^{\min\{s, \mathfrak{p}+1\}})$. Lipschitz continuity in the beta argument adds at most a constant multiple of the Brownian increment in (SA-II.2.10). Since $\min\{s, \mathfrak{p}+1\} > 1/2$, (SA-II.2.8) holds outside an $o(H^{-(d+1)/2})$ event with $a_h = h^{1/2} (\log(nT))^{1/2}$. The common argument proves part (ii).

Case (iii). Choose the local polynomial coefficients for the nonrandom beta trend as in case (i), and put the current Brownian level in the constant beta coefficient. The temporal intercept remainder is bounded by h^s , while the Lipschitz contribution from the beta argument is bounded by the beta increment. Its nonrandom component is of order $h^{\min\{s, 1\}}$, which is bounded by the claimed rate, and its Brownian component satisfies (SA-II.2.10). Therefore (SA-II.2.8) holds outside an $o(H^{-(d+1)/2})$ event with $a_h = \max\{h^s, h^{1/2} (\log(nT))^{1/2}\}$. The common argument proves part (iii). The Brownian component is

nowhere differentiable, so increasing $\mathbf{p}$ does not improve this component of the threshold.

Case (iv). Write $I_i(u) = \int_0^u W_i(v) dv$. For every fixed $M > 0$, the Brownian maximal and modulus inequalities give a fixed constant C_M such that the probability that either

$$\begin{aligned} \max_{i \in [n]} \sup_{u \in [0,1]} \|W_i(u)\| &> C_M (\log(nT))^{1/2}, \\ \max_{i \in [n]} \sup_{|u-v| \leq h} \|I_i(u) - I_i(v) - W_i(v)(u-v)\| &> C_M h^{3/2} (\log(nT))^{1/2} \end{aligned}$$

is bounded by a constant multiple of $(nT)^{-M}$. Take $M > (d+1)/2$.

If $\mathbf{p} = 0$, use the current levels of the beta and conditional intercept paths as the constant local-polynomial coefficients. Boundedness of the first derivatives of α gives (SA-II.2.8) with $a_h = h(\log(nT))^{1/2}$. If $\mathbf{p} \geq 1$, include the current slope $\sigma_i W_i(t/T)$ in the degree-one beta coefficient. For $g_i(u) = \alpha(u, \beta_i(u))$, also include

$$g'_i(u) = \partial_1 \alpha(u, \beta_i(u)) + \nabla_b \alpha(u, \beta_i(u))^\top \sigma_i W_i(u)$$

at $u = t/T$ in the degree-one intercept coefficient. The joint second-derivative bound on α , the Brownian bounds above, and $h \log(nT) = o(1)$ imply (SA-II.2.8) with $a_h = h^{3/2}(\log(nT))^{1/2}$. Combining the two polynomial orders gives the threshold in part (iv), and the common argument completes the proof. $\square$

SA-II.3 Second-Step Partitioning

Using the generic partition (portfolios) notation from Section SA-I.3 and the concrete design filtration $\{\mathcal{D}_t\}$ from Section SA-II.1.2, we now specialize the estimated partition (portfolio) to the marginal-quantile scheme used by the classical beta-sorted estimator. Recall that $\mathcal{B} = [\underline{\beta}, \bar{\beta}]$ is a rectangle. The partition is formed by tensor products across coordinates:

$$\hat{\Delta}_t = \otimes_{l \in [d]} \hat{\Delta}_{t,l}, \quad \hat{\Delta}_{t,l} = \left\{ [\hat{\tau}_{0t,l}, \hat{\tau}_{1t,l}), [\hat{\tau}_{1t,l}, \hat{\tau}_{2t,l}), \dots, [\hat{\tau}_{(J_{t,l}-1)t,l}, \hat{\tau}_{J_{t,l}t,l}] \right\}.$$

Here $\hat{\tau}_{jt,l}$ denotes a possibly random breakpoint and $J_{t,l}$ denotes the number of intervals for coordinate $l \in [d]$, so that $\bar{J}_t = \prod_{l \in [d]} J_{t,l}$.

For any first-step estimator $\hat{\beta}_{it} = (\hat{\beta}_{it,1}, \dots, \hat{\beta}_{it,d})^\top$, coordinate $l \in [d]$, and time $t = H+1, \dots, T$, define

$$\hat{F}_{\hat{\beta},t,l}(z) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}(\hat{\beta}_{it,l} \leq z) \quad \text{and} \quad \bar{F}_{\hat{\beta},t,l}(z) = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{P}(\hat{\beta}_{it,l} \leq z \mid \mathcal{D}_{t-1}).$$

Their generalized inverse functions are

$$\hat{F}_{\hat{\beta},t,l}^{-1}(\kappa) = \inf\{z \in \mathbb{R} : \hat{F}_{\hat{\beta},t,l}(z) \geq \kappa\} \quad \text{and} \quad \bar{F}_{\hat{\beta},t,l}^{-1}(\kappa) = \inf\{z \in \mathbb{R} : \bar{F}_{\hat{\beta},t,l}(z) \geq \kappa\},$$

for $\kappa \in (0, 1)$.

Our main inference results focus on partitions formed by tensor products of marginal empirical quantiles of the components of $\hat{\beta}_{it}$, separately on the negative and nonnegative sides of zero. For each $l \in [d]$ and

$t = H + 1, \dots, T$, define the split sample sizes

$$\hat{N}_{t,l}^- = \sum_{i=1}^{n_t} \mathbb{1}(\hat{\beta}_{it,l} < 0) \quad \text{and} \quad \hat{N}_{t,l}^+ = \sum_{i=1}^{n_t} \mathbb{1}(\hat{\beta}_{it,l} \geq 0).$$

On the event $\min\{\hat{N}_{t,l}^-, \hat{N}_{t,l}^+\} > 0$, define the split empirical distribution functions by

$$\hat{F}_{\hat{\beta},t,l,-}(z) = \frac{1}{\hat{N}_{t,l}^-} \sum_{i=1}^{n_t} \mathbb{1}(\hat{\beta}_{it,l} < 0) \mathbb{1}(\hat{\beta}_{it,l} \leq z) \quad \text{and} \quad \hat{F}_{\hat{\beta},t,l,+}(z) = \frac{1}{\hat{N}_{t,l}^+} \sum_{i=1}^{n_t} \mathbb{1}(\hat{\beta}_{it,l} \geq 0) \mathbb{1}(\hat{\beta}_{it,l} \leq z),$$

with generalized inverse functions

$$\hat{F}_{\hat{\beta},t,l,-}^{-1}(\kappa) = \inf\{z < 0 : \hat{F}_{\hat{\beta},t,l,-}(z) \geq \kappa\} \quad \text{and} \quad \hat{F}_{\hat{\beta},t,l,+}^{-1}(\kappa) = \inf\{z \geq 0 : \hat{F}_{\hat{\beta},t,l,+}(z) \geq \kappa\},$$

for $\kappa \in (0, 1)$. If either split sample is empty, define its empirical distribution and quantiles arbitrarily. The uniform empirical sign-share bound in the proof of Lemma SA-II.3.2 shows that this event has probability approaching zero.

Assumption SA-PA (Marginal-quantile-spaced partition). *For each $t = H + 1, \dots, T$, the partition $\hat{\Delta}_t = \otimes_{l \in [d]} \hat{\Delta}_{t,l}$ with*

$$\hat{\Delta}_{t,l} = \left\{ [\hat{\tau}_{0t,l}^-, \hat{\tau}_{1t,l}^-), [\hat{\tau}_{1t,l}^-, \hat{\tau}_{2t,l}^-), \dots, [\hat{\tau}_{(J_{t,l}^- - 1)t,l}^-, 0), [0, \hat{\tau}_{1t,l}^+), [\hat{\tau}_{1t,l}^+, \hat{\tau}_{2t,l}^+), \dots, [\hat{\tau}_{(J_{t,l}^+ - 1)t,l}^+, \hat{\tau}_{J_{t,l}^+}^+)] \right\}$$

where $\hat{\tau}_{0t,l}^- = \min_{i: \hat{\beta}_{it,l} < 0} \hat{\beta}_{it,l}$ and $\hat{\tau}_{J_{t,l}^+}^+ = \max_{i: \hat{\beta}_{it,l} \geq 0} \hat{\beta}_{it,l}$, $\hat{\tau}_{jt,l}^- = \hat{F}_{\hat{\beta},t,l,-}^{-1}(j/J_{t,l}^-)$ for $j \in [J_{t,l}^- - 1]$, $\hat{\tau}_{jt,l}^+ = \hat{F}_{\hat{\beta},t,l,+}^{-1}(j/J_{t,l}^+)$ for $j \in [J_{t,l}^+ - 1]$, and $J_{t,l}^- + J_{t,l}^+ = J_{t,l}$. Assume that $J_{t,l}^-, J_{t,l}^+ \asymp J$ for all $l \in [d]$ and some $J \rightarrow \infty$ as $n \rightarrow \infty$, and $J^d/n = o(1)$.

The corresponding reference partition is denoted by $\Delta_t = \otimes_{l \in [d]} \Delta_{t,l}$ and is nonrandom conditionally on $\mathcal{D}_{t-1}$. It uses the conditional analogs of the same split quantiles, and should be read as the conditional reference counterpart of the estimated quantile partition, rather than as a partition formed from the distribution of the true β_{it} . For each coordinate $l \in [d]$,

$$\Delta_{t,l} = \left\{ (\tau_{0t,l}^-, \tau_{1t,l}^-), [\tau_{1t,l}^-, \tau_{2t,l}^-), \dots, [\tau_{(J_{t,l}^- - 1)t,l}^-, 0), [0, \tau_{1t,l}^+), [\tau_{1t,l}^+, \tau_{2t,l}^+), \dots, [\tau_{(J_{t,l}^+ - 1)t,l}^+, \tau_{J_{t,l}^+}^+)] \right\},$$

where $\tau_{0t,l}^- = -\infty$ and $\tau_{J_{t,l}^+}^+ = +\infty$, $\tau_{jt,l}^- = \bar{F}_{\hat{\beta},t,l,-}^{-1}(j/J_{t,l}^-)$ for $j \in [J_{t,l}^- - 1]$, and $\tau_{jt,l}^+ = \bar{F}_{\hat{\beta},t,l,+}^{-1}(j/J_{t,l}^+)$ for $j \in [J_{t,l}^+ - 1]$, with

$$\begin{aligned} \bar{\pi}_{t,l,-} &= \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{P}(\hat{\beta}_{it,l} < 0 \mid \mathcal{D}_{t-1}), & \bar{\pi}_{t,l,+} &= \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{P}(\hat{\beta}_{it,l} \geq 0 \mid \mathcal{D}_{t-1}), \\ \bar{F}_{\hat{\beta},t,l,-}(z) &= \frac{1}{n_t \bar{\pi}_{t,l,-}} \sum_{i=1}^{n_t} \mathbb{P}(\hat{\beta}_{it,l} \leq z, \hat{\beta}_{it,l} < 0 \mid \mathcal{D}_{t-1}), & z < 0, \\ \bar{F}_{\hat{\beta},t,l,+}(z) &= \frac{1}{n_t \bar{\pi}_{t,l,+}} \sum_{i=1}^{n_t} \mathbb{P}(0 \leq \hat{\beta}_{it,l} \leq z \mid \mathcal{D}_{t-1}), & z \geq 0. \end{aligned}$$

Thus $\bar{F}_{\hat{\beta},t,l,-}$ and $\bar{F}_{\hat{\beta},t,l,+}$ are distribution functions on the negative and nonnegative sides, respectively. Their generalized inverses are defined over $z < 0$ and $z \geq 0$, respectively, as for the split empirical distribution

functions above. Accordingly, define marginal split-mixture densities

$$\begin{aligned}\bar{f}_{\hat{\beta},t,l,-}(z) &= \frac{1}{n_t \bar{\pi}_{t,l,-}} \sum_{i=1}^{n_t} f_{\hat{\beta}_{it,l}|\mathcal{D}_{t-1}}(z|\mathcal{D}_{t-1}) \mathbb{1}(z < 0) \quad \text{and} \\ \bar{f}_{\hat{\beta},t,l,+}(z) &= \frac{1}{n_t \bar{\pi}_{t,l,+}} \sum_{i=1}^{n_t} f_{\hat{\beta}_{it,l}|\mathcal{D}_{t-1}}(z|\mathcal{D}_{t-1}) \mathbb{1}(z \geq 0).\end{aligned}$$

These ratios are defined as zero if the corresponding pooled sign probability vanishes. Lemma SA-II.3.1(i) implies that $\bar{\pi}_{t,l,-}$ and $\bar{\pi}_{t,l,+}$ are uniformly bounded away from zero with probability approaching one, so this convention is asymptotically irrelevant. Cell membership, counts, and polynomial basis evaluation all use the raw generated regressor $\hat{\beta}_{it}$. The estimated partition therefore covers the coordinatewise range of the realized generated regressors, while the reference partition uses the corresponding complete conditional split distributions. The cells outside $\mathcal{B}^\circ$ may be unbounded, but they do not intersect the interior evaluation region with probability approaching one and are not used in any design or inverse-Gram claim below. The reference basis $p_t(\cdot)$ is defined from this Δ_t , and the reference Gram matrix Q_t is the conditional expectation of $p_t(\hat{\beta}_{it})p_t(\hat{\beta}_{it})^\top$ as in Section SA-I.3. For the verification with polynomial degree $\mathbf{q} \geq 0$, the basis normalization is cell-specific. For $j \in \mathcal{J}_t$, write the paired reference and estimated rectangles as $P_{jt} = \prod_{l \in [d]} [a_{jtl}, b_{jtl}]$ and $\hat{P}_{jt} = \prod_{l \in [d]} [\hat{a}_{jtl}, \hat{b}_{jtl}]$, with the endpoint and tie-breaking conventions inherited from the corresponding partitions. We set

$$\begin{aligned}c_{jt,l} &= \frac{a_{jtl} + b_{jtl}}{2}, & D_{jt} &= \text{diag}\{b_{jt1} - a_{jt1}, \dots, b_{jtd} - a_{jtd}\}, \\ \hat{c}_{jt,l} &= \frac{\hat{a}_{jtl} + \hat{b}_{jtl}}{2}, & \hat{D}_{jt} &= \text{diag}\{\hat{b}_{jt1} - \hat{a}_{jt1}, \dots, \hat{b}_{jtd} - \hat{a}_{jtd}\}.\end{aligned}$$

Thus, for b in the reference cell P_{jt} , the polynomial block is $\check{p}_{\mathbf{q}}(D_{jt}^{-1}(b - c_{jt}))$, while for b in the estimated cell $\hat{P}_{jt}$, the polynomial block is $\check{p}_{\mathbf{q}}(\hat{D}_{jt}^{-1}(b - \hat{c}_{jt}))$. The block entering the stacked basis is always the block of the cell containing the argument b . When $\mathbf{q} = 0$, this reduces to the cell indicator. When $\mathbf{q} \geq 1$, the monomials are therefore formed from the normalized coordinates of b within its own rectangle. For the cell containing the origin, the split construction sets the lower endpoint equal to zero in every coordinate, both for the reference and estimated partitions. Hence $D_{jt}^{-1}(0 - c_{jt}) = \hat{D}_{jt}^{-1}(0 - \hat{c}_{jt}) = (-1/2, \dots, -1/2)^\top$ on that cell, and therefore $\hat{p}_t(0) = p_t(0)$ for every $\mathbf{q} \geq 0$. This is the specialization of the affine scaling/rotation convention from Part I to the axis-aligned quantile rectangles used here. On the event where all side lengths are of order J^{-1} , the normalized basis is uniformly bounded and has derivative of order J in the original beta coordinates. The rest of this section collects the verification results for this random partitioning scheme and basis function.

Under Assumption SA-SEP, $\mathcal{J}_t$ contains precisely the estimated cells intersecting $\mathcal{B}^\circ$. Define the corresponding interior cell counts

$$\hat{n}_{\min}^\circ = \min_{H+1 \leq t \leq T} \min_{j \in \mathcal{J}_t} \hat{n}_{jt}, \quad \hat{n}_{\max}^\circ = \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \hat{n}_{jt}.$$

SA-II.3.1 Results

We present three results that establish the main properties of the quantile-based portfolio construction.

The first result establishes the density regularity of the generated regressor and translates it into density and spacing properties of the split reference quantile portfolios.

Lemma SA-II.3.1 (Generated-Regressor Density and Reference Quantile Spacing). *Suppose that Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold. Assume further that $T^{(3+d)/q}/H + \sqrt{\log T/H} = o(1)$. Then, with probability approaching one, the following statements hold:*

- (i) *The conditional density $f_{\hat{\beta}_{it}|\mathcal{D}_{t-1}}(\cdot|\mathcal{D}_{t-1})$ is uniformly bounded and bounded away from zero on $\mathcal{B}$, uniformly over $i \in [n_t]$ and $t = H+1, \dots, T$.*
- (ii) *The marginal split-mixture densities $\bar{f}_{\hat{\beta},t,l,-}$ and $\bar{f}_{\hat{\beta},t,l,+}$ are uniformly bounded and bounded away from zero on $[\beta_l, 0)$ and $[0, \bar{\beta}_l]$, respectively, uniformly over $l \in [d]$ and $t = H+1, \dots, T$.*

Consequently, the finite reference cells paired with cells in $\mathcal{J}_t$ satisfy

$$J^{-1} \lesssim_{\mathbb{P}} \min_{H+1 \leq t \leq T} \min_{j \in \mathcal{J}_t^\circ} \min_{l \in [d]} (b_{jtl} - a_{jtl}) \leq \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t^\circ} \max_{l \in [d]} (b_{jtl} - a_{jtl}) \lesssim_{\mathbb{P}} J^{-1},$$

where $\mathcal{J}_t^\circ = \{j \in \mathcal{J}_t : P_{jt} \text{ is bounded}\}$.

The next result compares the empirical breakpoints in $\hat{\Delta}_t$ with their conditional reference counterparts in Δ_t . The first two bounds control levels of the split quantiles, while the last two control adjacent spacings, which are needed when comparing cell boundaries.

Lemma SA-II.3.2 (Quantile Convergence). *Suppose that Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold. Assume further that $T^{(3+d)/q}/H + \sqrt{\log T/H} = o(1)$ and $\log T/n = o(1)$. Then, uniformly over the empirical cells intersecting $\mathcal{B}^\circ$ and their rank-paired reference cells,*

$$\begin{aligned} \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \max_{l \in [d]} \left(|\hat{a}_{jtl} - a_{jtl}| + |\hat{b}_{jtl} - b_{jtl}| \right) &\lesssim_{\mathbb{P}} \sqrt{\frac{\log T}{n}}, \\ \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \max_{l \in [d]} \left| (\hat{b}_{jtl} - \hat{a}_{jtl}) - (b_{jtl} - a_{jtl}) \right| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log T}{n}}. \end{aligned}$$

If, in addition, $J^2 \log T/n = o(1)$, then the estimated cell widths satisfy

$$J^{-1} \lesssim_{\mathbb{P}} \min_{H+1 \leq t \leq T} \min_{j \in \mathcal{J}_t} \min_{l \in [d]} (\hat{b}_{jtl} - \hat{a}_{jtl}) \leq \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \max_{l \in [d]} (\hat{b}_{jtl} - \hat{a}_{jtl}) \lesssim_{\mathbb{P}} J^{-1}.$$

With the quantile approximation in hand, we can verify the portfolio geometry, portfolio counts, and Gram behavior used by the subsequent results.

Lemma SA-II.3.3 (Generated Partition and Basis Moments). *Suppose that Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold. Assume further that*

$$\frac{T^{(3+d)/q}}{H} + \sqrt{\frac{\log T}{H}} = o(1), \quad \frac{J^d \log(nT)}{n} = o(1).$$

Then the following results hold.

1. *Generated partition coverage and geometry: $\mathcal{B}^\circ \cup \{\hat{\beta}_{it} : i \in [n_t]\}$ is contained in $\cup_{j \in [\bar{J}_t]} \hat{P}_{jt}$ uniformly over t w.p.a.1, and $\hat{\mathbf{R}}_{\mathcal{D}}^\circ \lesssim_{\mathbb{P}} J^{-1}$.*
2. *Cell counts:*

$$\frac{n}{J^d} \lesssim_{\mathbb{P}} \hat{n}_{\min}^\circ \leq \hat{n}_{\max}^\circ \lesssim_{\mathbb{P}} \frac{n}{J^d}.$$

3. The generated Gram blocks:

$$J^{-d} \lesssim_{\mathbb{P}} \lambda_{\min}(\hat{Q}_{jt}) \leq \lambda_{\max}(\hat{Q}_{jt}) \lesssim_{\mathbb{P}} J^{-d}$$

uniformly over $t = H + 1, \dots, T$ and interior cells $j \in \mathcal{J}_t$.

4. The reference Gram blocks:

$$J^{-d} \lesssim_{\mathbb{P}} \lambda_{\min}(Q_{jt}) \leq \lambda_{\max}(Q_{jt}) \lesssim_{\mathbb{P}} J^{-d}$$

uniformly over $t = H + 1, \dots, T$ and $j \in \mathcal{J}_t$.

5. The empirical true-exposure reference second moments:

$$\lambda_{\max} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} p_{jt}(\beta_{it}) p_{jt}(\beta_{it})^\top \right) \lesssim_{\mathbb{P}} J^{-d}$$

uniformly over $t = H + 1, \dots, T$ and $j \in \mathcal{J}_t$.

6. The comparison quantities from Lemma SA-I.3.1:

$$\hat{R}_Q \lesssim_{\mathbb{P}} R_Q = J^{-d} \sqrt{\frac{J^d \log(nT)}{n}} + J^{-(d-1)} \sqrt{\frac{\log T}{n}} + \frac{\log(nT)}{n},$$

and

$$\begin{aligned} \hat{R}_P \lesssim_{\mathbb{P}} R_P = J^{-d} \min \left\{ J \left(R_{FS} + \sqrt{\frac{\log T}{n}} \right), 1 \right\} + \sqrt{\frac{J^{-d} \min \left\{ J \left(R_{FS} + \sqrt{\frac{\log T}{n}} \right), 1 \right\} \log(nT)}{n}} \\ + \frac{\log(nT)}{n} + \mathbf{1}(\mathfrak{q} \geq 1) J^{-d} \min \left\{ J^2 \left(R_{FS} + \sqrt{\frac{\log T}{n}} \right)^2, 1 \right\}. \end{aligned}$$

SA-II.3.2 Discussion

- (i) The rate R_Q controls the inverse-Gram perturbation. It compares the empirical generated Gram matrix with the conditional reference Gram matrix of $\hat{\beta}_{it}$ evaluated on the reference partition. Its terms correspond to ordinary cross-sectional sampling variation within cells, movement of the empirical quantile endpoints, and empirical discretization. The condition $R_Q = o(J^{-d})$ is implied by

$$\frac{J^d \log(nT)}{n} = o(1) \quad \text{and} \quad \frac{J^2 \log T}{n} = o(1).$$

If these conditions hold, then the construction gives the measurability and coverage components of the partition-design condition (i), while Lemma SA-II.3.3 verifies the diameter, design, and Gram-convergence components in conditions (i) and (ii) of Lemma SA-I.3.1. The stochastic linearization for $\mathfrak{q} = 0$ can therefore be obtained through the comparison below.

- (ii) The rate R_P controls the basis-switching variance term $\mathcal{R}_2(\beta)$. It compares $\hat{p}_t(\hat{\beta}_{it})$ with $p_t(\beta_{it})$. The switch set is therefore a boundary strip between paired estimated and reference cells, with width

determined by both the first-step displacement $\widehat{\beta}_{it} - \beta_{it}$ and the endpoint quantile error. The condition $R_p = o(J^{-d})$ is implied by the same two displays above and by $JR_{FS} = o(1)$.

- (iii) For $q \geq 1$, R_p also contains a nonswitch polynomial-coordinate term. On observations that remain in paired cells, the indicators agree but the normalized coordinates differ because the estimated and reference cell endpoints are not identical and because $\widehat{\beta}_{it}$ differs from β_{it} . Bounded derivatives of the normalized basis convert the combined displacement into the final term in R_p .

SA-II.3.3 Proof of Lemma SA-II.3.1

Because $3 + d > 2$, the condition $T^{(3+d)/q}/H + \sqrt{\log T/H} = o(1)$ implies the rate condition in Lemma SA-II.2.2. Hence, on an event with probability approaching one, $\widehat{A}(t)$ is invertible and

$$1 \lesssim \lambda_{\min} \left(\frac{1}{H} \sum_{l=1}^H c_{itl} c_{itl}^\top \right) \leq \lambda_{\max} \left(\frac{1}{H} \sum_{l=1}^H c_{itl} c_{itl}^\top \right) \lesssim 1,$$

uniformly over $i \in [n_t]$ and $t = H + 1, \dots, T$, where

$$c_{itl} = S\widehat{A}(t)^{-1}X_l(t)K\left(\frac{l}{H}\right)\varsigma_{i(t-l)}.$$

It remains to verify the higher-order coefficient-moment bound used below. The uniform inverse bound for $\widehat{A}(t)$ and the boundedness of the kernel and $\varsigma_{i(t-l)}$ give

$$\max_{i \in [n_t]} \frac{1}{H} \sum_{l=1}^H \|c_{itl}\|^{3+d} \lesssim_{\mathbb{P}} \frac{1}{H} \sum_{l=1}^H \|X_l(t)\|^{3+d} \lesssim 1 + \frac{1}{H} \sum_{l=1}^H \|f_{t-l}\|^{3+d}.$$

The process $\{\|f_t\|^{3+d}\}$ has a finite moment of order $q/(3+d) > 2$ and inherits the geometric near-epoch dependence in Assumption SA-DGP(iv). Applying the same maximal Fuk–Nagaev argument as in Lemma SA-II.2.2, now with moment exponent $q/(3+d)$, yields

$$\max_{H+1 \leq t \leq T} \left| \frac{1}{H} \sum_{l=1}^H (\|f_{t-l}\|^{3+d} - \mathbb{E}[\|f_{t-l}\|^{3+d}]) \right| \lesssim_{\mathbb{P}} \frac{T^{(3+d)/q}}{H} + \sqrt{\frac{\log T}{H}} = o_{\mathbb{P}}(1).$$

Since $\sup_t \mathbb{E}[\|f_t\|^{3+d}] < \infty$, intersecting these events gives an event $\mathcal{A}$ with probability approaching one on which

$$\max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \frac{1}{H} \sum_{l=1}^H \|c_{itl}\|^{3+d} \lesssim 1$$

and the preceding inverse and eigenvalue bounds all hold. On this event, let

$$\mathcal{H}_{it}^o = \sigma \left((\alpha_{i(t-s)}, \beta_{i(t-s)}^\top, \varsigma_{i(t-s)}) : s \in [H] \right) \vee \mathcal{D}_{t-1}.$$

Conditional on $\mathcal{H}_{it}^\circ$, the only remaining randomness in the first-step variance component comes from the i.i.d. innovations $\{\eta_{i(t-s)} : s \in [H]\}$. Let $f_{\hat{\beta}_{it}|\mathcal{H}_{it}^\circ}(\cdot|\mathcal{H}_{it}^\circ)$ be the corresponding conditional density. Then

$$f_{\hat{\beta}_{it}|\mathcal{D}_{t-1}}(z|\mathcal{D}_{t-1}) = \mathbb{E}\left[f_{\hat{\beta}_{it}|\mathcal{H}_{it}^\circ}(z|\mathcal{H}_{it}^\circ)|\mathcal{D}_{t-1}\right] = H^{d/2}\mathbb{E}\left[f_{G_{it}}(\sqrt{H}(z - \check{\beta}_{it}))|\mathcal{D}_{t-1}\right],$$

where $G_{it} = H^{-1/2} \sum_{l=1}^H c_{itl} \eta_{i(t-l)}$ and $\check{\beta}_{it}$ is defined in (SA-II.2.2). Given $\mathcal{H}_{it}^\circ$, the coefficients c_{itl} and the signal $\check{\beta}_{it}$ are fixed, and the independence of the innovations in Assumption SA-DGP(ii) gives the displayed convolution representation. We record the needed properties of $f_{G_{it}}$ directly. Let φ_η denote the characteristic function of η_{it} . Conditional on $\mathcal{H}_{it}^\circ$,

$$\varphi_{G_{it}}(u) = \prod_{l=1}^H \varphi_\eta\left(\frac{c_{itl}^\top u}{\sqrt{H}}\right).$$

The condition $g_\eta^{(1)} \in L^1(\mathbb{R})$ implies polynomial decay of φ_η away from the origin, while the moment condition on η_{it} gives the usual expansion $\varphi_\eta(v) = 1 - v^2/2 + O(|v|^{2+\delta})$ for some $\delta > 0$ as $v \rightarrow 0$. Together with the eigenvalue bounds on $H^{-1} \sum_l c_{itl} c_{itl}^\top$ and the moment average bound on the c_{itl} 's, the standard three-region argument for products of characteristic functions applies: near the origin the product has Gaussian decay, on intermediate shells a fixed fraction of the projected arguments is bounded away from zero, and in the tails the decay from $g_\eta^{(1)} \in L^1(\mathbb{R})$ is applied to enough projected factors. Hence

$$\int_{\mathbb{R}^d} (1 + \|u\|) |\varphi_{G_{it}}(u)| du \lesssim 1.$$

Fourier inversion therefore gives a continuously differentiable density

$$f_{G_{it}}(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} \exp(-iu^\top x) \varphi_{G_{it}}(u) du$$

with $\sup_x \{f_{G_{it}}(x) + \|\nabla f_{G_{it}}(x)\|\} \lesssim 1$ uniformly.

The same Fourier calculation also gives a uniform lower bound near the origin. Indeed, for fixed r and $\|u\| \leq r$, the expansion of φ_η gives

$$\varphi_{G_{it}}(u) = \exp\left\{-\frac{1}{2H} \sum_{l=1}^H (c_{itl}^\top u)^2\right\} + o(1),$$

uniformly on $\|u\| \leq r$. The matrix inside the quadratic form has eigenvalues bounded and bounded away from zero on $\mathcal{A}$, so the real part of the Fourier integral over a sufficiently large fixed ball is close to the corresponding nondegenerate Gaussian integral. The tail outside that ball can be made uniformly small by the preceding integrability bound. Hence $f_{G_{it}}(0) \gtrsim 1$ uniformly, and boundedness of $\nabla f_{G_{it}}$ implies that $f_{G_{it}}(x) \gtrsim 1$ on a fixed neighborhood of zero. The same Fourier argument, with integration by parts, also gives a polynomial tail bound $f_{G_{it}}(x) \lesssim (1 + \|x\|)^{-d-1}$ uniformly. Let

$$\mathcal{A}_{it,\text{FS}} = \left\{ \|\check{\beta}_{it} - \beta_{it}\| \leq \mathbf{B}_{\text{FS}} \right\}.$$

By Assumption [SA-FSB](#) and the inclusion $\mathcal{A}_{it,\text{FS}}^c \subseteq \{\widehat{\mathbf{B}}_{\text{FS}} > \mathbf{B}_{\text{FS}}\}$,

$$\sqrt{H} \mathbf{B}_{\text{FS}} = o(1), \quad H^{d/2} \max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) \lesssim \sqrt{H} \mathbf{B}_{\text{FS}} + H^{-1/2} = o_{\mathbb{P}}(1).$$

We split the conditional convolution into its contributions on $\mathcal{A}_{it,\text{FS}}$ and $\mathcal{A}_{it,\text{FS}}^c$. Since $\sup_x f_{G_{it}}(x) \lesssim 1$, the latter contribution satisfies, uniformly over i, t , and $z \in \mathcal{B}$,

$$H^{d/2} \mathbb{E} \left[f_{G_{it}} \left(\sqrt{H}(z - \check{\beta}_{it}) \right) \mathbb{1}(\mathcal{A}_{it,\text{FS}}^c) \mid \mathcal{D}_{t-1} \right] \lesssim H^{d/2} \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) = o_{\mathbb{P}}(1). \quad (\text{SA-II.3.1})$$

On $\mathcal{A}_{it,\text{FS}}$, $\|\check{\beta}_{it} - \beta_{it}\| \leq \mathbf{B}_{\text{FS}}$. Hence, for every $r > 0$,

$$\mathbb{P}(\|\check{\beta}_{it} - z\| \leq r, \mathcal{A}_{it,\text{FS}} \mid \mathcal{D}_{t-1}) \leq \mathbb{P}(\|\beta_{it} - z\| \leq r + \mathbf{B}_{\text{FS}} \mid \mathcal{D}_{t-1}) \lesssim (r + \mathbf{B}_{\text{FS}})^d,$$

where the last inequality follows from the uniform upper bound on $f_{\beta_{it}}(\cdot \mid \mathcal{D}_{t-1})$. Decomposing $\mathcal{B}$ into annuli of width $H^{-1/2}$ around z , using $\sqrt{H} \mathbf{B}_{\text{FS}} = o(1)$ and the polynomial tail of $f_{G_{it}}$, shows that the contribution on $\mathcal{A}_{it,\text{FS}}$ is uniformly bounded. Together with [\(SA-II.3.1\)](#), this yields

$$\sup_{z \in \mathcal{B}} f_{\widehat{\beta}_{it} \mid \mathcal{D}_{t-1}}(z \mid \mathcal{D}_{t-1}) \lesssim_{\mathbb{P}} 1.$$

For the lower bound, fix $r_0 > 0$ small enough that $f_{G_{it}}(x)$ is uniformly bounded away from zero on $\{x : \|x\| \leq 2r_0\}$. Since $\sqrt{H} \mathbf{B}_{\text{FS}} = o(1)$, uniformly in $z \in \mathcal{B}$,

$$\begin{aligned} f_{\widehat{\beta}_{it} \mid \mathcal{D}_{t-1}}(z \mid \mathcal{D}_{t-1}) &\gtrsim H^{d/2} \mathbb{P}(\|\beta_{it} - z\| \leq r_0 H^{-1/2}, \mathcal{A}_{it,\text{FS}} \mid \mathcal{D}_{t-1}) \\ &\gtrsim H^{d/2} \left[\mathbb{P}(\|\beta_{it} - z\| \leq r_0 H^{-1/2} \mid \mathcal{D}_{t-1}) - \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) \right]. \end{aligned}$$

Because $\mathcal{B}$ is a rectangle, its intersection with a ball of radius $r_0 H^{-1/2}$ centered at any $z \in \mathcal{B}$ contains a cone whose volume is bounded below by a constant multiple of $H^{-d/2}$. The uniform lower bound on $f_{\beta_{it}}(\cdot \mid \mathcal{D}_{t-1})$ and Assumption [SA-FSB](#) therefore imply

$$\inf_{z \in \mathcal{B}} f_{\widehat{\beta}_{it} \mid \mathcal{D}_{t-1}}(z \mid \mathcal{D}_{t-1}) \gtrsim_{\mathbb{P}} 1.$$

This proves the full-vector density statement without requiring a density for the signal $\check{\beta}_{it}$ itself.

The marginal density statement follows by applying the preceding argument to each coordinate. The eigenvalue bounds for $H^{-1} \sum_{s=1}^H c_{its} c_{its}^\top$ bound each coordinate variance above and away from zero, and the exceptional-event bound above is stronger than needed for one coordinate. For the split mixtures,

$$\begin{aligned} \bar{f}_{\widehat{\beta},t,l,-}(z) &= \frac{1}{n_t \bar{\pi}_{t,l,-}} \sum_{i=1}^{n_t} f_{\widehat{\beta}_{it,l} \mid \mathcal{D}_{t-1}}(z \mid \mathcal{D}_{t-1}) \mathbb{1}(z < 0), \\ \bar{f}_{\widehat{\beta},t,l,+}(z) &= \frac{1}{n_t \bar{\pi}_{t,l,+}} \sum_{i=1}^{n_t} f_{\widehat{\beta}_{it,l} \mid \mathcal{D}_{t-1}}(z \mid \mathcal{D}_{t-1}) \mathbb{1}(z \geq 0). \end{aligned}$$

Because 0 is an interior point of $\mathcal{B}$ and the marginal densities are bounded and bounded away from zero on $[\underline{\beta}_l, \bar{\beta}_l]$, the denominators are bounded away from zero w.p.a.1. Thus the split densities are bounded and bounded away from zero on $[\underline{\beta}_l, 0)$ and $[0, \bar{\beta}_l]$, respectively. The separation of $\mathcal{B}^\circ$ from $\partial \mathcal{B}$ and $J_{t,l}^-, J_{t,l}^+ \asymp J$

imply that every finite reference endpoint paired with a cell in $\mathcal{J}_t$ lies in this density-regular region w.p.a.1. The displayed interior spacing bound therefore follows from the mean value theorem. $\square$

SA-II.3.4 Proof of Lemma SA-II.3.2

For $a \in \{-, +\}$, write $\mathcal{S}_- = (-\infty, 0)$ and $\mathcal{S}_+ = [0, \infty)$. Define

$$\begin{aligned}\hat{A}_{tla}(z) &= \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbf{1}(\hat{\beta}_{it,l} \in \mathcal{S}_a) \mathbf{1}(\hat{\beta}_{it,l} \leq z), \\ \bar{A}_{tla}(z) &= \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{P}(\hat{\beta}_{it,l} \in \mathcal{S}_a, \hat{\beta}_{it,l} \leq z | \mathcal{D}_{t-1}),\end{aligned}$$

and let $\hat{\pi}_{tla} = n_t^{-1} \sum_{i=1}^{n_t} \mathbf{1}(\hat{\beta}_{it,l} \in \mathcal{S}_a)$ and $\bar{\pi}_{tla} = n_t^{-1} \sum_{i=1}^{n_t} \mathbb{P}(\hat{\beta}_{it,l} \in \mathcal{S}_a | \mathcal{D}_{t-1})$. Then $\hat{F}_{\hat{\beta},t,l,a} = \hat{A}_{tla}/\hat{\pi}_{tla}$ and $\bar{F}_{\hat{\beta},t,l,a} = \bar{A}_{tla}/\bar{\pi}_{tla}$ on the corresponding side of zero.

Conditional on $\mathcal{D}_{t-1}$, $\hat{\beta}_{it}$ is independent over i . The classes of half-lines intersected with $\mathcal{S}_-$ and $\mathcal{S}_+$ have finite VC index and envelope one. Therefore the i.n.i.d. maximal inequality in Cattaneo et al. (2024, Lemma SA24), combined with conditional bounded-difference concentration and followed by a union bound over t , coordinates, and signs, gives

$$\begin{aligned}\max_{H+1 \leq t \leq T} \max_{l \in [d]} \max_{a \in \{-, +\}} \sup_{z \in \mathcal{S}_a} |\hat{A}_{tla}(z) - \bar{A}_{tla}(z)| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log T}{n}}, \\ \max_{H+1 \leq t \leq T} \max_{l \in [d]} \max_{a \in \{-, +\}} |\hat{\pi}_{tla} - \bar{\pi}_{tla}| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log T}{n}}.\end{aligned}$$

Lemma SA-II.3.1 and the fact that 0 lies in the interior of $\mathcal{B}$ imply $\bar{\pi}_{tla} \gtrsim_{\mathbb{P}} 1$ uniformly. Hence $\hat{\pi}_{tla} \gtrsim_{\mathbb{P}} 1$ uniformly and the ratio map gives

$$\begin{aligned}\max_{H+1 \leq t \leq T} \max_{l \in [d]} \sup_{z < 0} |\hat{F}_{\hat{\beta},t,l,-}(z) - \bar{F}_{\hat{\beta},t,l,-}(z)| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log T}{n}}, \\ \max_{H+1 \leq t \leq T} \max_{l \in [d]} \sup_{z \geq 0} |\hat{F}_{\hat{\beta},t,l,+}(z) - \bar{F}_{\hat{\beta},t,l,+}(z)| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log T}{n}}.\end{aligned}$$

Since $\log T/n = o(1)$, the uniform split-CDF bounds above, Assumption SA-SEP, and $J \rightarrow \infty$ imply that the first negative and last positive empirical intervals do not intersect $\mathcal{B}^\circ$ in any coordinate w.p.a.1. Hence all rank-paired reference cells indexed by $\mathcal{J}_t$ have finite endpoints on this event. For the probability indices defining cells in $\mathcal{J}_t$, the associated reference quantiles lie in the density-regular region identified in Lemma SA-II.3.1. The generalized inverse relation and the empirical quantile overshoot therefore give the first displayed bound in the lemma. The side-length bound follows from

$$\left| (\hat{b}_{jtl} - \hat{a}_{jtl}) - (b_{jtl} - a_{jtl}) \right| \leq |\hat{b}_{jtl} - b_{jtl}| + |\hat{a}_{jtl} - a_{jtl}|,$$

uniformly over $t, j \in \mathcal{J}_t$, and $l \in [d]$. Under $J^2 \log T/n = o(1)$, the right-hand side is $o_{\mathbb{P}}(J^{-1})$. Combining this result with the reference width bounds in Lemma SA-II.3.1 gives the asserted estimated width bounds. $\square$

SA-II.3.5 Proof of Lemma SA-II.3.3

Let $\hat{P}_{jt} = [\hat{\ell}_{jt}, \hat{u}_{jt}]$ denote a generic rectangle in $\hat{\Delta}_t$ and $P_{jt} = [\ell_{jt}, u_{jt}]$ the corresponding rectangle in the reference partition Δ_t . Let $\hat{\psi}_{jt}(b) = \hat{p}_{jt}(b)\hat{p}_{jt}(b)^\top$ and $\psi_{jt}(b) = p_{jt}(b)p_{jt}(b)^\top$. The condition $J^d \log(nT)/n = o(1)$ implies $\log T/n = o(1)$, so the argument in the proof of Lemma SA-II.3.2 ensures that all reference cells paired with $j \in \mathcal{J}_t$ have finite endpoints w.p.a.1. Lemma SA-II.3.1 gives reference side lengths of order J^{-1} . For cells indexed by $j \in \mathcal{J}_t$, the estimated side lengths satisfy the same order. To see this, apply the localized VC/Bernstein inequality to split intervals on each side of zero. Since the split densities are bounded and bounded away from zero, an interval with side length larger than a sufficiently large multiple of J^{-1} has conditional probability larger than a sufficiently large multiple of J^{-1} , while an interval with side length smaller than a sufficiently small multiple of J^{-1} has conditional probability smaller than a sufficiently small multiple of J^{-1} . The empirical split quantile intervals have empirical mass of order J^{-1} on their corresponding side, up to a discretization error of order n^{-1} , and $J^d \log(nT)/n = o(1)$ implies the localized stochastic error is $o_{\mathbb{P}}(J^{-1})$ uniformly. Hence the estimated and reference cells indexed by $\mathcal{J}_t$ belong w.p.a.1 to a uniformly quasi-uniform class of rectangles. Moreover, the first and last empirical order statistics lie outside $\mathcal{B}^\circ$ coordinatewise w.p.a.1: Assumption SA-SEP places at least one complete quantile cell between $\mathcal{B}^\circ$ and each support boundary, and every sample extremum lies weakly beyond the corresponding first or last interior empirical quantile. Thus the empirical partition covers $\mathcal{B}^\circ$ and, by construction, every realized $\hat{\beta}_{it}$. Assumption SA-SEP implies that every cell indexed by $j \in \mathcal{J}_t$ is separated from the outer boundary of $\mathcal{B}$ by more than the first-step smoothing scale, uniformly over t . The side-length bounds give

$$\hat{R}_D^\circ \lesssim_{\mathbb{P}} J^{-1}.$$

The classes generated by the entries of $\hat{\psi}_{jt}$ and ψ_{jt} are polynomial transformations of rectangle indicators. Hence they are VC-type with bounded envelopes and conditional variances of order J^{-d} . All empirical-process bounds below are first taken uniformly over deterministic quasi-uniform rectangles equipped with the midpoint and side-length normalization described above. The empirical quantile cells are then substituted into those uniform bounds.

The same maximal inequality applied only to the cell indicators indexed by $\mathcal{J}_t$ gives

$$\frac{n}{J^d} \lesssim_{\mathbb{P}} \hat{n}_{\min}^\circ \leq \hat{n}_{\max}^\circ \lesssim_{\mathbb{P}} \frac{n}{J^d}.$$

For a deterministic admissible rectangle P with its midpoint and side-length normalization, write $\psi_P(b) = p_q(b, P)p_q(b, P)^\top$. Conditional on $\mathcal{D}_{t-1}$, the maximal inequality gives

$$\max_{H+1 \leq t \leq T} \sup_P \left\| \frac{1}{n_t} \sum_{i=1}^{n_t} \left(\psi_P(\hat{\beta}_{it}) - \mathbb{E}[\psi_P(\hat{\beta}_{it}) | \mathcal{D}_{t-1}] \right) \right\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{nJ^d}}.$$

The supremum is over the deterministic quasi-uniform rectangle class containing the realized estimated cells w.p.a.1, so evaluating the display at $P = \hat{P}_{jt}$ controls the empirical part of $\hat{Q}_{jt}$. The conditional density of $\hat{\beta}_{it}$ is bounded and bounded away from zero on interior cells by Lemma SA-II.3.1. After the affine change of variables to cell-normalized coordinates, the corresponding population generated Gram block has eigenvalues of order J^{-d} because the monomials in $\check{p}_q$ are linearly independent on a fixed normalized rectangle. Since

$J^d \log(nT)/n = o(1)$, Weyl's inequality gives

$$J^{-d} \lesssim_{\mathbb{P}} \lambda_{\min}(\widehat{Q}_{jt}) \leq \lambda_{\max}(\widehat{Q}_{jt}) \lesssim_{\mathbb{P}} J^{-d}$$

uniformly over $t = H + 1, \dots, T$ and $j \in \mathcal{J}_t$. Multiplying by $n_t/\widehat{n}_{jt} \asymp_{\mathbb{P}} J^d$ gives the scaled Gram bound

$$1 \lesssim_{\mathbb{P}} \frac{n_t}{\widehat{n}_{jt}} \lambda_{\min}(\widehat{Q}_{jt}) \leq \frac{n_t}{\widehat{n}_{jt}} \lambda_{\max}(\widehat{Q}_{jt}) \lesssim_{\mathbb{P}} 1.$$

For the reference Gram blocks, recall that

$$Q_{jt} = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{E}[p_{jt}(\widehat{\beta}_{it}) p_{jt}(\widehat{\beta}_{it})^\top | \mathcal{D}_{t-1}].$$

After the change of variables $b = c_{jt} + D_{jt}z$, the normalized support for z is a rectangle with nonempty interior. The reference side-length bounds from Lemma SA-II.3.1, together with the conditional density bounds for $\widehat{\beta}_{it}$ in the same lemma, imply that the Jacobian is of order J^{-d} and that the weight attached to the monomial Gram matrix is bounded and bounded away from zero. Since the monomials up to degree $\mathfrak{q}$ are linearly independent on this rectangle,

$$J^{-d} \lesssim_{\mathbb{P}} \lambda_{\min}(Q_{jt}) \leq \lambda_{\max}(Q_{jt}) \lesssim_{\mathbb{P}} J^{-d}$$

uniformly over $t = H + 1, \dots, T$ and $j \in \mathcal{J}_t$.

For the empirical true-exposure reference second moments, apply the same VC-type maximal inequality to the class $\psi_{jt}(\beta_{it})$. Conditional on $\mathcal{D}_{t-1}$, Assumption SA-DGP(iii) gives independent unit histories and conditional densities for β_{it} that are bounded on $\mathcal{B}$. Since the reference cells have side lengths of order J^{-1} , the population block moments are of order J^{-d} . The empirical deviation is of order $\sqrt{\log(nT)/(nJ^d)}$, which is $o_{\mathbb{P}}(J^{-d})$ under $J^d \log(nT)/n = o(1)$. Hence

$$\lambda_{\max} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} p_{jt}(\beta_{it}) p_{jt}(\beta_{it})^\top \right) \lesssim_{\mathbb{P}} J^{-d}$$

uniformly over $t = H + 1, \dots, T$ and $j \in \mathcal{J}_t$.

It remains to bound $\widehat{R}_Q$. Add and subtract the population counterpart of the estimated rectangle, obtained by evaluating the deterministic-class map $P \mapsto n_t^{-1} \sum_i \mathbb{E}[\psi_P(\widehat{\beta}_{it}) | \mathcal{D}_{t-1}]$ at $P = \widehat{P}_{jt}$. The resulting empirical-process term is already bounded above by $\sqrt{\log(nT)/(nJ^d)}$. For the population-comparison term, compare the estimated and reference rectangles while holding the generated regressor fixed. The endpoint quantile bounds in Lemma SA-II.3.2 imply that the symmetric difference between paired rectangles has conditional mass $J^{-(d-1)} \sqrt{\log T/n}$, up to the empirical discretization term $\log(nT)/n$. When $\mathfrak{q} \geq 1$, the nonswitch change in normalized polynomial coordinates is of the same order: bounded derivatives of $\check{p}_{\mathfrak{q}}$ give a pointwise perturbation of order $J \sqrt{\log T/n}$ on a cell of mass J^{-d} . Combining these bounds gives $\widehat{R}_Q \lesssim_{\mathbb{P}} R_Q$.

For $\widehat{R}_P$, work on the high-probability event from Lemma SA-II.2.3 and Lemma SA-II.3.2 on which $\max_{i,t} \|\widehat{\beta}_{it} - \beta_{it}\| \lesssim R_{FS}$ and all paired endpoints differ by at most a constant multiple of $\sqrt{\log T/n}$. Decompose the full basis discrepancy into the generated-regressor displacement and the endpoint displacement:

$$\widehat{p}_{jt}(\widehat{\beta}_{it}) - p_{jt}(\beta_{it}) = \left\{ \widehat{p}_{jt}(\widehat{\beta}_{it}) - p_{jt}(\widehat{\beta}_{it}) \right\} + \left\{ p_{jt}(\widehat{\beta}_{it}) - p_{jt}(\beta_{it}) \right\}.$$

Any observation for which the corresponding cell indicators differ in either component belongs to a neighborhood of ∂P_{jt} (the boundary of P_{jt}) with radius proportional to $R_{\text{FS}} + \sqrt{\log T/n}$. When $J(R_{\text{FS}} + \sqrt{\log T/n}) \leq 1$, the density bounds give the boundary-strip probability $J^{-d} J(R_{\text{FS}} + \sqrt{\log T/n})$. Otherwise, the squared basis difference is supported on the union of the paired cells, whose mass is $O_{\mathbb{P}}(J^{-d})$. A Bernstein inequality in the first case and the cell-mass bound in the second give the first three terms in R_{P} .

On the complement of the switch set, the paired cell indicators agree. Both basis blocks are then evaluated in corresponding cells, and only the normalized polynomial coordinates can differ. For $\mathfrak{q} \geq 1$, bounded derivatives of the normalized polynomial basis, the first-step error bound, and the endpoint quantile bounds in Lemma SA-II.3.2 imply

$$\max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \max_{i \in [n_t]} \left\| \widehat{p}_{jt}(\widehat{\beta}_{it}) - p_{jt}(\beta_{it}) \right\| \mathbf{1}(\widehat{\beta}_{it} \in \widehat{P}_{jt}, \beta_{it} \in P_{jt}) \lesssim_{\mathbb{P}} \min \left\{ J \left(R_{\text{FS}} + \sqrt{\frac{\log T}{n}} \right), 1 \right\}.$$

The squared basis difference on the nonswitch set is therefore bounded by the corresponding squared displacement times the empirical mass of the paired cell, which is $\lesssim_{\mathbb{P}} J^{-d}$ uniformly. This gives the final term in R_{P} when $\mathfrak{q} \geq 1$ and proves $\widehat{R}_{\text{P}} \lesssim_{\mathbb{P}} R_{\text{P}}$. $\square$

SA-II.4 Stochastic Linearization

This section specializes the general stochastic linearization from Section SA-I.3 to the classical estimator defined in Sections SA-II.2 and SA-II.3. The first-step decomposition in (SA-II.2.2) separates the bias component $\check{\beta}_{it} - \beta_{it}$ from the first-step variance component, which determines the generated-regressor term $\mathcal{L}(\beta)$.

SA-II.4.1 Decomposition

Using the first-step decomposition in (SA-II.2.2), decompose

$$\mathcal{L}(\beta) = \mathcal{L}_{\text{B}}(\beta) + \mathcal{L}_{\text{V}}(\beta),$$

where

$$\begin{aligned} \mathcal{L}_{\text{B}}(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \widehat{p}_t(\beta)^\top \widehat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \widehat{p}_t(\widehat{\beta}_{it}) [\nabla \mu_t(\beta_{it})]^\top (\check{\beta}_{it} - \beta_{it}), \\ \mathcal{L}_{\text{V}}(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \widehat{p}_t(\beta)^\top \widehat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \widehat{p}_t(\widehat{\beta}_{it}) [\nabla \mu_t(\beta_{it})]^\top (\widehat{\beta}_{it} - \check{\beta}_{it}). \end{aligned}$$

$\mathcal{L}_{\text{B}}(\beta)$ is a bias term related to the bias of the first-step estimation, while $\mathcal{L}_{\text{V}}(\beta)$ contains both a bias component (due to the correlation between $\widehat{p}_t(\widehat{\beta}_{it})$ and $\varepsilon_{i(t-s)}$'s) and a variance component. Let $A(t) = \mathbb{E}[\widehat{A}(t)]$. The reference leading term for $\mathcal{L}_{\text{V}}(\beta)$ is

$$\mathcal{A}(\beta) = \frac{1}{T-H} \sum_{t=1}^T \sum_{i=1}^{n_t} w_{it}(\beta) \varepsilon_{it}$$

with

$$w_{it}(\beta) = \frac{1}{H} \sum_{s=\max\{1, H+1-t\}}^{\min\{H, T-t\}} \frac{1}{n_{t+s}} \widehat{p}_{t+s}(\beta)^\top Q_{t+s}^{-1} p_{t+s}(\beta_{i(t+s)}) [\nabla \mu_{t+s}(\beta_{i(t+s)})]^\top \\ \times K\left(\frac{s}{H}\right) S A(t+s)^{-1} X_s(t+s) \mathbf{1}(i \leq n_{t+s}).$$

Replacing β_{it} by $\widehat{\beta}_{it}$ can move the normalized polynomial coordinates even if both points remain in the same cell when $\mathfrak{q} \geq 1$. The proof below isolates the centered part of this smooth-interior movement and controls it by its uniform bound. We do not keep this component as a separate leading term: it is absorbed into the linear-approximation remainder. The remaining difference between $\mathcal{L}_V(\beta)$ and its reference approximation is driven by the conditional mean of the same-cell contribution, boundary switching, endpoint movement, and Gram perturbations.

SA-II.4.2 Results

The contribution of the first-step bias $\check{\beta}_{it} - \beta_{it}$ is a direct consequence of Assumption [SA-FSB](#). The preliminary lemma below controls both the conditional mean and conditional variance of the localized block-basis perturbation generated by the first-step variance term. For $\mathfrak{q} = 0$, this perturbation is exactly the cell-switching component. For $\mathfrak{q} \geq 1$, the same lemma also covers the smooth same-cell movement of the normalized polynomial coordinates.

Lemma SA-II.4.1 (Generated-Regressor Basis-Perturbation Bias and Variance). *Suppose that Assumptions [SA-DGP](#), [SA-FS](#), [SA-PA](#), and [SA-FSB](#) hold and that $s_\mu \geq 2$. Assume further that*

$$\frac{\max\{J^{2/(q-2)} T^{2/q}, T^{2(3+d)/q}\}}{H} = o(1).$$

Let $P = [\ell, u] \subset \mathcal{B}$ satisfy

$$\min_{k \in [d]} \min\{|\ell_k - \underline{\beta}_k|, |\bar{\beta}_k - u_k|\} \gg \frac{\log(nT)}{\sqrt{H}}, \quad J^{-1} \lesssim \min_{k \in [d]} (u_k - \ell_k) \leq \max_{k \in [d]} (u_k - \ell_k) \lesssim J^{-1}.$$

Then, the following results hold.

(a) *Conditional mean:*

$$\left\| \frac{1}{H} \sum_{s=1}^H \mathbb{E} \left[\left(p_{\mathfrak{q}}(\widehat{\beta}_{it}, P) - p_{\mathfrak{q}}(\beta_{it}, P) \right) [\nabla \mu_t(\beta_{it})]^\top S \widehat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varsigma_{i(t-s)} \eta_{i(t-s)} \middle| \mathcal{D}_{t-1} \right] \right\| \\ \lesssim_{\mathbb{P}} \frac{1}{H J^d} + \frac{\mathbf{B}_{\text{FS}}}{J^d}.$$

(b) *Conditional variance:*

$$\left\| \text{Var} \left[\frac{1}{H} \sum_{s=1}^H \left(p_{\mathfrak{q}}(\widehat{\beta}_{it}, P) - p_{\mathfrak{q}}(\beta_{it}, P) \right) [\nabla \mu_t(\beta_{it})]^\top S \widehat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varsigma_{i(t-s)} \eta_{i(t-s)} \middle| \mathcal{D}_{t-1} \right] \right\| \\ \lesssim_{\mathbb{P}} \frac{\log^d(nT)}{H J^{d-1} \max\{J, \sqrt{H}\}} + \left(\frac{1}{H J^d} + \frac{\mathbf{B}_{\text{FS}}}{J^d} \right)^2 + \mathbf{1}(\mathfrak{q} \geq 1) \frac{J^{2-d}}{H^2}.$$

These results hold uniformly over $i \in [n_t]$, $t = H + 1, \dots, T$, and rectangles P satisfying the conditions imposed above.

The next lemma refines only the generated-regressor component of Lemma SA-I.3.1. The remaining terms in the stochastic decomposition are already controlled by the general result in Part I. The refinement extracts the leading term $\mathcal{A}(\beta)$ and absorbs the smooth-interior polynomial movement that arises when $\mathfrak{q} \geq 1$ into the linear-approximation remainder.

Lemma SA-II.4.2 (Classical Generated-Regressor Linearization). *Suppose that Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold and that $s_\mu \geq 2$. Assume further that*

$$\begin{aligned} \frac{J^d \log(nT)}{n} &= o(1), \\ \frac{J^{d-1} \max\{J, \sqrt{H}\} \log^2(nT)}{\min\{n^{1-2/q} T^{1-4/q} h, n\}} &\lesssim 1, \quad \frac{\max\{J^{2/(q-2)} T^{2/q}, T^{2(3+d)/q}\}}{H} = o(1). \end{aligned} \quad (\text{SA-II.4.1})$$

Then, the bias term satisfies

$$\sup_{\beta \in \mathcal{B}^\circ} |\mathcal{L}_B(\beta)| \lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}},$$

and the variance terms satisfy

$$\sup_{\beta \in \mathcal{B}^\circ} |\mathcal{A}(\beta)| \lesssim_{\mathbb{P}} \sqrt{\frac{J^d \log(nT)}{nH}}, \quad |\mathcal{A}(0)| \lesssim_{\mathbb{P}} \sqrt{\frac{J^d}{nT}},$$

and

$$\sup_{\beta \in \mathcal{B}^\circ} |\mathcal{L}_V(\beta) - \mathcal{A}(\beta)| \lesssim_{\mathbb{P}} \mathbf{R}_{\text{LA}}$$

with

$$\begin{aligned} \mathbf{R}_{\text{LA}} &= \frac{1}{H} + \mathbf{B}_{\text{FS}} \\ &+ \sqrt{\frac{J^{2d} \log^{d+1}(nT)}{nH J^{d-1} \max\{J, \sqrt{H}\}}} + \sqrt{\frac{J^{d+1} \sqrt{\log T/n} \log(nT)}{nH}} + \min\{J^d \mathbf{R}_{\text{Q}}, 1\} \sqrt{\frac{J^d \log(nT)}{nH}} \\ &+ \mathbb{1}(\mathfrak{q} \geq 1) \sqrt{\frac{J^{d+2} \log(nT)}{nH^2}} + \mathbb{1}(\mathfrak{q} \geq 1) \sqrt{\frac{J^{d+2} \log T \log(nT)}{n^2 H}}. \end{aligned}$$

SA-II.4.3 Discussion

- (i) The comparison replaces the generated design in $\mathcal{L}_V(\beta)$ by the reference design. For $\mathfrak{q} = 0$, the within-cell basis is constant, so only boundary switching matters. For $\mathfrak{q} \geq 1$, movement inside a cell also changes the normalized polynomial coordinates. The proof isolates the centered product of this same-cell movement and the first-step shock multiplier, but the lemma absorbs it into $\mathbf{R}_{\text{LA}}$ using its uniform bound. This conservative treatment avoids relying on additional time-averaging for the same-cell component at the origin.

SA-II.4.4 Proof of Lemma SA-II.4.1(a)

The proof is separated into several steps. We first prove the switching-bias bound, which is the case $\mathbf{q} = 0$. The final step extends the same mean calculation to each component of the localized basis $p_{\mathbf{q}}(\cdot, P)$.

Step 1: Definitions and notation. For any generic rectangle of interest $P = [\ell, u]$ satisfying the boundary restrictions in the lemma, we write $\sqrt{H}(P - c) = [\sqrt{H}(\ell - c), \sqrt{H}(u - c)]$ to be the scaled shifted rectangle. For any vector v , v_{-j} denotes the subvector excluding the j -th entry; for any rectangle $\mathcal{R} = [\ell, u]$ in $\mathbb{R}^d$, $\mathcal{R}_{-j} = [\ell_{-j}, u_{-j}]$ in $\mathbb{R}^{d-1}$. Also write $\mathcal{B}_j = [\underline{\beta}_j, \bar{\beta}_j]$ for the j -th coordinate projection of $\mathcal{B}$.

Recall that

$$\check{\beta}_{it} = \frac{1}{H} \sum_{l=1}^H S\hat{A}(t)^{-1} X_l(t) K\left(\frac{l}{H}\right) (\alpha_{i(t-l)} + \beta_{i(t-l)}^\top f_{t-l})$$

with $\max_{i,t} \|\check{\beta}_{it} - \beta_{it}\| \leq \mathbf{B}_{\text{FS}}$ w.p.a.1 by Assumption SA-FSB, and that $\mathcal{E}_{t-1}$ contains the predictable paths, factor history, and volatility scales used in the conditioning arguments, but excludes the i.i.d. innovations η_{is} .

For each i and t , define the local regular event

$$\mathcal{A}_{it,\text{FS}} = \left\{ \|\check{\beta}_{it} - \beta_{it}\| \leq \mathbf{B}_{\text{FS}} \right\}.$$

Assumption SA-FSB and $\mathcal{A}_{it,\text{FS}}^c \subseteq \{\hat{\mathbf{B}}_{\text{FS}} > \mathbf{B}_{\text{FS}}\}$ give

$$H^{(d-1)/2} \max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) \lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}} + H^{-1}. \quad (\text{SA-II.4.2})$$

Because $\sqrt{H}(\mathbf{B}_{\text{FS}} + H^{-1}) = o(1)$, this also gives

$$H^{d/2} \max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) \lesssim_{\mathbb{P}} \sqrt{H} \mathbf{B}_{\text{FS}} + H^{-1/2} = o_{\mathbb{P}}(1). \quad (\text{SA-II.4.3})$$

Every $\mathcal{D}_{t-1}$ -conditional calculation below is split over $\mathcal{A}_{it,\text{FS}}$ and its complement. On the regular event, the signal displacement is bounded by the deterministic $\mathbf{B}_{\text{FS}}$. On the complement, the conditional convolution representations below, including their leave-one-out and leave-two-out versions, are bounded by the corresponding uniformly integrable density kernels multiplied by $H^{d/2} \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) = o_{\mathbb{P}}(1)$.

More explicitly, the uniform conditional density bound for β_{it} gives

$$\mathbb{P}(\mathcal{A}_{it,\text{FS}}^c, \beta_{it} \in P \mid \mathcal{D}_{t-1}) \lesssim_{\mathbb{P}} \min\{\mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}), J^{-d}\},$$

while conditional convolution on $\mathcal{A}_{it,\text{FS}}^c$ gives

$$\mathbb{P}(\mathcal{A}_{it,\text{FS}}^c, \hat{\beta}_{it} \in P \mid \mathcal{D}_{t-1}) \lesssim_{\mathbb{P}} \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) \min\left\{1, \frac{H^{d/2}}{J^d}\right\} = o_{\mathbb{P}}(J^{-d}).$$

These two bounds cover respectively $J < \sqrt{H}$ and $J \geq \sqrt{H}$. For the conditional mean, the leave-one-out Taylor coefficient and conditional convolution give the sharper exceptional-event bound

$$\frac{1}{\sqrt{H}} \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) \min\left\{1, \frac{H^{d/2}}{J^d}\right\}$$

$$\leq \frac{1}{J^d} \min \left\{ \frac{J^d}{\sqrt{H}}, H^{(d-1)/2} \right\} \mathbb{P}(\mathcal{A}_{it,\text{FS}}^c \mid \mathcal{D}_{t-1}) \lesssim_{\mathbb{P}} \frac{\mathbf{B}_{\text{FS}} + H^{-1}}{J^d} = \frac{\mathbf{B}_{\text{FS}}}{J^d} + \frac{1}{HJ^d},$$

by (SA-II.4.2). For the conditional variance, the two preceding cell-probability bounds imply that the exceptional-event contribution is absorbed by $\{J^{d-1} \max(J, \sqrt{H})\}^{-1}$. Thus the drift terms in the calculations below are evaluated on $\mathcal{A}_{it,\text{FS}}$ and are controlled by $\mathbf{B}_{\text{FS}}$, without treating $\hat{\mathbf{B}}_{\text{FS}}$ as $\mathcal{D}_{t-1}$ -measurable. To avoid repeating event indicators, the $\mathcal{A}_{it,\text{FS}}$ indicator is suppressed in the regular-event displacement bounds below; the complementary contribution is always controlled by the bounds just displayed.

Write

$$\begin{aligned} c_{itl} &= S\hat{A}(t)^{-1} X_l(t) K\left(\frac{l}{H}\right) \varsigma_{i(t-l)} \in \mathbb{R}^d, \\ c_{ts} &= S\hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \in \mathbb{R}^d. \end{aligned}$$

Define $G_{its} = \frac{1}{\sqrt{H}} \sum_{\substack{l=1 \\ l \neq s}}^H S\hat{A}(t)^{-1} X_l(t) K\left(\frac{l}{H}\right) \varsigma_{i(t-l)} \eta_{i(t-l)}$ and $\delta_{its} = \frac{1}{\sqrt{H}} c_{its} \eta_{i(t-s)}$. Let

$$F_{G_{its}}(z) = \mathbb{P} \left(\frac{1}{\sqrt{H}} \sum_{\substack{l=1 \\ l \neq s}}^H S\hat{A}(t)^{-1} X_l(t) K\left(\frac{l}{H}\right) \varsigma_{i(t-l)} \eta_{i(t-l)} \leq z \mid \mathcal{E}_{t-1}, \eta_{i(t-s)} \right)$$

be the conditional CDF of G_{its} , with $f_{G_{its}}(z)$ the corresponding conditional density. Its existence follows from the same characteristic-function argument as in Lemma SA-II.3.1. For any rectangle $[\ell, u]$, $F_{G_{its}}([\ell, u]) = \mathbb{P}(G_{its} \in [\ell, u] \mid \mathcal{E}_{t-1}, \eta_{i(t-s)}) = \mathbb{P}(G_{its} \in [\ell, u] \mid \mathcal{E}_{t-1})$ by independence of $\eta_{i(t-s)}$. Use $f_{\beta_{it}}(\cdot \mid \mathcal{D}_{t-1})$ to denote the conditional density of β_{it} given $\mathcal{D}_{t-1}$, which is possibly heterogeneous over i and t .

By independence, the (conditional) characteristic function of G_{its} can be written as

$$\varphi_{G_{its}}(z) = \prod_{\substack{l=1 \\ l \neq s}}^H \varphi_{\eta}(c_{itl}^\top z / \sqrt{H}), \quad \text{for } z \in \mathbb{R}^d$$

where $\varphi_{\eta}(z) = \mathbb{E}[e^{iz\eta}]$ is the characteristic function of a generic random variable η that follows the same distribution as η_{it} . By the inversion formula,

$$f_{G_{its}}(z) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-iu^\top z} \varphi_{G_{its}}(u) \, du, \quad \text{for } z \in \mathbb{R}^d.$$

Its derivative is

$$\nabla f_{G_{its}}(z) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (-iu) e^{-iu^\top z} \varphi_{G_{its}}(u) \, du.$$

Before we prove the main results, we need some properties of the density $f_{G_{its}}$. We summarize them in the following claim: with probability approaching one, uniformly over $i \in [n_t]$, $t = H+1, \dots, T$, $s \in [H]$, and $k = 0, 1, 2, 3$,

$$\sup_{z \in \mathbb{R}^d} \|\nabla^k f_{G_{its}}(z)\|_\infty \lesssim 1, \quad \|\nabla^k f_{G_{its}}(z)\|_\infty \lesssim \|z\|^{-d-1} \quad \text{for } \|z\| \geq 1. \quad (\text{SA-II.4.4})$$

We will prove this claim later in Step 5.

Step 2: Key expansion.

$$\begin{aligned}
& \mathbb{E} \left[\left(\mathbf{1}(\hat{\beta}_{it} \in P) - \mathbf{1}(\beta_{it} \in P) \right) (\nabla \mu_t(\beta_{it}))^\top \left(S \hat{A}(t)^{-1} X_s(t) \right) K\left(\frac{s}{H}\right) \varsigma_{i(t-s)} \eta_{i(t-s)} \middle| \mathcal{D}_{t-1} \right] \\
&= \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} \mathbb{E} \left[\mathbf{1}(\hat{\beta}_{it} \in P) \eta_{i(t-s)} \middle| \mathcal{E}_{t-1} \right] \middle| \mathcal{D}_{t-1} \right] \\
&= \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} \mathbb{E} \left[\mathbf{1} \left(\frac{1}{\sqrt{H}} \sum_{\substack{l=1 \\ l \neq s}}^H c_{itl} \eta_{i(t-l)} \in \sqrt{H} \left(P - \check{\beta}_{it} - \frac{1}{H} c_{its} \eta_{i(t-s)} \right) \right) \eta_{i(t-s)} \right. \right. \\
&\quad \left. \left. \middle| \mathcal{E}_{t-1}, \eta_{i(t-s)} \right] \middle| \mathcal{D}_{t-1} \right] \\
&= \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} \eta_{i(t-s)} \left(F_{G_{its}} \left(\sqrt{H} (P - \check{\beta}_{it}) \right) \right. \right. \\
&\quad \left. \left. + F_{G_{its}} \left(\sqrt{H} (P - \check{\beta}_{it} - \frac{1}{H} c_{its} \eta_{i(t-s)}) \right) - F_{G_{its}} \left(\sqrt{H} (P - \check{\beta}_{it}) \right) \right) \middle| \mathcal{D}_{t-1} \right] \\
&= \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} \eta_{i(t-s)} \left(F_{G_{its}} \left(\sqrt{H} (P - \check{\beta}_{it} - \frac{1}{H} c_{its} \eta_{i(t-s)}) \right) - F_{G_{its}} \left(\sqrt{H} (P - \check{\beta}_{it}) \right) \right) \middle| \mathcal{D}_{t-1} \right]
\end{aligned}$$

by the independence between $\eta_{i(t-s)}$ and variables in $\mathcal{E}_{t-1}$.

Now, by the density bound (SA-II.4.4), we further have

$$\begin{aligned}
& \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} \mathbb{E} \left[\mathbf{1}(\hat{\beta}_{it} \in P) \eta_{i(t-s)} \middle| \mathcal{E}_{t-1} \right] \middle| \mathcal{D}_{t-1} \right] \\
&= \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} \eta_{i(t-s)} \int_{\sqrt{H}(P-\check{\beta}_{it})} (f_{G_{its}}(z - \delta_{its}) - f_{G_{its}}(z)) \, dz \middle| \mathcal{D}_{t-1} \right] \\
&= \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} \eta_{i(t-s)} \left[-\delta_{its}^\top \int_{\sqrt{H}(P-\check{\beta}_{it})} \nabla f_{G_{its}}(z) \, dz \right. \right. \\
&\quad \left. \left. + \delta_{its}^\top \int_{\sqrt{H}(P-\check{\beta}_{it})} \int_0^1 \left(\nabla f_{G_{its}}(z) - \nabla f_{G_{its}}(z - v\delta_{its}) \right) \, dv \, dz \right] \middle| \mathcal{D}_{t-1} \right] \\
&= -\frac{1}{\sqrt{H}} \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} c_{its}^\top \int_{\sqrt{H}(P-\check{\beta}_{it})} \nabla f_{G_{its}}(z) \, dz \middle| \mathcal{D}_{t-1} \right] \\
&\quad + \frac{1}{H} \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} c_{its}^\top \mathbb{E} \left[\eta_{i(t-s)}^3 \int_{\sqrt{H}(P-\check{\beta}_{it})} \int_0^1 (1-v) \nabla^2 f_{G_{its}}(z - v\delta_{its}) c_{its} \, dv \, dz \middle| \mathcal{E}_{t-1} \right] \middle| \mathcal{D}_{t-1} \right] \\
&= -\frac{1}{\sqrt{H}} I_{1,s} + \frac{1}{H} I_{2,s},
\end{aligned}$$

where $I_{1,s}$ and $I_{2,s}$ denote the corresponding terms.

Step 3: Bound $I_{1,s}$.

$$\begin{aligned}
& \left| \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} c_{its}^\top \int_{\sqrt{H}(P-\check{\beta}_{it})} \nabla f_{G_{its}}(z) \, dz \middle| \mathcal{D}_{t-1} \right] \right| \\
&= \left| \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} c_{its}^\top \varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P-\check{\beta}_{it})} \nabla f_{G_{its}}(z) \, dz \middle| \mathcal{D}_{t-1} \right] \right| \\
&= |c_{ts}^\top M c_{ts}| \\
&\leq \|c_{ts}\|^2 \|M\|.
\end{aligned}$$

where

$$M = \mathbb{E} \left[\varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P-\check{\beta}_{it})} \nabla f_{G_{its}}(z) dz (\nabla \mu_t(\beta_{it}))^\top \middle| \mathcal{D}_{t-1} \right].$$

It is enough to consider each entry of M , since d is fixed.

A typical (j, k) -th entry of M is

$$\begin{aligned} & \mathbb{E} \left[\varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P-\check{\beta}_{it})} \nabla_j f_{G_{its}}(z) dz \nabla_k \mu_t(\beta_{it}) \middle| \mathcal{D}_{t-1} \right] \\ &= \mathbb{E} \left[\varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P-\beta_{it})} \nabla_j f_{G_{its}}(z) dz \nabla_k \mu_t(\beta_{it}) \middle| \mathcal{D}_{t-1} \right] \\ &+ \mathbb{E} \left[\varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P-\check{\beta}_{it})} \nabla_j f_{G_{its}}(z) dz \nabla_k \mu_t(\beta_{it}) - \varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P-\beta_{it})} \nabla_j f_{G_{its}}(z) dz \nabla_k \mu_t(\beta_{it}) \middle| \mathcal{D}_{t-1} \right]. \end{aligned}$$

For the integration by parts in b_j , write $f_{G_{its}}(z) \equiv f_{G_{its}}(z_j; z_{-j})$ and define

$$m(b_j; z_{-j}, b_{-j}) = \int_0^{\sqrt{H}(u_j - \ell_j)} f_{G_{its}}(z + \sqrt{H}(\ell - b)) dz_j.$$

On the one hand,

$$\begin{aligned} & \mathbb{E} \left[\varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P-\beta_{it})} \nabla_j f_{G_{its}}(z) dz \nabla_k \mu_t(\beta_{it}) \middle| \mathcal{D}_{t-1} \right] \\ &= \varsigma_{i(t-s)}^2 \int_{\mathcal{B}} \left(\int_0^{\sqrt{H}(u-\ell)} \nabla_j f_{G_{its}}(z + \sqrt{H}(\ell - b)) dz \right) \\ &\quad \times \nabla_k \mu_t(b) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) db \\ &= \varsigma_{i(t-s)}^2 \int_{\mathcal{B}_{-j}} \int_0^{\sqrt{H}(u_j - \ell_j)} \int_{\mathcal{B}_j} \left(\int_0^{\sqrt{H}(u_j - \ell_j)} \nabla_j f_{G_{its}}(z + \sqrt{H}(\ell - b)) dz_j \right) \\ &\quad \times \nabla_k \mu_t(b) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) db_j dz_{-j} db_{-j} \\ &= -\frac{\varsigma_{i(t-s)}^2}{\sqrt{H}} \int_{\mathcal{B}_{-j}} \int_0^{\sqrt{H}(u_j - \ell_j)} m(b_j; z_{-j}, b_{-j}) \nabla_k \mu_t(b) \\ &\quad \times f_{\beta_{it}}(b | \mathcal{D}_{t-1}) \Big|_{\underline{\beta}_{-j}}^{\bar{\beta}_j} dz_{-j} db_{-j} \\ &+ \frac{\varsigma_{i(t-s)}^2}{\sqrt{H}} \int_{\mathcal{B}_{-j}} \int_0^{\sqrt{H}(u_j - \ell_j)} \int_{\mathcal{B}_j} \left(\int_0^{\sqrt{H}(u_j - \ell_j)} f_{G_{its}}(z + \sqrt{H}(\ell - b)) dz_j \right) \\ &\quad \times \nabla_j \left(\nabla_k \mu_t(b) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) \right) db_j dz_{-j} db_{-j} \\ &= O\left(\frac{1}{\sqrt{H}J^d}\right) \\ &+ \frac{\varsigma_{i(t-s)}^2}{\sqrt{H}} \int_{\mathcal{B}_{-j}} \int_0^{\sqrt{H}(u_j - \ell_j)} \int_{\mathcal{B}_j} \left(\int_0^{\sqrt{H}(u_j - \ell_j)} f_{G_{its}}(z + \sqrt{H}(\ell - b)) dz_j \right) \\ &\quad \times \nabla_j \left(\nabla_k \mu_t(b) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) \right) db_j dz_{-j} db_{-j} \\ &= O\left(\frac{1}{\sqrt{H}J^d}\right) + \frac{\varsigma_{i(t-s)}^2}{\sqrt{H}} \int_{\sqrt{H}(P-\mathcal{B})} f_{G_{its}}(z) \left(\int_{\mathcal{B} \cap (P-z/\sqrt{H})} \nabla_j \left(\nabla_k \mu_t(b) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) \right) db \right) dz \end{aligned}$$

$$\lesssim_{\mathbb{P}} \frac{1}{\sqrt{H}J^d}$$

where the inner integral in the last display is $\lesssim_{\mathbb{P}} J^{-d}$ uniformly. The above calculation uses the bounded derivatives of μ_t and the bounded Lipschitz conditional density $f_{\beta_{it}}(\cdot|\mathcal{D}_{t-1})$; the integration-by-parts step is understood coordinatewise using the corresponding bounded weak derivative. We also use that $[\ell, u]$ is at least of order $\log(nT)/\sqrt{H}$ away from the boundary of $\mathcal{B}$. To see the boundary term, define also

$$\begin{aligned} \tilde{m}(b_j, z_{-j}) &= \int_0^{\sqrt{H}(u_j - \ell_j)} f_{G_{its}}(z_j + \sqrt{H}(\ell_j - b_j); z_{-j}) dz_j. \\ &- \frac{1}{\sqrt{H}} \int_{\mathcal{B}_{-j}} \int_0^{\sqrt{H}(u_j - \ell_j)} m(b_j; z_{-j}, b_{-j}) \nabla_k \mu_t(b) f_{\beta_{it}}(b|\mathcal{D}_{t-1}) \Big|_{\underline{\beta}_j}^{\bar{\beta}_j} dz_{-j} db_{-j} \\ &= \frac{1}{\sqrt{H}} \int_{\sqrt{H}(P_{-j} - \mathcal{B}_{-j})} \int_{\mathcal{B}_{-j} \cap (P_{-j} - z_{-j}/\sqrt{H})} \left[\tilde{m}(\underline{\beta}_j, z_{-j}) \nabla_k \mu_t(\underline{\beta}_j, b_{-j}) f_{\beta_{it}}(\underline{\beta}_j, b_{-j}|\mathcal{D}_{t-1}) \right. \\ &\quad \left. - \tilde{m}(\bar{\beta}_j, z_{-j}) \nabla_k \mu_t(\bar{\beta}_j, b_{-j}) f_{\beta_{it}}(\bar{\beta}_j, b_{-j}|\mathcal{D}_{t-1}) \right] db_{-j} dz_{-j} \\ &\lesssim \frac{1}{\sqrt{H}} \left(\int_{\sqrt{H}(P_{-j} - \mathcal{B}_{-j})} \int_0^{\sqrt{H}(u_j - \ell_j)} f_{G_{its}}(z_j + \sqrt{H}(\ell_j - \underline{\beta}_j); z_{-j}) dz_j dz_{-j} \right. \\ &\quad \left. + \int_{\sqrt{H}(P_{-j} - \mathcal{B}_{-j})} \int_0^{\sqrt{H}(u_j - \ell_j)} f_{G_{its}}(z_j + \sqrt{H}(\ell_j - \bar{\beta}_j); z_{-j}) dz_j dz_{-j} \right) \frac{1}{J^{d-1}} \\ &\lesssim \frac{1}{\sqrt{H}J^{d-1}} \mathbb{P}(|G_{its,j}| > \tau_n | \mathcal{E}_{t-1}) \lesssim_{\mathbb{P}} \frac{1}{\sqrt{H}J^d}, \end{aligned}$$

where $G_{its,j}$ denotes the j -th coordinate of G_{its} and $\tau_n \gtrsim \log(nT)$ denotes the distance between $[\ell, u]$ and the boundary of the support of β_{it} . The last inequality follows from Bernstein's inequality and the usual truncation argument: on an $\mathcal{E}_{t-1}$ -measurable event that happens w.p.a.1, each innovation is truncated at M and recentered before applying Bernstein, while Chebyshev's inequality is applied to the centered complementary tail sum.

$$\mathbb{P}(|G_{its,j}| \geq \tau_n | \mathcal{E}_{t-1}) \leq \exp\left(-\frac{0.5\tau_n^2}{M_0 + \tau_n M/3\sqrt{H}}\right) + \frac{1}{\tau_n^2 M^{q-2}} \lesssim J^{-1} \quad \text{for } M \asymp J^{\frac{1}{q-2}}.$$

because $J^{1/(q-2)}T^{1/q}/\sqrt{H} = o(1)$ and $\tau_n \gtrsim \log(nT)$.

On the other hand, we will use a partitioning argument to bound the remainder in $I_{1,s}$. Specifically, if $J \gtrsim H^{1/2}$, we can introduce a multi-index $m \in \mathbb{Z}^d$ and $\hbar \asymp H^{-1/2}$ and define $\mathcal{B}_m = \{b \in \mathcal{B} : b_j \in [\ell_j + m_j \hbar, \ell_j + (m_j + 1)\hbar], j \in [d]\}$. Denote this collection of subrectangles by $\{\mathcal{B}_m : m \in \mathcal{M}\}$ for an index set $\mathcal{M}$. Since $\mathcal{B}$ is a bounded rectangle, we need at most a constant multiple of $H^{d/2}$ such subrectangles to cover the entire $\mathcal{B}$, i.e., $|\mathcal{M}| \lesssim H^{d/2}$. Then, we have

$$\begin{aligned} &\mathbb{E} \left[\varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P - \check{\beta}_{it})} \nabla_j f_{G_{its}}(z) dz \nabla_k \mu_t(\beta_{it}) - \varsigma_{i(t-s)}^2 \int_{\sqrt{H}(P - \beta_{it})} \nabla_j f_{G_{its}}(z) dz \nabla_k \mu_t(\beta_{it}) \Big| \mathcal{D}_{t-1} \right] \\ &= \varsigma_{i(t-s)}^2 \mathbb{E} \left[\int_{\sqrt{H}(P - \beta_{it})} \left(\nabla_j f_{G_{its}}(z + \sqrt{H}(\beta_{it} - \check{\beta}_{it})) - \nabla_j f_{G_{its}}(z) \right) dz \nabla_k \mu_t(\beta_{it}) \Big| \mathcal{D}_{t-1} \right] \end{aligned}$$

$$\begin{aligned}
&\lesssim \sum_{m \in \mathcal{M}} \mathbb{E} \left[\mathbf{1}(\beta_{it} \in \mathcal{B}_m) \left| \int_{\sqrt{H}(P-\beta_{it})} \left(\nabla_j f_{G_{its}}(z + \sqrt{H}(\beta_{it} - \check{\beta}_{it})) - \nabla_j f_{G_{its}}(z) \right) dz \right| \middle| \mathcal{D}_{t-1} \right] \\
&\lesssim \frac{\sqrt{H} \mathbf{B}_{\text{FS}}}{J^d}.
\end{aligned}$$

In the last step we use the following fact: for the first term,

$$\begin{aligned}
&\sum_{m \in \mathcal{M}} \mathbb{E} \left[\mathbf{1}(\beta_{it} \in \mathcal{B}_m) \left| \int_{\sqrt{H}(P-\beta_{it})} \left(\nabla_j f_{G_{its}}(z - \sqrt{H}(\check{\beta}_{it} - \beta_{it})) - \nabla_j f_{G_{its}}(z) \right) dz \right| \middle| \mathcal{D}_{t-1} \right] \\
&= \sum_{m \in \mathcal{M}} \mathbb{E} \left[\mathbf{1}(\beta_{it} \in \mathcal{B}_m) \left| \int_{\sqrt{H}(P-\beta_{it})} [\nabla(\nabla_j f_{G_{its}})(\xi_z)]^\top (\sqrt{H}(\beta_{it} - \check{\beta}_{it})) dz \right| \middle| \mathcal{D}_{t-1} \right] \\
&\lesssim \sum_{m \in \mathcal{M}} (1 + \|m\|)^{-d-1} (\sqrt{H}/J)^d \sqrt{H} \mathbf{B}_{\text{FS}} \mathbb{E} \left[\mathbf{1}(\beta_{it} \in \mathcal{B}_m) \middle| \mathcal{D}_{t-1} \right] \lesssim \frac{H^{d/2}}{J^d} \frac{1}{H^{d/2}} \sqrt{H} \mathbf{B}_{\text{FS}}
\end{aligned}$$

where ξ_z is some point between z and $z - \sqrt{H}(\check{\beta}_{it} - \beta_{it})$ and we use the tail bound for $\nabla^2 f_{G_{its}}$.

Now, consider the case $J = o(H^{1/2})$. We slightly redefine the partition: divide the whole support $\mathcal{B}$ into rectangles of length proportional to $H^{-1/2}$, labelling these regions by a multi-index

$$m = (m_1, m'_1, m_2, m'_2, \dots, m_d, m'_d) \in \mathbb{Z}^{2d}.$$

Define $\mathcal{B}_m$ as the set of $b \in \mathcal{B}$ such that $\sqrt{H}(b_k - \ell_k) \in [m_k, m_k + 1]$ and $\sqrt{H}(b_k - u_k) \in [m'_k, m'_k + 1]$ for all $k \in [d]$. That is, when a rectangle is associated with a multi-index with some m_k or m'_k large in absolute value, this region is far away from the facets of the hyper-rectangle $[\ell, u]$. Let $\mathcal{M}$ be the collection of possible indices m . Again, $|\mathcal{M}| \lesssim H^{d/2}$. For each $\mathcal{B}_m$, let $\mathcal{J}_m = \{j \in [d] : \exists b \in \mathcal{B}_m \text{ s.t. } b_j \in [\ell_j, u_j]\}$ and $\mathcal{J}_m^c = [d] \setminus \mathcal{J}_m$. For $\beta_{it} \in \mathcal{B}_m$, we have

$$\int_{\sqrt{H}(P-\beta_{it})} \|\nabla(\nabla_j f_{G_{its}})(\xi_z)\|_\infty dz \lesssim \frac{1}{\prod_{j \in \mathcal{J}_m^c} (1 + |\min\{m_j, m'_j\}|)^{1+1/d}} \times \left(\frac{\sqrt{H}}{J} \right)^{|\mathcal{J}_m^c|}.$$

Note that for a fixed $j_0 \in [d]$, all $\mathcal{B}_m$'s such that $|\mathcal{J}_m| = j_0$ can be classified into $\lesssim (\sqrt{H}/J)^{j_0}$ groups, each containing $\lesssim \sqrt{H}$ subrectangles. Repeating the previous argument, we have

$$\begin{aligned}
&\sum_{m \in \mathcal{M}} \mathbb{E} \left[\mathbf{1}(\beta_{it} \in \mathcal{B}_m) \left| \int_{\sqrt{H}(P-\beta_{it})} [\nabla(\nabla_j f_{G_{its}})(\xi_z)]^\top (\sqrt{H}(\beta_{it} - \check{\beta}_{it})) dz \right| \middle| \mathcal{D}_{t-1} \right] \\
&\lesssim \sum_{j_0=1}^d \left(\frac{\sqrt{H}}{J} \right)^{j_0} \times \left(\frac{\sqrt{H}}{J} \right)^{d-j_0} \sqrt{H} \mathbf{B}_{\text{FS}} \times \frac{1}{H^{d/2}} \lesssim \frac{\sqrt{H} \mathbf{B}_{\text{FS}}}{J^d}.
\end{aligned}$$

Then, we conclude that

$$\frac{1}{H} \sum_{s=1}^H I_{1,s} \lesssim \mathbb{P} \frac{1}{\sqrt{H} J^d} + \frac{\sqrt{H} \mathbf{B}_{\text{FS}}}{J^d}.$$

Step 4: Bound $I_{2,s}$. We use a similar argument:

$$\begin{aligned} & \mathbb{E} \left[(\nabla \mu_t(\beta_{it}))^\top c_{its} c_{its}^\top \mathbb{E} \left[\eta_{i(t-s)}^3 \int_{\sqrt{H}(P-\check{\beta}_{it})} \int_0^1 (1-v) \nabla^2 f_{G_{its}}(z - v\delta_{its}) c_{its} dv dz \middle| \mathcal{E}_{t-1} \right] \middle| \mathcal{D}_{t-1} \right] \\ &= c_{ts}^\top \mathbb{E} \left[\eta_{i(t-s)}^3 \varsigma_{i(t-s)}^3 \int_{\sqrt{H}(P-\check{\beta}_{it})} \int_0^1 (1-v) \nabla^2 f_{G_{its}}(z - v\delta_{its}) c_{ts} dv dz (\nabla \mu_t(\beta_{it}))^\top \middle| \mathcal{D}_{t-1} \right] c_{ts}. \end{aligned}$$

Again, we only need to bound the operator norm of the matrix in the middle. Furthermore,

$$\begin{aligned} & \text{vec} \left(\mathbb{E} \left[\eta_{i(t-s)}^3 \varsigma_{i(t-s)}^3 \int_{\sqrt{H}(P-\check{\beta}_{it})} \int_0^1 (1-v) \nabla^2 f_{G_{its}}(z - v\delta_{its}) c_{ts} dv dz (\nabla \mu_t(\beta_{it}))^\top \middle| \mathcal{D}_{t-1} \right] \right) \\ &= \mathbb{E} \left[\nabla \mu_t(\beta_{it}) \otimes \eta_{i(t-s)}^3 \varsigma_{i(t-s)}^3 \int_{\sqrt{H}(P-\check{\beta}_{it})} \int_0^1 (1-v) \nabla^2 f_{G_{its}}(z - v\delta_{its}) dv dz \middle| \mathcal{D}_{t-1} \right] c_{ts}. \end{aligned}$$

We only need to bound each entry of the expected matrix (a $d^2 \times d$ matrix). Consider a generic entry: for some j, j', j'' ,

$$\begin{aligned} & \mathbb{E} \left[\nabla_j \mu_t(\beta_{it}) \eta_{i(t-s)}^3 \varsigma_{i(t-s)}^3 \int_{\sqrt{H}(P-\check{\beta}_{it})} \int_0^1 (1-v) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) dv dz \middle| \mathcal{D}_{t-1} \right] \\ &= \varsigma_{i(t-s)}^3 \int_{\mathcal{B}} \nabla_j \mu_t(b) \int_{\sqrt{H}(P-b)} \mathbb{E} \left[\eta_{i(t-s)}^3 \int_0^1 (1-v) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) dv \middle| \mathcal{E}_{t-1} \right] dz f_{\beta_{it}}(b | \mathcal{D}_{t-1}) db \\ &+ \mathbb{E} \left[\nabla_j \mu_t(\beta_{it}) \left(\varsigma_{i(t-s)}^3 \int_{\sqrt{H}(P-\check{\beta}_{it})} \eta_{i(t-s)}^3 \int_0^1 (1-v) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) dv dz \right. \right. \\ &\quad \left. \left. - \varsigma_{i(t-s)}^3 \int_{\sqrt{H}(P-\beta_{it})} \eta_{i(t-s)}^3 \int_0^1 (1-v) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) dv dz \right) \middle| \mathcal{D}_{t-1} \right]. \end{aligned}$$

Let g_η denote the Lebesgue density of η_{it} and let $M_0 > 0$ be fixed. For the first term,

$$\begin{aligned} & \int_{\mathcal{B}} \nabla_j \mu_t(b) \int_{\sqrt{H}(P-b)} \mathbb{E} \left[\eta_{i(t-s)}^3 \int_0^1 (1-v) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) dv \middle| \mathcal{E}_{t-1} \right] dz f_{\beta_{it}}(b | \mathcal{D}_{t-1}) db \\ &= \int_{\mathbb{R}} \eta^3 g_\eta(\eta) \int_0^1 (1-v) \int_{\mathcal{B}} \int_{\sqrt{H}(P-b)} \nabla_j \mu_t(b) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) dz db \times \\ &\quad \left(\mathbb{1}(\|c_{ts}\| |\eta| \leq M_0 \sqrt{H}) + \mathbb{1}(\|c_{ts}\| |\eta| > M_0 \sqrt{H}) \right) dv d\eta \\ &\lesssim_{\mathbb{P}} J^{-d}. \end{aligned}$$

Note that for $\|c_{ts}\| |\eta| \leq M_0 \sqrt{H}$, $\|v\delta_{its}\| \lesssim \|c_{ts}\| |\eta| / \sqrt{H} \lesssim 1$, and thus in the last line we apply the previous partitioning argument again to obtain

$$\begin{aligned} & \left| \int_{\mathcal{B}} \int_{\sqrt{H}(P-b)} \nabla_j \mu_t(b) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) dz db \mathbb{1}(\|c_{ts}\| |\eta| \leq M_0 \sqrt{H}) \right| \\ &\lesssim \int_{\mathcal{B}} \int_{\sqrt{H}(P-b)} |\nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its})| db dz \mathbb{1}(\|c_{ts}\| |\eta| \leq M_0 \sqrt{H}) \\ &= \sum_{m \in \mathcal{M}} \int_{\mathcal{B}_m} \int_{\sqrt{H}(P-b)} |\nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its})| \mathbb{1}(\|c_{ts}\| |\eta| \leq M_0 \sqrt{H}) db dz \\ &\lesssim J^{-d}. \end{aligned}$$

On the other hand,

$$\begin{aligned}
& \int_{\mathcal{B}} \nabla_j \mu_t(b) \int_{\sqrt{H}(P-b)} \mathbb{E} \left[\eta_{i(t-s)}^3 \int_0^1 (1-v) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) dv \right. \\
& \quad \left. \times \mathbf{1}(\|c_{ts}\| |\eta_{i(t-s)}| > M_0 \sqrt{H}) \middle| \mathcal{E}_{t-1} \right] dz f_{\beta_{it}}(b | \mathcal{D}_{t-1}) db \\
& \lesssim \mathbb{E}[|\eta_{i(t-s)}|^{3+d}] \|c_{ts}\|^d H^{-d/2} \times \left(\frac{\sqrt{H}}{J} \right)^d \\
& \lesssim \|c_{ts}\|^d J^{-d}.
\end{aligned}$$

In addition, repeating the previous argument in Step 3 yields

$$\begin{aligned}
& \mathbb{E} \left[\nabla_j \mu_t(\beta_{it}) \left(\varsigma_{i(t-s)}^3 \int_{\sqrt{H}(P-\beta_{it})} \eta_{i(t-s)}^3 \int_0^1 (1-v) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) dv dz \right. \right. \\
& \quad \left. \left. - \varsigma_{i(t-s)}^3 \int_{\sqrt{H}(P-\beta_{it})} \eta_{i(t-s)}^3 \int_0^1 (1-v) \nabla_{j',j''}^2 f_{G_{its}}(z - v\delta_{its}) dv dz \right) \middle| \mathcal{D}_{t-1} \right] \\
& \lesssim \frac{\sqrt{H} \mathbf{B}_{\text{FS}}}{J^d}.
\end{aligned}$$

Therefore, we conclude that

$$\frac{1}{H} \sum_{s=1}^H I_{2,s} \lesssim_{\mathbb{P}} \frac{1}{J^d}.$$

Also, note that in the above calculation we need $\frac{1}{H} \sum_{s=1}^H \|c_{ts}\|^{3+d} \lesssim_{\mathbb{P}} 1$.

Step 5: Density bounds for G_{its} . We prove (SA-II.4.4).

Step 5.1: Factor moments and projection lower bound. The density bounds use two properties, both holding with probability approaching one. First,

$$\max_{H+1 \leq t \leq T} \frac{1}{H} \sum_{s=1}^H |f_{t-s}|^m \lesssim 1, \quad m = 3 + d.$$

This is guaranteed by the imposed moment conditions and the maximal Fuk–Nagaev inequality described in Lemma SA-II.2.2. The process $|f_{t-s}|^m$ is $L_{q/m}$ -NED with $q/m > 2$, so

$$\max_{H+1 \leq t \leq T} \left| \frac{1}{H} \sum_{s=1}^H (|f_{t-s}|^m - \mathbb{E}[|f_{t-s}|^m]) \right| = o_{\mathbb{P}}(1)$$

by the same argument as in the proof of Lemma SA-II.2.2. Second, for $v \in \mathbb{R}^d$, $m_v = |\{l : |v^\top c_{itl}| \geq a\|v\|\}| \geq \pi_0(H-1)$ for a constant $a > 0$ and a positive sequence π_0 with $\pi_0 H \rightarrow \infty$. To see this, note that $\frac{1}{H} \sum_{l=1}^H |v^\top c_{itl}|^2 / \|v\|^2 \geq \lambda_{\min}(\frac{1}{H} \sum_{l=1}^H c_{itl} c_{itl}^\top) \geq \underline{\lambda}$ for some $\underline{\lambda}$, w.p.a.1. Let $a = \sqrt{\underline{\lambda}/2}$. We have

$$\frac{1}{H-1} \sum_{\substack{l=1 \\ l \neq s}}^H |v^\top c_{itl}|^2 / \|v\|^2 \leq a^2 \left(1 - \frac{m_v}{H-1} \right) + \frac{\bar{c}_f^2 m_v}{H-1} = a^2 + \frac{(\bar{c}_f^2 - a^2) m_v}{H-1},$$

which implies that $\frac{m_u}{H-1} \geq \pi_0$ with $\pi_0 = \frac{\lambda/2}{\bar{c}_f^2 - \lambda/2}$. Given the moment condition imposed, $\bar{c}_f \lesssim_{\mathbb{P}} T^{1/q}$. Therefore, under the rate condition $T^{2/q}/H = o(1)$, we can guarantee $\pi_0 H \rightarrow \infty$.

Step 5.2: Uniform boundedness of density derivatives. Use ∇^α to denote the partial derivative of order $\alpha \in \mathbb{Z}_+^d$. Note that

$$\sup_{z \in \mathbb{R}^d} |\nabla^\alpha f_{G_{its}}(z)| = \sup_{z \in \mathbb{R}^d} \left| \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} (-iu)^\alpha e^{-iu^\top z} \varphi_{G_{its}}(u) du \right| \lesssim \int_{\mathbb{R}^d} \|u\|^{|\alpha|} |\varphi_{G_{its}}(u)| du.$$

The assumption $g_\eta^{(1)} \in L^1(\mathbb{R})$ implies that $\varphi_\eta(u) \rightarrow 0$ and $|\varphi_\eta(u)| \lesssim |u|^{-1}$ as $|u| \rightarrow \infty$. Together with the second-order expansion of φ_η at the origin, there are constants $c_\eta, C_\eta > 0$ and $B_\eta < \infty$ such that, for all $x \in \mathbb{R}$,

$$|\varphi_\eta(x)| \leq \exp\{-c_\eta \min(x^2, 1)\}, \quad |\varphi_\eta(x)| \leq \frac{C_\eta}{|x|} \quad \text{if } |x| \geq B_\eta.$$

Let $\bar{c}_f = \max_{itl} \|c_{itl}\|$ and $z_l(u) = c_{itl}^\top u / \sqrt{H}$. For $\|u\| \leq B\sqrt{H}$ with fixed $B < \infty$, the first bound gives

$$|\varphi_{G_{its}}(u)| \leq \exp \left\{ -c_\eta \sum_{\substack{l=1 \\ l \neq s}}^H \min(z_l(u)^2, 1) \right\}.$$

Since $\lambda_{\min}(H^{-1} \sum_{\substack{l=1 \\ l \neq s}}^H c_{itl} c_{itl}^\top) \gtrsim 1$ w.p.a.1 and $\max_l z_l(u)^2 \leq \bar{c}_f^2 \|u\|^2 / H$,

$$\sum_{\substack{l=1 \\ l \neq s}}^H \min(z_l(u)^2, 1) \gtrsim \frac{\sum_{\substack{l=1 \\ l \neq s}}^H z_l(u)^2}{1 + \max_{\substack{l=1 \\ l \neq s}}^H z_l(u)^2} \gtrsim \frac{\|u\|^2}{1 + \bar{c}_f^2 \|u\|^2 / H} \gtrsim \min \left\{ \|u\|^2, \frac{H}{\bar{c}_f^2} \right\}.$$

Hence

$$\int_{\|u\| \leq B\sqrt{H}} \|u\|^{|\alpha|} |\varphi_{G_{its}}(u)| du \lesssim 1 + H^{(d+|\alpha|)/2} \exp\{-cH/\bar{c}_f^2\} \lesssim_{\mathbb{P}} 1,$$

using $\bar{c}_f \lesssim_{\mathbb{P}} T^{1/q}$ and the imposed rate condition $T^{2(3+d)/q}/H = o(1)$.

It remains to consider $\|u\| > B\sqrt{H}$ for a fixed B large enough. Write $v = u/\sqrt{H}$ and let $\mathcal{I}(u) = \{l : |v^\top c_{itl}| \geq a\|v\|\}$. By Step 5.1, $|\mathcal{I}(u)| \geq \pi_0 H$ for some $a > 0$ and a positive sequence π_0 satisfying $\pi_0 H \rightarrow \infty$ w.p.a.1. For $l \in \mathcal{I}(u)$, $|z_l(u)| \geq a\|v\|$. Choosing B so that $aB \geq B_\eta$, the tail bound on φ_η yields

$$\int_{\|u\| > B\sqrt{H}} \|u\|^{|\alpha|} |\varphi_{G_{its}}(u)| du \lesssim H^{\frac{d+|\alpha|}{2}} \int_{\|v\| > B} \|v\|^{|\alpha|} \left(\frac{C_\eta}{a\|v\|} \right)^{\pi_0 H} dv \lesssim_{\mathbb{P}} 1,$$

for $\pi_0 H$ sufficiently large. This proves the boundedness of the density and its derivatives.

Step 5.3: Polynomial tail bound for density derivatives. Fix $\|z\| \geq 1$ and choose $r \in [d]$ such that $|z_r| \geq \|z\|/\sqrt{d}$. Let $\gamma = (d+1)e_r$ and define $\Psi(u) = (-iu)^\alpha \varphi_{G_{its}}(u)$. By integration by parts in the r -th coordinate,

$$|z_r|^{d+1} |f_{G_{its}}^{(\alpha)}(z)| \leq \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \left| \frac{\partial^{d+1}}{\partial u_r^{d+1}} \Psi(u) \right| du.$$

Thus it suffices to show that the last integral is finite uniformly, because then $|f_{G_{its}}^{(\alpha)}(z)| \lesssim \|z\|^{-d-1}$.

Note that

$$[u^\alpha \varphi_{G_{its}}(u)]^{(\gamma)} = \sum_{\gamma_1 \leq \gamma} \binom{\gamma}{\gamma_1} (u^\alpha)^{(\gamma_1)} (\varphi_{G_{its}}(u))^{(\gamma - \gamma_1)}$$

where $\gamma_1 \leq \gamma$ means $\gamma_{1,j} \leq \gamma_j$ for every $j \in [d]$ and $\binom{\gamma}{\gamma_1} = \prod_{j \in [d]} \binom{\gamma_j}{\gamma_{1,j}}$. The derivative of u^α is still polynomial in u . For the derivative of $\varphi_{G_{its}}$, the product rule, the mean-zero expansion of φ_η around the origin, and the moment condition on η_{it} imply the following envelope for every fixed multi-index τ with $|\tau| \leq d + 1 + |\alpha|$:

$$|\varphi_{G_{its}}^{(\tau)}(u)| \lesssim_{\mathbb{P}} (1 + \|u\|)^{|\tau|} \max_{\substack{A \subset \{1, \dots, H\} \\ |A| \leq |\tau|}} \prod_{l \notin A} \left| \varphi_\eta \left(\frac{u^\top c_{itl}}{\sqrt{H}} \right) \right|.$$

The finite set A only removes a fixed number of factors from the product, so the bounds in Step 5.2 continue to hold after replacing H by $H - |A|$. Hence, for a constant $M_{\alpha d} < \infty$ that depends only on d and $|\alpha|$,

$$|\Psi^{(\gamma)}(u)| \lesssim_{\mathbb{P}} (1 + \|u\|)^{M_{\alpha d}} \max_{\substack{A \subset \{1, \dots, H\} \\ |A| \leq d + 1 + |\alpha|}} \prod_{l \notin A} \left| \varphi_\eta \left(\frac{u^\top c_{itl}}{\sqrt{H}} \right) \right|.$$

On $\|u\| \leq B\sqrt{H}$, the same Gaussian bound as in Step 5.2 applies, with H reduced by at most a fixed constant; on $\|u\| > B\sqrt{H}$, Step 5.1 gives at least $\pi_0 H$ minus a fixed constant projected arguments larger than $a\|u\|/\sqrt{H}$. Therefore, for some fixed $\bar{C} < \infty$,

$$\int_{\mathbb{R}^d} |\Psi^{(\gamma)}(u)| du \lesssim_{\mathbb{P}} 1 + H^{(d+M_{\alpha d})/2} \int_{\|v\| > B} (1 + \|v\|)^{M_{\alpha d}} \left(\frac{C_\eta}{a\|v\|} \right)^{\pi_0 H - \bar{C}} dv \lesssim_{\mathbb{P}} 1.$$

This proves the polynomial tail bound in (SA-II.4.4).

Step 6: Localized-basis extension. Fix a scalar component of $p_q(\cdot, P)$; the vector statement follows because the number of components is fixed. Conditional on $\mathcal{E}_{t-1}$ and $\eta_{i(t-s)}$, write

$$\hat{\beta}_{it} = \check{\beta}_{it} + H^{-1/2} G_{its} + H^{-1} c_{its} \eta_{i(t-s)}.$$

The term containing $p_q(\beta_{it}, P)$ has conditional mean zero after multiplication by $\eta_{i(t-s)}$. For the generated-regressor term, define the truncated moment functional

$$M_P(a; b_0) = \int_P p_q(v, P) H^{d/2} f_{G_{its}}(\sqrt{H}(v - b_0 - a)) dv.$$

Then the conditional expectation is the expectation, over β_{it} and $\eta_{i(t-s)}$, of a bounded predictable multiplier times

$$\eta_{i(t-s)} \left[M_P \left(H^{-1} c_{its} \eta_{i(t-s)}; \check{\beta}_{it} \right) - M_P(0; \check{\beta}_{it}) \right].$$

For each fixed component, $M_P(a; b_0)$ is differentiable in a and

$$\nabla_a M_P(a; b_0) = -H^{(d+1)/2} \int_P p_q(v, P) \nabla f_{G_{its}}(\sqrt{H}(v - b_0 - a)) dv.$$

Thus the first-order expansion in a differentiates the density $f_{G_{its}}$, not the normalized polynomial basis. Equivalently,

$$M_P\left(H^{-1}c_{its}\eta_{i(t-s)}; \check{\beta}_{it}\right) - M_P(0; \check{\beta}_{it}) = H^{-1}\eta_{i(t-s)}c_{its}^\top \int_0^1 \nabla_a M_P\left(rH^{-1}c_{its}\eta_{i(t-s)}; \check{\beta}_{it}\right) dr.$$

To make explicit why no derivative of the basis appears, let $a_k(b) = [\nabla\mu_t(b)]_k f_{\beta_{it}}(b|\mathcal{D}_{t-1})$ denote a generic bounded smooth multiplier. For any coordinate r ,

$$\begin{aligned} & H^{(d+1)/2} \int_{\mathcal{B}} \int_P p_{\mathbf{q}}(v, P) \partial_r f_{G_{its}}\left(\sqrt{H}(v-b-a)\right) a_k(b) dv db \\ &= H^{d/2} \int_P p_{\mathbf{q}}(v, P) \int_{\mathcal{B}} f_{G_{its}}\left(\sqrt{H}(v-b-a)\right) \partial_r a_k(b) db dv + \mathcal{R}_{\partial, P}. \end{aligned}$$

The boundary term $\mathcal{R}_{\partial, P}$ is negligible uniformly over the cells in the lemma because P is separated from the boundary of $\mathcal{B}$ by more than $\log(nT)/\sqrt{H}$ and the same innovation truncation used above controls the tails of G_{its} . The main term is $\lesssim_{\mathbb{P}} J^{-d}$ because $p_{\mathbf{q}}(\cdot, P)$, $\partial_r a_k$, and the conditional convolution density are uniformly bounded and $\text{vol}(P) \asymp J^{-d}$. Multiplying by the increment $H^{-1}c_{its}\eta_{i(t-s)}$ gives $\lesssim_{\mathbb{P}} (HJ^d)^{-1}$, with no additional derivative factor in J . Replacing $\check{\beta}_{it}$ by β_{it} in the density argument gives the same deterministic-drift term as before, namely $\lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}}/J^d$. Averaging over s gives the displayed bound for the localized basis. When $\mathbf{q} = 0$, this argument coincides with the switching-bias calculation above. $\square$

SA-II.4.5 Proof of Lemma SA-II.4.1(b)

The notation is as in part (a). We first prove the switching-variance bound, which is the case $\mathbf{q} = 0$, and then add the same-cell polynomial-coordinate movement that arises when $\mathbf{q} \geq 1$. As in (SA-II.4.3), every $\mathcal{D}_{t-1}$ -conditional moment below is split over $\mathcal{A}_{it, \text{FS}}$ and $\mathcal{A}_{it, \text{FS}}^c$. The complement is absorbed by the displayed baseline variance terms using Assumption SA-FSB; on $\mathcal{A}_{it, \text{FS}}$, the signal displacement is bounded by $\mathbf{B}_{\text{FS}}$. Define $\tilde{c}_{its} = [\nabla\mu_t(\beta_{it})]^\top c_{its}$. Let $\tilde{P} = \sqrt{H}[\ell - \check{\beta}_{it}, u - \check{\beta}_{it}]$ and $\delta_{its} = H^{-1/2}c_{its}\eta_{i(t-s)}$.

Step 1: Switching summand variances. Note that

$$\begin{aligned} & \mathbb{V}\text{ar} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \sum_{s=1}^H \tilde{c}_{its}\eta_{i(t-s)} \middle| \mathcal{D}_{t-1} \right] \\ &= \sum_{s=1}^H \mathbb{V}\text{ar} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \tilde{c}_{its}\eta_{i(t-s)} \middle| \mathcal{D}_{t-1} \right] \\ &+ \sum_{\substack{s, s'=1 \\ s \neq s'}}^H \text{Cov} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \tilde{c}_{its}\eta_{i(t-s)}, \left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \tilde{c}_{its'}\eta_{i(t-s')} \middle| \mathcal{D}_{t-1} \right] \end{aligned}$$

For the variance of each summand, first note the crude bound

$$\mathbb{V}\text{ar} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \tilde{c}_{its}\eta_{i(t-s)} \middle| \mathcal{D}_{t-1} \right] \leq \mathbb{E} \left[\tilde{c}_{its}^2 \eta_{i(t-s)}^2 \middle| \mathcal{D}_{t-1} \right] \lesssim \|c_{ts}\|^2,$$

using that the squared indicator difference is bounded by one. The last inequality uses the bounded gradient and scale to obtain $|\tilde{c}_{its}| \lesssim \|c_{ts}\|$, together with the $\mathcal{D}_{t-1}$ -measurability of c_{ts} and $\mathbb{E}[\eta_{i(t-s)}^2 | \mathcal{D}_{t-1}] = 1$. The sharper bound uses the leave-one-out expansion rather than multiplying dependent conditional expectations.

On the other hand, we also have the following bound

$$\begin{aligned}
& \mathbb{V}\text{ar} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \tilde{c}_{its} \eta_{i(t-s)} \middle| \mathcal{D}_{t-1} \right] \\
& \lesssim \|c_{ts}\|^2 \mathbb{E} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right)^2 \eta_{i(t-s)}^2 \middle| \mathcal{D}_{t-1} \right] \\
& = \|c_{ts}\|^2 \mathbb{E} \left[\mathbb{E} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right)^2 \eta_{i(t-s)}^2 \middle| \mathcal{E}_{t-1}, \eta_{i(t-s)} \right] \middle| \mathcal{D}_{t-1} \right] \\
& = \|c_{ts}\|^2 \mathbb{E} \left[\eta_{i(t-s)}^2 \left(F_{G_{its}}(\tilde{P} - \delta_{its}) - 2\mathbb{1}(\beta_{it} \in P) F_{G_{its}}(\tilde{P} - \delta_{its}) + \mathbb{1}(\beta_{it} \in P) \right) \middle| \mathcal{D}_{t-1} \right] \\
& = \|c_{ts}\|^2 \mathbb{E} \left[\eta_{i(t-s)}^2 \left[\left(F_{G_{its}}(\tilde{P}) \right. \right. \right. \\
& \quad \left. \left. \left. + \int_{\tilde{P}} \left(-\nabla f_{G_{its}}(z)^\top \delta_{its} + \frac{1}{2} \delta_{its}^\top \nabla^2 f_{G_{its}}(\xi_{its}) \delta_{its} \right) dz \right) (1 - 2\mathbb{1}(\beta_{it} \in P)) + \mathbb{1}(\beta_{it} \in P) \right] \middle| \mathcal{D}_{t-1} \right] \\
& \lesssim \|c_{ts}\|^2 \mathbb{E} \left[\eta_{i(t-s)}^2 \left(F_{G_{its}}(\tilde{P})(1 - 2\mathbb{1}(\beta_{it} \in P)) + \mathbb{1}(\beta_{it} \in P) \right) \middle| \mathcal{D}_{t-1} \right] \\
& \quad - \frac{\|c_{ts}\|^2}{\sqrt{H}} \mathbb{E} \left[\int_{\tilde{P}} \nabla f_{G_{its}}(z)^\top dz c_{its} (1 - 2\mathbb{1}(\beta_{it} \in P)) \eta_{i(t-s)}^3 \middle| \mathcal{D}_{t-1} \right] \\
& \quad + \frac{\|c_{ts}\|^2}{2H} \mathbb{E} \left[c_{its}^\top \int_{\tilde{P}} \nabla^2 f_{G_{its}}(\xi_{its}) c_{its} dz (1 - 2\mathbb{1}(\beta_{it} \in P)) \eta_{i(t-s)}^4 \middle| \mathcal{D}_{t-1} \right] \\
& \lesssim \|c_{ts}\|^2 \mathbb{E} \left[\left(F_{G_{its}}(\tilde{P})(1 - 2\mathbb{1}(\beta_{it} \in P)) + \mathbb{1}(\beta_{it} \in P) \right) \middle| \mathcal{D}_{t-1} \right] \\
& \quad + \left(\frac{\|c_{ts}\|^3 \mathbb{E}[\|\eta_{it}\|^3]}{\sqrt{H} J^d} + \frac{\|c_{ts}\|^4}{H J^d} \right) (1 + \sqrt{H} \mathbf{B}_{\text{FS}})
\end{aligned}$$

with ξ_{its} some point between z and $z - \delta_{its}$, and the last step follows by the same argument in Step 4 of the proof of part (a).

Now, for the leading term, we apply the partitioning argument again. Specifically, we partition the whole support $\mathcal{B}$ of β_{it} into subrectangles of length proportional to $H^{-1/2}$. Recall that we label these subrectangles by a multi-index $m = (m_1, m'_1, m_2, m'_2, \dots, m_d, m'_d) \in \mathbb{Z}^{2d}$. Define $\mathcal{B}_m$ as the set of $b \in \mathcal{B}$ such that $\sqrt{H}(b_k - \ell_k) \in [m_k, m_k + 1]$ and $\sqrt{H}(b_k - u_k) \in [m'_k, m'_k + 1]$ for all $k \in [d]$. That is, when a rectangle is associated with a multi-index with some m_k or m'_k large in absolute value, this region is far away from the facets of the hyper-rectangle $[\ell, u]$. Let $\mathcal{M}$ be the collection of possible indices m . Note that $|\mathcal{M}| \lesssim H^{d/2}$. Let

$$\begin{aligned}
\mathcal{M}_0 = & \left\{ m \in \mathcal{M} : \forall k, m_k m'_k < 0, \text{ and } \exists l, |m_l| \leq \tau_n \text{ or } |m'_l| \leq \tau_n \right\} \\
& \cup \left\{ m \in \mathcal{M} : \exists k, m_k m'_k \geq 0; |m_l| \leq \tau_n \text{ or } |m'_l| \leq \tau_n \text{ whenever } m_l m'_l \geq 0 \right\}
\end{aligned}$$

for $\tau_n \asymp \log(nT)$. That is, $\cup_{m \in \mathcal{M}_0} \mathcal{B}_m$ is the “neighborhood” of every facet of P . Then,

$$\begin{aligned}
& \mathbb{E} \left[F_{G_{its}}(\tilde{P})(1 - 2\mathbb{1}(\beta_{it} \in P)) + \mathbb{1}(\beta_{it} \in P) \middle| \mathcal{D}_{t-1} \right] \\
& = \sum_{m \in \mathcal{M}} \int_{\mathcal{B}_m} \left(F_{G_{its}}(\sqrt{H}[\ell - b, u - b])(1 - 2\mathbb{1}(b \in P)) + \mathbb{1}(b \in P) \right) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) db \\
& \quad + \mathbb{E} \left[\left(F_{G_{its}}(\sqrt{H}[\ell - \check{\beta}_{it}, u - \check{\beta}_{it}]) - F_{G_{its}}(\sqrt{H}[\ell - \beta_{it}, u - \beta_{it}]) \right) (1 - 2\mathbb{1}(\beta_{it} \in P)) \middle| \mathcal{D}_{t-1} \right] \\
& = \sum_{m \in \mathcal{M}_0} \int_{\mathcal{B}_m} \left(F_{G_{its}}(\sqrt{H}[\ell - b, u - b])(1 - 2\mathbb{1}(b \in P)) + \mathbb{1}(b \in P) \right) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) db
\end{aligned}$$

$$\begin{aligned}
& + \sum_{m \in \mathcal{M}_0} \mathbb{E} \left[\mathbb{1}(\beta_{it} \in \mathcal{B}_m) \left(F_{G_{its}}(\sqrt{H}[\ell - \check{\beta}_{it}, u - \check{\beta}_{it}]) - F_{G_{its}}(\sqrt{H}[\ell - \beta_{it}, u - \beta_{it}]) \right) \right. \\
& \quad \left. \times (1 - 2\mathbb{1}(\beta_{it} \in P)) \right] \Big| \mathcal{D}_{t-1} \Big] \\
& + \sum_{m \in \mathcal{M} \setminus \mathcal{M}_0} \int_{\mathcal{B}_m} \left(F_{G_{its}}(\sqrt{H}[\ell - b, u - b])(1 - 2\mathbb{1}(b \in P)) + \mathbb{1}(b \in P) \right) f_{\beta_{it}}(b | \mathcal{D}_{t-1}) db \\
& + \sum_{m \in \mathcal{M} \setminus \mathcal{M}_0} \mathbb{E} \left[\mathbb{1}(\beta_{it} \in \mathcal{B}_m) \left(F_{G_{its}}(\sqrt{H}[\ell - \check{\beta}_{it}, u - \check{\beta}_{it}]) - F_{G_{its}}(\sqrt{H}[\ell - \beta_{it}, u - \beta_{it}]) \right) \right. \\
& \quad \left. \times (1 - 2\mathbb{1}(\beta_{it} \in P)) \right] \Big| \mathcal{D}_{t-1} \Big] \\
& \lesssim_{\mathbb{P}} \frac{\tau_n^d}{J^{d-1} \max\{J, \sqrt{H}\}} + \mathbb{P}(\|G_{its}\| > \tau_n | \mathcal{E}_{t-1}),
\end{aligned}$$

Here

$$\sqrt{H} \|\check{\beta}_{it} - \beta_{it}\| \leq \sqrt{H} B_{\text{FS}} = o(1) \quad \text{on } \mathcal{A}_{it, \text{FS}},$$

and the complementary contribution is absorbed by (SA-II.4.3). The neighborhood of the facets of P has conditional mass bounded by the smaller of the boundary-strip order $J^{-(d-1)} \tau_n^d / \sqrt{H}$ and the whole-cell order $\tau_n^d J^{-d}$. Whenever b lies in $\mathcal{B}_m$ with $m \notin \mathcal{M}_0$, either $F_{G_{its}}(\sqrt{H}[\ell - b, u - b]) \leq \mathbb{P}(\|G_{its}\|_{\infty} > \tau_n | \mathcal{E}_{t-1}) \lesssim H^{-d/2}$ ($b \notin P$) or $1 - F_{G_{its}}(\sqrt{H}[\ell - b, u - b]) \leq \mathbb{P}(\|G_{its}\|_{\infty} > \tau_n | \mathcal{E}_{t-1}) \lesssim H^{-d/2}$ ($b \in P$) w.p.a.1 (apply the Bernstein inequality as in Step 3 of the proof of part (a), but now we set $M = H^{\frac{d}{2(q-2)}}$). Combining all results together, we conclude that the first term in the variance is $\lesssim_{\mathbb{P}} \tau_n^d / (J^{d-1} \max\{J, \sqrt{H}\})$.

Step 2: Switching summand covariances. Now, consider the covariance between each pair of summands. For $s \neq s'$,

$$\begin{aligned}
& \text{Cov} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \tilde{c}_{its} \eta_{i(t-s)}, \left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \tilde{c}_{its'} \eta_{i(t-s')} \right] \Big| \mathcal{D}_{t-1} \Big] \\
& = \mathbb{E} \left[\left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right)^2 \tilde{c}_{its} \tilde{c}_{its'} \eta_{i(t-s)} \eta_{i(t-s')} \right] \Big| \mathcal{D}_{t-1} \Big] - \mathbf{m}_{its} \mathbf{m}_{its'} \\
& = \mathbb{E} \left[\tilde{c}_{its} \tilde{c}_{its'} \eta_{i(t-s)} \eta_{i(t-s')} \left[\left(F_{G_{itss'}}(\tilde{P}) \right. \right. \right. \\
& \quad \left. \left. + \int_{\tilde{P}} \left(-\nabla f_{G_{itss'}}(z)^\top \delta_{itss'} + \delta_{itss'}^\top \int_0^1 (1-v) \nabla^2 f_{G_{itss'}}(z - v \delta_{itss'}) dv \delta_{itss'} \right) dz \right) (1 - 2\mathbb{1}(\beta_{it} \in P)) \right. \right. \\
& \quad \left. \left. + \mathbb{1}(\beta_{it} \in P) \right] \right] \Big| \mathcal{D}_{t-1} \Big] - \mathbf{m}_{its} \mathbf{m}_{its'}
\end{aligned}$$

where $\mathbf{m}_{its} = \mathbb{E}[(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P)) \tilde{c}_{its} \eta_{i(t-s)} | \mathcal{D}_{t-1}]$, $F_{G_{itss'}}$ and $f_{G_{itss'}}$, analogous to $F_{G_{its}}$ and $f_{G_{its}}$, are the distribution and density of $\frac{1}{\sqrt{H}} \sum_{l=1}^H c_{itl} \eta_{i(t-l)}$ conditional on $\sigma(\eta_{i(t-s)}, \eta_{i(t-s')}) \vee \mathcal{E}_{t-1}$, and $\delta_{itss'} = \frac{1}{\sqrt{H}} (c_{its} \eta_{i(t-s)} + c_{its'} \eta_{i(t-s')})$. The density bounds from Step 5 of the proof of part (a) apply to $f_{G_{itss'}}$ as well, because removing a fixed number of summands preserves the rolling-window Gram lower bound and the characteristic-function product bounds. When the diagonal and off-diagonal pieces are recombined, the mean-product contribution to the variance of the normalized sum is $-(H^{-1} \sum_{s=1}^H \mathbf{m}_{its})^2$. This term is

nonpositive. Part (a) of Lemma SA-II.4.1 bounds its absolute value by

$$\left| \frac{1}{H} \sum_{s=1}^H \mathbf{m}_{its} \right|^2 \lesssim_{\mathbb{P}} \left(\frac{1}{HJ^d} + \frac{\mathbf{B}_{\text{FS}}}{J^d} \right)^2,$$

which is the squared-mean term retained in the final display.

Since for any $s \neq s'$,

$$\mathbb{E}[\eta_{i(t-s)}\eta_{i(t-s')}|\mathcal{E}_{t-1}] = 0, \quad \mathbb{E}[\eta_{i(t-s)}^2\eta_{i(t-s')}|\mathcal{E}_{t-1}] = \mathbb{E}[\eta_{i(t-s)}^2|\mathcal{E}_{t-1}] = 0,$$

all Taylor terms that are linear in one of the two innovations vanish after multiplication by $\eta_{i(t-s)}\eta_{i(t-s')}$; the second-order Hessian term remains because it contains products such as $\eta_{i(t-s)}^2\eta_{i(t-s')}^2$. Thus the leading term simplifies to

$$\begin{aligned} & \frac{1}{H} \mathbb{E} \left[(c_{its}\eta_{i(t-s)} + c_{its'}\eta_{i(t-s')})^\top \tilde{c}_{its}\tilde{c}_{its'}\eta_{i(t-s)}\eta_{i(t-s')} \times \right. \\ & \quad \left. \int_{\tilde{P}} \int_0^1 (1-v) \nabla^2 f_{G_{itss'}}(z - v\delta_{itss'}) \, dv \, dz (1 - 2\mathbb{1}(\beta_{it} \in P)) (c_{its}\eta_{i(t-s)} + c_{its'}\eta_{i(t-s')}) \Big| \mathcal{D}_{t-1} \right]. \end{aligned}$$

Consider a generic Hessian element $f_{G_{itss'}}^{(\alpha)}$ with $|\alpha| = 2$. Choose a coordinate $j \in [d]$ such that $\alpha_j \geq 1$. Writing $P = [\ell, u]$ and letting $(x, z_{-j})_j$ denote the vector with j -th coordinate x and remaining coordinates z_{-j} , the fundamental theorem of calculus gives

$$\begin{aligned} \int_{\sqrt{H}(P - \check{\beta}_{it})} f_{G_{itss'}}^{(\alpha)}(z - v\delta_{itss'}) \, dz &= \int_{\sqrt{H}(P_{-j} - \check{\beta}_{it,-j})} \left[f_{G_{itss'}}^{(\alpha - e_j)}((\sqrt{H}(u_j - \check{\beta}_{it,j}), z_{-j})_j - v\delta_{itss'}) \right. \\ & \quad \left. - f_{G_{itss'}}^{(\alpha - e_j)}((\sqrt{H}(\ell_j - \check{\beta}_{it,j}), z_{-j})_j - v\delta_{itss'}) \right] dz_{-j}. \end{aligned}$$

The bracket contains derivatives of order one, evaluated on two facets of the rectangle. Keep these facet terms inside the outer conditional expectation. The density and tail bounds, together with the same innovation truncation, give

$$\begin{aligned} & \mathbb{E} \left[(1 + |\eta_{i(t-s)}| + |\eta_{i(t-s')}|)^4 \times \sup_{v \in [0,1]} \left| \int_{\sqrt{H}(P - \check{\beta}_{it})} f_{G_{itss'}}^{(\alpha)}(z - v\delta_{itss'}) \, dz \right| \Big| \mathcal{D}_{t-1} \right] \\ & \lesssim_{\mathbb{P}} \begin{cases} \tau_n^d / (J^{d-1} \sqrt{H}), & J \leq \sqrt{H}, \\ J^{-d}, & J > \sqrt{H}, \end{cases} \\ & \lesssim_{\mathbb{P}} \frac{\tau_n^d}{J^{d-1} \max\{J, \sqrt{H}\}}. \end{aligned}$$

For the first case, integration over β_{it} localizes its coordinate normal to each facet. For the second, Fubini's theorem is applied to the Hessian integral before separating the two facets. Hence the leading term in the previous expansion is $\lesssim_{\mathbb{P}} \tau_n^d / [HJ^{d-1} \max\{J, \sqrt{H}\}]$.

Combining all these results yields the desired bound:

$$\mathbb{V}\text{ar} \left[\frac{1}{H} \sum_{s=1}^H \left(\mathbb{1}(\hat{\beta}_{it} \in P) - \mathbb{1}(\beta_{it} \in P) \right) \tilde{c}_{its}\eta_{i(t-s)} \Big| \mathcal{D}_{t-1} \right]$$

$$\begin{aligned}
&\lesssim_{\mathbb{P}} H^{-2} \left[H \times \frac{\log^d(nT)}{J^{d-1} \max\{\sqrt{H}, J\}} + H^2 \times \frac{\log^d(nT)}{H J^{d-1} \max\{J, \sqrt{H}\}} \right] + \left(\frac{1}{H J^d} + \frac{\mathbf{B}_{\text{FS}}}{J^d} \right)^2 \\
&\lesssim_{\mathbb{P}} \frac{\log^d(nT)}{H J^{d-1} \max\{J, \sqrt{H}\}} + \left(\frac{1}{H J^d} + \frac{\mathbf{B}_{\text{FS}}}{J^d} \right)^2.
\end{aligned}$$

The imposed conditions guarantee that, uniformly over $t = H + 1, \dots, T$, $\frac{1}{H} \sum_{s=1}^H \|c_{ts}\|^4 \lesssim 1$ and $\frac{1}{H^2} \sum_{\substack{s, s'=1 \\ s \neq s'}}^H \|c_{ts'}\|^2 \|c_{ts}\|^2 \lesssim 1$ w.p.a.1.

Step 3: Localized-basis extension. Fix a scalar component of $p_{\mathbf{q}}(\cdot, P)$. Write

$$p_{\mathbf{q}}(\widehat{\beta}_{it}, P) - p_{\mathbf{q}}(\beta_{it}, P) = \Delta_{itP}^{\text{bd}} + \Delta_{itP}^{\text{in}},$$

where Δ_{itP}^{bd} is supported on the event that exactly one of $\widehat{\beta}_{it}$ and β_{it} belongs to P , and Δ_{itP}^{in} is the remaining same-cell coordinate movement. Because $p_{\mathbf{q}}(\cdot, P)$ is bounded, the boundary part is controlled by the switching calculation above and the squared mean bound from part (a). This gives $\log^d(nT)/\{H J^{d-1} \max\{J, \sqrt{H}\}\} + \{1/(H J^d) + \mathbf{B}_{\text{FS}}/J^d\}^2$. When $\mathbf{q} = 0$, the basis is constant on P and $\Delta_{itP}^{\text{in}} = 0$. When $\mathbf{q} \geq 1$, the normalized basis has derivative of order J , so on the same-cell event

$$\|\Delta_{itP}^{\text{in}}\| \lesssim J \|\widehat{\beta}_{it} - \beta_{it}\| \mathbf{1}(\beta_{it} \in P).$$

This derivative bound is used only for the centered variance. From (SA-II.2.2),

$$\widehat{\beta}_{it} - \check{\beta}_{it} = \frac{1}{H} \sum_{s=1}^H S \widehat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)}.$$

Conditional Rosenthal bounds for the martingale sum, together with the moment bounds on the local-polynomial weights, give $\mathbb{E}[\|\widehat{\beta}_{it} - \check{\beta}_{it}\|^2 | \mathcal{E}_{t-1}] \lesssim_{\mathbb{P}} H^{-1}$ and $\mathbb{E}[\|\widehat{\beta}_{it} - \check{\beta}_{it}\|^4 | \mathcal{E}_{t-1}] \lesssim_{\mathbb{P}} H^{-2}$, uniformly over i and t . On the intersection of the same-cell event with $\mathcal{A}_{it, \text{FS}}$,

$$\|\Delta_{itP}^{\text{in}}\| \lesssim J(\|\widehat{\beta}_{it} - \check{\beta}_{it}\| + \mathbf{B}_{\text{FS}}) \mathbf{1}(\beta_{it} \in P)$$

and the complementary contribution is absorbed by (SA-II.4.3) as above. Using the conditional independence of the innovations in $\widehat{\beta}_{it} - \check{\beta}_{it}$ from β_{it} given $\mathcal{E}_{t-1}$, and the regular-event bound above, we obtain

$$\mathbb{E} \left[\|\Delta_{itP}^{\text{in}}\|^2 \|\widehat{\beta}_{it} - \check{\beta}_{it}\|^2 \mathbf{1}(\mathcal{A}_{it, \text{FS}}) \mid \mathcal{D}_{t-1} \right] \lesssim_{\mathbb{P}} J^2 \mathbb{P}(\beta_{it} \in P | \mathcal{D}_{t-1}) (H^{-2} + \mathbf{B}_{\text{FS}}^2 H^{-1}) \lesssim_{\mathbb{P}} \frac{J^{2-d}}{H^2}.$$

On $\mathcal{A}_{it, \text{FS}}^c$, boundedness of the localized basis, the conditional second-moment bound for $\widehat{\beta}_{it} - \check{\beta}_{it}$, and the first cell-probability bound above give

$$\mathbb{E} \left[\|\Delta_{itP}^{\text{in}}\|^2 \|\widehat{\beta}_{it} - \check{\beta}_{it}\|^2 \mathbf{1}(\mathcal{A}_{it, \text{FS}}^c) \mid \mathcal{D}_{t-1} \right] \lesssim_{\mathbb{P}} \frac{1}{H} \min\{\mathbb{P}(\mathcal{A}_{it, \text{FS}}^c | \mathcal{D}_{t-1}), J^{-d}\} \lesssim_{\mathbb{P}} \frac{1}{H J^{d-1} \max\{J, \sqrt{H}\}}.$$

Therefore, for the same-cell centered component,

$$\mathbb{V} \text{ar} \left[\Delta_{itP}^{\text{in}} [\nabla \mu_t(\beta_{it})]^\top \frac{1}{H} \sum_{s=1}^H S \widehat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)} \mid \mathcal{D}_{t-1} \right]$$

$$\leq \mathbb{E} \left[\left\| \Delta_{itP}^{\text{in}} \right\|^2 \left\| \frac{1}{H} \sum_{s=1}^H S\hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)} \right\|^2 \middle| \mathcal{D}_{t-1} \right] \lesssim \frac{1}{H J^{d-1} \max\{J, \sqrt{H}\}} + \frac{J^{2-d}}{H^2}.$$

Combining the boundary and interior pieces gives the displayed localized-basis variance bound. $\square$

SA-II.4.6 Proof of Lemma SA-II.4.2

We first record the direct bound for $\mathcal{L}_{\mathbf{B}}(\beta)$, and then focus on the expansion of $\mathcal{L}_{\mathbf{V}}(\beta)$.

By Lemma SA-II.3.3, with probability approaching one, $\|\hat{Q}_t^{-1}\| \lesssim J^d$ and $\|Q_t^{-1}\| \lesssim J^d$ uniformly over $t = H+1, \dots, T$. Since the gradients are uniformly bounded under Assumption SA-DGP, and since the local-polynomial weights have bounded absolute sum under the Gram eigenvalue bound,

$$|\mathcal{L}_{\mathbf{B}}(\beta)| \lesssim \mathbb{P} \max_{H+1 \leq t \leq T} \max_{i \in [n_t]} \|\check{\beta}_{it} - \beta_{it}\| \lesssim \mathbb{P} \mathbf{B}_{\text{FS}}.$$

Step 1: Bounds for $\mathcal{A}(\beta)$ and $\mathcal{A}(0)$. The term $\mathcal{A}(\beta)$ is the re-indexed version of the reference-basis expression with the reference Gram matrix Q_t . It can be written as

$$\mathcal{A}(\beta) = \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^\top Q_t^{-1} \left[\frac{1}{n_t} \sum_{i=1}^{n_t} p_t(\beta_{it}) [\nabla \mu_t(\beta_{it})]^\top \frac{1}{H} \sum_{s=1}^H K(s/H) S\hat{A}(t)^{-1} X_s(t) \varepsilon_{i(t-s)} \right]$$

The summands in the bracketed double sum are centered and independent conditional on $\mathcal{E}_{t-1}$, are supported on reference cells, and have conditional variance of order $J^{-d}/(nH)$ per time period. The evaluation block and inverse Gram matrix only select and weight elements of the same rectangle-polynomial class; the operator norm contributes at most the factor J^d . Since this class is VC-type with bounded envelope, a standard maximal inequality gives

$$\sup_{\beta \in \mathcal{B}^\circ} |\mathcal{A}(\beta)| \lesssim \mathbb{P} \left(\frac{J^d \log(nT)}{nH} \right)^{1/2}.$$

For the special case $\beta = 0$, the partition convention and boundary tie-breaking assign 0 to a unique cell at each date; write its label as $j_0(u)$ at date u . For compactness, write $P_{0,u} = P_{j_0(u),u}$, $p_{0,u} = p_{j_0(u),u}$, and $Q_{0,u} = Q_{j_0(u),u}$. By the split-at-zero construction and the midpoint and side-length normalization in Section SA-II.3, $\hat{p}_{j_0(u),u}(0) = p_{0,u}(0)$ for every $\mathbf{q} \geq 0$. Let $\mathcal{S}_t = \{s \in [H] : H+1 \leq t+s \leq T\}$. Thus,

$$\begin{aligned} \mathcal{A}(0) &= \frac{1}{T-H} \sum_{t=1}^T \sum_{i=1}^{n_t} \frac{1}{H} \sum_{s \in \mathcal{S}_t} \frac{1}{n_{t+s}} p_{0,t+s}(0)^\top Q_{0,t+s}^{-1} \\ &\quad \times p_{0,t+s}(\beta_{i(t+s)}) [\nabla \mu_{t+s}(\beta_{i(t+s)})]^\top S\hat{A}(t+s)^{-1} K\left(\frac{s}{H}\right) X_s(t+s) \mathbf{1}(i \leq n_{t+s}) \varsigma_{it} \eta_{it}. \end{aligned}$$

The display is obtained by re-indexing the rolling-window shocks: the primitive innovation η_{it} enters future first-step estimates at dates $t+s$. After the zero-cell identity above, all coefficients in $\mathcal{A}(0)$ are $\mathcal{E}_\infty$ -measurable; conditional on $\mathcal{E}_\infty$, the primitive innovations are independent over (i, t) and have mean zero. For the variance display, write $\Xi_s(u) = S\hat{A}(u)^{-1} K(s/H) X_s(u)$. Jensen's inequality over the rolling-window

average and the bounds $\|Q_{0,u}^{-1}\| \lesssim_{\mathbb{P}} J^d$ and $\|p_{0,u}(0)\| \|p_{0,u}(\beta_{iu})\| \lesssim \mathbf{1}(\beta_{iu} \in P_{0,u})$ give

$$\begin{aligned}
\mathbb{V}\text{ar}[\mathcal{A}(0)|\mathcal{E}_{\infty}] &\lesssim \frac{1}{T^2} \sum_{t=1}^T \sum_{i=1}^{n_t} \left(\frac{1}{H} \sum_{s \in \mathcal{S}_t} \frac{1}{n_{t+s}} p_{0,t+s}(0)^{\top} Q_{0,t+s}^{-1} p_{0,t+s}(\beta_{i(t+s)}) [\nabla \mu_{t+s}(\beta_{i(t+s)})]^{\top} \right. \\
&\quad \left. \times \Xi_s(t+s) \mathbf{1}(i \leq n_{t+s}) \right)^2 \\
&\lesssim \frac{J^{2d}}{T^2} \sum_{t=1}^T \sum_{i=1}^{n_t} \frac{1}{H} \sum_{s \in \mathcal{S}_t} \frac{1}{n_{t+s}^2} \mathbf{1}(\beta_{i(t+s)} \in P_{0,t+s}) \mathbf{1}(i \leq n_{t+s}) \|\Xi_s(t+s)\|^2 \\
&\lesssim \frac{J^{2d}}{T^2} \sum_{u=H+1}^T \frac{1}{n_u^2} \sum_{i=1}^{n_u} \mathbf{1}(\beta_{iu} \in P_{0,u}) \frac{1}{H} \sum_{s=1}^H \|\Xi_s(u)\|^2 \\
&\lesssim_{\mathbb{P}} \frac{J^{2d}}{nT} \frac{1}{J^d}.
\end{aligned}$$

The last line uses the true-exposure reference second-moment bound from Lemma SA-II.3.3 and the local-polynomial Gram bound. Then, $\mathcal{A}(0) \lesssim_{\mathbb{P}} \sqrt{J^d/(nT)}$.

Step 2: First-step Gram replacement and inverse-Gram perturbation. Let $\mathcal{A}^{\dagger}(\beta)$ denote the same re-indexed expression as $\mathcal{A}(\beta)$, but with $A(t+s)^{-1}$ replaced by $\hat{A}(t+s)^{-1}$. The mean-square version of Lemma SA-II.2.2, the resolvent identity, and the bound for $\mathcal{A}(\beta)$ in Step 1 give

$$\sup_{\beta \in \mathcal{B}^{\circ}} |\mathcal{A}^{\dagger}(\beta) - \mathcal{A}(\beta)| \lesssim_{\mathbb{P}} \left(\frac{T^{2/q}}{H} + \frac{1}{\sqrt{H}} \right) \sqrt{\frac{J^d \log(nT)}{nH}} = o_{\mathbb{P}} \left(\frac{1}{H} \right),$$

where the last equality follows from (SA-II.4.1) and $J^d \log(nT)/n = o(1)$. Thus this replacement error is absorbed by the H^{-1} term in the residual bound.

Next, replace the generated inverse Gram $\hat{Q}_t^{-1}$ in $\mathcal{L}_V(\beta)$ by the reference inverse Gram Q_t^{-1} . By Lemma SA-II.3.3, $\max_{H+1 \leq t \leq T} \|\hat{Q}_t^{-1} - Q_t^{-1}\| \lesssim_{\mathbb{P}} J^{2d} R_{\mathbf{Q}}$.

The cost of this replacement is $\min\{J^d R_{\mathbf{Q}}, 1\} \sqrt{J^d \log(nT)/(nH)}$. Indeed, if $J^d R_{\mathbf{Q}} \lesssim 1$, the resolvent identity gives an additional relative factor $J^d R_{\mathbf{Q}}$ compared to the centered reference-basis bound in Step 1; otherwise, the crude bound by the same centered reference-basis scale gives the truncation at one. In the remaining steps we work after this inverse-Gram replacement and add this perturbation cost at the end.

Step 3: Generated-regressor switch term. For a fixed admissible partition $\Delta = \{P_j : j \in [\bar{J}_t]\}$ equipped with its cell normalizations, let $p(\cdot; \Delta)$ denote the corresponding stacked local-polynomial basis. Consider the following two terms:

$$\begin{aligned}
\mathcal{T}_{\mathbf{G}}(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^{\top} Q_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} [p(\hat{\beta}_{it}; \hat{\Delta}_t) - p(\beta_{it}; \hat{\Delta}_t)] [\nabla \mu_t(\beta_{it})]^{\top} (\hat{\beta}_{it} - \check{\beta}_{it}), \\
\mathcal{T}_{\mathbf{P}}(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \hat{p}_t(\beta)^{\top} Q_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} [p(\beta_{it}; \hat{\Delta}_t) - p(\beta_{it}; \Delta_t)] [\nabla \mu_t(\beta_{it})]^{\top} (\hat{\beta}_{it} - \check{\beta}_{it}).
\end{aligned}$$

Consider the generated-regressor switch term $\mathcal{T}_{\mathbf{G}}(\beta)$. For any $\beta \in \mathcal{B}^{\circ}$ (corresponding to an index set $\mathcal{J}_t$ that excludes bins close to the boundary), let $\mathcal{P}^{\circ}$ denote the collection of quasi-uniform rectangles $P =$

$[\ell, u] \subset \mathcal{B}$ with $\text{diam}(P) \asymp J^{-1}$ and $\min_{k \in [d]} \min\{|\ell_k - \underline{\beta}_k|, |\bar{\beta}_k - u_k|\} \gg \log(nT)/\sqrt{H}$.

$$\begin{aligned}
|\mathcal{T}_{\mathbf{g}}(\beta)| &\lesssim_{\mathbb{P}} \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \left\| \frac{J^d}{n} \frac{1}{H} \sum_{i=1}^{n_t} \sum_{s=1}^H [p_{\mathbf{q}}(\hat{\beta}_{it}; \hat{P}_{jt}) - p_{\mathbf{q}}(\beta_{it}; \hat{P}_{jt})] \right. \\
&\quad \times [\nabla \mu_t(\beta_{it})]^\top S \hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)} \left. \right\| \\
&\lesssim_{\mathbb{P}} \max_{H+1 \leq t \leq T} \sup_{P \in \mathcal{P}^\circ} \left\| \frac{J^d}{nH} \sum_{i=1}^{n_t} \sum_{s=1}^H [p(\hat{\beta}_{it}; P) - p(\beta_{it}; P)] [\nabla \mu_t(\beta_{it})]^\top \right. \\
&\quad \times S \hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)} \left. \right\| \\
&\lesssim \sup_{t, P} \left\| \frac{J^d}{nH} \sum_{i=1}^{n_t} \sum_{s=1}^H \mathbb{E}[g_{its} | \mathcal{D}_{t-1}] \right\| + \sup_{t, P} \left\| \frac{J^d}{nH} \sum_{i=1}^{n_t} \sum_{s=1}^H (g_{its} - \mathbb{E}[g_{its} | \mathcal{D}_{t-1}]) \right\|
\end{aligned}$$

where

$$g_{its} = [p_{\mathbf{q}}(\hat{\beta}_{it}; P) - p_{\mathbf{q}}(\beta_{it}; P)] [\nabla \mu_t(\beta_{it})]^\top S \hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)},$$

and the shorthand $\sup_{t, P}$ in the preceding display ranges over $H+1 \leq t \leq T$ and $P \in \mathcal{P}^\circ$.

By Lemma SA-II.4.1(a),

$$\max_{H+1 \leq t \leq T} \sup_{P \in \mathcal{P}^\circ} \left| \frac{J^d}{nH} \sum_{i=1}^{n_t} \sum_{s=1}^H \mathbb{E}[g_{its} | \mathcal{D}_{t-1}] \right| \lesssim_{\mathbb{P}} \frac{1}{H} + \mathbf{B}_{\text{FS}}.$$

On the other hand, conditional on $\mathcal{D}_{t-1}$, the second term is a sum of terms independent over i with zero means. By Lemma SA-II.4.1(b),

$$\mathbb{V}\text{ar} \left[\frac{J^d}{nH} \sum_{i=1}^{n_t} \sum_{s=1}^H g_{its} \mid \mathcal{D}_{t-1} \right] \lesssim_{\mathbb{P}} \frac{J^{2d}}{n} \left[\frac{\log^d(nT)}{H J^{d-1} \max\{J, \sqrt{H}\}} + \left(\frac{1}{H J^d} + \frac{\mathbf{B}_{\text{FS}}}{J^d} \right)^2 + \mathbb{1}(\mathbf{q} \geq 1) \frac{J^{2-d}}{H^2} \right].$$

Define

$$\begin{aligned}
g_i(t, P) &= \frac{1}{H} \sum_{s=1}^H g_{its}, \\
\mathcal{S} &= \{g(\cdot; t, P) : P \in \mathcal{P}^\circ, H+1 \leq t \leq T\}.
\end{aligned}$$

Here $g(\cdot; t, P)$ is the function that evaluates to $g_i(t, P)$ on the full input history for asset i , including the common quantities and all observations entering the rolling window. The envelope satisfies

$$\begin{aligned}
\sup_{t, P} \|g_i(t, P)\| &\lesssim \sup_{H+1 \leq t \leq T} \left\| \frac{1}{H} \sum_{s=1}^H X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)} \right\|, \\
\max_{i \in [n]} \sup_{t, P} |g_i(t, P)| &\lesssim_{\mathbb{P}} \frac{\sqrt{\log(nT)}}{\sqrt{H}} + \frac{(nT^2)^{1/q} \log(nT)}{H}.
\end{aligned}$$

The class indexed by P is VC-type with a finite exponent because it is generated by rectangular cell indicators

and bounded polynomial basis functions on normalized cells. To transfer this covering bound to the rolling-window summands, fix t and a scalar basis coordinate $r \in [K_q]$. Let

$$b_{P,r}(x, z) = p_{q,r}(x, P) - p_{q,r}(z, P),$$

$$M_{it} = [\nabla \mu_t(\beta_{it})]^\top \frac{1}{H} \sum_{s=1}^H S \hat{A}(t)^{-1} X_s(t) K(s/H) \varepsilon_{i(t-s)},$$

where $p_{q,r}$ is the r th coordinate of p_q . The corresponding coordinate of $g_i(t, P)$ is exactly $M_{it} b_{P,r}(\hat{\beta}_{it}, \beta_{it})$, and M_{it} does not depend on P . On the full input-history space, denote the functions corresponding to M_{it} and $b_{P,r}(\hat{\beta}_{it}, \beta_{it})$ by M and $B_{P,r}$. For any probability measure $\mathbb{Q}$ with $0 < \mathbb{E}_{\mathbb{Q}}[M^2] < \infty$, define the probability measure

$$d\mathbb{Q}_M = \frac{M^2}{\mathbb{E}_{\mathbb{Q}}[M^2]} d\mathbb{Q}.$$

Then, for any $P, \tilde{P} \in \mathcal{P}^\circ$,

$$\|M(B_{P,r} - B_{\tilde{P},r})\|_{L_2(\mathbb{Q})} = \|M\|_{L_2(\mathbb{Q})} \|B_{P,r} - B_{\tilde{P},r}\|_{L_2(\mathbb{Q}_M)}.$$

The bounded basis-difference class $\{B_{P,r} : P \in \mathcal{P}^\circ\}$ has a polynomial covering bound uniformly over probability measures, with an exponent depending only on the fixed d and q . The identity transfers this bound to $\{MB_{P,r} : P \in \mathcal{P}^\circ\}$ with envelope $C|M|$, where C bounds the basis differences. If $\mathbb{E}_{\mathbb{Q}}[M^2] = 0$, the multiplied class is zero $\mathbb{Q}$ -almost surely. Applying the argument to the fixed number of basis coordinates establishes the asserted VC-type bound.

The other index t can only take a finite number $(T - H)$ values. Taking the union over these dates multiplies the covering bound by at most $T - H$. The concentration argument is applied for each fixed t conditional on $\mathcal{D}_{t-1}$, using the conditional independence across i stated above. Given these facts, by the maximal inequality for i.n.i.d. empirical processes in [Cattaneo et al. \(2024, Lemma SA24\)](#),

$$|\mathcal{T}_G(\beta)| \lesssim_{\mathbb{P}} \left(\frac{1}{H} + \mathbf{B}_{\text{FS}} \right) + \sqrt{\frac{J^{2d} \log^{d+1}(nT)}{nH J^{d-1} \max\{J, \sqrt{H}\}}} + \mathbf{1}(q \geq 1) \sqrt{\frac{J^{d+2} \log(nT)}{nH^2}},$$

under the restriction

$$\frac{J^{d-1} \max\{J, \sqrt{H}\} \log^2(nT)}{\min\{n^{1-2/q} T^{1-4/q} h, n\}} \lesssim 1.$$

This restriction ensures that

$$\frac{\log^{3/2}(nT)}{\sqrt{n}} + \frac{(nT^2)^{1/q} \log^2(nT)}{\sqrt{nH}} \lesssim (J^{d-1} \max\{J, \sqrt{H}\})^{-1/2} \log^{(d+1)/2}(nT).$$

Step 4: Partition switch term. For $\mathcal{T}_P(\beta)$, by Lemma [SA-II.3.2](#), for all $\beta \in \mathcal{B}^\circ$,

$$|\mathcal{T}_P(\beta)| \lesssim_{\mathbb{P}} \max_{H+1 \leq t \leq T} \max_{j \in J_t} \left\| \frac{J^d}{n} \frac{1}{H} \sum_{i=1}^{n_t} \sum_{s=1}^H [p_q(\beta_{it}; \hat{P}_{jt}) - p_q(\beta_{it}; P_{jt})] \right\|$$

$$\begin{aligned}
& \times \left\| [\nabla \mu_t(\beta_{it})]^\top S \hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)} \right\| \\
& \lesssim \max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \sup_{\tilde{P} \in \mathcal{P}_{jt}} \left\| \frac{J^d}{n} \frac{1}{H} \sum_{i=1}^{n_t} \sum_{s=1}^H [p_{\mathbf{q}}(\beta_{it}; \tilde{P}_{jt}) - p_{\mathbf{q}}(\beta_{it}; P_{jt})] [\nabla \mu_t(\beta_{it})]^\top \right. \\
& \quad \left. \times S \hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)} \right\|
\end{aligned}$$

where, by Lemma SA-II.3.2, we take the supremum over $\mathcal{P}_{jt} = \{[a, b] : \max\{\|a - \ell\|, \|b - u\|\} \leq \sqrt{\log T/n}\}$ for a generic cell $P_{jt} = [\ell, u]$. Again, conditional on $\mathcal{E}_{t-1}$, the summands are centered, independent over i , and have zero cross terms across different lags by the innovation orthogonality. The symmetric difference between $\tilde{P}$ and P_{jt} has $\mathcal{D}_{t-1}$ -conditional probability at most of order $J^{-(d-1)} \sqrt{\log T/n}$, uniformly over $\tilde{P} \in \mathcal{P}_{jt}$. A VC count bound conditional on $\mathcal{D}_{t-1}$ and a union over dates give

$$\max_{H+1 \leq t \leq T} \max_{j \in \mathcal{J}_t} \sup_{\tilde{P} \in \mathcal{P}_{jt}} \sum_{i=1}^{n_t} \mathbb{1}(\beta_{it} \in \tilde{P} \triangle P_{jt}) \lesssim_{\mathbb{P}} n J^{-(d-1)} \sqrt{\frac{\log T}{n}} + \log(nT).$$

The same argument and $J^d \log(nT)/n = o(1)$ bound the true-beta count in each reference cell by $O_{\mathbb{P}}(n/J^d)$, uniformly over dates and cells. Conditional on $\mathcal{E}_{t-1}$, the realized counts are fixed; the strip and reference-cell count bounds control the boundary and coordinate-normalization contributions, respectively. Since $H^{-1} \sum_{s=1}^H \|S \hat{A}(t)^{-1} X_s(t) K(\frac{s}{H})\|^2 \lesssim_{\mathbb{P}} 1$, the comparison of boundaries and the coordinate-normalization component between two cells gives the following variance bound:

$$\begin{aligned}
& \frac{J^{2d}}{n^2 H^2} \sum_{i=1}^{n_t} \sum_{s=1}^H \| [p_{\mathbf{q}}(\beta_{it}; \tilde{P}_{jt}) - p_{\mathbf{q}}(\beta_{it}; P_{jt})] \|^2 \left\| S \hat{A}(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \right\|^2 \\
& \lesssim_{\mathbb{P}} \frac{J^{d+1} \sqrt{\log T/n}}{nH} + \frac{J^{2d} \log(nT)}{n^2 H} + \mathbb{1}(\mathbf{q} \geq 1) \frac{J^{d+2} \log T}{n^2 H}.
\end{aligned}$$

In addition, the class of local polynomial basis functions is of VC-type with a finite exponent, and the multiplier argument from Step 3 applies because the rolling-score multiplier does not depend on the perturbed cell. Using the kernel-weighted rolling-score envelope, datewise truncation, and bounded empirical-process concentration together with Cattaneo et al. (2024, Lemma SA24), then taking a union over dates and retaining the envelope terms, we conclude that

$$\begin{aligned}
|\mathcal{T}_{\mathbf{P}}(\beta)| & \lesssim_{\mathbb{P}} \sqrt{\frac{J^{d+1} \sqrt{\log T/n} \log(nT)}{nH}} + \mathbb{1}(\mathbf{q} \geq 1) \sqrt{\frac{J^{d+2} \log T \log(nT)}{n^2 H}} \\
& \quad + \frac{J^d}{n} \left[\frac{\log^{3/2}(nT)}{\sqrt{H}} + \frac{(nT^2)^{1/q} \log^2(nT)}{H} \right],
\end{aligned}$$

and this bound holds uniformly over $\beta \in \mathcal{B}^\circ$. The additional boundary-count variance term contributes at most $J^d \log(nT)/(n\sqrt{H})$ after concentration and is covered by the displayed envelope terms.

Step 5: Combine the bounds. Multiplying the final inequality in Step 3 by $J^d/\sqrt{nH}$ gives

$$\frac{J^d}{n} \left[\frac{\log^{3/2}(nT)}{\sqrt{H}} + \frac{(nT^2)^{1/q} \log^2(nT)}{H} \right] \lesssim \sqrt{\frac{J^{2d} \log^{d+1}(nT)}{nH J^{d-1} \max\{J, \sqrt{H}\}}}.$$

Thus the extra terms retained in Step 4 are absorbed by the existing generated-regressor switch component of R_{LA} . Combining the first-step Gram replacement and inverse-Gram perturbation from Step 2, the generated-regressor switch bound from Step 3, and the partition-switch bound from Step 4 proves

$$\sup_{\beta \in \mathcal{B}^\circ} |\mathcal{L}_V(\beta) - \mathcal{A}(\beta)| \lesssim_{\mathbb{P}} R_{\text{LA}}.$$

□

SA-II.5 Convergence Rate

We now combine the first-step and partitioning results from Sections SA-II.2 and SA-II.3 with the general results from Part I. The first result is the direct specialization of the general beta-sorted portfolio convergence theorem to the classical construction. The second uses the stochastic linearization and is sharper at each fixed evaluation point.

SA-II.5.1 Results

Corollary SA-II.5.1 (General Beta-Sorted Portfolios).

Suppose that Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold and that $s_\mu \geq 1$. Assume further that

$$\begin{aligned} \frac{T^{(3+d)/q}}{H} + \sqrt{\frac{\log T}{H}} &= o(1), & R_{\text{FS}} &= o(1), \\ \frac{J^d \log(nT)}{n} &= o(1), & \frac{(nT)^{2/q} J^d \log(nT)}{n} &\lesssim 1. \end{aligned}$$

Then, for each fixed $\beta \in \mathcal{B}^\circ$ and any $\ell_n \rightarrow \infty$,

$$|\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H)| \lesssim_{\mathbb{P}} \ell_n \left[R_{\text{FS}} + J^{-\min\{s_\mu, \mathfrak{q}+1\}} + \sqrt{\frac{J^d \log(nT)}{n}} + \frac{\|\beta\|}{\sqrt{T}} \right].$$

The next corollary uses the stochastic linearization to recover the time-series averaging in the leading sampling terms. The generated-regressor correction $\mathcal{A}(\beta)$ is the only additional term extracted from $\mathcal{L}_V(\beta)$; the nonconstant-basis movement that appears when $\mathfrak{q} \geq 1$ is controlled uniformly and absorbed by R_{LA} in Lemma SA-II.4.2. To state the leading first-step correction compactly, define

$$R_{\text{A}}(\beta) = \begin{cases} \sqrt{\frac{J^d \log(nT)}{nH}} & \text{if } \beta \neq 0 \\ \sqrt{\frac{J^d}{nT}} & \text{if } \beta = 0 \end{cases}.$$

This rate collects the pointwise bound for the first-step correction $\mathcal{A}(\beta)$ obtained in Lemma SA-II.4.2. The Gram-perturbation and basis-switching pieces enter through the explicit $R_{\mathfrak{q}}$ and $R_{\mathfrak{p}}$ terms displayed below. Under the stronger idiosyncratic-scale localization condition $JR_{\text{FS}} = o(1)$, these terms are absorbed by $\sqrt{J^d/(nT)}$; fixed nonzero- β inference only needs them to be negligible at the factor scale.

Corollary SA-II.5.2 (Classical Pointwise Convergence Rate). *Suppose that Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold and that $s_\mu \geq 2$. Assume further that*

$$\begin{aligned} \frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}} &= o(1), \quad R_{\text{FS}} = o(1), \\ \frac{J^d \log(nT)}{n} &= o(1), \quad \frac{J^2 \log T}{n} = o(1), \\ \frac{J^{d-1} \max\{J, \sqrt{H}\} \log^2(nT)}{\min\{n^{1-2/q} T^{1-4/q} h, n\}} &\lesssim 1, \quad \frac{\max\{J^{2/(q-2)} T^{2/q}, T^{2(3+d)/q}\}}{H} = o(1). \end{aligned}$$

Then, for each fixed $\beta \in \mathcal{B}^\circ$,

$$\begin{aligned} |\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H)| &\lesssim_{\mathbb{P}} \frac{\|\beta\|}{\sqrt{T}} + \sqrt{\frac{J^d}{nT}} + \sqrt{\frac{J^{3d} R_{\text{q}}^2}{nT}} \\ &\quad + \sqrt{\frac{J^{2d} R_{\text{p}}}{nT}} + R_{\text{A}}(\beta) + R_{\text{LA}} + J^{-\min\{s_\mu, \text{q}+1\}} + \frac{R_{\text{FS}}}{\sqrt{T}} + R_{\text{FS}}^2 \\ &\quad + \mathbb{1}(\text{q} = 0) \frac{1}{J\sqrt{T}}. \end{aligned}$$

SA-II.5.2 Proof of Corollary SA-II.5.1

By the first displayed rate condition, Lemma SA-II.2.3 gives

$$\hat{R}_{\text{FS}} \lesssim_{\mathbb{P}} R_{\text{FS}}.$$

Because $R_{\text{FS}} = o(1)$ and $\mathcal{B}^+$ is a fixed neighborhood of $\mathcal{B}$, the first-step estimates lie in $\mathcal{B}^+$ uniformly over i and t , w.p.a.1.

Next consider the local partition conditions along the cells in $\mathcal{J}_t$. The first-step bias condition, the Gram condition for the rolling-window design, and $J^d \log(nT)/n = o(1)$ imply the assumptions of Lemmas SA-II.3.1, SA-II.3.2, and SA-II.3.3. These results imply that the estimated partition covers $\mathcal{B}^\circ$ and all raw generated regressors, and that the reference and estimated cells indexed by $\mathcal{J}_t$ have side lengths of order J^{-1} . Consequently,

$$\hat{R}_{\text{D}}^\circ \lesssim_{\mathbb{P}} J^{-1}.$$

The same lemma gives the required counts only over the relevant interior cells:

$$\frac{n}{J^d} \lesssim_{\mathbb{P}} \hat{n}_{\min}^\circ \leq \hat{n}_{\max}^\circ \lesssim_{\mathbb{P}} \frac{n}{J^d}.$$

Consequently,

$$\frac{(nT)^{2/q} \log(nT)}{\hat{n}_{\min}^\circ} \lesssim_{\mathbb{P}} \frac{(nT)^{2/q} J^d \log(nT)}{n} \lesssim 1.$$

The scaled Gram conclusion in Lemma SA-II.3.3 gives the required eigenvalue bounds uniformly over $j \in \mathcal{J}_t$. The pointwise proof of Theorem SA-I.2.1 is therefore applicable cellwise to the block containing each fixed $\beta \in \mathcal{B}^\circ$; its argument never uses an unselected tail block. Applying that local argument and substituting $\hat{R}_{\text{FS}} \lesssim_{\mathbb{P}} R_{\text{FS}}$, $\hat{R}_{\text{D}}^\circ \lesssim_{\mathbb{P}} J^{-1}$, and the displayed cell-count chain gives the displayed pointwise bound. $\square$

SA-II.5.3 Proof of Corollary SA-II.5.2

By the first displayed rate condition, Lemma SA-II.2.3 gives

$$\widehat{\mathbf{R}}_{\text{FS}} \lesssim_{\mathbb{P}} \mathbf{R}_{\text{FS}}.$$

Since $J^d \log(nT)/n = o(1)$ implies $J \log(nT)/n = o(1)$, the displayed rate conditions imply the conditions of Lemma SA-II.3.3. Thus,

$$\widehat{\mathbf{R}}_{\mathbf{Q}} \lesssim_{\mathbb{P}} \mathbf{R}_{\mathbf{Q}}, \quad \widehat{\mathbf{R}}_{\mathbf{P}} \lesssim_{\mathbb{P}} \mathbf{R}_{\mathbf{P}}, \quad \widehat{\mathbf{R}}_{\mathbf{D}}^{\circ} \lesssim_{\mathbb{P}} J^{-1},$$

and the generated Gram bounds hold. The parts of the proof of Lemma SA-I.3.1 that keep the generated design fixed give

$$\begin{aligned} |\mathcal{G}(\beta)| &\lesssim_{\mathbb{P}} \frac{\|\beta\|}{\sqrt{T}}, & |\mathcal{B}(\beta)| &\lesssim_{\mathbb{P}} J^{-\min\{s_{\mu}, \mathbf{q}+1\}}, & |\mathcal{R}_3(\beta)| &\lesssim_{\mathbb{P}} \mathbf{1}(\mathbf{q} = 0) \frac{1}{J\sqrt{T}}, \\ |\mathcal{R}_4(\beta)| &\lesssim_{\mathbb{P}} \frac{\mathbf{R}_{\text{FS}}}{\sqrt{T}}, & |\mathcal{R}_5(\beta)| &\lesssim_{\mathbb{P}} \mathbf{R}_{\text{FS}}^2. \end{aligned}$$

Lemma SA-II.4.2 gives

$$|\mathcal{L}_{\mathbf{B}}(\beta)| \lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}}, \quad |\mathcal{A}(\beta)| \lesssim_{\mathbb{P}} \mathbf{R}_{\mathbf{A}}(\beta),$$

and the $\mathcal{L}_{\mathbf{B}}(\beta)$ bound is absorbed by $\mathbf{R}_{\mathbf{LA}}$ because $\mathbf{R}_{\mathbf{LA}}$ contains $\mathbf{B}_{\text{FS}}$. The design-comparison argument in Lemma SA-I.3.1, using the generated and reference Gram bounds verified above, yields

$$|\mathcal{E}(\beta)| \lesssim_{\mathbb{P}} \sqrt{\frac{J^d}{nT}}, \quad |\mathcal{R}_1(\beta)| \lesssim_{\mathbb{P}} \sqrt{\frac{J^{3d} \mathbf{R}_{\mathbf{Q}}^2}{nT}}, \quad |\mathcal{R}_2(\beta)| \lesssim_{\mathbb{P}} \sqrt{\frac{J^{2d} \mathbf{R}_{\mathbf{P}}}{nT}}.$$

The remaining generated-regressor residual is exactly the linear-approximation error controlled in Lemma SA-II.4.2: $|\mathcal{L}_{\mathbf{V}}(\beta) - \mathcal{A}(\beta)| \lesssim_{\mathbb{P}} \mathbf{R}_{\mathbf{LA}}$. Combining these displays with the algebraic decomposition

$$\begin{aligned} \widehat{\mu}(\beta) - \bar{\mu}_T(\beta; H) &= \mathcal{G}(\beta) + \mathcal{E}(\beta) - \mathcal{L}(\beta) + \mathcal{B}(\beta) \\ &\quad + \mathcal{R}_1(\beta) + \mathcal{R}_2(\beta) + \mathcal{R}_3(\beta) + \mathcal{R}_4(\beta) - \mathcal{R}_5(\beta), \end{aligned}$$

and $\mathcal{L} = \mathcal{L}_{\mathbf{B}} + \mathcal{L}_{\mathbf{V}}$ gives the stated rate. □

SA-II.6 Pointwise-in- β Distributional Approximation

We now isolate the numerator for the distributional approximation:

$$\bar{\mathcal{J}}(\beta) = \mathcal{G}(\beta) + \mathcal{E}(\beta) - \mathcal{A}(\beta) = \frac{1}{T-H} \sum_{t=H+1}^T \beta^{\top} \tilde{f}_t + \frac{1}{T-H} \sum_{t=1}^T \sum_{i=1}^{n_t} \varsigma_{it} \eta_{it} W_{it}(\beta), \quad (\text{SA-II.6.1})$$

where the direct second-step and first-step primitive-shock weights are

$$W_{it}(\beta) = \frac{1}{n_t} \widehat{p}_t(\beta)^{\top} Q_t^{-1} p_t(\beta_{it}) \mathbf{1}(t \geq H+1) - w_{it}(\beta).$$

Therefore, we obtain

$$\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H) = \bar{\mathcal{S}}(\beta) + \mathcal{R}(\beta),$$

where

$$\begin{aligned} \mathcal{R}(\beta) &= \mathcal{B}(\beta) + \mathcal{R}_1(\beta) + \mathcal{R}_2(\beta) + \mathcal{R}_3(\beta) + \mathcal{R}_4(\beta) - \mathcal{R}_5(\beta) \\ &\quad - \mathcal{L}_B(\beta) - \left[\mathcal{L}_V(\beta) - \mathcal{A}(\beta) \right]. \end{aligned} \tag{SA-II.6.2}$$

Thus (SA-II.6.2) lists all residual terms explicitly: the smoothing bias, the design-comparison and Taylor remainders, the first-step bias contribution, and the residual generated-regressor approximation error from Lemma SA-II.4.2. The centered nonconstant-basis term that appears when $\mathfrak{q} \geq 1$ is included in that last residual and controlled uniformly by R_{LA} .

For every evaluation point $\beta \in \mathcal{B}^\circ$, define the baseline normalization as the sum of the factor variance and the reference idiosyncratic scale:

$$\begin{aligned} \Sigma(\beta) &= \Sigma_f(\beta) + \Sigma_\varepsilon(\beta), \\ \Sigma_f(\beta) &= \frac{1}{(T-H)^2} \sum_{t=H+1}^T \beta^\top \mathbb{V}\text{ar}(f_t \mid \mathcal{G}_{t-1}) \beta, \\ \Sigma_\varepsilon(\beta) &= \frac{1}{(T-H)^2} \sum_{t=1}^T \sum_{i=1}^{n_t} \varsigma_{it}^2(W_{it}(\beta))^2, \end{aligned}$$

where $W_{it}(\beta)$ is the generated-regressor-adjusted reference weight. At an anchored evaluation point satisfying $\hat{p}_t(\beta) = p_t(\beta)$, the weights $W_{it}(\beta)$ are measurable with respect to the innovation-free environment, and $\Sigma_\varepsilon(\beta)$ is the predictable conditional variance of the idiosyncratic leading term. At a generic nonanchored evaluation point, $\Sigma_\varepsilon(\beta)$ is used only as a reference quadratic scale. The proofs below control the corresponding stochastic term through the separate bounds for $\mathcal{E}(\beta)$ and $\mathcal{A}(\beta)$.

SA-II.6.1 Results

The first result isolates the Gaussian approximation for the leading term. The proof treats the origin and nonzero evaluation points separately, but the conclusion is stated as a single Kolmogorov-distance approximation. Away from the origin, the factor variance is nondegenerate under Assumption SA-DGP(iv); the only additional variance condition is needed at the origin, where the factor component vanishes.

Lemma SA-II.6.1 (Gaussian Approximation for the Leading Terms). *Suppose that Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold. Assume further that*

$$\begin{aligned} \frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}} &= o(1), \\ \frac{J^d \log(nT)}{n} &= o(1), \\ \frac{J^{d-1} \max\{J, \sqrt{H}\} \log^2(nT)}{\min\{n^{1-2/q} T^{1-4/q} h, n\}} &\lesssim 1, \quad \frac{\max\{J^{2/(q-2)} T^{2/q}, T^{2(3+d)/q}\}}{H} = o(1). \end{aligned}$$

If the fixed evaluation point is $\beta \neq 0$, assume additionally that

$$\frac{J^d \log(nT)}{nh} = o(1).$$

If the fixed evaluation point is $\beta = 0$, assume additionally that

$$\frac{nT}{J^d} \Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} 1.$$

Then, for each fixed $\beta \in \mathcal{B}^\circ$,

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\bar{\mathcal{S}}(\beta)}{\sqrt{\Sigma(\beta)}} \leq x \right) - \Phi(x) \right| \rightarrow 0.$$

The next theorem combines the Gaussian approximation for the leading term with the pointwise stochastic linearization error. We state the conclusion directly in Kolmogorov distance, so the approximation can be read as the infeasible size control for a pointwise normal test. Let

$$\mathbf{R}_{\text{CLT}} = J^{-\min\{s_\mu, q+1\}} + \sqrt{\frac{J^{3d} \mathbf{R}_{\mathbf{Q}}^2}{nT}} + \sqrt{\frac{J^{2d} \mathbf{R}_{\mathbf{P}}}{nT}} + \mathbf{R}_{\text{LA}} + \frac{\mathbf{R}_{\text{FS}}}{\sqrt{T}} + \mathbf{R}_{\text{FS}}^2 + \mathbf{1}(q=0) \frac{1}{J\sqrt{T}}.$$

This is the non-leading pointwise remainder after removing the leading components $\mathcal{G}(\beta)$, $\mathcal{E}(\beta)$, and $\mathcal{A}(\beta)$.

Theorem SA-II.6.2 (Distributional Approximation for the Infeasible t-Statistic). *Suppose that the conditions of Lemmas SA-II.4.2 and SA-II.6.1 hold. For fixed $\beta \in \mathcal{B}^\circ$, suppose further that*

$$\mathbf{R}_{\text{FS}} = o(1), \quad \frac{J^2 \log T}{n} = o(1), \quad \frac{\mathbf{R}_{\text{CLT}}}{\sqrt{\Sigma(\beta)}} = o_{\mathbb{P}}(1).$$

Then

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H)}{\sqrt{\Sigma(\beta)}} \leq x \right) - \Phi(x) \right| \rightarrow 0.$$

SA-II.6.2 Discussion

- (i) *Normalization.* The theorem imposes the high-level normalization condition $\mathbf{R}_{\text{CLT}}/\sqrt{\Sigma(\beta)} = o_{\mathbb{P}}(1)$ directly. For each fixed $\beta \neq 0$, Assumption SA-DGP(iv) gives $\Sigma(\beta) \asymp_{\mathbb{P}} T^{-1}$, so it is enough that

$$\sqrt{T} \mathbf{R}_{\text{CLT}} \rightarrow 0.$$

At the origin, Lemma SA-II.6.1 imposes $(nT/J^d) \Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} 1$, and the leading-weight upper bound used below gives $\Sigma(0) \asymp_{\mathbb{P}} J^d/(nT)$. Thus it is enough that

$$\sqrt{\frac{nT}{J^d}} \mathbf{R}_{\text{CLT}} \rightarrow 0.$$

- (ii) *Polynomial order.* Higher-order second-step polynomials reduce the smoothing-bias component in the CLT remainder: $J^{-\min\{s_\mu, q+1\}}$. Under the factor nondegeneracy condition in Assumption SA-

DGP(iv), the relevant bias restrictions are

$$\begin{aligned} \frac{T}{J^{2 \min\{s_\mu, \mathfrak{q}+1\}}} &\rightarrow 0 && \text{when } \beta \neq 0, \\ \frac{nT}{J^{d+2 \min\{s_\mu, \mathfrak{q}+1\}}} &\rightarrow 0 && \text{when } \beta = 0. \end{aligned}$$

The stronger restriction at the origin appears because the factor component is absent and $\Sigma(0) \asymp_{\mathbb{P}} J^d/(nT)$. At the origin, the sufficient rate conditions stated here are incompatible with $\mathfrak{q} = 0$.

SA-II.6.3 Proof of Lemma SA-II.6.1

Decompose the leading term into its factor and idiosyncratic components:

$$\begin{aligned} \bar{\mathcal{S}}(\beta) &= \bar{\mathcal{S}}_f(\beta) + \bar{\mathcal{S}}_\varepsilon(\beta), \\ \bar{\mathcal{S}}_f(\beta) &= \frac{1}{T-H} \sum_{t=H+1}^T \beta^\top \tilde{f}_t, \\ \bar{\mathcal{S}}_\varepsilon(\beta) &= \frac{1}{T-H} \sum_{t=1}^T \sum_{i=1}^{n_t} \varsigma_{it} \eta_{it} W_{it}(\beta). \end{aligned}$$

The displayed rate conditions imply the conditions of Lemma SA-II.3.3. Work on the high-probability event from that lemma on which the relevant cell counts, Gram matrices, and local-polynomial basis norms are uniformly controlled. By local support of the partition basis and the quasi-uniform cell-count bounds, define, only for this proof,

$$C_{it}(\beta) = \frac{1}{n_t} \hat{p}_t(\beta)^\top Q_t^{-1} p_t(\beta_{it}) \mathbb{1}(t \geq H+1).$$

Then

$$\sum_{i=1}^{n_t} |C_{it}(\beta)|^m \lesssim_{\mathbb{P}} \frac{J^{(m-1)d}}{n^{m-1}}, \quad m = 2, 3,$$

uniformly in t . For the rolling-window weights, Jensen's inequality and the same local-support argument give, for each t ,

$$\sum_{i=1}^{n_t} |w_{it}(\beta)|^m \lesssim_{\mathbb{P}} \frac{J^{(m-1)d}}{n^{m-1}} (1 + \|f_t\|^m), \quad m = 2, 3.$$

It follows from $T^{-1} \sum_{t=1}^T (1 + \|f_t\|^m) = O_{\mathbb{P}}(1)$ that

$$\frac{1}{T} \sum_{t=1}^T \sum_{i=1}^{n_t} |w_{it}(\beta)|^m \lesssim_{\mathbb{P}} \frac{J^{(m-1)d}}{n^{m-1}}, \quad m = 2, 3.$$

Since ς_{it} is uniformly bounded, it follows that, for each fixed $\beta \in \mathcal{B}^\circ$,

$$\frac{1}{T} \sum_{t=1}^T \sum_{i=1}^{n_t} |\varsigma_{it} W_{it}(\beta)|^3 \lesssim_{\mathbb{P}} \frac{J^{2d}}{n^2} \quad \text{and} \quad \Sigma_\varepsilon(\beta) = \frac{1}{(T-H)^2} \sum_{t=1}^T \sum_{i=1}^{n_t} \varsigma_{it}^2 (W_{it}(\beta))^2 \lesssim_{\mathbb{P}} \frac{J^d}{nT}.$$

At the origin, the split partition fixes the relevant cell containing zero, and the midpoint and side-length normalization gives $\widehat{p}_t(0) = p_t(0)$. Thus the leading weights at the origin are already expressed in reference zero-cell coordinates and are $\mathcal{E}_\infty$ -measurable. Conditional on $\mathcal{E}_\infty$, the shocks $\{\eta_{it}\}$ are independent, mean zero, and have uniformly bounded third moments. Therefore the conditional Berry–Esseen inequality gives

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\bar{\mathcal{J}}_\varepsilon(0)}{\sqrt{\Sigma_\varepsilon(0)}} \leq x \mid \mathcal{E}_\infty \right) - \Phi(x) \right| \lesssim_{\mathbb{P}} \frac{TJ^{2d}/n^2}{((T-H)^2\Sigma_\varepsilon(0))^{3/2}}.$$

At $\beta = 0$, the variance nondegeneracy assumption gives $(T-H)^2\Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} TJ^d/n$, and the displayed bound is $\lesssim_{\mathbb{P}} \sqrt{J^d/(nT)} = o_{\mathbb{P}}(1)$. Hence

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\bar{\mathcal{J}}_\varepsilon(0)}{\sqrt{\Sigma_\varepsilon(0)}} \leq x \right) - \Phi(x) \right| \rightarrow 0.$$

Since $\bar{\mathcal{J}}_f(0) = 0$ and $\Sigma(0) = \Sigma_\varepsilon(0)$, the claim follows for $\beta = 0$.

Now take $\beta \neq 0$. The process $\{\beta^\top \tilde{f}_t\}$ is a martingale-difference sequence with respect to $\{\mathcal{G}_t\}$. Define the deterministic reference variance

$$\Sigma_{f,0}(\beta) := \frac{\beta^\top \Omega_f \beta}{T-H}.$$

By (SA-II.1.1), the sum of the conditional variances of the normalized martingale differences satisfies

$$\frac{1}{(T-H)\beta^\top \Omega_f \beta} \sum_{t=H+1}^T \mathbb{V}\text{ar}(\beta^\top \tilde{f}_t \mid \mathcal{G}_{t-1}) \rightarrow_{\mathbb{P}} 1.$$

For every $\epsilon > 0$, the conditional fourth-moment bound verifies the Lindeberg condition using the deterministic threshold $\epsilon \sqrt{(T-H)\beta^\top \Omega_f \beta}$:

$$\begin{aligned} & \frac{1}{(T-H)\beta^\top \Omega_f \beta} \sum_{t=H+1}^T \mathbb{E} \left[|\beta^\top \tilde{f}_t|^2 \mathbf{1} \left(|\beta^\top \tilde{f}_t| > \epsilon \sqrt{(T-H)\beta^\top \Omega_f \beta} \right) \mid \mathcal{G}_{t-1} \right] \\ & \lesssim_{\mathbb{P}} \frac{\sum_{t=H+1}^T \mathbb{E}[|\beta^\top \tilde{f}_t|^4 \mid \mathcal{G}_{t-1}]}{(T-H)^2(\beta^\top \Omega_f \beta)^2} \lesssim_{\mathbb{P}} (T-H)^{-1} = o_{\mathbb{P}}(1). \end{aligned}$$

The martingale central limit theorem and Polya's theorem therefore give

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\bar{\mathcal{J}}_f(\beta)}{\sqrt{\Sigma_{f,0}(\beta)}} \leq x \right) - \Phi(x) \right| \rightarrow 0.$$

Finally,

$$\frac{\Sigma_f(\beta)}{\Sigma_{f,0}(\beta)} = \frac{\beta^\top \left[(T-H)^{-1} \sum_{t=H+1}^T \mathbb{V}\text{ar}(f_t \mid \mathcal{G}_{t-1}) \right] \beta}{\beta^\top \Omega_f \beta} \rightarrow_{\mathbb{P}} 1.$$

Slutsky's lemma and continuity of the standard normal cdf thus imply

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\bar{\mathcal{J}}_f(\beta)}{\sqrt{\Sigma_f(\beta)}} \leq x \right) - \Phi(x) \right| \rightarrow 0.$$

For the idiosyncratic component, use the exact decomposition $\bar{\mathcal{J}}_\varepsilon(\beta) = \mathcal{E}(\beta) - \mathcal{A}(\beta)$. The direct idiosyncratic bound from Lemma SA-I.3.1 and the uniform first-step-correction bound from Lemma SA-II.4.2 give

$$|\mathcal{E}(\beta)| \lesssim_{\mathbb{P}} \sqrt{\frac{J^d}{n(T-H)}}, \quad |\mathcal{A}(\beta)| \lesssim_{\mathbb{P}} \sqrt{\frac{J^d \log(nT)}{nH}}.$$

Because $\Sigma_f(\beta) \asymp_{\mathbb{P}} (T-H)^{-1}$, the displayed rate conditions imply

$$\frac{\mathcal{E}(\beta)}{\sqrt{\Sigma_f(\beta)}} = o_{\mathbb{P}}(1), \quad \frac{\mathcal{A}(\beta)}{\sqrt{\Sigma_f(\beta)}} = o_{\mathbb{P}}(1),$$

and hence $\bar{\mathcal{J}}_\varepsilon(\beta)/\sqrt{\Sigma_f(\beta)} = o_{\mathbb{P}}(1)$. Separately, the reference idiosyncratic scale is negligible relative to the factor variance because

$$\frac{\Sigma_\varepsilon(\beta)}{\Sigma_f(\beta)} \lesssim_{\mathbb{P}} \frac{J^d/(nT)}{1/T} = \frac{J^d}{n} = o(1).$$

Thus $\bar{\mathcal{J}}_\varepsilon(\beta)/\sqrt{\Sigma(\beta)} = o_{\mathbb{P}}(1)$ and $\Sigma_f(\beta)/\Sigma(\beta) = 1 + o_{\mathbb{P}}(1)$. Hence full joint convergence of the factor and idiosyncratic components is not needed for fixed $\beta \neq 0$; Slutsky's argument only requires the factor approximation and the negligibility of the idiosyncratic component. Since the standard normal distribution is continuous, the preceding Kolmogorov approximation is preserved under this $o_{\mathbb{P}}(1)$ perturbation, and hence

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\bar{\mathcal{J}}(\beta)}{\sqrt{\Sigma(\beta)}} \leq x \right) - \Phi(x) \right| \rightarrow 0.$$

This gives the stated conclusion for $\beta \neq 0$. □

SA-II.6.4 Proof of Theorem SA-II.6.2

By the exact decomposition at the beginning of this section,

$$\frac{\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H)}{\sqrt{\Sigma(\beta)}} = \frac{\bar{\mathcal{J}}(\beta)}{\sqrt{\Sigma(\beta)}} + \frac{\mathcal{R}(\beta)}{\sqrt{\Sigma(\beta)}}.$$

The componentwise bounds from Lemma SA-I.3.1, the partition bounds from Lemma SA-II.3.3, and the generated-regressor refinement in Lemma SA-II.4.2 imply

$$|\mathcal{R}(\beta)| \lesssim_{\mathbb{P}} R_{\text{CLT}}.$$

Here the Gram-perturbation and basis-switching pieces are not reabsorbed into the local-support idiosyncratic scale; they are part of R_{CLT} through the explicit $R_{\mathbf{Q}}$ and $R_{\mathbf{P}}$ terms in its definition. The normalization condition $R_{\text{CLT}}/\sqrt{\Sigma(\beta)} = o_{\mathbb{P}}(1)$ therefore removes the full remainder directly. Lemma SA-II.6.1 gives the

Kolmogorov approximation for the leading terms, and the standard perturbation argument for continuous limiting cdfs gives the displayed result. The sufficient conditions for $\beta \neq 0$ and $\beta = 0$ follow from $\Sigma(\beta) \asymp_{\mathbb{P}} T^{-1}$ and $\Sigma(0) \asymp_{\mathbb{P}} J^d/(nT)$, respectively. $\square$

SA-II.7 Pointwise-in- β Variance Estimation

This section constructs feasible estimators of the components of the baseline normalization in Section SA-II.6. Recall that $\Sigma(\beta) = \Sigma_f(\beta) + \Sigma_\varepsilon(\beta)$. The idiosyncratic variance result has two parts. At fixed evaluation points whose basis coordinates are anchored in the estimated and reference partitions, the weights $W_{it}(\beta)$ are $\mathcal{E}_\infty$ -measurable, $\Sigma(\beta)$ is the predictable quadratic variation of the leading sum in (SA-II.6.1), and the factor-idiosyncratic cross term vanishes by Assumption SA-DGP(ii)(a). At a generic nonanchored point, $\Sigma_\varepsilon(\beta)$ remains a reference scale. For fixed $\beta \neq 0$, only a local-support upper bound for the jackknife estimator is needed because the factor variance is of order T^{-1} .

Let $m_t = \mathbb{E}[f_t \mid \mathcal{G}_{t-1}]$, and let $\hat{m}_t$ be a $\mathcal{G}_{t-1}$ -measurable predictor of m_t . For instance, $\hat{m}_t$ can be a recursive past sample mean of $\{f_s : s < t\}$ when $\mathbb{E}[f_t \mid \mathcal{G}_{t-1}] = \mu_f$ for every t and the recursive mean is consistent for μ_f , or a fitted predictive model using only information in $\mathcal{G}_{t-1}$ when richer factor dynamics are imposed. Define the estimated factor innovation $\hat{U}_t = f_t - \hat{m}_t$. The factor-variance estimator is

$$\hat{\Sigma}_f(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \beta^\top \hat{U}_t \hat{U}_t^\top \beta.$$

For a fixed evaluation point, we propose a bin-restricted delete- i jackknife estimator. At an anchored point it estimates $\Sigma_\varepsilon(\beta)$ at the idiosyncratic scale; at a generic nonzero point we use only its order bound. Let $\hat{M}_t^{(-i)}(\beta)$ denote the second-step estimator with unit i removed at time t , holding the partition $\hat{\Delta}_t$ fixed. Define the leave-one-out pseudo-value

$$\hat{d}_{it}(\beta) = \hat{M}_t^{(-i)}(\beta) - \hat{M}_t(\beta),$$

with the convention that $\hat{d}_{it}(\beta) = 0$ when unit i is absent from the cross-section at time t . The bin-restricted jackknife estimator is

$$\hat{\Sigma}_\varepsilon(\beta) = \frac{1}{(T-H)^2} \sum_{i=1}^n \sum_{t=H+1}^T \sum_{s=H+1}^T \hat{d}_{it}(\beta) \hat{d}_{is}(\beta) \mathbf{1}(|t-s| \leq H),$$

where $[n]$ is the common index set from the general setup and absent units have zero pseudo-values. The Sherman-Morrison formula for deleting the i -th observation from the period- t least-squares normal equations gives

$$\hat{d}_{it}(\beta) = -\hat{C}_{it}(\beta) \hat{\varepsilon}_{it}, \quad \hat{C}_{it}(\beta) = \frac{1}{n_t} \frac{\hat{p}_t(\beta)^\top \hat{Q}_t^{-1} \hat{p}_t(\hat{\beta}_{it})}{1 - n_t^{-1} \hat{p}_t(\hat{\beta}_{it})^\top \hat{Q}_t^{-1} \hat{p}_t(\hat{\beta}_{it})}, \quad \hat{\varepsilon}_{it} = R_{it} - \hat{M}_t(\hat{\beta}_{it}).$$

When $\mathbf{q} = 0$, this reduces to the more familiar bin-count expression

$$\hat{\Sigma}_\varepsilon(\beta) = \frac{1}{(T-H)^2} \sum_{i=1}^n \sum_{t=H+1}^T \sum_{s=H+1}^T \frac{\mathbf{1}_{it}^{(\hat{\beta})} \mathbf{1}_{is}^{(\hat{\beta})}}{(\hat{n}_{j_0(t)t} - 1)(\hat{n}_{j_0(s)s} - 1)} \hat{\varepsilon}_{it} \hat{\varepsilon}_{is} \mathbf{1}(|t-s| \leq H),$$

where $\mathbf{1}_{it}^{(\hat{\beta})} = \mathbf{1}(\hat{\beta}_{it} \in \hat{P}_{j_0(t)t})$ and $j_0(t)$ denotes the cell containing the evaluation point β at time t .

Combining the two components, the feasible variance estimator is

$$\hat{\Sigma}(\beta) = \hat{\Sigma}_f(\beta) + \hat{\Sigma}_\varepsilon(\beta).$$

Recall that for $\beta = 0$, $\Sigma_f(0) = 0$ and $\hat{\Sigma}(0) = \hat{\Sigma}_\varepsilon(0)$ suffices; for each fixed $\beta \neq 0$, $\Sigma_f(\beta) \asymp T^{-1}$.

SA-II.7.1 Results

The first result controls the factor component of the variance estimator and isolates the cost of replacing the predictable factor mean by a fitted predictor.

Lemma SA-II.7.1 (Factor Variance Estimation). *Suppose that Assumption SA-DGP(iv) holds and $H \leq cT$ for some fixed $c \in (0, 1)$. If $\hat{m}_t$ is $\mathcal{G}_{t-1}$ -measurable and*

$$\frac{1}{T-H} \sum_{t=H+1}^T \|m_t - \hat{m}_t\|^2 = o_{\mathbb{P}}(T),$$

then, for each fixed $\beta \in \mathcal{B}$,

$$\hat{\Sigma}_f(\beta) - \Sigma_f(\beta) - \mathcal{B}_F(\beta) = \begin{cases} o_{\mathbb{P}}(\|\beta\|^2 T^{-1}), & \beta \neq 0, \\ 0, & \beta = 0, \end{cases}$$

where

$$\mathcal{B}_F(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \beta^\top (m_t - \hat{m}_t)(m_t - \hat{m}_t)^\top \beta \geq 0$$

is the squared factor-prediction bias.

The next result first gives, at every fixed evaluation point, the upper bound needed to make the entire idiosyncratic component negligible relative to the factor variance. This bound does not require the selected block to be anchored. The split-at-zero construction is anchored at $\beta = 0$, and the generalized construction in Section SA-II.9 is anchored at each prespecified contrast point. At these points the result further gives sharp additive consistency at the idiosyncratic scale.

Lemma SA-II.7.2 (Bin-Restricted Jackknife). *Fix $\beta \in \mathcal{B}^\circ$. Suppose Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold, and that $s_\mu \geq 2$. Assume further that*

$$\begin{aligned} \frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}} &= o(1), \\ \frac{J^2 \log T}{n} &= o(1), \\ \frac{J^{d-1} \max\{J, \sqrt{H}\} \log^2(nT)}{\min\{n^{1-2/q} T^{1-4/q} h, n\}} &\lesssim 1, \quad \frac{\max\{J^{2/(q-2)} T^{2/q}, T^{2(3+d)/q}\}}{H} = o(1). \end{aligned}$$

Assume further that

$$H \log(nT) (J^{-2 \min\{s_\mu, q+1\}} + R_{\text{FS}}^4 + B_{\text{FS}}^2) + \frac{\log(nT)}{H} + \frac{J^d \log^{d+2}(nT)}{n} = o(1). \quad (\text{SA-II.7.1})$$

Then the following conclusions hold.

(i) For every fixed $\beta \in \mathcal{B}^\circ$,

$$|\widehat{\Sigma}_\varepsilon(\beta)| = O_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

(ii) If, in addition, $JR_{\text{FS}} = o(1)$ and

$$\widehat{p}_t(\beta) = p_t(\beta), \quad t = H+1, \dots, T, \quad (\text{SA-II.7.2})$$

then

$$\widehat{\Sigma}_\varepsilon(\beta) = \Sigma_\varepsilon(\beta) + o_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

Combining the two pieces gives the feasible variance estimator used in the pointwise Gaussian approximation.

Theorem SA-II.7.3 (Feasible Variance Estimation). *Fix $\beta \in \mathcal{B}^\circ$. Suppose Assumptions [SA-DGP](#), [SA-FS](#), [SA-PA](#), [SA-SEP](#), and [SA-FSB](#) hold, and that $s_\mu \geq 2$. Assume that $\widehat{m}_t$ is $\mathcal{G}_{t-1}$ -measurable,*

$$\frac{1}{T-H} \sum_{t=H+1}^T \|m_t - \widehat{m}_t\|^2 = o_{\mathbb{P}}(T),$$

and that the following rate restrictions hold:

$$\begin{aligned} \frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}} &= o(1), \\ \frac{J^2 \log T}{n} &= o(1), \\ \frac{J^{d-1} \max\{J, \sqrt{H}\} \log^2(nT)}{\min\{n^{1-2/q} T^{1-4/q} h, n\}} &\lesssim 1, \quad \frac{\max\{J^{2/(q-2)} T^{2/q}, T^{2(3+d)/q}\}}{H} = o(1), \\ H \log(nT) (J^{-2 \min\{s_\mu, q+1\}} + R_{\text{FS}}^4 + B_{\text{FS}}^2) + \frac{\log(nT)}{H} + \frac{J^d \log^{d+2}(nT)}{n} &= o(1). \end{aligned}$$

Then

$$\widehat{\Sigma}(\beta) - \Sigma(\beta) - \mathcal{B}_{\text{F}}(\beta) = o_{\mathbb{P}}(T^{-1}).$$

For $\beta = 0$, if, in addition, equation [\(SA-II.7.2\)](#) holds, $JR_{\text{FS}} = o(1)$, and $(nT/J^d)\Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} 1$, then

$$\widehat{\Sigma}(0) - \Sigma(0) = o_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

SA-II.7.2 Fama-MacBeth Variance Estimator

The Fama-MacBeth variance estimator is widely used because it treats the time-series dispersion of the per-period second-step estimates as the sampling variance. Its target generally differs from the baseline normalization in Section SA-II.6, which equals the predictable variance of the leading sum at anchored evaluation points. Recall that

$$\mathcal{B}(\beta) = \frac{1}{T-H} \sum_{t=H+1}^T \mathcal{B}_t(\beta),$$

where $\mathcal{B}_t(\beta)$ is the period- t smoothing-bias component from Section SA-II.4. The classical Fama-MacBeth estimator and its oracle bias-corrected variant are

$$\begin{aligned} \widehat{\Sigma}_{\text{FM}}(\beta) &= \frac{1}{(T-H)^2} \sum_{t=H+1}^T (\widehat{M}_t(\beta) - \widehat{\mu}(\beta))^2, \\ \widetilde{\Sigma}_{\text{FM}}(\beta) &= \frac{1}{(T-H)^2} \sum_{t=H+1}^T \left[(\widehat{M}_t(\beta) - \mathcal{B}_t(\beta)) - (\widehat{\mu}(\beta) - \mathcal{B}(\beta)) \right]^2. \end{aligned}$$

The second estimator is an oracle benchmark because $\mathcal{B}_t(\beta)$ depends on the unknown function $\mu_t(\cdot)$.

For the comparison at the origin, define the direct oracle idiosyncratic term

$$\mathcal{E}_{\circ,t}(\beta) = \sum_{i=1}^{n_t} \varsigma_{it} \eta_{it} \left(\frac{1}{n_t} \widehat{p}_t(\beta)^\top Q_t^{-1} p_t(\beta_{it}) \right),$$

and its pre-shock conditional variance

$$\Sigma_{\varepsilon,\circ}(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathbb{E}[\mathcal{E}_{\circ,t}(\beta)^2 \mid \mathcal{C}_t] = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \varsigma_{it}^2 \left(\frac{1}{n_t} \widehat{p}_t(\beta)^\top Q_t^{-1} p_t(\beta_{it}) \right)^2.$$

Here $\mathcal{C}_t$, defined in Section SA-II.1.3, fixes the current innovation-free scales and the past generated design while excluding the current primitive innovations. This is the variance of the direct cross-sectional idiosyncratic component. It differs from $\Sigma_\varepsilon(\beta)$ in Section SA-II.6, whose weight $W_{it}(\beta)$ also subtracts the first-step generated-regressor correction.

The following theorem makes the comparison precise by characterizing the probability limit of the Fama-MacBeth estimator. Define

$$\mathbf{R}_{\text{FM}} = \mathbf{R}_{\text{LA}} + \frac{\log(nT)}{H} + \sqrt{\frac{J^{2d} \mathbf{R}_{\text{P}}}{n}} + \frac{\mathbb{1}(\mathbf{q} = 0)}{J} + \mathbf{R}_{\text{FS}}^2.$$

Theorem SA-II.7.4 (Fama-MacBeth Variance Estimation).

Suppose Assumptions SA-DGP, SA-FS, SA-PA, SA-SEP, and SA-FSB hold, and that $s_\mu \geq 2$. Assume further that

$$\begin{aligned} \frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}} &= o(1), & \mathbf{R}_{\text{FS}} &= o(1), \\ \frac{J^2 \log T}{n} &= o(1), \end{aligned}$$

$$\frac{J^{d-1} \max\{J, \sqrt{H}\} \log^2(nT)}{\min\{n^{1-2/q} T^{1-4/q} h, n\}} \lesssim 1, \quad \frac{\max\{J^{2/(q-2)} T^{2/q}, T^{2(3+d)/q}\}}{H} = o(1).$$

The anchored-basis condition is required for the origin statement below and holds automatically under Assumption SA-PA. If, in addition,

$$J^{-\min\{s_\mu, q+1\}} + R_{\text{FM}} = o(1),$$

then, for each fixed $\beta \neq 0$ in $\mathcal{B}^\circ$,

$$\widehat{\Sigma}_{\text{FM}}(\beta) = \Sigma_f(\beta) + \Sigma_\mu(\beta) + o_{\mathbb{P}}(T^{-1}),$$

where

$$\Sigma_\mu(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T (\mu_t(\beta) - \bar{\mu}_T(\beta; H))^2.$$

For $\beta = 0$, if, in addition, $JR_{\text{FS}} = o(1)$,

$$\left| \widetilde{\Sigma}_{\text{FM}}(0) - \Sigma_{\varepsilon,0}(0) - \Sigma_\mu(0) \right| \lesssim_{\mathbb{P}} \frac{R_{\text{FM}}^2}{T} + \left(\sqrt{\frac{J^d}{nT}} + \frac{R_{\text{FM}}}{\sqrt{T}} \right) \sqrt{\Sigma_\mu(0)} + o_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

SA-II.7.3 Discussion

- (i) *Truncated jackknife.* The estimator $\widehat{\Sigma}_\varepsilon(\beta)$ uses the delete- i jackknife perturbations only for time pairs with $|t - s| \leq H$. After the rolling-window first-step correction is isolated, pairs farther apart do not share primitive idiosyncratic shocks and are conditionally mean zero. The truncation therefore keeps the serial dependence induced by the first-step window while reducing the number of nonzero pairs from order T^2 to order TH .
- (ii) *Scale.* At every fixed evaluation point, the selected-block argument gives $|\widehat{\Sigma}_\varepsilon(\beta)| = O_{\mathbb{P}}(J^d/(nT))$. For fixed $\beta \neq 0$, this is sufficient: $J^d/n = o(1)$ makes both the estimator and the reference idiosyncratic scale negligible relative to the first-order factor variance. At an anchored evaluation point, Lemma SA-II.7.2 further gives additive consistency for the idiosyncratic variance at the scale $J^d/(nT)$. At $\beta = 0$, this is the relevant first-order variance scale, so relative consistency additionally requires the nondegeneracy condition $(nT/J^d)\Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} 1$.
- (iii) *Factor prediction.* The factor variance estimator separates sampling variation from prediction bias. Lemma SA-II.7.1 controls the centered estimation error, while $\mathcal{B}_{\text{F}}(\beta)$ records the bias from using $\widehat{m}_t$ in place of m_t . Thus relative consistency of $\widehat{\Sigma}(\beta)$ for fixed $\beta \neq 0$ requires this prediction bias to be negligible relative to $\Sigma(\beta)$; otherwise the estimator is conservative.
- (iv) *Approximation bias in the jackknife.* The jackknife condition (SA-II.7.1) uses the actual second-step approximation order $J^{-\min\{s_\mu, q+1\}}$. Up to the displayed $\log(nT)$ factor, it gives the familiar HJ^{-2} restriction for $q = 0$, and HJ^{-4} for a degree-one second step when $s_\mu \geq 2$. This variance-estimation restriction is distinct from the same-cell nonconstant-basis term absorbed by R_{LA} in the stochastic linearization.

(v) *Primitive conditions.* The displayed conditions $J^2 \log T/n = o(1)$ and $T^{2/q}/H + \sqrt{\log T/H} = o(1)$ verify the common partition and first-step moment requirements. The additional condition $J\mathbf{R}_{\text{FS}} = o(1)$ in the anchored case verifies the comparison with a single reference block, including $\Delta_{\text{JK}} = o(1)$ and $\mathbf{R}_{\text{JKMD}} = o(1)$. The selected-block upper bound does not make that comparison and does not require this stronger localization.

(vi) *Fama-MacBeth.* Theorem SA-II.7.4 shows that the Fama-MacBeth variance has a different target. For fixed $\beta \neq 0$, it includes the time-series variation $\Sigma_\mu(\beta)$ of the conditional mean path. At $\beta = 0$, even the oracle bias-corrected version targets $\Sigma_{\varepsilon,0}(0) + \Sigma_\mu(0)$, where $\Sigma_{\varepsilon,0}(0)$ is based on the direct second-step idiosyncratic weights rather than the generated-regressor-adjusted weights in $\Sigma_\varepsilon(0)$. The origin comparison uses (SA-II.7.35) and conditions on current innovation-free information and earlier primitive shocks. It does not require the sharp jackknife rate conditions. The extra condition $J - \min\{s_\mu, q+1\} + \mathbf{R}_{\text{FM}} = o(1)$ is only needed for the fixed nonzero- β probability limit.

SA-II.7.4 Proof of Lemma SA-II.7.1

Let $\tilde{\Sigma}_f(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \beta^\top \tilde{f}_t \tilde{f}_t^\top \beta$, and recall $\tilde{f}_t = f_t - \mathbb{E}[f_t \mid \mathcal{G}_{t-1}]$. Decompose $\hat{\Sigma}_f(\beta) = \tilde{\Sigma}_f(\beta) + \mathcal{B}_F(\beta) + \mathcal{V}_F(\beta)$, where

$$\mathcal{V}_F(\beta) = \frac{2}{(T-H)^2} \sum_{t=H+1}^T \beta^\top \tilde{f}_t (m_t - \hat{m}_t)^\top \beta.$$

Conditional on $\mathcal{F}_{t-1}$, the summands in $\mathcal{V}_F(\beta)$ form a martingale-difference sequence, and their predictable quadratic variation is bounded by

$$\begin{aligned} \frac{4\|\beta\|^4}{(T-H)^4} \sum_{t=H+1}^T \|m_t - \hat{m}_t\|^2 \|\text{Var}(f_t \mid \mathcal{F}_{t-1})\| &\lesssim_{\mathbb{P}} \frac{\|\beta\|^4}{(T-H)^4} \sum_{t=H+1}^T \|m_t - \hat{m}_t\|^2 \\ &= \|\beta\|^4 o_{\mathbb{P}}(T^{-2}). \end{aligned}$$

Assumption SA-DGP(iv) and conditional Cauchy-Schwarz give $\max_{1 \leq t \leq T} \|\text{Var}(f_t \mid \mathcal{F}_{t-1})\| \lesssim_{\mathbb{P}} 1$. The Lenglart-Rebolledo inequality, applied after localization, gives $\mathcal{V}_F(\beta) = \|\beta\|^2 o_{\mathbb{P}}(T^{-1})$, while $\mathcal{V}_F(0) = 0$ by construction. It remains to compare $\tilde{\Sigma}_f(\beta)$ with $\Sigma_f(\beta)$. Write $\xi_t = (\beta^\top \tilde{f}_t)^2 - \mathbb{E}[(\beta^\top \tilde{f}_t)^2 \mid \mathcal{G}_{t-1}]$, so that $\{\xi_t, \mathcal{G}_t\}$ is a martingale difference sequence and

$$\tilde{\Sigma}_f(\beta) - \Sigma_f(\beta) = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \xi_t.$$

By the unconditional higher-moment bound in Assumption SA-DGP(iv), $\sup_t \mathbb{E}[\|\tilde{f}_t\|^4] \lesssim 1$, so $\mathbb{E}[\xi_t^2] \leq \mathbb{E}[(\beta^\top \tilde{f}_t)^4] \lesssim \|\beta\|^4$. The variance of $\tilde{\Sigma}_f(\beta) - \Sigma_f(\beta)$ satisfies

$$\frac{1}{(T-H)^4} \sum_{t=H+1}^T \mathbb{E}[\xi_t^2] \lesssim \frac{\|\beta\|^4}{(T-H)^3}.$$

For $\beta \neq 0$, applying Chebyshev's inequality gives

$$\left| \widetilde{\Sigma}_f(\beta) - \Sigma_f(\beta) \right| \lesssim_{\mathbb{P}} \frac{\|\beta\|^2}{T^{3/2}}.$$

For $\beta = 0$, both $\widetilde{\Sigma}_f(0)$ and $\Sigma_f(0)$ vanish identically, so the result is immediate. $\square$

SA-II.7.5 Preliminary Results for Proof of Lemma SA-II.7.2

We next record the unit-level residual bounds used in the jackknife proof. They are stated separately because the aggregate linearization in Lemmas SA-I.3.1 and SA-II.4.2 does not by itself control the banded products of fitted residuals. Put $T_0 = T - H$ and, for $B = C$ or $B = \widehat{C}$, write

$$\mathcal{H}_B(U, V) = \frac{1}{T_0^2} \sum_{i=1}^n \sum_{t=H+1}^T \sum_{s=H+1}^T B_{it}(\beta) B_{is}(\beta) U_{it} V_{is} \mathbf{1}(|t-s| \leq H),$$

where coefficients and arrays are set to zero for invalid unit–date pairs. The map $\mathcal{H}_B$ is only a convenient signed bilinear form; the flat band matrix $[\mathbf{1}(|t-s| \leq H)]_{t,s}$ is not positive semidefinite, so $\mathcal{H}_B$ is not used as an inner product.

For any fixed β , define

$$v_{it} = \frac{1}{H} \sum_{l=1}^H K\left(\frac{l}{H}\right) SA(t)^{-1} X_l(t) \varepsilon_{i,t-l},$$

$$Y_{it} = \varepsilon_{it} - \nabla \mu_t(\beta_{it})^\top v_{it}.$$

When β is anchored, also define

$$C_{it}(\beta) = \frac{1}{n_t} p_t(\beta)^\top Q_t^{-1} p_t(\beta_{it}).$$

For $l \in [H]$, define, whenever $t+l \in \{H+1, \dots, T\}$ and $i \in [n_{t+l}]$,

$$a_{itl}(\beta) = \nabla \mu_{t+l}(\beta_{i,t+l})^\top K\left(\frac{l}{H}\right) SA(t+l)^{-1} X_l(t+l),$$

and set it equal to zero otherwise. On the high-probability event from Lemma SA-II.3.3, local support gives, for every fixed β ,

$$\begin{aligned} \max_{t \in [T]} \max_{i \in [n]} |\widehat{C}_{it}(\beta)| &\lesssim_{\mathbb{P}} \frac{J^d}{n}, \\ \max_{t \in [T]} \sum_{i=1}^n |\widehat{C}_{it}(\beta)|^m &\lesssim_{\mathbb{P}} \frac{J^{(m-1)d}}{n^{m-1}}, \quad m = 2, 3, \end{aligned} \tag{SA-II.7.3}$$

because $\widehat{p}_t(\beta)$ selects only one well-conditioned interior block and its normalized polynomial coordinates are bounded. The same bounds and scalar Cauchy–Schwarz give, uniformly over $t, s \in [T]$,

$$\sum_{i=1}^n \widehat{C}_{it}(\beta)^2 \widehat{C}_{is}(\beta)^2 \lesssim_{\mathbb{P}} \frac{J^{3d}}{n^3}, \quad \sum_{i=1}^n |\widehat{C}_{it}(\beta) \widehat{C}_{is}(\beta)| \lesssim_{\mathbb{P}} \frac{J^d}{n}. \tag{SA-II.7.4}$$

Under (SA-II.7.2), all these bounds also hold with $\widehat{C}_{it}(\beta)$ replaced by $C_{it}(\beta)$. For a period- t vector $z = (z_1, \dots, z_{n_t})^\top$ and a row satisfying $\widehat{C}_{it}(\beta) \neq 0$, let

$$(\widehat{\Pi}_t z)_i = \widehat{p}_t(\widehat{\beta}_{it})^\top \widehat{Q}_t^{-1} \frac{1}{n_t} \sum_{k=1}^{n_t} \widehat{p}_t(\widehat{\beta}_{kt}) z_k,$$

and set $(\widehat{\Pi}_t z)_i = 0$ on the unused rows. Thus only the interior block selected by the jackknife weight is inverted below. Recall $M_t(b) = \mu_t(b) + b^\top \widetilde{f}_t$ and, finally, write

$$\begin{aligned} \delta_{it} &= \widehat{\beta}_{it} - \beta_{it}, \\ \mathcal{T}_{it}^{(\mu)} &= \mu_t(\widehat{\beta}_{it}) - \mu_t(\beta_{it}) - \nabla \mu_t(\beta_{it})^\top \delta_{it}. \end{aligned}$$

On every row with $\widehat{C}_{it}(\beta) \neq 0$, the following is an exact definition of the residual remainder; set $\widetilde{\mathcal{R}}_{it} = 0$ on the unused rows:

$$\begin{aligned} \widehat{\varepsilon}_{it} &= Y_{it} + \widetilde{\mathcal{R}}_{it}, \quad \widehat{C}_{it}(\beta) \neq 0, \\ \widetilde{\mathcal{R}}_{it} &= \left[(I - \widehat{\Pi}_t) M_t(\widehat{\beta}_{\cdot t}) \right]_i - (\widehat{\Pi}_t \varepsilon_{\cdot t})_i - \nabla \mu_t(\beta_{it})^\top (\delta_{it} - v_{it}) \\ &\quad - \widetilde{f}_t^\top \delta_{it} - \mathcal{T}_{it}^{(\mu)} + \left\{ \widehat{\Pi}_t \left[\nabla \mu_t(\beta_{\cdot t})^\top \delta_{\cdot t} + \widetilde{f}_t^\top \delta_{\cdot t} + \mathcal{T}_{\cdot t}^{(\mu)} \right] \right\}_i. \end{aligned} \tag{SA-II.7.5}$$

Here $M_t(\widehat{\beta}_{\cdot t})$ and the expressions inside $\widehat{\Pi}_t$ denote the corresponding period- t vectors. Number the six terms in the second equality of (SA-II.7.5) in their displayed order. We use the following exact split. The block $\widetilde{\mathcal{R}}_{it}^{(\text{md})}$ contains term (2); the part of term (3) obtained by replacing $\delta_{it} - v_{it}$ with the summation in (SA-II.7.9); and, after writing $\delta_{it} = v_{it} + (\delta_{it} - v_{it})$, the two parts of term (4) containing v_{it} and that same summation. The block $\widetilde{\mathcal{R}}_{it}^{(\text{pr})}$ contains all the remaining pieces. Thus $\widetilde{\mathcal{R}}_{it} = \widetilde{\mathcal{R}}_{it}^{(\text{md})} + \widetilde{\mathcal{R}}_{it}^{(\text{pr})}$. The current-shock pieces of the first block are handled by ordering dates. Its remaining piece and the second block are handled by direct squared-sum bounds.

Define

$$\begin{aligned} \Delta_{\text{JK}} &= \frac{J^d}{n} + J^{2d} \mathbf{R}_{\mathbf{Q}}^2 + J^d \mathbf{R}_{\mathbf{P}}, \\ \mathbf{R}_{\text{JKMD}} &= \frac{J^d}{n} + J^{2d} \mathbf{R}_{\mathbf{Q}}^2 + J^d \mathbf{R}_{\mathbf{P}} + \frac{1}{H} + \frac{1}{\sqrt{T}} + \left(\frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}} \right)^2, \\ \mathbf{R}_{\text{JKPR}} &= H \left(\mathbf{R}_{\mathbf{FS}}^4 + J^{-2 \min\{s_\mu, q+1\}} + \mathbf{B}_{\mathbf{FS}}^2 \right) + \frac{J^d \log^{d+1}(nT)}{n} + \frac{1}{H}. \end{aligned}$$

Lemma SA-II.7.5 (Anchored Residual Arrays). *Suppose that the assumptions and rate conditions of Lemma SA-II.7.2 hold and that $J\mathbf{R}_{\mathbf{FS}} = o(1)$. For every fixed evaluation point satisfying (SA-II.7.2),*

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} (\widehat{C}_{it}(\beta) - C_{it}(\beta))^2 \left(\varepsilon_{it}^2 + \frac{1}{H} \sum_{l=1}^H a_{i,t-l,l}(\beta)^2 \varepsilon_{i,t-l}^2 \right) \lesssim_{\mathbb{P}} T \frac{J^d}{n} \Delta_{\text{JK}}, \tag{SA-II.7.6}$$

and

$$|\mathcal{H}_{\widehat{C}}(Y, Y) - \mathcal{H}_C(Y, Y)| \lesssim_{\mathbb{P}} \frac{J^d}{nT} \Delta_{\text{JK}}^{1/4}, \tag{SA-II.7.7}$$

$$\begin{aligned}
\left| \mathcal{H}_{\widehat{C}}(Y, \widetilde{\mathcal{R}}^{(\text{md})}) \right| &\lesssim_{\mathbb{P}} \frac{J^d}{nT} R_{\text{JKMD}}^{1/2}, \\
\left| \mathcal{H}_{\widehat{C}}(\widetilde{\mathcal{R}}^{(\text{md})}, \widetilde{\mathcal{R}}^{(\text{md})}) \right| &\lesssim_{\mathbb{P}} \frac{J^d}{nT} R_{\text{JKMD}}, \\
\left| \mathcal{H}_{\widehat{C}}(Y, \widetilde{\mathcal{R}}^{(\text{pr})}) \right| &\lesssim_{\mathbb{P}} \frac{J^d}{nT} \sqrt{R_{\text{JKPR}} \log(nT)}, \\
\left| \mathcal{H}_{\widehat{C}}(\widetilde{\mathcal{R}}^{(\text{pr})}, \widetilde{\mathcal{R}}^{(\text{pr})}) \right| &\lesssim_{\mathbb{P}} \frac{J^d}{nT} R_{\text{JKPR}}, \\
\left| \mathcal{H}_{\widehat{C}}(\widetilde{\mathcal{R}}^{(\text{md})}, \widetilde{\mathcal{R}}^{(\text{pr})}) \right| &\lesssim_{\mathbb{P}} \frac{J^d}{nT} \sqrt{R_{\text{JKMD}} R_{\text{JKPR}}}.
\end{aligned} \tag{SA-II.7.8}$$

Proof. Step 1: exact residual expansion. On every row with $\widehat{C}_{it}(\beta) \neq 0$,

$$\widehat{\varepsilon}_{it} = \left[(I - \widehat{\Pi}_t) \{M_t(\beta_{\cdot t}) + \varepsilon_{\cdot t}\} \right]_i.$$

The unused rows never enter $\mathcal{H}_{\widehat{C}}$. Write

$$M_t(\beta_{it}) = M_t(\widehat{\beta}_{it}) - \nabla \mu_t(\beta_{it})^\top \delta_{it} - \widetilde{f}_t^\top \delta_{it} - \mathcal{T}_{it}^{(\mu)}.$$

Substitution and the definition of Y_{it} give (SA-II.7.5). Moreover, the exact first-step decomposition gives

$$\delta_{it} - v_{it} = \check{\beta}_{it} - \beta_{it} + \frac{1}{H} \sum_{l=1}^H K\left(\frac{l}{H}\right) S\{\widehat{A}(t)^{-1} - A(t)^{-1}\} X_l(t) \varepsilon_{i,t-l}. \tag{SA-II.7.9}$$

Thus the first-step bias and the error from replacing $A(t)^{-1}$ by $\widehat{A}(t)^{-1}$ enter the remainder separately. If $\mathbf{q} \geq 1$, polynomial reproduction makes the factor part of $[(I - \widehat{\Pi}_t)M_t(\widehat{\beta}_{\cdot t})]_i$ identically zero. If $\mathbf{q} = 0$, that part is the within-cell deviation of $\widehat{\beta}_{it}^\top \widetilde{f}_t$ and is bounded by $J^{-1} \|\widetilde{f}_t\|$. Together with (SA-II.7.9), this verifies that the two blocks defined after (SA-II.7.5) contain every part of the remainder exactly once.

Step 2: shock-weighted generated-design bounds. Put

$$\omega_{it} = \varepsilon_{it}^2 + \frac{1}{H} \sum_{l=1}^H a_{i,t-l,l}(\beta)^2 \varepsilon_{i,t-l}^2.$$

We first make the leave-out construction explicit. For a set $\mathcal{I}$ containing one or two distinct primitive-shock indices, define

$$\widehat{\beta}_{kr}^{[-\mathcal{I}]} = \widehat{\beta}_{kr} - \sum_{(i,u) \in \mathcal{I}} \frac{\mathbf{1}(k=i, 1 \leq r-u \leq H)}{H} K\left(\frac{r-u}{H}\right) S\widehat{A}(r)^{-1} X_{r-u}(r) \varepsilon_{iu}. \tag{SA-II.7.10}$$

Every partition, basis, Gram block, leverage, and coefficient carrying the superscript $[-\mathcal{I}]$ is rebuilt from the entire trajectory in (SA-II.7.10). Thus a removed shock is deleted from all future first-step estimates whose rolling windows contain it. Conditional on $\mathcal{E}_\infty$ and the remaining innovations, the rebuilt objects do not depend on the removed shock or pair.

Let j_t be the anchored block containing β . Apply the deterministic quasi-uniform rectangle bounds used in Lemma SA-II.3.3 to the leave-out trajectories before substituting the empirical cells. The conditional

convolution and boundary-strip calculations in Lemmas [SA-II.3.1](#) and [SA-II.4.1](#) then give

$$\begin{aligned} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \omega_{it} \left\{ \|\widehat{p}_{j_t t}(\widehat{\beta}_{it})\|^2 + \|p_{j_t t}(\beta_{it})\|^2 \right\} &\lesssim_{\mathbb{P}} \frac{Tn}{J^d}, \\ \sum_{t=H+1}^T \sum_{i=1}^{n_t} \omega_{it} \|\widehat{p}_{j_t t}(\widehat{\beta}_{it}) - p_{j_t t}(\beta_{it})\|^2 &\lesssim_{\mathbb{P}} TnR_{\mathbb{P}}. \end{aligned} \quad (\text{SA-II.7.11})$$

To pass from a leave-out empirical partition to the original one, restore the removed observation at every affected date. Each marginal empirical quantile moves by at most one rank. At each affected date, only a fixed number of observations can cross a boundary of the selected block. The conditional moment bounds used above show that their total weighted contribution over all dates is $O_{\mathbb{P}}(T \log(nT))$. After division by Tn , this is of order $\log(nT)/n$, which is already included in $R_{\mathbb{Q}}$ and $R_{\mathbb{P}}$. For $\mathfrak{q} \geq 1$, the nonswitch coordinate displacement is handled by the order- J derivative of the normalized polynomial basis. The weights containing $a_{i,t-l,l}(\beta)^2$ are controlled by the factor fourth-moment and NED time-average bounds. This proves [\(SA-II.7.11\)](#) without replacing the boundary probability by a fractional power.

For completeness, set

$$\tilde{C}_{it}(\beta) = \frac{1}{n_t} p_t(\beta)^\top \widehat{Q}_t^{-1} \widehat{p}_t(\widehat{\beta}_{it}), \quad h_{it} = \frac{1}{n_t} \widehat{p}_t(\widehat{\beta}_{it})^\top \widehat{Q}_t^{-1} \widehat{p}_t(\widehat{\beta}_{it}).$$

On the selected block, the exact decomposition

$$\begin{aligned} \widehat{C}_{it}(\beta) - C_{it}(\beta) &= \frac{\tilde{C}_{it}(\beta) h_{it}}{1 - h_{it}} \\ &\quad + \frac{1}{n_t} p_t(\beta)^\top (\widehat{Q}_t^{-1} - Q_t^{-1}) \widehat{p}_t(\widehat{\beta}_{it}) \\ &\quad + \frac{1}{n_t} p_t(\beta)^\top Q_t^{-1} \{\widehat{p}_t(\widehat{\beta}_{it}) - p_t(\beta_{it})\} \end{aligned}$$

separates leverage, inverse-Gram, and basis perturbations. The leverage bound, the resolvent identity, and [\(SA-II.7.11\)](#) yield

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} \omega_{it} (\widehat{C}_{it}(\beta) - C_{it}(\beta))^2 \lesssim_{\mathbb{P}} T \frac{J^d}{n} \left\{ \frac{J^d}{n} + J^{2d} R_{\mathbb{Q}}^2 + J^d R_{\mathbb{P}} \right\}.$$

This is [\(SA-II.7.6\)](#).

To bound the terms in $\mathcal{H}_C(Y, Y)$ that contain $\varepsilon_{iu} \varepsilon_{iv}$ with $u \neq v$ in the proof of Lemma [SA-II.7.2](#), define

$$\begin{aligned} \Gamma_{iuv}(C) &= \sum_{r=H+1}^T \sum_{s=H+1}^T C_{ir}(\beta) C_{is}(\beta) \left\{ \mathbb{1}(u=r) - \frac{\mathbb{1}(1 \leq r-u \leq H)}{H} a_{iu, r-u}(\beta) \right\} \\ &\quad \times \left\{ \mathbb{1}(v=s) - \frac{\mathbb{1}(1 \leq s-v \leq H)}{H} a_{iv, s-v}(\beta) \right\} \mathbb{1}(|r-s| \leq H). \end{aligned}$$

Then

$$T_0^2 \mathcal{H}_C(Y, Y) = \sum_{i=1}^n \sum_{u=1}^T \sum_{v=1}^T \Gamma_{iuv}(C) \varepsilon_{iu} \varepsilon_{iv}.$$

We now prove (SA-II.7.7). It bounds the difference between $\mathcal{H}_{\hat{C}}(Y, Y)$ and $\mathcal{H}_C(Y, Y)$. Write $\ell_{it} = \nabla \mu_t(\beta_{it})^\top v_{it}$, so $Y_{it} = \varepsilon_{it} - \ell_{it}$.

Squared-sum bounds for the rolling term. Use the exact leverage, inverse-Gram, and basis decomposition displayed immediately before (SA-II.7.6), now with unit weights. It gives

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} (\hat{C}_{it}(\beta) - C_{it}(\beta))^2 \lesssim_{\mathbb{P}} T \frac{J^d}{n} \Delta_{JK}, \quad \max_{i,t} |\hat{C}_{it}(\beta) - C_{it}(\beta)| \lesssim_{\mathbb{P}} \frac{J^d}{n}.$$

It follows that

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} (\hat{C}_{it}(\beta) - C_{it}(\beta))^4 \lesssim_{\mathbb{P}} T \frac{J^{3d}}{n^3} \Delta_{JK}.$$

We next control ℓ_{it} for observations that receive nonzero weight. Anchoring makes the reference and estimated bases select the same cell index at β . On the high-probability events in Lemmas SA-II.2.3, SA-II.3.2, and SA-II.3.3, the cell-endpoint error and the maximum first-step error are both $o(J^{-1})$. Hence, if $C_{it}(\beta)$ or $\hat{C}_{it}(\beta)$ is nonzero, β_{it} lies in the reference cell containing β or in one of a fixed number of neighboring reference cells. The union of these cells is $\mathcal{E}_\infty$ -measurable and contains $O_{\mathbb{P}}(n/J^d)$ units at each date.

We can therefore sum over this union without conditioning on the random selection induced by $\hat{C}_{it}(\beta)$. The bounded gradient, the conditional fourth moment of ε_{it} , and independence of the primitive shocks give

$$\mathbb{E}[\ell_{it}^4 \mid \mathcal{E}_\infty] \lesssim_{\mathbb{P}} \frac{1}{H^4} \left[\left\{ \sum_{l=1}^H (1 + \|f_{t-l}\|^2) \right\}^2 + \sum_{l=1}^H (1 + \|f_{t-l}\|^4) \right].$$

To sum this bound over the union and over time, use $(\sum_l x_l)^2 \leq H \sum_l x_l^2$ and the time-average fourth-moment bound for the factors. Conditional Markov's inequality then gives

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} \ell_{it}^4 \mathbf{1}(|C_{it}(\beta)| + |\hat{C}_{it}(\beta)| > 0) \lesssim_{\mathbb{P}} \frac{Tn}{J^d H^2}.$$

The weights vanish outside the same union. Their pointwise bounds imply

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} (C_{it}(\beta)^4 + \hat{C}_{it}(\beta)^4) \lesssim_{\mathbb{P}} T \frac{J^{3d}}{n^3}.$$

Scalar Cauchy–Schwarz now gives the first bound below. For the second, use the earlier fourth-power bound for $\hat{C}_{it}(\beta) - C_{it}(\beta)$ in the same way:

$$\begin{aligned} \sum_{t=H+1}^T \sum_{i=1}^{n_t} (C_{it}(\beta)^2 + \hat{C}_{it}(\beta)^2) \ell_{it}^2 &\lesssim_{\mathbb{P}} T \frac{J^d}{nH}, \\ \sum_{t=H+1}^T \sum_{i=1}^{n_t} (\hat{C}_{it}(\beta) - C_{it}(\beta))^2 \ell_{it}^2 &\lesssim_{\mathbb{P}} T \frac{J^d}{nH} \Delta_{JK}^{1/2}. \end{aligned} \tag{SA-II.7.12}$$

Current-shock windows. At date t , the weights $C_{it}(\beta)$ and $\hat{C}_{it}(\beta)$ depend only on the innovation-free environment and innovations dated before t . Thus their products with ε_{it} are martingale differences when the

dates are ordered. The same is true after replacing either weight by their difference. Successive conditioning, (SA-II.7.3), and the unweighted comparison above give

$$\begin{aligned} \sum_{i=1}^n \sum_{s=H+1}^T \left(\sum_{\substack{H+1 \leq t \leq T \\ |t-s| \leq H}} B_{it}(\beta) \varepsilon_{it} \right)^2 &\lesssim_{\mathbb{P}} TH \frac{J^d}{n}, \quad B = C, \hat{C}, \\ \sum_{i=1}^n \sum_{s=H+1}^T \left(\sum_{\substack{H+1 \leq t \leq T \\ |t-s| \leq H}} (\hat{C}_{it}(\beta) - C_{it}(\beta)) \varepsilon_{it} \right)^2 &\lesssim_{\mathbb{P}} TH \frac{J^d}{n} \Delta_{\text{JK}}. \end{aligned} \quad (\text{SA-II.7.13})$$

For unequal dates, condition at the later date; the cross term then has conditional mean zero. Each diagonal term is counted only $O(H)$ times.

In the remaining displays, the common argument β is suppressed. We first compare the current-shock terms. For $B = C, \hat{C}$, order the off-diagonal sum by its later date. Its summed conditional variance is at most the squared pointwise weight bound times the corresponding one-sided sum of squares. The same successive-conditioning calculation as (SA-II.7.13) bounds this sum of squares at the same order. Hence

$$\left| \mathcal{H}_B(\varepsilon, \varepsilon) - \frac{1}{T_0^2} \sum_{i,t} B_{it}^2 \varepsilon_{it}^2 \right| \lesssim_{\mathbb{P}} \sqrt{\frac{HJ^{3d}}{n^3 T^3}}.$$

For the diagonal part, use $\hat{C}_{it}^2 - C_{it}^2 = 2C_{it}(\hat{C}_{it} - C_{it}) + (\hat{C}_{it} - C_{it})^2$. Equation (SA-II.7.6) and scalar Cauchy–Schwarz give

$$|\mathcal{H}_{\hat{C}}(\varepsilon, \varepsilon) - \mathcal{H}_C(\varepsilon, \varepsilon)| \lesssim_{\mathbb{P}} \frac{J^d}{nT} \left\{ \Delta_{\text{JK}}^{1/2} + \Delta_{\text{JK}} + \sqrt{\frac{HJ^d}{nT}} \right\}. \quad (\text{SA-II.7.14})$$

For the current–rolling terms, use

$$\hat{C}_{it} \hat{C}_{is} - C_{it} C_{is} = (\hat{C}_{it} - C_{it}) \hat{C}_{is} + C_{it} (\hat{C}_{is} - C_{is}).$$

Aggregate each current-shock window before applying scalar Cauchy–Schwarz. Equations (SA-II.7.12) and (SA-II.7.13) give

$$\begin{aligned} &|\mathcal{H}_{\hat{C}}(\varepsilon, \ell) - \mathcal{H}_C(\varepsilon, \ell)| \\ &\leq \frac{1}{T_0^2} \left\{ \sum_{i,s} \hat{C}_{is}^2 \ell_{is}^2 \right\}^{1/2} \left\{ \sum_{i,s} \left(\sum_{|t-s| \leq H} (\hat{C}_{it} - C_{it}) \varepsilon_{it} \right)^2 \right\}^{1/2} \\ &\quad + \frac{1}{T_0^2} \left\{ \sum_{i,s} (\hat{C}_{is} - C_{is})^2 \ell_{is}^2 \right\}^{1/2} \left\{ \sum_{i,s} \left(\sum_{|t-s| \leq H} C_{it} \varepsilon_{it} \right)^2 \right\}^{1/2} \\ &\lesssim_{\mathbb{P}} \frac{J^d}{nT} \left(\Delta_{\text{JK}}^{1/2} + \Delta_{\text{JK}}^{1/4} \right). \end{aligned} \quad (\text{SA-II.7.15})$$

Finally, use the order- H absolute row-sum bound rather than treating the signed flat band as positive

semidefinite. This gives

$$\begin{aligned} & |\mathcal{H}_{\hat{C}}(\ell, \ell) - \mathcal{H}_C(\ell, \ell)| \\ & \lesssim \frac{H}{T_0^2} \left\{ \sum_{i,t} (\hat{C}_{it} - C_{it})^2 \ell_{it}^2 \right\}^{1/2} \left[\left\{ \sum_{i,t} \hat{C}_{it}^2 \ell_{it}^2 \right\}^{1/2} + \left\{ \sum_{i,t} C_{it}^2 \ell_{it}^2 \right\}^{1/2} \right] \lesssim_{\mathbb{P}} \frac{J^d}{nT} \Delta_{\text{JK}}^{1/4}. \end{aligned} \quad (\text{SA-II.7.16})$$

Expand $Y = \varepsilon - \ell$ and combine (SA-II.7.14)–(SA-II.7.16). Here $J^d/n \leq \Delta_{\text{JK}}$ and $H \leq T$. Moreover, $\Delta_{\text{JK}} = o(1)$. Hence all the displayed terms are bounded by $\Delta_{\text{JK}}^{1/4}$. This proves (SA-II.7.7).

Step 3: current-shock and inverse-matrix terms. Here and in Step 4, the term numbers refer to the six terms in the second equality of (SA-II.7.5). This step controls $\tilde{\mathcal{R}}^{(\text{md})}$: term (2), the part of term (3) obtained from the summation in (SA-II.7.9), and the two parts of term (4) containing v_{it} and that same summation. For this proof only, write $\tilde{\mathcal{R}}^{(\text{md})} = \tilde{\mathcal{R}}^{(\text{md}, \text{cur})} + \tilde{\mathcal{R}}^{(\text{md}, \text{lag})}$. $\tilde{\mathcal{R}}^{(\text{md}, \text{cur})}$ contains term (2) and the two parts of term (4). Each part of term (4) contains the date- t factor innovation multiplying a coefficient fixed before that innovation. $\tilde{\mathcal{R}}^{(\text{md}, \text{lag})}$ is the stated part of term (3).

We first give the calculation for term (2). Conditional on $\mathcal{C}_t$, $\hat{C}_{it}(\beta)$ and $\hat{\Pi}_t$ are fixed, whereas the current idiosyncratic innovations are independent and centered. The selected least-squares projection has fixed dimension and does not increase the squared norm. Together with (SA-II.7.3), this gives

$$\sum_{i=1}^{n_t} \mathbb{E} \left[\hat{C}_{it}(\beta)^2 \{(\hat{\Pi}_t \varepsilon_t)_i\}^2 \mid \mathcal{C}_t \right] \lesssim_{\mathbb{P}} \left(\frac{J^d}{n} \right)^2. \quad (\text{SA-II.7.17})$$

The diagonal part of its banded quadratic form is therefore $O_{\mathbb{P}}(J^{2d}/(n^2T))$. For unequal dates, order each pair by its later date. The earlier projected error and both weights are then fixed, while the later projected error has conditional mean zero. After division by T_0^4 , the sum of conditional variances is $O_{\mathbb{P}}(\frac{H}{T^3}(\frac{J^d}{n})^4)$. Thus the off-diagonal part is also $O_{\mathbb{P}}(J^{2d}/(n^2T))$, since $H \leq T$.

For the two parts of term (4), the factor innovation is common across units. Order unequal dates as above and condition on the past at the later date. The conditional mean of the later factor innovation is zero. The second bound in (SA-II.7.4) controls the sum over units of the weighted coefficient multiplying that innovation. The conditional second moment of v_{it} is of order H^{-1} . Therefore, the time-diagonal part of the banded product for $-\tilde{f}_t^\top v_{it}$ is bounded by $J^d/(nTH)$. For unequal dates, first sum over the lags shared by the two rolling windows and then take the conditional variance. The factor fourth-moment bound yields $O_{\mathbb{P}}((J^d/n)T^{-3/2})$ for the off-diagonal part. Relative to the common factor $J^d/(nT)$, these are the H^{-1} and $T^{-1/2}$ contributions in $\mathbf{R}_{\text{JKMD}}$. For the coefficient containing $\hat{A}(t)^{-1} - A(t)^{-1}$, use $\hat{A}(t)^{-1} - A(t)^{-1} = \hat{A}(t)^{-1}\{A(t) - \hat{A}(t)\}A(t)^{-1}$ and Lemma SA-II.2.2. This gives the additional factor $\{T^{2/q}/H + \sqrt{\log T/H}\}^2$. Combining these calculations with (SA-II.7.17) gives

$$\begin{aligned} & \sum_{i,s} \left(\sum_{|t-s| \leq H} \hat{C}_{it}(\beta) \tilde{\mathcal{R}}_{it}^{(\text{md}, \text{cur})} \right)^2 \lesssim_{\mathbb{P}} TH \frac{J^d}{n} \mathbf{R}_{\text{JKMD}}, \\ & \left| \mathcal{H}_{\hat{C}}(\tilde{\mathcal{R}}^{(\text{md}, \text{cur})}, \tilde{\mathcal{R}}^{(\text{md}, \text{cur})}) \right| \lesssim_{\mathbb{P}} \frac{J^d}{nT} \mathbf{R}_{\text{JKMD}}, \\ & \left| \mathcal{H}_{\hat{C}}(\varepsilon, \tilde{\mathcal{R}}^{(\text{md}, \text{cur})}) \right| \lesssim_{\mathbb{P}} \frac{J^d}{nT} \mathbf{R}_{\text{JKMD}}^{1/2}. \end{aligned} \quad (\text{SA-II.7.18})$$

Consider the part of term (3) in (SA-II.7.5) obtained by substituting the summation in (SA-II.7.9). This is precisely $\tilde{\mathcal{R}}_{it}^{(\text{md}, \text{lag})}$. On the enlarged-support event used in Step 2, at most $O_{\mathbb{P}}(n/J^d)$ units enter at each date and $\max_i \hat{C}_{it}(\beta)^2 \lesssim_{\mathbb{P}} (J^d/n)^2$. The inverse-matrix identity and Lemma SA-II.2.2 bound the squared matrix difference by $\{T^{2/q}/H + \sqrt{\log T/H}\}^2$. Conditional on $\mathcal{E}_{\infty}$, the rolling sum has second moment of order H^{-1} . Conditional Markov's inequality therefore yields

$$\begin{aligned} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \hat{C}_{it}(\beta)^2 \{\tilde{\mathcal{R}}_{it}^{(\text{md}, \text{lag})}\}^2 &\lesssim_{\mathbb{P}} \left(\frac{J^d}{n}\right)^2 T \frac{n}{J^d} \frac{1}{H} \left(\frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}}\right)^2 \\ &= T \frac{J^d}{nH} \left(\frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}}\right)^2 \lesssim_{\mathbb{P}} T \frac{J^d}{nH} R_{\text{JKMD}}. \end{aligned} \quad (\text{SA-II.7.19})$$

This calculation sums conditional second moments over dates and therefore does not require different rolling windows to be independent.

Cauchy–Schwarz within each lag window and (SA-II.7.19) give the same squared-window bound as the first line of (SA-II.7.18). Hence

$$\sum_{i,s} \left(\sum_{|t-s| \leq H} \hat{C}_{it}(\beta) \tilde{\mathcal{R}}_{it}^{(\text{md})} \right)^2 \lesssim_{\mathbb{P}} TH \frac{J^d}{n} R_{\text{JKMD}}. \quad (\text{SA-II.7.20})$$

The band has at most $2H + 1$ entries in each row. Thus (SA-II.7.19) bounds the lag–lag product. Scalar Cauchy–Schwarz, using the current-part window bound and the lagged squared sum, bounds the current–lag product. Together with (SA-II.7.18), this gives the second line below. For ε paired with the current array, use the third line of (SA-II.7.18). For ε paired with the lagged array, use (SA-II.7.13) and Cauchy–Schwarz. For ℓ , use (SA-II.7.12) with the corresponding current or lagged squared-sum bound. Therefore

$$\begin{aligned} \left| \mathcal{H}_{\hat{C}}(Y, \tilde{\mathcal{R}}^{(\text{md})}) \right| &\lesssim_{\mathbb{P}} \frac{J^d}{nT} R_{\text{JKMD}}^{1/2}, \\ \left| \mathcal{H}_{\hat{C}}(\tilde{\mathcal{R}}^{(\text{md})}, \tilde{\mathcal{R}}^{(\text{md})}) \right| &\lesssim_{\mathbb{P}} \frac{J^d}{nT} R_{\text{JKMD}}. \end{aligned} \quad (\text{SA-II.7.21})$$

Step 4: terms controlled by squared sums. This step controls $\tilde{\mathcal{R}}^{(\text{pr})}$: term (1), the first-step-bias parts of terms (3) and (4), term (5), and every projected piece in term (6). We first bound these terms before forming their banded products. Polynomial reproduction and local approximation bound term (1) by $J^{-2 \min\{s_{\mu}, q+1\}}$ after squaring. The first-step-bias parts of terms (3) and (4) contribute B_{FS}^2 , and the second-order Taylor remainder in term (5) contributes R_{FS}^4 .

For the projected rolling terms in (6), we bound the conditional mean induced by sorting on the estimated betas. First consider $z_{it} = \nabla \mu_t(\beta_{it})^{\top} (\hat{\beta}_{it} - \check{\beta}_{it})$ for $i \in [n_t]$ and $t = H + 1, \dots, T$. For a deterministic admissible rectangle $P \in \mathcal{P}^{\circ}$, with $\mathcal{P}^{\circ}$ as in the proof of Lemma SA-II.4.2, define the vector

$$Z_t(P) = \frac{J^d}{n_t} \sum_{i=1}^{n_t} p_q(\hat{\beta}_{it}; P) z_{it}.$$

For the reference basis, conditioning first on the innovation-free environment makes the rolling term centered.

Applying Lemma SA-II.4.1(a) then gives

$$\max_{H+1 \leq t \leq T} \sup_{P \in \mathcal{P}^\circ} \|\mathbb{E}[Z_t(P) \mid \mathcal{D}_{t-1}]\| \lesssim_{\mathbb{P}} H^{-1} + \mathbf{B}_{\text{FS}}.$$

This mean need not have order H^{-1} alone.

We next bound the centered part before substituting the empirical cell. Conditional on $\mathcal{D}_{t-1}$, the unit histories and primitive innovations entering $Z_t(P)$ are independent over i . The reference-basis term has conditional second moment of order J^{-d}/H . Since $J\mathbf{R}_{\text{FS}} = o(1)$ implies $J/\sqrt{H} = o(1)$, Lemma SA-II.4.1(b), together with $\mathbf{B}_{\text{FS}} = o(H^{-1/2})$, bounds the conditional second moment of each generated-basis summand by $\log^d(nT)/(HJ^d)$. The rectangle and normalized-polynomial class is VC-type with fixed exponent. The conditional L_2 maximal bound, obtained by combining the empirical-process and Hoffmann–Jørgensen inequalities in Cattaneo et al. (2024, Lemmas SA24 and SA41), gives

$$\mathbb{E} \left[\sup_{P \in \mathcal{P}^\circ} \|Z_t(P) - \mathbb{E}[Z_t(P) \mid \mathcal{D}_{t-1}]\|^2 \mid \mathcal{D}_{t-1} \right] \lesssim_{\mathbb{P}} \frac{J^d \log^{d+1}(nT)}{nH} + \frac{J^{2d} \log^2(nT)}{n^2} \mathbb{E} \left[\max_{i \in [n_t]} |z_{it}|^2 \mid \mathcal{D}_{t-1} \right].$$

Localize the $\mathcal{D}_{t-1}$ -measurable constants in these conditional bounds and the first-step Gram bound. The empirical-cell and second-step inverse-Gram events are used only in the pathwise projection bound below. Independent-innovation truncation gives

$$\mathbb{E} \left[\max_{i \in [n_t]} |z_{it}|^2 \mid \mathcal{D}_{t-1} \right] \lesssim \frac{\log(nT)}{H} + \frac{n^{2/q} \log^2(nT)}{H^2} \left\{ \sum_{l=1}^H (1 + \|f_{t-l}\|)^q \right\}^{2/q}.$$

To verify this bound, truncate each coordinate's primitive summands at $H^{-1}\{n \sum_{l=1}^H (1 + \|f_{t-l}\|)^q\}^{1/q}$ and center the truncated and discarded parts separately. Bernstein's inequality bounds the squared maximum of the truncated sums by the displayed right-hand side. Independence over lags and the q th innovation moment bound the squared L_2 norm of the maximum discarded sum by the squared truncation level. The time average of the last factor is $O_{\mathbb{P}}(H^{2/q})$ by the factor moment bound and concavity. Thus the time average of the envelope term is bounded by

$$\frac{J^{2d}}{n^2} \left\{ \frac{\log^3(nT)}{H} + \frac{(nH)^{2/q} \log^4(nT)}{H^2} \right\} \lesssim \frac{J^d \log^{d+1}(nT)}{nH}.$$

After division by the right-hand side, the two ratios are bounded by constants times $(J/\sqrt{H}) \log^{-d}(nT)$ and $(J/\sqrt{H})(H/T^2)^{2/q} \log^{1-d}(nT)$, respectively. This follows from the common ratio restriction, using each of its two denominator branches. Taking expectations after localization and applying Markov's inequality to the nonnegative sum over dates now give

$$\sum_{t=H+1}^T \sup_{P \in \mathcal{P}^\circ} \|Z_t(P) - \mathbb{E}[Z_t(P) \mid \mathcal{D}_{t-1}]\|^2 \lesssim_{\mathbb{P}} \frac{T J^d \log^{d+1}(nT)}{nH}.$$

This is a bound for a sum of squares; it does not require independence between rolling windows.

The estimated selected cell belongs to $\mathcal{P}^\circ$ w.p.a.1. Bounded basis coordinates, the selected inverse-Gram

bound, and (SA-II.7.3) imply, pathwise on the same events,

$$\sum_{i=1}^{n_t} \widehat{C}_{it}(\beta)^2 |(\widehat{\Pi}_t z_t)_i|^2 \lesssim \frac{J^d}{n} \sup_{P \in \mathcal{P}_o} \|Z_t(P)\|^2.$$

Separate $Z_t(P)$ into its conditional mean and centered part in this inequality. Squaring the mean bound retains $\mathbf{B}_{\text{FS}}^2$ in addition to H^{-2} .

To replace $\widehat{\beta}_{it} - \check{\beta}_{it}$ by v_{it} , use the inverse-Gram remainder in (SA-II.7.9). Projection contraction bounds the weighted squared norm by a constant times $(J^d/n)^2$ times the sum of squared unprojected terms over the innovation-free enlarged selected support used in Step 2. This support contains $O_{\mathbb{P}}(n/J^d)$ units at each date, giving

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} \widehat{C}_{it}(\beta)^2 \left| \left\{ \widehat{\Pi}_t [\nabla \mu_t(\beta_t)^\top (\widehat{\beta}_{\cdot t} - \check{\beta}_{\cdot t} - v_{\cdot t})] \right\}_i \right|^2 \lesssim_{\mathbb{P}} \frac{TJ^d}{nH^2}.$$

For the last bound, conditional on $\mathcal{E}_\infty$ the rolling innovation sum has covariance bounded by a constant times H^{-1} on the uniform Gram event. The resolvent identity and the time-averaged squared Gram bound at the end of the proof of Lemma SA-II.2.2 give

$$\frac{1}{T_0} \sum_{t=H+1}^T \|\widehat{A}(t)^{-1} - A(t)^{-1}\|^2 \lesssim_{\mathbb{P}} H^{-1}.$$

Conditional Markov's inequality and the support count then establish the displayed projected inverse-Gram bound. Combining the mean, centered, and Gram-replacement bounds proves

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} \widehat{C}_{it}(\beta)^2 \left| \left\{ \widehat{\Pi}_t [\nabla \mu_t(\beta_t)^\top v_{\cdot t}] \right\}_i \right|^2 \lesssim_{\mathbb{P}} T \frac{J^d}{n} \left\{ \frac{J^d \log^{d+1}(nT)}{nH} + \frac{1}{H^2} + \mathbf{B}_{\text{FS}}^2 \right\}.$$

For each coordinate of $\widehat{\beta}_{it} - \check{\beta}_{it}$, the same argument applies: the basis-perturbation mean and variance calculations use a constant coordinate multiplier in place of $\nabla \mu_t(\beta_{it})$. The resulting projected arrays and weights are $\mathcal{F}_{t-1}$ -measurable. Conditioning at date t and using the conditional factor moment bound therefore gives the same displayed order for $\{\widehat{\Pi}_t [\widehat{f}_t^\top v_{\cdot t}]\}_i$. The projected inverse-Gram terms, with or without the current factor, are covered by the inverse-Gram bound above. Finally, bounded absolute row sums of the selected projection give contributions $TJ^d \mathbf{B}_{\text{FS}}^2/n$ and $TJ^d \mathbf{R}_{\text{FS}}^4/n$ for the projected first-step-bias and Taylor terms, respectively; use the same conditional factor argument for the factor-times-bias term. The square of their sum is at most a fixed constant times the sum of their squares. Hence

$$\begin{aligned} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \widehat{C}_{it}(\beta)^2 \{\widetilde{\mathcal{R}}_{it}^{(\text{pr})}\}^2 &\lesssim_{\mathbb{P}} T \frac{J^d}{n} \left\{ J^{-2 \min\{s_\mu, q+1\}} + \mathbf{B}_{\text{FS}}^2 + \mathbf{R}_{\text{FS}}^4 + \frac{J^d \log^{d+1}(nT)}{nH} + \frac{1}{H^2} \right\} \\ &= T \frac{J^d}{nH} \mathbf{R}_{\text{JKPR}}. \end{aligned} \tag{SA-II.7.22}$$

Since $2|xy| \leq x^2 + y^2$ and each date has at most $2H + 1$ partners in the band, this gives

$$\left| \mathcal{H}_{\widehat{C}}(\widetilde{\mathcal{R}}^{(\text{pr})}, \widetilde{\mathcal{R}}^{(\text{pr})}) \right| \lesssim_{\mathbb{P}} \frac{J^d}{nT} \mathbf{R}_{\text{JKPR}}. \tag{SA-II.7.23}$$

Move the factor indexed by s outside the inner sum and apply scalar Cauchy–Schwarz.

Equation (SA-II.7.22) controls the first factor below. Equation (SA-II.7.20) controls the second:

$$\begin{aligned} & \left| \mathcal{H}_{\widehat{C}}(\widetilde{\mathcal{R}}^{(\text{md})}, \widetilde{\mathcal{R}}^{(\text{pr})}) \right| \\ & \leq \frac{1}{T_0^2} \left\{ \sum_{i,s} \widehat{C}_{is}(\beta)^2 \{ \widetilde{\mathcal{R}}_{is}^{(\text{pr})} \}^2 \right\}^{1/2} \left\{ \sum_{i,s} \left(\sum_{|t-s| \leq H} \widehat{C}_{it}(\beta) \widetilde{\mathcal{R}}_{it}^{(\text{md})} \right)^2 \right\}^{1/2} \lesssim_{\mathbb{P}} \frac{J^d}{nT} \sqrt{R_{\text{JKMD}} R_{\text{JKPR}}}. \end{aligned} \quad (\text{SA-II.7.24})$$

Equation (SA-II.7.13) controls the lag windows of ε . Cauchy–Schwarz within each window and (SA-II.7.12) give the same bound for ℓ . Thus

$$\sum_{i,s} \left(\sum_{|t-s| \leq H} \widehat{C}_{it}(\beta) Y_{it} \right)^2 \lesssim_{\mathbb{P}} TH \frac{J^d}{n}.$$

Applying the same scalar Cauchy–Schwarz bound with Y in place of $\widetilde{\mathcal{R}}^{(\text{md})}$ first gives the right-hand side below without the factor $\sqrt{\log(nT)}$. We retain the weaker displayed bound to match the sufficient condition in (SA-II.7.1):

$$\left| \mathcal{H}_{\widehat{C}}(Y, \widetilde{\mathcal{R}}^{(\text{pr})}) \right| \lesssim_{\mathbb{P}} \frac{J^d}{nT} \sqrt{R_{\text{JKPR}} \log(nT)}. \quad (\text{SA-II.7.25})$$

Equations (SA-II.7.21), (SA-II.7.23), (SA-II.7.24), and (SA-II.7.25) prove (SA-II.7.8). $\square$

Lemma SA-II.7.6 (Selected-Block Residual Bound). *Suppose that the assumptions and common rate conditions stated before parts (i)–(ii) of Lemma SA-II.7.2 hold. For every fixed $\beta \in \mathcal{B}^\circ$, without imposing (SA-II.7.2),*

$$\left| \mathcal{H}_{\widehat{C}}(\widehat{\varepsilon}, \widehat{\varepsilon}) \right| \lesssim_{\mathbb{P}} \frac{J^d}{nT}.$$

Proof. On the rows used by $\widehat{C}_{it}(\beta)$, the exact identity in Step 1 of the preceding proof can be written as

$$\widehat{\varepsilon}_{it} = e_{it} + r_{it}, \quad e_{it} = \left[(I - \widehat{\Pi}_t) \varepsilon_{\cdot t} \right]_i, \quad r_{it} = \left[(I - \widehat{\Pi}_t) M_t(\beta_{\cdot t}) \right]_i,$$

and all three quantities are set to zero on unused rows. Conditional on $\mathcal{C}_t$, the selected empirical block, $\widehat{C}_{it}(\beta)$, and $\widehat{\Pi}_t$ are fixed, whereas the current innovations entering e_{it} are cross-sectionally independent and centered.

We first record two weighted squared-sum bounds. Projection contraction, the uniform cell-count and inverse-Gram bounds, and conditional second moments give

$$\sum_{t=H+1}^T \sum_{i=1}^{n_t} \widehat{C}_{it}(\beta)^2 e_{it}^2 \lesssim_{\mathbb{P}} T \frac{J^d}{n}. \quad (\text{SA-II.7.26})$$

For r_{it} , polynomial reproduction removes the factor part of $M_t(\widehat{\beta}_{it})$ when $\mathbf{q} \geq 1$, while local polynomial approximation controls its conditional-mean part. When $\mathbf{q} = 0$, the entire within-cell variation is of order

J^{-1} . Apply the first-step decomposition (SA-II.7.9) inside r_{it} . The local polynomial residual contributes $J^{-2 \min\{s_\mu, q+1\}}$ after squaring. The linear first-step term has selected-cell mean square of order H^{-1} , its bias contributes $\mathbf{B}_{\text{FS}}^2$, and the Taylor remainder contributes $\mathbf{R}_{\text{FS}}^4$. For the projected rolling terms, the coarser H^{-1} order suffices here. On the selected block at each date $t = H+1, \dots, T$, projection contraction and $\max_{i \in [n_t]} \hat{C}_{it}(\beta)^2 \lesssim_{\mathbb{P}} (J^d/n)^2$ bound the weighted squared norm of a projection by a constant times $(J^d/n)^2$ times the unprojected block's squared norm. The selected-cell rolling moment bound just used therefore also bounds the projected rolling contribution by $O_{\mathbb{P}}(TJ^d/(nH))$. Bounded absolute projection row sums handle the bias and Taylor pieces, and conditioning at date t handles current factor multipliers. This argument does not use the sharp anchored projected bound or impose $J\mathbf{R}_{\text{FS}} = o(1)$. These bounds hold uniformly over the deterministic quasi-uniform rectangle class, so they may be evaluated at the realized selected block. Hence

$$\begin{aligned} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \hat{C}_{it}(\beta)^2 r_{it}^2 &\lesssim_{\mathbb{P}} T \frac{J^d}{n} \left\{ \frac{1}{H} + J^{-2 \min\{s_\mu, q+1\}} + \mathbf{R}_{\text{FS}}^4 + \mathbf{B}_{\text{FS}}^2 + \frac{J^d \log^{d+1}(nT)}{nH} + \frac{1}{H^2} \right\} \\ &\lesssim_{\mathbb{P}} T \frac{J^d}{nH}. \end{aligned} \quad (\text{SA-II.7.27})$$

The first term in braces covers both the unprojected and projected rolling first-step terms. The remaining nonnegative terms are bounded by $\mathbf{R}_{\text{JKPR}}/H$, so the last inequality follows from (SA-II.7.1). This calculation does not compare the cell containing β with a reference cell. The underlying weighted moment bounds hold uniformly over the deterministic quasi-uniform rectangle class before the realized empirical block is selected.

The current-innovation array $\{\hat{C}_{it}(\beta)e_{it}\}$ is a chronological martingale-difference array. Martingale orthogonality and (SA-II.7.26) therefore give

$$\sum_{s=H+1}^T \sum_{i=1}^n \left(\sum_{\substack{H+1 \leq t \leq T \\ |t-s| \leq H}} \hat{C}_{it}(\beta)e_{it} \right)^2 \lesssim_{\mathbb{P}} TH \frac{J^d}{n}. \quad (\text{SA-II.7.28})$$

In the displays below, $\sum_{s,i}$ uses these same ranges, and invalid unit-date pairs continue to contribute zero. This aggregation is important: bounding the individual time pairs first would lose a factor H . By scalar Cauchy-Schwarz after the lag aggregation, (SA-II.7.27) and (SA-II.7.28) imply

$$|\mathcal{H}_{\hat{C}}(e, r)| \leq \frac{1}{T_0^2} \left\{ \sum_{s,i} \hat{C}_{is}(\beta)^2 r_{is}^2 \right\}^{1/2} \left\{ \sum_{s,i} \left(\sum_{|t-s| \leq H} \hat{C}_{it}(\beta)e_{it} \right)^2 \right\}^{1/2} \lesssim_{\mathbb{P}} \frac{J^d}{nT}.$$

The absolute band bound and (SA-II.7.27) likewise give

$$|\mathcal{H}_{\hat{C}}(r, r)| \lesssim \frac{H}{T_0^2} \sum_{t,i} \hat{C}_{it}(\beta)^2 r_{it}^2 \lesssim_{\mathbb{P}} \frac{J^d}{nT}.$$

Finally, split $\mathcal{H}_{\hat{C}}(e, e)$ into its time-diagonal and off-diagonal parts. Conditional second moments and (SA-II.7.26) show that the diagonal is $O_{\mathbb{P}}(J^d/(nT))$. For the off-diagonal part, martingale orthogonality, the

leverage bound, and (SA-II.7.4) give

$$|\text{off-diagonal part of } \mathcal{H}_{\widehat{C}}(e, e)| = O_{\mathbb{P}}\left(\sqrt{\frac{HJ^{3d}}{n^3T^3}}\right) = o_{\mathbb{P}}\left(\frac{J^d}{nT}\right),$$

where the last equality uses $H \leq T$ and $J^d/n = o(1)$. Expanding $\mathcal{H}_{\widehat{C}}(e + r, e + r)$ and combining the preceding bounds proves the result. The proof controls only magnitude and therefore does not require the selected block to be innovation-free. $\square$

SA-II.7.6 Proof of Lemma SA-II.7.2

Step 1: common reduction. By the Sherman–Morrison identity above and the convention for invalid unit–date pairs, $\widehat{\Sigma}_{\varepsilon}(\beta) = \mathcal{H}_{\widehat{C}}(\widehat{\varepsilon}, \widehat{\varepsilon})$. Expanding (SA-II.7.5) in the symmetric bilinear form gives

$$\begin{aligned} \widehat{\Sigma}_{\varepsilon}(\beta) &= \mathcal{H}_{\widehat{C}}(Y, Y) + 2\mathcal{H}_{\widehat{C}}(Y, \widetilde{\mathcal{R}}^{(\text{md})}) + 2\mathcal{H}_{\widehat{C}}(Y, \widetilde{\mathcal{R}}^{(\text{pr})}) \\ &\quad + \mathcal{H}_{\widehat{C}}(\widetilde{\mathcal{R}}^{(\text{md})}, \widetilde{\mathcal{R}}^{(\text{md})}) + 2\mathcal{H}_{\widehat{C}}(\widetilde{\mathcal{R}}^{(\text{md})}, \widetilde{\mathcal{R}}^{(\text{pr})}) + \mathcal{H}_{\widehat{C}}(\widetilde{\mathcal{R}}^{(\text{pr})}, \widetilde{\mathcal{R}}^{(\text{pr})}). \end{aligned}$$

Step 2: rough bound for every fixed evaluation point. For any fixed $\beta \in \mathcal{B}^{\circ}$, do not impose either $JR_{\text{FS}} = o(1)$ or (SA-II.7.2). Apply Lemma SA-II.7.6 directly to the common reduction in Step 1. It gives

$$|\widehat{\Sigma}_{\varepsilon}(\beta)| \lesssim_{\mathbb{P}} \frac{J^d}{nT}.$$

The proof of Lemma SA-II.6.1 establishes $\sum_{t,i} \varsigma_{it}^2 W_{it}(\beta)^2 \lesssim_{\mathbb{P}} TJ^d/n$. By the definition of $\Sigma_{\varepsilon}(\beta)$ following (SA-II.6.1), this gives $\Sigma_{\varepsilon}(\beta) \lesssim_{\mathbb{P}} J^d/(nT)$. The triangle inequality therefore yields

$$\widehat{\Sigma}_{\varepsilon}(\beta) - \Sigma_{\varepsilon}(\beta) = O_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

Finally, $J^d/n = o(1)$ under (SA-II.7.1), so the last display is $o_{\mathbb{P}}(T^{-1})$. This argument controls only the magnitude of the realized selected block and therefore requires neither anchoring nor the stronger localization $JR_{\text{FS}} = o(1)$.

Step 3: anchored evaluation points. Suppose (SA-II.7.2) holds and $JR_{\text{FS}} = o(1)$. Re-indexing the primitive shocks in Y_{it} gives the reference weight

$$W_{it}(\beta) = C_{it}(\beta) - \frac{1}{H} \sum_{l=1}^H C_{i,t+l}(\beta) a_{iul}(\beta). \quad (\text{SA-II.7.29})$$

This is exactly the weight $W_{it}(\beta)$ defined in Section SA-II.6, and it is $\mathcal{E}_{\infty}$ -measurable. Consequently, Lemma SA-II.7.5 yields

$$\begin{aligned} \widehat{\Sigma}_{\varepsilon}(\beta) &= \mathcal{H}_C(Y, Y) + \mathcal{R}_{\text{JK}}^{\star}(\beta), \\ |\mathcal{R}_{\text{JK}}^{\star}(\beta)| &\lesssim_{\mathbb{P}} \frac{J^d}{nT} \rho_{\text{JK}}, \\ \rho_{\text{JK}} &= \Delta_{\text{JK}}^{1/4} + R_{\text{JKMD}}^{1/2} + R_{\text{JKMD}} + R_{\text{JKPR}} + \sqrt{R_{\text{JKPR}} \log(nT)}. \end{aligned} \quad (\text{SA-II.7.30})$$

The mixed term in (SA-II.7.8) is absorbed here by $\sqrt{ab} \leq (a+b)/2$. Lemma SA-II.3.3, $J^2 \log T/n = o(1)$, $JR_{\text{FS}} = o(1)$, and $T^{2/q}/H + \sqrt{\log T/H} = o(1)$ imply $\Delta_{\text{JK}} = o(1)$ and $R_{\text{JKMD}} = o(1)$, while (SA-II.7.1) implies $R_{\text{JKPR}} \log(nT) = o(1)$. Hence $\rho_{\text{JK}} = o(1)$.

It remains to identify the leading part of $\mathcal{H}_C(Y, Y)$. After re-indexing its primitive shocks, write

$$\begin{aligned}\mathcal{H}_C(Y, Y) &= \mathcal{D}_{\text{JK}}^\circ(\beta) + \mathcal{O}_{\text{JK}}^\circ(\beta), \\ \mathcal{D}_{\text{JK}}^\circ(\beta) &= \frac{1}{(T-H)^2} \sum_{i=1}^n \sum_{t=1}^T W_{it}(\beta)^2 \varepsilon_{it}^2, \\ \mathcal{O}_{\text{JK}}^\circ(\beta) &:= \mathcal{H}_C(Y, Y) - \mathcal{D}_{\text{JK}}^\circ(\beta).\end{aligned}$$

The lag restriction is automatically satisfied between any two appearances of the same primitive shock, so its coefficient is precisely $W_{it}(\beta)^2$. Every term in $\mathcal{O}_{\text{JK}}^\circ(\beta)$ is instead a product of two distinct primitive shocks.

Conditional on $\mathcal{E}_\infty$, these shocks are independent and centered, and therefore

$$\mathbb{E}[\mathcal{D}_{\text{JK}}^\circ(\beta) \mid \mathcal{E}_\infty] = \Sigma_\varepsilon(\beta), \quad \mathbb{E}[\mathcal{O}_{\text{JK}}^\circ(\beta) \mid \mathcal{E}_\infty] = 0.$$

The local-support bounds in (SA-II.7.3), Jensen's inequality over the rolling window, and $T^{-1} \sum_t (1 + \|f_t\|^4) = O_{\mathbb{P}}(1)$ give

$$\sum_{i=1}^n \sum_{t=1}^T W_{it}(\beta)^2 \lesssim_{\mathbb{P}} T \frac{J^d}{n}, \quad \sum_{i=1}^n \sum_{t=1}^T W_{it}(\beta)^4 \lesssim_{\mathbb{P}} T \frac{J^{3d}}{n^3}.$$

The conditional fourth-moment bound thus gives

$$\mathcal{D}_{\text{JK}}^\circ(\beta) - \Sigma_\varepsilon(\beta) = O_{\mathbb{P}}\left(\sqrt{\frac{J^{3d}}{n^3 T^3}}\right). \quad (\text{SA-II.7.31})$$

For the off-diagonal term, (SA-II.7.4), Jensen's inequality, and $T^{-1} \sum_t (1 + \|f_t\|^4) = O_{\mathbb{P}}(1)$ give the following bound on the sum of the squared coefficients:

$$\sum_{i=1}^n \sum_{u \neq v} \Gamma_{iuv}(C)^2 \lesssim_{\mathbb{P}} TH \frac{J^{3d}}{n^3}.$$

Conditional orthogonality of distinct-shock products therefore gives

$$\mathcal{O}_{\text{JK}}^\circ(\beta) = O_{\mathbb{P}}\left(\sqrt{\frac{HJ^{3d}}{n^3 T^3}}\right). \quad (\text{SA-II.7.32})$$

Combining the preceding displays preserves the explicit leading expansion

$$\begin{aligned}\hat{\Sigma}_\varepsilon(\beta) - \Sigma_\varepsilon(\beta) &= \frac{1}{(T-H)^2} \sum_{i=1}^n \sum_{t=1}^T W_{it}(\beta)^2 (\varepsilon_{it}^2 - \varsigma_{it}^2) \\ &\quad + \mathcal{O}_{\text{JK}}^\circ(\beta) + \mathcal{R}_{\text{JK}}^\star(\beta).\end{aligned} \quad (\text{SA-II.7.33})$$

Since $H \leq T$ and $J^d/n = o(1)$, the two leading fluctuations in (SA-II.7.31)–(SA-II.7.32) are $o_{\mathbb{P}}(J^d/(nT))$; the remainder has the same conclusion by (SA-II.7.30). Thus, for every fixed anchored evaluation point—both

$\beta = 0$ and an anchored $\beta \neq 0$ —

$$\widehat{\Sigma}_\varepsilon(\beta) = \Sigma_\varepsilon(\beta) + o_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

□

SA-II.7.7 Proof of Theorem SA-II.7.3

For $\beta \neq 0$, part (i) of Lemma SA-II.7.2 and $\Sigma_\varepsilon(\beta) \lesssim_{\mathbb{P}} J^d/(nT)$ give

$$\widehat{\Sigma}_\varepsilon(\beta) - \Sigma_\varepsilon(\beta) = O_{\mathbb{P}}\left(\frac{J^d}{nT}\right) = o_{\mathbb{P}}(T^{-1}).$$

Lemma SA-II.7.1 then gives

$$\widehat{\Sigma}(\beta) - \Sigma(\beta) - \mathcal{B}_{\mathbf{F}}(\beta) = o_{\mathbb{P}}(T^{-1}).$$

For $\beta = 0$, $\Sigma_f(0) = \widehat{\Sigma}_f(0) = \mathcal{B}_{\mathbf{F}}(0) = 0$. The additive claim follows directly from part (ii) of Lemma SA-II.7.2. If $(nT/J^d)\Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} 1$, then $\Sigma(0) = \Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} J^d/(nT)$, so the additive error is also $o_{\mathbb{P}}(\Sigma(0))$. □

SA-II.7.8 Proof of Theorem SA-II.7.4

Fix an evaluation point β and let u_t be the periodwise estimation error entering the Fama–MacBeth dispersion, with $\bar{u} = (T - H)^{-1} \sum_{t=H+1}^T u_t$. For any variance target V_u , adding and subtracting $\mu_t(\beta)$ and $\bar{\mu}_T(\beta; H)$ gives

$$\left| \frac{1}{(T - H)^2} \sum_{t=H+1}^T \left[u_t - \bar{u} + \mu_t(\beta) - \bar{\mu}_T(\beta; H) \right]^2 - V_u - \Sigma_\mu(\beta) \right| \leq \mathcal{V}_{\mathbf{FM},1} + \mathcal{V}_{\mathbf{FM},2} + \mathcal{V}_{\mathbf{FM},3},$$

where

$$\begin{aligned} \mathcal{V}_{\mathbf{FM},1} &= \left| \frac{1}{(T - H)^2} \sum_{t=H+1}^T u_t^2 - V_u \right|, \\ \mathcal{V}_{\mathbf{FM},2} &= \frac{\bar{u}^2}{T - H}, \\ \mathcal{V}_{\mathbf{FM},3} &= \left| \frac{2}{(T - H)^2} \sum_{t=H+1}^T u_t (\mu_t(\beta) - \bar{\mu}_T(\beta; H)) \right|. \end{aligned}$$

Step 1: The case $\beta \neq 0$. Take $u_t = \widehat{M}_t(\beta) - \mu_t(\beta)$ and $V_u = \Sigma_f(\beta) + \Sigma_\varepsilon(\beta)$. The left-hand side in the common display is $|\widehat{\Sigma}_{\mathbf{FM}}(\beta) - \Sigma_f(\beta) - \Sigma_\varepsilon(\beta) - \Sigma_\mu(\beta)|$.

For $\mathcal{V}_{\mathbf{FM},1}$, the per-period linearization in Lemma SA-II.4.2 gives

$$\widehat{M}_t(\beta) - \mu_t(\beta) = \beta^\top \widetilde{f}_t + \mathcal{J}_{\mathbf{FM},t}(\beta) + \mathcal{R}_{\mathbf{FM},t}(\beta),$$

where $\mathcal{J}_{\mathbf{FM},t}(\beta)$ collects the idiosyncratic leading terms and $\mathcal{R}_{\mathbf{FM},t}(\beta)$ collects the remaining linearization

errors. The same weight calculations as in the proof of Lemma [SA-II.7.2](#) give

$$\frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathcal{J}_{\text{FM},t}(\beta)^2 \lesssim_{\mathbb{P}} \frac{J^d}{nT} = o_{\mathbb{P}}(T^{-1}),$$

and the periodwise residual bounds apply. Since

$$J^{-\min\{s_{\mu}, q+1\}} + \mathbf{R}_{\text{FM}} = o(1),$$

they give

$$\frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathcal{R}_{\text{FM},t}(\beta)^2 = o_{\mathbb{P}}(T^{-1}).$$

Hence all terms involving $\mathcal{J}_{\text{FM},t}(\beta)$ or $\mathcal{R}_{\text{FM},t}(\beta)$ are $o_{\mathbb{P}}(T^{-1})$ by Cauchy–Schwarz. For the factor square,

$$\frac{1}{(T-H)^2} \sum_{t=H+1}^T \left[(\beta^{\top} \tilde{f}_t)^2 - \text{Var}(\beta^{\top} f_t \mid \mathcal{G}_{t-1}) \right] = o_{\mathbb{P}}(T^{-1}),$$

because the summands are martingale differences with conditional second moments bounded by Assumption [SA-DGP\(iv\)](#). Since $\Sigma_{\varepsilon}(\beta) \lesssim_{\mathbb{P}} J^d/(nT) = o(T^{-1})$, we obtain

$$\mathcal{V}_{\text{FM},1} = o_{\mathbb{P}}\left(\frac{1}{T}\right),$$

for $\beta \neq 0$.

Next, the aggregate expansion gives $\widehat{\mu}(\beta) - \bar{\mu}_T(\beta; H) = o_{\mathbb{P}}(1)$: the factor term is $O_{\mathbb{P}}(T^{-1/2})$, the idiosyncratic term is $O_{\mathbb{P}}(\sqrt{J^d/(nT)}) = o_{\mathbb{P}}(T^{-1/2})$, and the residual terms are $o_{\mathbb{P}}(1)$ under the displayed rate condition. Therefore

$$\mathcal{V}_{\text{FM},2} = o_{\mathbb{P}}\left(\frac{1}{T}\right).$$

Finally, for $\mathcal{V}_{\text{FM},3}$, write

$$\mathcal{V}_{\text{FM},3} = \left| \frac{2}{(T-H)^2} \sum_{t=H+1}^T (\widehat{M}_t(\beta) - \mu_t(\beta)) \mu_t(\beta) - \frac{2\bar{\mu}_T(\beta; H)}{T-H} (\widehat{\mu}(\beta) - \bar{\mu}_T(\beta; H)) \right|.$$

For the first term, by the linearization in Lemma [SA-II.4.2](#), the leading term in $\widehat{M}_t(\beta) - \mu_t(\beta)$ is a martingale difference $\beta^{\top} \tilde{f}_t$, and thus the corresponding component in the first term of $\mathcal{V}_{\text{FM},3}$ is

$$\frac{2}{(T-H)^2} \sum_{t=H+1}^T \beta^{\top} \tilde{f}_t \mu_t(\beta).$$

Since $\mu_t(\beta)$ is uniformly bounded, the predictable quadratic variation of this term is bounded by

$$\frac{\|\beta\|^2}{(T-H)^4} \sum_{t=H+1}^T \mu_t(\beta)^2 \mathbb{E}[\|\tilde{f}_t\|^2 \mid \mathcal{G}_{t-1}] \lesssim \frac{1}{T^3},$$

so that by the Lenglart–Rebolledo inequality,

$$\frac{2}{(T-H)^2} \sum_{t=H+1}^T \beta^\top \tilde{f}_t \mu_t(\beta) \lesssim_{\mathbb{P}} T^{-3/2}.$$

The idiosyncratic leading term and the remaining linearization errors contribute $o_{\mathbb{P}}(T^{-1})$ by the same predictable-variation and Cauchy–Schwarz bounds used for $\mathcal{V}_{\text{FM},1}$, plus the fact that the deterministic periodwise residual scale is $o(1)$.

For the second term in $\mathcal{V}_{\text{FM},3}(\beta)$, since $\bar{\mu}_T(\beta; H) \lesssim_{\mathbb{P}} 1$ and $\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H) = o_{\mathbb{P}}(1)$ by the preceding aggregate expansion,

$$\frac{\bar{\mu}_T(\beta; H)}{T-H} (\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H)) = o_{\mathbb{P}}(T^{-1}).$$

Thus,

$$\mathcal{V}_{\text{FM},3} = o_{\mathbb{P}}\left(\frac{1}{T}\right).$$

Combining the bounds for $\mathcal{V}_{\text{FM},1}$, $\mathcal{V}_{\text{FM},2}$, and $\mathcal{V}_{\text{FM},3}$, and using $\Sigma_{\varepsilon}(\beta) \lesssim_{\mathbb{P}} J^d/(nT) = o(T^{-1})$, yields the desired result for $\beta \neq 0$.

Step 2: The case $\beta = 0$. Consider the bias-corrected estimator $\tilde{\Sigma}_{\text{FM}}(0)$. The periodwise version of Lemmas [SA-I.3.1](#) and [SA-II.4.2](#), with the smoothing-bias component $\mathcal{B}_t(0)$ removed exactly, gives

$$\widehat{M}_t(0) - \mathcal{B}_t(0) - \mu_t(0) = \mathcal{E}_{\text{o},t}(0) + \mathcal{R}_{\text{FM},t}, \quad (\text{SA-II.7.34})$$

The remainder $\mathcal{R}_{\text{FM},t}$ is detailed below. Equation [\(SA-II.7.35\)](#) will give the time-averaged squared bound used in the proof. The rate $\mathbf{R}_{\text{FM}}$ collects the periodwise versions of the residual terms in the stochastic linearization: the first-step bias and smooth-interior generated-regressor terms in $\mathbf{R}_{\text{LA}}$, the first-step time-series approximation term $\log(nT)/H$, and, when $\mathbf{q} = 0$, the curvature and endpoint-comparison terms.

In the common decomposition, take

$$u_t = \widehat{M}_t(0) - \mathcal{B}_t(0) - \mu_t(0), \quad V_u = \Sigma_{\varepsilon,\text{o}}(0).$$

Then $\bar{u} = \hat{\mu}(0) - \mathcal{B}(0) - \bar{\mu}_T(0; H)$, and the left-hand side is $|\tilde{\Sigma}_{\text{FM}}(0) - \Sigma_{\varepsilon,\text{o}}(0) - \Sigma_{\mu}(0)|$.

For $\mathcal{V}_{\text{FM},1}$, insert [\(SA-II.7.34\)](#). Expanding the square gives

$$\begin{aligned} & \frac{1}{(T-H)^2} \sum_{t=H+1}^T (\widehat{M}_t(0) - \mathcal{B}_t(0) - \mu_t(0))^2 \\ &= \frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathcal{E}_{\text{o},t}(0)^2 + \frac{2}{(T-H)^2} \sum_{t=H+1}^T \mathcal{E}_{\text{o},t}(0) \mathcal{R}_{\text{FM},t} + \frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathcal{R}_{\text{FM},t}^2. \end{aligned}$$

At the origin, $\hat{p}_t(0) = p_t(0)$, so the periodwise leading weights below use the same reference zero-cell coordinates as Lemma [SA-II.4.2](#). The periodwise weights and volatility scales are $\mathcal{E}_{\infty}$ -measurable; the primitive

innovations in $\mathcal{E}_{o,t}(0)$ are conditionally independent over i with mean zero and unit variance. Hence

$$\mathbb{E}[\mathcal{E}_{o,t}(0)^2 \mid \mathcal{E}_\infty] = \sum_{i=1}^{n_t} \varsigma_{it}^2 \left(\frac{1}{n_t} p_t(0)^\top Q_t^{-1} p_t(\beta_{it}) \right)^2.$$

Therefore,

$$\frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathcal{E}_{o,t}(0)^2 = \Sigma_{\varepsilon,o}(0) + o_{\mathbb{P}}\left(\frac{J^d}{nT}\right),$$

where the deviation follows from conditional independence given $\mathcal{E}_\infty$ and the same fourth-moment calculation used for the jackknife variance terms. Also, $\Sigma_{\varepsilon,o}(0) \lesssim_{\mathbb{P}} J^d/(nT)$. We next control the cross product without applying Cauchy–Schwarz to the full remainder. Write the periodwise summands of the linearization remainders as follows:

$$\begin{aligned} \mathcal{L}_{\mathbf{B},t}(0) &= \hat{p}_t(0)^\top \hat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) [\nabla \mu_t(\beta_{it})]^\top (\check{\beta}_{it} - \beta_{it}), \\ \mathcal{L}_{\mathbf{V},t}(0) &= \hat{p}_t(0)^\top \hat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) [\nabla \mu_t(\beta_{it})]^\top (\hat{\beta}_{it} - \check{\beta}_{it}), \\ \mathcal{A}_t(0) &= p_t(0)^\top Q_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} p_t(\beta_{it}) [\nabla \mu_t(\beta_{it})]^\top \frac{1}{H} \sum_{s=1}^H S A(t)^{-1} X_s(t) K\left(\frac{s}{H}\right) \varepsilon_{i(t-s)}. \end{aligned}$$

The five periodwise remainders from Lemma [SA-I.3.1](#) are

$$\begin{aligned} \mathcal{R}_{1,t}(0) &= \hat{p}_t(0)^\top \left(\hat{Q}_t^{-1} - Q_t^{-1} \right) \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) \varepsilon_{it}, \\ \mathcal{R}_{2,t}(0) &= \hat{p}_t(0)^\top Q_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} (\hat{p}_t(\hat{\beta}_{it}) - p_t(\beta_{it})) \varepsilon_{it}, \\ \mathcal{R}_{3,t}(0) &= \hat{p}_t(0)^\top \hat{Q}_t^{-1} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) \hat{\beta}_{it}^\top \right) \tilde{f}_t, \\ \mathcal{R}_{4,t}(0) &= \hat{p}_t(0)^\top \hat{Q}_t^{-1} \left(\frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) (\beta_{it} - \hat{\beta}_{it})^\top \right) \tilde{f}_t, \\ \mathcal{R}_{5,t}(0) &= \frac{1}{2} \hat{p}_t(0)^\top \hat{Q}_t^{-1} \frac{1}{n_t} \sum_{i=1}^{n_t} \hat{p}_t(\hat{\beta}_{it}) (\hat{\beta}_{it} - \beta_{it})^\top \nabla^2 \mu_t(\tilde{\beta}_{it}) (\hat{\beta}_{it} - \beta_{it}). \end{aligned}$$

Thus the remainder in [\(SA-II.7.34\)](#) is

$$\mathcal{R}_{\mathbf{FM},t} = -\mathcal{A}_t(0) - \mathcal{L}_{\mathbf{B},t}(0) - (\mathcal{L}_{\mathbf{V},t}(0) - \mathcal{A}_t(0)) + \sum_{j=1}^4 \mathcal{R}_{j,t}(0) - \mathcal{R}_{5,t}(0).$$

The periodwise bounds in Lemmas [SA-I.3.1](#) and [SA-II.4.2](#), with the sharper origin calculation for $\mathcal{A}_t(0)$,

imply

$$\frac{1}{T-H} \sum_{t=H+1}^T \left[\mathcal{A}_t(0)^2 + \mathcal{L}_{B,t}(0)^2 + (\mathcal{L}_{V,t}(0) - \mathcal{A}_t(0))^2 + \sum_{j=1}^5 \mathcal{R}_{j,t}(0)^2 \right] \lesssim_{\mathbb{P}} R_{\text{FM}}^2 + o_{\mathbb{P}}\left(\frac{J^d}{n}\right). \quad (\text{SA-II.7.35})$$

We spell out the argument for $\mathcal{L}_{B,t}(0)$. This term is $\mathcal{C}_t$ -measurable, while $\mathbb{E}[\mathcal{E}_{o,t}(0) \mid \mathcal{C}_t] = 0$. Hence the predictable quadratic variation of the cross product satisfies

$$\frac{1}{(T-H)^4} \sum_{t=H+1}^T \mathbb{E}[\mathcal{E}_{o,t}(0)^2 \mathcal{L}_{B,t}(0)^2 \mid \mathcal{C}_t] \lesssim_{\mathbb{P}} \frac{1}{(T-H)^4} \frac{J^d}{n} \sum_{t=H+1}^T \mathcal{L}_{B,t}(0)^2.$$

Therefore,

$$\begin{aligned} \left| \frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathcal{E}_{o,t}(0) \mathcal{L}_{B,t}(0) \right| &\lesssim_{\mathbb{P}} \frac{1}{T^{3/2}} \sqrt{\frac{J^d}{n}} \left(\frac{1}{T-H} \sum_{t=H+1}^T \mathcal{L}_{B,t}(0)^2 \right)^{1/2} \\ &\lesssim_{\mathbb{P}} \frac{J^d}{nT^2} + \frac{1}{T(T-H)} \sum_{t=H+1}^T \mathcal{L}_{B,t}(0)^2 \\ &= o_{\mathbb{P}}\left(\frac{J^d}{nT}\right) + O_{\mathbb{P}}\left(\frac{1}{T(T-H)} \sum_{t=H+1}^T \mathcal{L}_{B,t}(0)^2\right), \end{aligned}$$

where the last line uses Young's inequality. To bound the components of $\sum_{t=H+1}^T \mathcal{E}_{o,t}(0) \mathcal{R}_{\text{FM},t}$ that contain $\mathcal{A}_t(0)$, $\mathcal{L}_{V,t}(0) - \mathcal{A}_t(0)$, $\mathcal{R}_{3,t}(0)$, $\mathcal{R}_{4,t}(0)$, and $\mathcal{R}_{5,t}(0)$, apply the preceding predictable-variation argument used for $\mathcal{L}_{B,t}(0)$ after conditioning on $\mathcal{E}_{\infty}$ and the primitive shocks dated before t . Use Cauchy-Schwarz for any part without a conditionally mean-zero shock. Summing over these displayed components and using (SA-II.7.35) gives a contribution bounded by $R_{\text{FM}}^2/T + o_{\mathbb{P}}(J^d/(nT))$.

It remains to handle the two design-comparison terms. For $\mathcal{R}_{1,t}(0)$, Cauchy-Schwarz after the inverse-Gram decomposition and the Gram-convergence bound from Lemma SA-II.3.3 give $o_{\mathbb{P}}(J^d/(nT))$. For $\mathcal{R}_{2,t}(0)$, open the product with $\mathcal{E}_{o,t}(0)$ into diagonal and off-diagonal pieces. The predictable diagonal piece is

$$\frac{2}{(T-H)^2} \sum_{t=H+1}^T \sum_{i=1}^{n_t} \varsigma_{it}^2 \left(\frac{1}{n_t} p_t(0)^{\top} Q_t^{-1} p_t(\beta_{it}) \right) \left(\frac{1}{n_t} p_t(0)^{\top} Q_t^{-1} (\hat{p}_t(\hat{\beta}_{it}) - p_t(\beta_{it})) \right),$$

and is bounded by the switch-probability and cell-count rates entering R_{FM}^2/T . The centered diagonal remainder and the off-diagonal piece are conditionally mean-zero sums and are $o_{\mathbb{P}}(J^d/(nT))$ by the same predictable-variation calculation as above. Therefore,

$$\left| \frac{2}{(T-H)^2} \sum_{t=H+1}^T \mathcal{E}_{o,t}(0) \mathcal{R}_{\text{FM},t} \right| \lesssim_{\mathbb{P}} \frac{R_{\text{FM}}^2}{T} + o_{\mathbb{P}}\left(\frac{J^d}{nT}\right),$$

and

$$\frac{1}{(T-H)^2} \sum_{t=H+1}^T \mathcal{R}_{\text{FM},t}^2 \lesssim_{\mathbb{P}} \frac{R_{\text{FM}}^2}{T} + o_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

Hence, we conclude that

$$\mathcal{V}_{\text{FM},1} \lesssim_{\mathbb{P}} \frac{R_{\text{FM}}^2}{T} + o_{\mathbb{P}}\left(\frac{J^d}{nT}\right).$$

For $\mathcal{V}_{\text{FM},2}$, the aggregate deviation satisfies $\widehat{\mu}(0) - \mathcal{B}(0) - \bar{\mu}_T(0; H) \lesssim_{\mathbb{P}} \sqrt{J^d/(nT)} + R_{\text{FM}}$ by (SA-II.7.34), so

$$\mathcal{V}_{\text{FM},2} \lesssim_{\mathbb{P}} \frac{J^d}{nT^2} + \frac{R_{\text{FM}}^2}{T} = o_{\mathbb{P}}\left(\frac{J^d}{nT}\right) + \frac{R_{\text{FM}}^2}{T}.$$

For $\mathcal{V}_{\text{FM},3}$, by Cauchy-Schwarz,

$$\mathcal{V}_{\text{FM},3}^2 \lesssim \left(\frac{1}{(T-H)^2} \sum_{t=H+1}^T (\widehat{M}_t(0) - \mathcal{B}_t(0) - \mu_t(0))^2 \right) \left(\frac{1}{(T-H)^2} \sum_{t=H+1}^T (\mu_t(0) - \bar{\mu}_T(0; H))^2 \right),$$

where the first factor is $\lesssim_{\mathbb{P}} J^d/(nT) + R_{\text{FM}}^2/T$ by (SA-II.7.34), and the second equals $\Sigma_{\mu}(0)$. Thus,

$$\mathcal{V}_{\text{FM},3} \lesssim_{\mathbb{P}} \left(\sqrt{\frac{J^d}{nT}} + \frac{R_{\text{FM}}}{\sqrt{T}} \right) \sqrt{\Sigma_{\mu}(0)}.$$

Combining the bounds for $\mathcal{V}_{\text{FM},1}$, $\mathcal{V}_{\text{FM},2}$, and $\mathcal{V}_{\text{FM},3}$ gives the asserted bound for $\widetilde{\Sigma}_{\text{FM}}(0) - \Sigma_{\varepsilon,\circ}(0) - \Sigma_{\mu}(0)$. $\square$

SA-II.8 Uniform-in- β Inference

The pointwise-in- β Gaussian approximation in Theorem SA-II.6.2 and the feasible variance estimators above give the usual confidence intervals at each fixed nonsingular evaluation point. This section converts those pointwise intervals into a family of confidence intervals whose coverage is controlled uniformly over $\beta \in \mathcal{B}^\circ$, where $\mathcal{B}^\circ$ is the interior evaluation region used in the stochastic-linearization results. The result is not a simultaneous coverage statement for the entire function; it is a uniform statement about the coverage probability of the interval indexed by the evaluation point.

The only nonstandard feature is the transition in the variance as β approaches the origin. Away from zero, the factor variance is first order and $\Sigma(\beta) \asymp_{\mathbb{P}} \|\beta\|^2/T$. At the origin, the factor component vanishes and the idiosyncratic variance is at the local-support scale $J^d/(nT)$. We therefore use the usual feasible standard error outside a shrinking neighborhood of the origin and a conservative variance floor inside that neighborhood. The floor argument deliberately avoids using the pointwise origin Gaussian approximation, so no lower bound on $\Sigma_{\varepsilon}(0)$ is needed in this section. Throughout this section, take $\mathcal{B}^\circ$ to contain a fixed neighborhood of 0.

SA-II.8.1 Result

We impose the stochastic-linearization and factor-CLT conditions used above, but the variance requirements are weaker than full pointwise jackknife consistency. For the factor component, we use the factor-variance estimator condition in Lemma SA-II.7.1. For the idiosyncratic component, the conservative floor only needs the local-support upper bound verified in the proof of Theorem SA-II.8.1. Thus this section does not import the primitive jackknife localization condition $JR_{\text{FS}} = o(1)$. We also do not impose the origin-specific lower-

bound condition $(nT/J^d)\Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} 1$ from Lemma SA-II.6.1; the conservative floor below replaces that requirement. Set

$$R_{\text{UI}} = \sqrt{T - H} R_{\text{CLT}}.$$

The rate R_{UI} is the threshold scale obtained by comparing the pointwise linearization remainder with the factor standard deviation. By definition, it includes the same-cell nonconstant-basis component in R_{LA} when $\mathfrak{q} \geq 1$. The first-step correction $\mathcal{A}(\beta)$ is part of the adjusted idiosyncratic leading term, but the uniform bound for this term is the stronger scale from Lemma SA-II.4.2. Define

$$R_{\text{A}}^\circ = \sqrt{T - H} \left(\frac{J^d \log(nT)}{nH} \right)^{1/2}.$$

Thus $R_{\text{A}}^\circ/\sqrt{T - H}$ is the uniform scale for $\mathcal{A}(\beta)$ over $\mathcal{B}^\circ$. The factor-variance estimation error is handled differently: it is a quadratic form in β and is therefore controlled relative to $\Sigma_f(\beta)$ by the factor-prediction condition in Lemma SA-II.7.1.

Let $\delta > 0$ be a possibly data-dependent factor-variance threshold. The estimated factor variance $\widehat{\Sigma}_f(\beta)$ is invariant to nonsingular rescalings of the beta coordinates and factor normalization. The associated normalized factor-risk scale is $\sqrt{(T - H)\widehat{\Sigma}_f(\beta)}$. The condition below compares all error scales that can bind with the corresponding boundary scale $\sqrt{(T - H)\delta}$. We require δ to satisfy

$$(T - H)\delta = o_{\mathbb{P}}(1), \quad \frac{R_{\text{UI}} + R_{\text{A}}^\circ}{\sqrt{(T - H)\delta}} = o_{\mathbb{P}}(1) \quad (\text{SA-II.8.1})$$

Let $z_{1-\alpha/2}$ denote the $1 - \alpha/2$ quantile of the standard normal distribution. The uniform-in- β $100(1 - \alpha)\%$ confidence interval is

$$[\widehat{L}_T(\beta), \widehat{U}_T(\beta)] = \widehat{\mu}(\beta) \pm z_{1-\alpha/2} \sqrt{\widehat{\Sigma}_{\text{UI}}(\beta)},$$

where

$$\widehat{\Sigma}_{\text{UI}}(\beta) = \begin{cases} \widehat{\Sigma}(\beta) & \text{if } \widehat{\Sigma}_f(\beta) > \delta \\ \max\{\widehat{\Sigma}(\beta), \widehat{\Sigma}_\delta\} & \text{if } \widehat{\Sigma}_f(\beta) \leq \delta \end{cases},$$

and, when the factor-variance level set is nonempty,

$$\widehat{\Sigma}_\delta = \sup_{\substack{b \in \mathcal{B}^\circ \\ \widehat{\Sigma}_f(b) = \delta}} \max\{\widehat{\Sigma}(b), \delta\} = \delta + \sup_{\substack{b \in \mathcal{B}^\circ \\ \widehat{\Sigma}_f(b) = \delta}} \max\{\widehat{\Sigma}_\varepsilon(b), 0\}.$$

The floor $\widehat{\Sigma}_\delta$ adds the threshold factor variance δ to the largest nonnegative feasible idiosyncratic component on the boundary of the shrinking factor-variance neighborhood. On the negligible complement event where the level set is empty, set $\widehat{\Sigma}_\delta = \delta$; this convention has no asymptotic effect. The level set is nonempty w.p.a.1 because $\widehat{\Sigma}_f(0) = 0$, $\widehat{\Sigma}_f(\cdot)$ is continuous, and $\mathcal{B}^\circ$ contains a fixed neighborhood of zero. The uniform quadratic-form bound used below, factor nondegeneracy, and $\mathcal{B}_{\text{F}}(b) \geq 0$ give $\widehat{\Sigma}_f(b) \gtrsim_{\mathbb{P}} (T - H)^{-1}$ uniformly on any fixed small sphere around zero. Since $(T - H)\delta = o_{\mathbb{P}}(1)$, continuity then gives a point inside the sphere at which $\widehat{\Sigma}_f(b) = \delta$.

For a family of random variables $\{X_n(\beta) : \beta \in \mathcal{I}_n\}$ and a positive deterministic or data-dependent rate r_n , write $X_n(\beta) = o_{\mathbb{P}, \mathcal{I}_n}(r_n)$ if, for every $\epsilon > 0$,

$$\sup_{\beta \in \mathcal{I}_n} \mathbb{P}(|X_n(\beta)| > \epsilon r_n) \rightarrow 0,$$

and $X_n(\beta) \lesssim_{\mathbb{P}, \mathcal{I}_n} r_n$ if

$$\lim_{M \rightarrow \infty} \limsup_{n, T \rightarrow \infty} \sup_{\beta \in \mathcal{I}_n} \mathbb{P}(|X_n(\beta)| > M r_n) = 0.$$

These are pointwise-uniform probability statements; they are the right notion for the infimum of pointwise coverage probabilities below and do not assert a realized supremum bound unless one is displayed explicitly.

Theorem SA-II.8.1 (Uniform Coverage of Confidence Intervals). *Assume that the conditions of Corollary SA-II.5.2 hold, that (SA-II.7.1) holds, and that the factor-variance estimator satisfies the conditions of Lemma SA-II.7.1. If δ satisfies (SA-II.8.1), then*

$$\liminf_{n, T \rightarrow \infty} \inf_{\beta \in \mathcal{B}^\circ} \mathbb{P}(\bar{\mu}_T(\beta; H) \in [\hat{L}_T(\beta), \hat{U}_T(\beta)]) \geq 1 - \alpha.$$

The theorem does not require a pointwise Gaussian approximation at $\beta = 0$, and the origin-specific lower-bound condition $(nT/J^d)\Sigma_\varepsilon(0) \gtrsim_{\mathbb{P}} 1$ is not imposed. Inside the factor-variance neighborhood $\hat{\Sigma}_f(\beta) \leq \delta$, the interval is calibrated using the boundary variance floor $\hat{\Sigma}_\delta$. The condition (SA-II.8.1), together with local support of the portfolio weights, ensures that the idiosyncratic leading term, the uniform first-step correction, and the full linearization remainder are negligible relative to that boundary standard-error scale.

SA-II.8.2 Proof of Theorem SA-II.8.1

We first record the uniform consequences of the linearization and variance bounds. The pointwise Gaussian approximation at the origin is not used.

The stochastic linearization in Lemmas SA-I.3.1 and SA-II.4.2, together with Section SA-II.6, gives

$$\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H) = \mathcal{G}(\beta) + \bar{\mathcal{J}}_\varepsilon(\beta) + \mathcal{R}(\beta), \quad \bar{\mathcal{J}}_\varepsilon(\beta) = \mathcal{E}(\beta) - \mathcal{A}(\beta).$$

Thus the first-step correction $\mathcal{A}(\beta)$ is not part of the non-leading remainder, but it must be controlled at its own uniform scale. The rate R_{CLT} controls only $\mathcal{R}(\beta)$, while Lemma SA-II.4.2 controls $\mathcal{A}(\beta)$:

$$|\mathcal{R}(\beta)| \lesssim_{\mathbb{P}, \mathcal{B}^\circ} R_{\text{CLT}},$$

$$|\mathcal{A}(\beta)| \lesssim_{\mathbb{P}, \mathcal{B}^\circ} \frac{R_{\mathbf{A}}^\circ}{\sqrt{T - H}}.$$

The proofs of Lemmas SA-II.6.1 and SA-II.7.6 take their local-support and selected-block bounds uniformly over the deterministic quasi-uniform rectangle class before the block containing β is selected. Their constants therefore do not depend on β , so

$$\Sigma_\varepsilon(\beta) + |\hat{\Sigma}_\varepsilon(\beta)| \lesssim_{\mathbb{P}, \mathcal{B}^\circ} \frac{J^d}{n(T - H)}.$$

This is dominated by $(R_{\mathbf{A}}^\circ)^2/(T - H)$ under the maintained rolling-window scaling. Hence (SA-II.8.1) and

the definitions of R_{UI} and R_{A}° imply

$$\begin{aligned}\mathcal{R}(\beta) &= o_{\mathbb{P}, \mathcal{B}^\circ}(\sqrt{\delta}), \\ \mathcal{A}(\beta) &= o_{\mathbb{P}, \mathcal{B}^\circ}(\sqrt{\delta}), \\ \Sigma_\varepsilon(\beta) + |\widehat{\Sigma}_\varepsilon(\beta)| &= o_{\mathbb{P}, \mathcal{B}^\circ}(\delta).\end{aligned}$$

The bound for $\mathcal{A}(\beta)$ is the uniform first-step-correction bound from Lemma SA-II.4.2. For the direct idiosyncratic term $\mathcal{E}(\beta)$, condition period by period on the field $\mathcal{C}_t$ from Section SA-II.1.3. Its coefficients are fixed under this conditioning, and its predictable variance is the direct variance $\Sigma_{\varepsilon, \circ}(\beta)$ defined in the Fama–MacBeth variance subsection above, not the adjusted reference scale $\Sigma_\varepsilon(\beta)$. The same local-support calculation used for $\mathcal{E}(\beta)$ in Lemma SA-I.3.1 gives

$$\Sigma_{\varepsilon, \circ}(\beta) \lesssim_{\mathbb{P}, \mathcal{B}^\circ} \frac{J^d}{n(T-H)} = o_{\mathbb{P}, \mathcal{B}^\circ}(\delta).$$

Because δ may be data dependent, choose a deterministic sequence $a_n \rightarrow \infty$ such that $\mathbb{P}(\delta < a_n J^d / [n(T-H)]) \rightarrow 0$. The preceding conditional-variance bound and conditional Chebyshev’s inequality at the deterministic threshold then imply that for every $\epsilon > 0$,

$$\sup_{\beta \in \mathcal{B}^\circ} \mathbb{P}\left(|\mathcal{E}(\beta)| > \epsilon \sqrt{\delta}\right) \leq \mathbb{P}\left(\delta < \frac{a_n J^d}{n(T-H)}\right) + \sup_{\beta \in \mathcal{B}^\circ} \mathbb{P}\left(|\mathcal{E}(\beta)| > \epsilon \sqrt{\frac{a_n J^d}{n(T-H)}}\right) \rightarrow 0,$$

or equivalently $\mathcal{E}(\beta) = o_{\mathbb{P}, \mathcal{B}^\circ}(\sqrt{\delta})$. Here the supremum is taken over the pointwise probabilities after applying Chebyshev’s inequality; no realized supremum of $\Sigma_{\varepsilon, \circ}(\beta)$ or $\Sigma_\varepsilon(\beta)$ is invoked. Consequently, $\mathcal{S}_\varepsilon(\beta) = \mathcal{E}(\beta) - \mathcal{A}(\beta) = o_{\mathbb{P}, \mathcal{B}^\circ}(\sqrt{\delta})$.

Next, record the uniform quadratic-form version of Lemma SA-II.7.1. Write

$$\widehat{\Sigma}_f(\beta) - \Sigma_f(\beta) - \mathcal{B}_F(\beta) = \beta^\top \Delta_F \beta,$$

where

$$\Delta_F = \frac{1}{(T-H)^2} \sum_{t=H+1}^T \left\{ \widetilde{f}_t \widetilde{f}_t^\top - \mathbb{E}[\widetilde{f}_t \widetilde{f}_t^\top \mid \mathcal{G}_{t-1}] \right\} + \frac{2}{(T-H)^2} \sum_{t=H+1}^T \widetilde{f}_t (m_t - \widehat{m}_t)^\top.$$

Because d is fixed and $\mathcal{B}^\circ$ is bounded, the entrywise martingale bounds in the proof of Lemma SA-II.7.1 and the factor-prediction condition in Lemma SA-II.7.1 give

$$\|\Delta_F\| = o_{\mathbb{P}}((T-H)^{-1}).$$

Then, by the factor nondegeneracy condition in Assumption SA-DGP(iv), we have

$$\omega_n := \sup_{\substack{\beta \in \mathcal{B}^\circ \\ \beta \neq 0}} \frac{|\widehat{\Sigma}_f(\beta) - \Sigma_f(\beta) - \mathcal{B}_F(\beta)|}{\Sigma_f(\beta)} = o_{\mathbb{P}}(1),$$

while the numerator is zero at $\beta = 0$. Since $\mathcal{B}_F(\beta) \geq 0$, this implies

$$\left(\Sigma_f(\beta) - \widehat{\Sigma}_f(\beta)\right)_+ \leq \omega_n \Sigma_f(\beta),$$

where $(a)_+ := \max\{a, 0\}$ for any $a \in \mathbb{R}$. Combining this display with $|\widehat{\Sigma}_\varepsilon(\beta)| = o_{\mathbb{P}, \mathcal{B}^\circ}(\delta)$ and $\widehat{\Sigma}(\beta) = \widehat{\Sigma}_f(\beta) + \widehat{\Sigma}_\varepsilon(\beta)$ gives the pointwise-uniform lower bound

$$\left(1 - \frac{\widehat{\Sigma}_{\text{UI}}(\beta)}{\max\{\Sigma_f(\beta), \delta\}}\right)_+ = o_{\mathbb{P}, \mathcal{B}^\circ}(1). \quad (\text{SA-II.8.2})$$

To see this, consider the two regions used in the construction. If $\widehat{\Sigma}_f(\beta) > \delta$, then $\widehat{\Sigma}_{\text{UI}}(\beta) = \widehat{\Sigma}(\beta)$ and the centered factor-variance bound gives

$$\widehat{\Sigma}_{\text{UI}}(\beta) \geq (1 - \omega_n)\Sigma_f(\beta) - o_{\mathbb{P}, \mathcal{B}^\circ}(\delta), \quad \widehat{\Sigma}_{\text{UI}}(\beta) \geq \delta - o_{\mathbb{P}, \mathcal{B}^\circ}(\delta).$$

Thus (SA-II.8.2) holds on the outer region. If $\widehat{\Sigma}_f(\beta) \leq \delta$, since $\widehat{\Sigma}_f(\beta) \geq (1 - \omega_n)\Sigma_f(\beta)$, $\Sigma_f(\beta) \leq (1 + o_{\mathbb{P}}(1))\delta$ uniformly on this region. Also, $\widehat{\Sigma}_{\text{UI}}(\beta) \geq \widehat{\Sigma}_\delta \geq \delta$, where the last inequality follows from the definition of the variance floor. This gives the same ratio lower bound on the inner region. In particular,

$$\left(1 - \frac{\widehat{\Sigma}_{\text{UI}}(\beta)}{\delta}\right)_+ = o_{\mathbb{P}, \mathcal{B}^\circ}(1). \quad (\text{SA-II.8.3})$$

Next, write

$$\mathcal{G}(\beta) = \beta^\top \frac{1}{T - H} \sum_{t=H+1}^T \widetilde{f}_t.$$

Let

$$S_T := \sum_{t=H+1}^T \widetilde{f}_t, \quad V_T := \sum_{t=H+1}^T \mathbb{V}\text{ar}(\widetilde{f}_t \mid \mathcal{G}_{t-1}).$$

Then

$$\Sigma_f(\beta) = \frac{\beta^\top V_T \beta}{(T - H)^2}, \quad \left\| \frac{V_T}{T - H} - \Omega_f \right\| = o_{\mathbb{P}}(1)$$

by (SA-II.1.1).

For $u \in \mathbb{S}^{d-1}$, define the deterministically normalized process

$$Z_T^0(u) := \frac{u^\top S_T}{\sqrt{(T - H)u^\top \Omega_f u}}.$$

For each fixed u , the sum of the conditional variances of its martingale increments converges in probability to one:

$$\frac{u^\top V_T u}{(T - H)u^\top \Omega_f u} \rightarrow_{\mathbb{P}} 1.$$

The conditional Lyapunov bound is uniform over directions because $\lambda_{\min}(\Omega_f) > 0$:

$$\sup_{u \in \mathbb{S}^{d-1}} \frac{\sum_{t=H+1}^T \mathbb{E}[|u^\top \tilde{f}_t|^4 \mid \mathcal{G}_{t-1}]}{(T-H)^2 (u^\top \Omega_f u)^2} \lesssim_{\mathbb{P}} (T-H)^{-1} = o_{\mathbb{P}}(1).$$

The scalar martingale CLT therefore gives

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(Z_T^0(u) \leq x) - \Phi(x)| \rightarrow 0$$

for each fixed u .

We now make this approximation uniform over directions. Fix $\epsilon > 0$ and let $\mathcal{N}_\epsilon$ be a finite ϵ -net of $\mathbb{S}^{d-1}$. The preceding display holds uniformly over the fixed finite set $\mathcal{N}_\epsilon$. Positive definiteness of Ω_f and martingale orthogonality give

$$\sup_{\|u-v\| \leq \epsilon} |Z_T^0(u) - Z_T^0(v)| \lesssim \epsilon \frac{\|S_T\|}{\sqrt{T-H}} = O_{\mathbb{P}}(\epsilon), \quad \frac{\|S_T\|}{\sqrt{T-H}} = O_{\mathbb{P}}(1).$$

The finite-net argument and normal anticoncentration thus imply

$$\sup_{u \in \mathbb{S}^{d-1}} \sup_{x \in \mathbb{R}} |\mathbb{P}(Z_T^0(u) \leq x) - \Phi(x)| \rightarrow 0. \quad (\text{SA-II.8.4})$$

Finally, stabilization gives

$$\sup_{u \in \mathbb{S}^{d-1}} \left| \frac{u^\top V_T u}{(T-H)u^\top \Omega_f u} - 1 \right| \leq \frac{\|V_T/(T-H) - \Omega_f\|}{\lambda_{\min}(\Omega_f)} = o_{\mathbb{P}}(1).$$

Since $\sup_{u \in \mathbb{S}^{d-1}} |Z_T^0(u)| \lesssim \|S_T\|/\sqrt{T-H} = O_{\mathbb{P}}(1)$,

$$\sup_{u \in \mathbb{S}^{d-1}} \left| \frac{u^\top S_T}{\sqrt{u^\top V_T u}} - Z_T^0(u) \right| = o_{\mathbb{P}}(1).$$

Combining this uniform replacement with (SA-II.8.4) yields

$$\sup_{u \in \mathbb{S}^{d-1}} \sup_{x \in \mathbb{R}} \left| \mathbb{P}\left(\frac{u^\top S_T}{\sqrt{u^\top V_T u}} \leq x\right) - \Phi(x) \right| \rightarrow 0.$$

Thus the factor Gaussian approximation is uniform after writing $\beta = \|\beta\|u$; at $\beta = 0$, $\mathcal{G}(0) = 0$. Since $\max\{\Sigma_f(\beta), \delta\} \geq \Sigma_f(\beta)$, the event with the conservative normalizer contains the event with the true factor normalizer for every realization of δ . Therefore, for any fixed $c \geq 0$,

$$\inf_{\beta \in \mathcal{B}^\circ} \mathbb{P}\left(|\mathcal{G}(\beta)| \leq c\sqrt{\max\{\Sigma_f(\beta), \delta\}}\right) \geq 2\Phi(c) - 1 + o(1).$$

This statement does not condition on the data-dependent inner or outer region; the conservative scale $\max\{\Sigma_f(\beta), \delta\}$ dominates the true factor variance for every β and every realization.

Finally, (SA-II.8.3) and $\bar{\mathcal{J}}_\varepsilon(\beta) + \mathcal{R}(\beta) = o_{\mathbb{P}, \mathcal{B}^\circ}(\sqrt{\delta})$ give

$$\frac{|\bar{\mathcal{J}}_\varepsilon(\beta) + \mathcal{R}(\beta)|}{\sqrt{\hat{\Sigma}_{\text{UI}}(\beta)}} = o_{\mathbb{P}, \mathcal{B}^\circ}(1).$$

To assemble the coverage event, fix $\eta > 0$ with $\eta < z_{1-\alpha/2}$. The probability of the following event tends to one, uniformly over $\beta \in \mathcal{B}^\circ$ in the pointwise sense:

$$|\bar{\mathcal{J}}_\varepsilon(\beta) + \mathcal{R}(\beta)| \leq \eta \sqrt{\hat{\Sigma}_{\text{UI}}(\beta)} \quad \text{and} \quad \hat{\Sigma}_{\text{UI}}(\beta) \geq (1 - \eta) \max\{\Sigma_f(\beta), \delta\}.$$

On this event,

$$\left\{ |\mathcal{G}(\beta)| \leq (z_{1-\alpha/2} - \eta) \sqrt{1 - \eta} \sqrt{\max\{\Sigma_f(\beta), \delta\}} \right\} \subseteq \left\{ |\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H)| \leq z_{1-\alpha/2} \sqrt{\hat{\Sigma}_{\text{UI}}(\beta)} \right\}.$$

The factor bound above with $c = (z_{1-\alpha/2} - \eta) \sqrt{1 - \eta}$ gives

$$\inf_{\beta \in \mathcal{B}^\circ} \mathbb{P} \left(|\hat{\mu}(\beta) - \bar{\mu}_T(\beta; H)| \leq z_{1-\alpha/2} \sqrt{\hat{\Sigma}_{\text{UI}}(\beta)} \right) \geq 2\Phi \left((z_{1-\alpha/2} - \eta) \sqrt{1 - \eta} \right) - 1 + o(1).$$

Letting $\eta \downarrow 0$ yields the claimed lower bound $1 - \alpha$. The boundary $\hat{\Sigma}_f(\beta) = \delta$ is covered by the inner formula in the definition of $\hat{\Sigma}_{\text{UI}}(\beta)$, so no separate boundary argument is needed. The displayed lower bound is the claimed uniform coverage. $\square$

SA-II.9 Factor-Cancelling Linear Contrasts

This section develops inference for finite linear contrasts that eliminate the factor component in the pointwise stochastic expansion. Let K_Ψ be fixed, let $\mathbf{b}_1, \dots, \mathbf{b}_{K_\Psi} \in \mathcal{B}^\circ$, and let $a_1, \dots, a_{K_\Psi}$ be fixed real weights. The contrast of interest is

$$\hat{\Psi} = \sum_{k=1}^{K_\Psi} a_k \hat{\mu}(\mathbf{b}_k), \quad \bar{\Psi}_T = \sum_{k=1}^{K_\Psi} a_k \bar{\mu}_T(\mathbf{b}_k; H).$$

The key restriction is the factor-cancelling condition

$$\sum_{k=1}^{K_\Psi} a_k \mathbf{b}_k = 0. \tag{SA-II.9.1}$$

For example, the butterfly spread corresponds to $K_\Psi = 3$, $(a_1, a_2, a_3) = (1, -2, 1)$, and $\mathbf{b}_2 = (\mathbf{b}_1 + \mathbf{b}_3)/2$.

SA-II.9.1 Setup and Linearization

For any scalar term $A(\cdot)$, write its contrast as

$$A_\Psi = \sum_{k=1}^{K_\Psi} a_k A(\mathbf{b}_k).$$

Applying this operator to the pointwise linearization in Section SA-II.6 gives

$$\widehat{\Psi} - \bar{\Psi}_T = \bar{\mathcal{J}}_{\varepsilon\Psi} + \mathcal{R}_\Psi, \quad (\text{SA-II.9.2})$$

because

$$\bar{\mathcal{J}}_{f\Psi} = \frac{1}{T-H} \sum_{t=H+1}^T \left(\sum_{k=1}^{K_\Psi} a_k \mathbf{b}_k \right)^\top \tilde{f}_t = 0.$$

The idiosyncratic leading term can be written as

$$\bar{\mathcal{J}}_{\varepsilon\Psi} = \frac{1}{T-H} \sum_{t=1}^T \sum_{i=1}^{n_t} \varsigma_{it} \eta_{it} W_{it,\Psi}, \quad W_{it,\Psi} = \sum_{k=1}^{K_\Psi} a_k W_{it}(\mathbf{b}_k).$$

Since K_Ψ is fixed, the leading-weight bounds used in Lemma SA-II.6.1 imply

$$|\bar{\mathcal{J}}_{\varepsilon\Psi}| \lesssim_{\mathbb{P}} \sqrt{\frac{J^d}{nT}}.$$

Because K_Ψ and the contrast weights are fixed, define the deterministic rate

$$\mathbf{R}_\Psi := \left(\sum_{k=1}^{K_\Psi} |a_k| \right) \mathbf{R}_{\text{CLT}}.$$

The triangle inequality and the common pointwise remainder bound give

$$|\mathcal{R}_\Psi| \lesssim_{\mathbb{P}} \mathbf{R}_\Psi.$$

The condition needed for idiosyncratic-rate inference is

$$\sqrt{\frac{nT}{J^d}} \mathbf{R}_\Psi \rightarrow 0. \quad (\text{SA-II.9.3})$$

In particular, the second-step smoothing-bias component requires

$$\frac{nT}{J^{d+2 \min\{s_\mu, \mathfrak{q}+1\}}} \rightarrow 0,$$

the same smoothing-bias restriction as in the pointwise $\beta = 0$ theory. Thus, $\mathfrak{q} = 0$ cannot satisfy the sufficient rate conditions for factor-cancelling contrast inference.

SA-II.9.2 Evaluation-Point-Anchored Partitioning

For factor-cancelling contrasts, the second-step partition is constructed so that the evaluation points themselves are fixed marginal breakpoints. This is the analogue of the split-at-zero construction in Section SA-II.3: the cell containing each evaluation point has a nonrandom index, because each coordinate of the evaluation point is assigned to the subsample immediately to its right. For notational simplicity, we state the construction under the convention that the evaluation points have distinct coordinates in every dimension. If several evaluation points share a coordinate in some dimension, the same construction uses the sorted

unique coordinates in that dimension; the number of fixed breakpoints, and hence the number of marginal subsamples, is then smaller than K_Ψ and $K_\Psi + 1$, respectively.

We describe the construction first. For each coordinate $l \in [d]$, order the fixed evaluation-point coordinates as

$$\underline{\beta}_l < \mathbf{b}_{\pi_l(1),l} < \dots < \mathbf{b}_{\pi_l(K_\Psi),l} < \bar{\beta}_l,$$

where π_l is a permutation of $[K_\Psi]$. Set $\gamma_{0,l} = \underline{\beta}_l$, $\gamma_{r,l} = \mathbf{b}_{\pi_l(r),l}$ for $r = 1, \dots, K_\Psi$, and $\gamma_{K_\Psi+1,l} = \bar{\beta}_l$. These fixed breakpoints split coordinate l into the $K_\Psi + 1$ subsamples

$$\mathcal{I}_{r,l} = [\gamma_{r,l}, \gamma_{r+1,l}), \quad r = 0, \dots, K_\Psi - 1, \quad \mathcal{I}_{K_\Psi,l} = [\gamma_{K_\Psi,l}, \gamma_{K_\Psi+1,l}],$$

with ties at a fixed breakpoint assigned to the interval on the right. Thus, if $\mathbf{b}_{k,l} = \gamma_{r,l}$, then $\mathbf{b}_{k,l} \in \mathcal{I}_{r,l}$.

Within each subsample, form marginal empirical quantile-spaced intervals of the generated regressor. Specifically, for $t = H + 1, \dots, T$, $l \in [d]$, and $r = 0, \dots, K_\Psi$, let

$$\begin{aligned} \hat{N}_{t,l,r} &= \sum_{i=1}^{n_t} \mathbb{1}(\hat{\beta}_{it,l} \in \mathcal{I}_{r,l}), \\ \hat{F}_{\hat{\beta},t,l,r}(z) &= \frac{1}{\hat{N}_{t,l,r}} \sum_{i=1}^{n_t} \mathbb{1}(\hat{\beta}_{it,l} \in \mathcal{I}_{r,l}) \mathbb{1}(\hat{\beta}_{it,l} \leq z), \quad z \in \mathcal{I}_{r,l}, \end{aligned}$$

on the event $\hat{N}_{t,l,r} > 0$, with generalized inverse on $\mathcal{I}_{r,l}$. Given integers $J_{t,l,r}$, define the empirical breakpoints by

$$\hat{\tau}_{0t,l,r} = \gamma_{r,l}, \quad \hat{\tau}_{J_{t,l,r}t,l,r} = \gamma_{r+1,l}, \quad \hat{\tau}_{jt,l,r} = \hat{F}_{\hat{\beta},t,l,r}^{-1}\left(\frac{j}{J_{t,l,r}}\right), \quad j = 1, \dots, J_{t,l,r} - 1.$$

The marginal partition $\hat{\Delta}_{t,l}$ is the ordered collection of intervals generated by these breakpoints over $r = 0, \dots, K_\Psi$, and the full partition is the tensor product $\hat{\Delta}_t = \otimes_{l \in [d]} \hat{\Delta}_{t,l}$.

Assumption SA-PA-G (Evaluation-point-anchored partition). $\mathbf{b}_1, \dots, \mathbf{b}_{K_\Psi}$ are fixed evaluation points with distinct coordinates in each dimension and lie strictly inside $\mathcal{B}^\circ$. For every $t = H + 1, \dots, T$, the estimated partition $\hat{\Delta}_t$ is formed by the evaluation-point-based marginal-quantile construction described above, with $J_{t,l,r} \asymp J$ uniformly over t, l, r , $J \rightarrow \infty$, and $J^d/n = o(1)$.

In this scenario, the reference partition is the conditional counterpart of the same construction. For each t, l, r , define

$$\begin{aligned} \bar{\pi}_{t,l,r} &= \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{P}(\hat{\beta}_{it,l} \in \mathcal{I}_{r,l} \mid \mathcal{D}_{t-1}), \\ \varrho_{it,l,r} &= \frac{\mathbb{P}(\hat{\beta}_{it,l} \in \mathcal{I}_{r,l} \mid \mathcal{D}_{t-1})}{\bar{\pi}_{t,l,r}}, \\ \bar{F}_{\hat{\beta},t,l,r}(z) &= \frac{1}{n_t} \sum_{i=1}^{n_t} \varrho_{it,l,r} \mathbb{P}(\hat{\beta}_{it,l} \leq z \mid \hat{\beta}_{it,l} \in \mathcal{I}_{r,l}, \mathcal{D}_{t-1}), \quad z \in \mathcal{I}_{r,l}. \end{aligned}$$

Let

$$\tau_{0t,l,r} = \gamma_{r,l}, \quad \tau_{J_{t,l,r}t,l,r} = \gamma_{r+1,l}, \quad \tau_{jt,l,r} = \bar{F}_{\hat{\beta},t,l,r}^{-1} \left(\frac{j}{J_{t,l,r}} \right), \quad j = 1, \dots, J_{t,l,r} - 1.$$

The reference marginal partition $\Delta_{t,l}$ and tensor-product partition Δ_t are formed from these conditional breakpoints. Cell normalizations, raw-regressor membership conventions, and reference-basis definitions are the same as in Section SA-II.3.

Under Assumption SA-PA-G, each evaluation point $\mathbf{b}_k$ belongs to a cell with a deterministic tensor-product label. On that cell, every coordinate of $\mathbf{b}_k$ is a fixed lower endpoint of both the estimated and reference cell. Therefore the midpoint and side-length normalization gives

$$\hat{p}_t(\mathbf{b}_k) = p_t(\mathbf{b}_k), \quad k = 1, \dots, K_\Psi,$$

for every polynomial degree $q \geq 0$, exactly as the split-at-zero construction gives $\hat{p}_t(0) = p_t(0)$ in Section SA-II.3.

SA-II.9.3 Result

The relevant variance is the variance of the factor-cancelling idiosyncratic leading term,

$$\Sigma_{\varepsilon\Psi} = \frac{1}{(T-H)^2} \sum_{t=1}^T \sum_{i=1}^{n_t} \varsigma_{it}^2 W_{it,\Psi}^2.$$

The bin-restricted jackknife from Section SA-II.7 extends by applying the same contrast to the delete- i pseudo-values:

$$\hat{\Sigma}_{\varepsilon\Psi} = \frac{1}{(T-H)^2} \sum_{i=1}^n \sum_{t=H+1}^T \sum_{s=H+1}^T \Delta_{it\Psi} \Delta_{is\Psi} \mathbb{1}(|t-s| \leq H),$$

where

$$\Delta_{it\Psi} = \sum_{k=1}^{K_\Psi} a_k \Delta_{it}(\mathbf{b}_k), \quad \Delta_{it}(\beta) = \widehat{M}_t^{(-i)}(\beta) - \widehat{M}_t(\beta).$$

Theorem SA-II.9.1 (Factor-Cancelling Contrast Inference).

Let K_Ψ be fixed, $\mathbf{b}_1, \dots, \mathbf{b}_{K_\Psi} \in \mathcal{B}^\circ$ be fixed evaluation points, and $a_1, \dots, a_{K_\Psi}$ be fixed contrast weights satisfying (SA-II.9.1). Suppose Assumptions SA-DGP, SA-FS, SA-PA-G, SA-SEP, and SA-FSB hold, and that $s_\mu \geq 2$. Assume that the rate conditions (SA-II.4.1), (SA-II.7.1), and (SA-II.9.3) hold, that

$$\frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}} = o(1),$$

and assume further that $JR_{\text{FS}} = o(1)$ and $J^2 \log T/n = o(1)$. If

$$\frac{nT}{J^d} \Sigma_{\varepsilon\Psi} \gtrsim_{\mathbb{P}} 1,$$

then

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\widehat{\Psi} - \bar{\Psi}_T}{\sqrt{\widehat{\Sigma}_{\varepsilon\Psi}}} \leq x \right) - \Phi(x) \right| \rightarrow 0,$$

and

$$\widehat{\Sigma}_{\varepsilon\Psi} = \Sigma_{\varepsilon\Psi} + o_{\mathbb{P}} \left(\frac{J^d}{nT} \right).$$

SA-II.9.4 Discussion

- (i) *Confidence interval.* The theorem gives both the Gaussian approximation and relative consistency of $\widehat{\Sigma}_{\varepsilon\Psi}$ under the nondegeneracy condition. Therefore Slutsky's lemma implies that

$$\widehat{\Psi} \pm z_{1-\alpha/2} \sqrt{\widehat{\Sigma}_{\varepsilon\Psi}}$$

has asymptotic coverage $1 - \alpha$.

- (ii) *Butterfly spreads.* The butterfly contrast is the special case $\widehat{\Psi} = \widehat{\mu}(\mathbf{b}_1) - 2\widehat{\mu}(\mathbf{b}_2) + \widehat{\mu}(\mathbf{b}_3)$ with $\mathbf{b}_2 = (\mathbf{b}_1 + \mathbf{b}_3)/2$. The factor component vanishes because $\mathbf{b}_1 - 2\mathbf{b}_2 + \mathbf{b}_3 = 0$, so no factor variance estimator is needed.
- (iii) *Pairwise differences.* The pairwise difference $\widehat{D} = \widehat{\mu}(\mathbf{b}_2) - \widehat{\mu}(\mathbf{b}_1)$ does not cancel the factor component:

$$\mathcal{J}_{fD} = \frac{1}{T-H} \sum_{t=H+1}^T (\mathbf{b}_2 - \mathbf{b}_1)^\top \widetilde{f}_t.$$

Its factor variance contribution is

$$\frac{1}{(T-H)^2} (\mathbf{b}_2 - \mathbf{b}_1)^\top \sum_{t=H+1}^T \mathbb{V}\text{ar}(\widetilde{f}_t \mid \mathcal{G}_{t-1}) (\mathbf{b}_2 - \mathbf{b}_1).$$

A two-point difference with weights $(-1, 1)$ cancels the factor term only in the degenerate case $\mathbf{b}_2 = \mathbf{b}_1$, so a nontrivial two-point difference is outside Theorem [SA-II.9.1](#).

- (iv) *Other difference-in-differences.* A quadruple contrast with weights $(1, -1, -1, 1)$ also removes the factor component whenever $\mathbf{b}_1 + \mathbf{b}_4 = \mathbf{b}_2 + \mathbf{b}_3$. It targets the difference-in-differences restriction

$$\bar{\mu}_T(\mathbf{b}_1; H) - \bar{\mu}_T(\mathbf{b}_2; H) = \bar{\mu}_T(\mathbf{b}_3; H) - \bar{\mu}_T(\mathbf{b}_4; H).$$

The theorem applies by taking $K_\Psi = 4$ and the corresponding weights.

SA-II.9.5 Proof of Theorem [SA-II.9.1](#)

By ([SA-II.9.1](#)), the factor component in the pointwise linearization cancels exactly, giving ([SA-II.9.2](#)). Under ([SA-II.9.3](#)), the residual satisfies

$$|\mathcal{R}_\Psi| = o_{\mathbb{P}} \left(\sqrt{\frac{J^d}{nT}} \right).$$

It remains to verify the Gaussian approximation and the jackknife variance consistency for the combined idiosyncratic weights.

The partition, Gram, and basis-switching bounds from Section SA-II.3 carry over under Assumption SA-PA-G: the number of fixed subsamples is finite, and the interior fixed breakpoints and density bounds give each subsample positive conditional mass, so the marginal quantile-spacing and cell-count arguments apply with constants depending only on K_Ψ .

Under Assumption SA-PA-G, the anchored construction gives $\hat{p}_t(\mathbf{b}_k) = p_t(\mathbf{b}_k)$ for every contrast point. Thus the combined reference weights $W_{it,\Psi}$ are $\mathcal{E}_\infty$ -measurable by the same argument as in Lemma SA-II.7.2. Conditional on $\mathcal{E}_\infty$, $\mathcal{J}_{\varepsilon\Psi}$ is a sum of independent mean-zero primitive shocks with weights $W_{it,\Psi}/(T-H)$. Since the number of contrast points is fixed, the single-point local-support bounds, Jensen's inequality over the rolling windows, and $T^{-1} \sum_{t=1}^T (1 + \|f_t\|^3) = O_{\mathbb{P}}(1)$ imply

$$\sum_{t=1}^T \sum_{i=1}^{n_t} W_{it,\Psi}^2 \lesssim_{\mathbb{P}} \frac{TJ^d}{n}, \quad \sum_{t=1}^T \sum_{i=1}^{n_t} |W_{it,\Psi}|^3 \lesssim_{\mathbb{P}} \frac{TJ^{2d}}{n^2}.$$

The conditional Berry–Esseen bound therefore gives

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\frac{\mathcal{J}_{\varepsilon\Psi}}{\sqrt{\Sigma_{\varepsilon\Psi}}} \leq x \mid \mathcal{E}_\infty \right) - \Phi(x) \right| \lesssim_{\mathbb{P}} \frac{TJ^{2d}/n^2}{((T-H)^2 \Sigma_{\varepsilon\Psi})^{3/2}} = o_{\mathbb{P}}(1),$$

where the last equality uses $(nT/J^d)\Sigma_{\varepsilon\Psi} \gtrsim_{\mathbb{P}} 1$ and $J^d/n = o(1)$. Removing the conditioning gives the infeasible Gaussian approximation for $\mathcal{J}_{\varepsilon\Psi}$, and the residual bound above transfers it to $\hat{\Psi} - \bar{\Psi}_T$.

For the variance estimator, apply the proof of Lemma SA-II.7.2 to the combined delete- i perturbation $\Delta_{it\Psi}$. Each diagonal, off-diagonal, and remainder term in that proof is quadratic in the relevant weights. Replacing $W_{it}(\beta)$ by the fixed finite sum $W_{it,\Psi} = \sum_{k=1}^{K_\Psi} a_k W_{it}(\mathbf{b}_k)$ only changes the constants in the second- and fourth-power sums and the squared-coefficient bound in that proof. The same argument therefore yields

$$\hat{\Sigma}_{\varepsilon\Psi} = \Sigma_{\varepsilon\Psi} + o_{\mathbb{P}} \left(\frac{J^d}{nT} \right).$$

This proves the theorem. □

SA-II.10 Fixed-Grid Test of Linearity

SA-II.10.1 Testing Problem

We test whether the conditional mean $\mathbb{E}[R_{it} | \mathcal{F}_{t-1}]$ is linear in β_{it} at every date. Specifically, the null hypothesis is

$$\mathbf{H}_0^{\text{LIN}} : \quad \mu_t(b) = a_t + c_t^\top b, \quad b \in \mathcal{B}, \quad t = 1, \dots, T, \quad (\text{SA-II.10.1})$$

for some a_t and c_t that may vary over time and are $\mathcal{G}_{t-1}$ -measurable. Under the null, realized returns satisfy

$$R_{it} = a_t + \varpi_t^\top \beta_{it} + \varepsilon_{it}, \quad \varpi_t = c_t + \tilde{f}_t.$$

Thus, the cross-sectional return slope is ϖ_t , not c_t : the linear fit must absorb the realized factor innovation.

The test uses two adjacent first-step windows. The partition at date t is formed from $\widehat{\beta}_{it}$, whose window ends at $t - 1$. The test outcome is dated $t + H$, and $\widehat{\beta}_{i(t+H)}$ uses dates $t, \dots, t + H - 1$. The two windows do not overlap. The linear fit uses a generic instrument available before the future window; it need not be $\widehat{\beta}_{it}$.

For $t = H + 1, \dots, T - H$, define the future-window conditioning field

$$\mathcal{K}_t = \mathcal{E}_\infty \vee \sigma(\eta_{jr} : j \in [n], r \leq t - 1).$$

To accommodate an unbalanced panel, for each base date $t = H + 1, \dots, T - H$, let $\mathcal{I}_t^{\text{LIN}} \subseteq [n]$ collect the units observed throughout both rolling windows and at the outcome date $t + H$, and write $n_t^{\text{LIN}} = |\mathcal{I}_t^{\text{LIN}}|$. Thus both $\widehat{\beta}_{it}$ and $\widehat{\beta}_{i(t+H)}$, as well as $R_{i(t+H)}$, are available for every $i \in \mathcal{I}_t^{\text{LIN}}$. All date- t cross-sectional objects in this section, including the partition and Gram matrix, are constructed on $\mathcal{I}_t^{\text{LIN}}$. We require $\mathcal{I}_t^{\text{LIN}}$ to be measurable with respect to the conditioning field $\mathcal{K}_t$, so sample availability does not select on the future primitive innovations, and assume that, for some fixed $c > 0$,

$$\min_{H+1 \leq t \leq T-H} n_t^{\text{LIN}} \geq cn$$

with probability approaching one. The DGP, partition, and Gram conditions invoked below are transferred to these matched cross-sections through the following random-missingness condition. For this purpose, $\{(\alpha_{ir}, \beta_{ir}, \varsigma_{ir}, \eta_{ir}) : i \in [n], r \in [T]\}$ is understood as a latent full-data array satisfying Assumption [SA-DGP](#) before selection. The auxiliary variables entering $\mathcal{E}_\infty$ are also understood to include the innovation-free availability process.

Assumption SA-LIN-M (Persistent random matched-sample availability). *Let $D_{it}^{\text{LIN}} = \mathbb{1}(i \in \mathcal{I}_t^{\text{LIN}})$ and let $\mathcal{Z}_t^{\text{LIN}}$ be the sigma-field generated by the full-data variables*

$$\{f_r : t - H \leq r \leq t + H\} \quad \text{and} \quad \{(\alpha_{ir}, \beta_{ir}, \varsigma_{ir}, \eta_{ir}) : i \in [n], t - H \leq r \leq t + H\}.$$

Uniformly over $t = H + 1, \dots, T - H$:

- (i) *The availability vector $(D_{1t}^{\text{LIN}}, \dots, D_{nt}^{\text{LIN}})$ is $\mathcal{K}_t$ -measurable. Given $\mathcal{D}_{t-1}$, this vector has mutually independent coordinates and is independent of $\mathcal{Z}_t^{\text{LIN}}$.*
- (ii) *For some fixed $\pi_0 > 0$, $\min_{i \in [n]} \mathbb{P}(D_{it}^{\text{LIN}} = 1 \mid \mathcal{D}_{t-1}) \geq \pi_0$ almost surely.*

Define the matched design field

$$\mathcal{D}_{t-1}^{\text{LIN}} = \mathcal{D}_{t-1} \vee \sigma(D_{1t}^{\text{LIN}}, \dots, D_{nt}^{\text{LIN}}).$$

Conditional on this field, the retained old-window predictable histories and old first-step estimates remain independent over i and retain their original conditional laws. Future-window primitive shocks remain independent conditional on $\mathcal{K}_t$. Although $\mathcal{D}_{t-1}^{\text{LIN}}$ need not be contained in $\mathcal{F}_{t-1}$, it is contained in $\mathcal{K}_t$. The matched versions of Lemmas [SA-II.3.1](#), [SA-II.3.2](#), and [SA-II.3.3](#) therefore follow by the same conditional arguments, with n_t replaced by n_t^{LIN} and with the reference partitions conditioning on $\mathcal{D}_{t-1}^{\text{LIN}}$. Whenever a sum is written over the common index set $[n]$, the contribution of unit i is set to zero if $i \notin \mathcal{I}_t^{\text{LIN}}$.

SA-II.10.2 IV Fit and Test Statistic

For $t = H + 1, \dots, T - H$, define the future regressor

$$\widehat{x}_{i(t+H)} = (1, \widehat{\beta}_{i(t+H)}^\top)^\top.$$

Let $\zeta_{it} \in \mathbb{R}^{d+1}$ be a generic instrument, whose first coordinate is one to accommodate the intercept. Set

$$\widehat{M}_t = \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \zeta_{it} \widehat{x}_{i(t+H)}^\top, \quad M_t^\circ = \mathbb{E}[\widehat{M}_t \mid \mathcal{K}_t]. \quad (\text{SA-II.10.2})$$

We impose the following high-level conditions on the instrument.

Assumption SA-LIN-IV (High-level instrument). *The following conditions hold uniformly over $t = H + 1, \dots, T - H$:*

(i) **Predetermination.** ζ_{it} is $\mathcal{F}_{t-1}$ -measurable for every $i \in \mathcal{I}_t^{\text{LIN}}$, and hence is $\mathcal{K}_t$ -measurable.

(ii) **Moment and leverage regularity.**

$$\max_{H+1 \leq t \leq T-H} \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \|\zeta_{it}\|^4 \lesssim_{\mathbb{P}} 1, \quad \max_{H+1 \leq t \leq T-H} \max_{i \in \mathcal{I}_t^{\text{LIN}}} \|\zeta_{it}\| \lesssim_{\mathbb{P}} 1.$$

(iii) **Relevance and concentration.**

$$\max_{H+1 \leq t \leq T-H} \{\|M_t^\circ\| + \|(M_t^\circ)^{-1}\|\} \lesssim_{\mathbb{P}} 1, \quad \max_{H+1 \leq t \leq T-H} \|\widehat{M}_t - M_t^\circ\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{n}}.$$

Predetermination is the exclusion condition. It ensures that the instrument is fixed before the innovations entering $\widehat{\beta}_{i(t+H)}$ and $R_{i(t+H)}$ are observed. Moment and leverage regularity control both aggregate moments and cell-specific instrument leverage in the covariance proof. The first bound in part (iii) is the relevance condition for the intercept and realized cross-sectional slope; the second is a high-level uniform cross-sectional concentration condition on the sample instrument-regressor matrix.

One natural choice is $\zeta_{it} = (1, \widehat{\beta}_{it}^\top)^\top$. Lemma SA-II.10.4 below verifies Assumption SA-LIN-IV for this choice under an oracle old-future beta relevance condition and the primitive first-step conditions. Other predetermined characteristics or generated instruments may be used whenever Assumption SA-LIN-IV holds.

Given this generic instrument ζ_{it} , define

$$\widehat{\theta}_{t+H}^{\text{IV}} = \widehat{M}_t^{-1} \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \zeta_{it} R_{i(t+H)}. \quad (\text{SA-II.10.3})$$

Write $\widehat{\theta}_{t+H}^{\text{IV}} = (\widehat{a}_{t+H}, \widehat{\omega}_{t+H}^\top)^\top$ and

$$\widehat{e}_{i(t+H)} = R_{i(t+H)} - \widehat{x}_{i(t+H)}^\top \widehat{\theta}_{t+H}^{\text{IV}}, \quad i \in \mathcal{I}_t^{\text{LIN}}, \quad t = H + 1, \dots, T - H.$$

Set $\widehat{e}_{i(t+H)} = 0$ when $i \notin \mathcal{I}_t^{\text{LIN}}$.

For any evaluation point $b \in \mathcal{B}^\circ$, define the projection weight

$$\hat{w}_{it}(b) = \frac{1}{n_t^{\text{LIN}}} \hat{p}_t(b)^\top \hat{Q}_t^{-1} \hat{p}_t(\hat{\beta}_{it}).$$

Set $\hat{w}_{it}(b) = 0$ when $i \notin \mathcal{I}_t^{\text{LIN}}$. When $\mathbf{q} = 0$, this is the usual equal weight in the cell containing b . The test statistic is

$$\hat{\nu}(b) = \frac{1}{T-2H} \sum_{t=H+1}^{T-H} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \hat{w}_{it}(b) \hat{e}_{i(t+H)}. \quad (\text{SA-II.10.4})$$

More generally, for distinct evaluation points $\mathbf{b}_1, \dots, \mathbf{b}_{K_{\text{LIN}}} \in \mathcal{B}^\circ$ with K_{LIN} fixed, define

$$\hat{\boldsymbol{\nu}}_{\text{LIN}} = (\hat{\nu}(\mathbf{b}_1), \dots, \hat{\nu}(\mathbf{b}_{K_{\text{LIN}}}))^\top.$$

Let Σ_{LIN} denote the predictable covariance matrix of the fixed-grid leading term under the null. The variance-stabilization condition below makes it asymptotically equivalent to an $\mathcal{E}_\infty$ -measurable reference covariance $\bar{\Sigma}_{\text{LIN}}$. Let $\hat{\Sigma}_{\text{LIN}}$ denote the feasible estimator of the predictable covariance; these objects are defined explicitly below. The Wald test statistic is

$$\mathbf{W}_{\text{LIN}} := \hat{\boldsymbol{\nu}}_{\text{LIN}}^\top \hat{\Sigma}_{\text{LIN}}^{-1} \hat{\boldsymbol{\nu}}_{\text{LIN}}. \quad (\text{SA-II.10.5})$$

Its null validity is established in Theorem [SA-II.10.3](#).

SA-II.10.3 Decomposition

From [\(SA-II.2.2\)](#), write the centered first-step error as

$$v_{it} = \hat{\beta}_{it} - \check{\beta}_{it} = \frac{1}{H} \sum_{s=1}^H w_s(t) \varepsilon_{i(t-s)}.$$

Under the null, let $\theta_t = (a_t, \varpi_t^\top)^\top$. The future residual has the exact decomposition

$$\begin{aligned} \hat{e}_{i(t+H)} &= \varepsilon_{i(t+H)} - \varpi_{t+H}^\top v_{i(t+H)} - \varpi_{t+H}^\top (\check{\beta}_{i(t+H)} - \beta_{i(t+H)}) \\ &\quad - \hat{x}_{i(t+H)}^\top (\hat{\theta}_{t+H}^{\text{IV}} - \theta_{t+H}). \end{aligned} \quad (\text{SA-II.10.6})$$

Substitution into [\(SA-II.10.4\)](#) gives

$$\hat{\nu}(b) = \mathcal{S}_{\text{LIN}}(b) + \mathcal{B}_{\text{LIN}}(b) + \mathcal{R}_{\text{LIN}}(b), \quad (\text{SA-II.10.7})$$

where

$$\begin{aligned} \mathcal{S}_{\text{LIN}}(b) &= \frac{1}{T-2H} \sum_{t=H+1}^{T-H} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \hat{w}_{it}(b) [\varepsilon_{i(t+H)} - \varpi_{t+H}^\top v_{i(t+H)}], \\ \mathcal{B}_{\text{LIN}}(b) &= -\frac{1}{T-2H} \sum_{t=H+1}^{T-H} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \hat{w}_{it}(b) \varpi_{t+H}^\top (\check{\beta}_{i(t+H)} - \beta_{i(t+H)}), \end{aligned}$$

$$\mathcal{R}_{\text{LIN}}(b) = -\frac{1}{T-2H} \sum_{t=H+1}^{T-H} \left(\sum_{i \in \mathcal{I}_t^{\text{LIN}}} \hat{w}_{it}(b) \hat{x}_{i(t+H)}^\top \right) (\hat{\theta}_{t+H}^{\text{IV}} - \theta_{t+H}).$$

The first term is centered and determines the limiting distribution. The second is the smooth first-step bias. The third is the error from the IV fit.

Note that the forward split removes the leave-in bias from the first term. By Assumption [SA-LIN-M\(i\)](#), the matched set is also fixed conditional on $\mathcal{K}_t$. Thus $\hat{w}_{it}(b)$ and ζ_{it} are fixed, while $v_{i(t+H)}$ uses innovations dated $t, \dots, t+H-1$ and $\varepsilon_{i(t+H)}$ is dated $t+H$. Hence

$$\mathbb{E}[\hat{w}_{it}(b)v_{i(t+H)} \mid \mathcal{K}_t] = 0, \quad (\text{SA-II.10.8})$$

$$\mathbb{E}[\hat{w}_{it}(b)\varepsilon_{i(t+H)} \mid \mathcal{K}_t] = 0. \quad (\text{SA-II.10.9})$$

By Assumption [SA-LIN-IV\(i\)](#), the same argument gives

$$\mathbb{E}[\zeta_{it}v_{i(t+H)}^\top \mid \mathcal{K}_t] = 0, \quad \mathbb{E}[\zeta_{it}\varepsilon_{i(t+H)} \mid \mathcal{K}_t] = 0. \quad (\text{SA-II.10.10})$$

Therefore the generic IV score is centered at the true realized slope, up to the smooth first-step bias.

For the limiting distribution, set $\hat{w}_{it}(b) = 0$ when $t \notin \{H+1, \dots, T-H\}$ or $i \notin \mathcal{I}_t^{\text{LIN}}$. Re-indexing the shocks in $\mathcal{S}_{\text{LIN}}(b)$ gives

$$\mathcal{S}_{\text{LIN}}(b) = \frac{1}{T-2H} \sum_{r=1}^T \sum_{i=1}^n W_{ir}(b) \varepsilon_{ir}, \quad (\text{SA-II.10.11})$$

where

$$W_{ir}(b) = \hat{w}_{i,r-H}(b) - \frac{1}{H} \sum_{s=1}^H \hat{w}_{i,r+s-H}(b) \varpi_{r+s}^\top w_s(r+s). \quad (\text{SA-II.10.12})$$

Invalid terms are set to zero. Every coefficient on ε_{ir} uses a partition whose first-step window ends before r . Hence $W_{ir}(b)$ is predictable. For $b, b' \in \mathcal{B}^\circ$, define

$$\Sigma_{\text{LIN}}(b, b') = \frac{1}{(T-2H)^2} \sum_{r=1}^T \sum_{i=1}^n \varsigma_{ir}^2 W_{ir}(b) W_{ir}(b'). \quad (\text{SA-II.10.13})$$

Let

$$\Sigma_{\text{LIN}} = (\Sigma_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_l))_{k,l \in [K_{\text{LIN}}]},$$

which is the terminal predictable covariance of the fixed-grid leading term under the null.

SA-II.10.4 Results

The first lemma gives the infeasible fixed-grid Gaussian limit for a generic instrument satisfying Assumption [SA-LIN-IV](#).

Lemma SA-II.10.1 (Forward-Split Asymptotic Normality). *Suppose Assumptions [SA-DGP](#), [SA-FS](#), [SA-PA](#), [SA-LIN-M](#), [SA-LIN-IV](#), [SA-SEP](#), and [SA-FSB](#) hold. Assume further that:*

$$(i) \quad T^{(3+d)/q}/H + \sqrt{\log T/H} = o(1).$$

$$(ii) \quad J^d \log(nT)/n = o(1).$$

(iii) The following rate restrictions hold:

$$\frac{T - 2H}{nH} = o(1), \quad (\text{SA-II.10.14})$$

$$\sqrt{\frac{n(T - 2H)}{J^d}} \left(\mathbf{B}_{\text{FS}} + \frac{\log(nT)}{n} \right) = o(1). \quad (\text{SA-II.10.15})$$

(iv) Variance stabilization. There exists an $\mathcal{E}_\infty$ -measurable positive-definite matrix $\bar{\Sigma}_{\text{LIN}}$ such that

$$\lambda_{\min} \left(\frac{n(T - 2H)}{J^d} \bar{\Sigma}_{\text{LIN}} \right) \gtrsim_{\mathbb{P}} 1, \quad \lambda_{\max} \left(\frac{n(T - 2H)}{J^d} \bar{\Sigma}_{\text{LIN}} \right) \lesssim_{\mathbb{P}} 1, \quad (\text{SA-II.10.16})$$

$$\left\| \bar{\Sigma}_{\text{LIN}}^{-1/2} \Sigma_{\text{LIN}} \bar{\Sigma}_{\text{LIN}}^{-1/2} - I_{K_{\text{LIN}}} \right\| = o_{\mathbb{P}}(1). \quad (\text{SA-II.10.17})$$

Then under the null hypothesis (SA-II.10.1),

$$\Sigma_{\text{LIN}}^{-1/2} \hat{\nu}_{\text{LIN}} \rightsquigarrow \mathcal{N}(0, I_{K_{\text{LIN}}}).$$

For feasible inference, define

$$\hat{\Sigma}_{\text{LIN}}(b, b') = \frac{1}{(T - 2H)^2} \sum_{i=1}^n \sum_{t=H+1}^{T-H} \sum_{s=H+1}^{T-H} \hat{w}_{it}(b) \hat{w}_{is}(b') \hat{e}_{i(t+H)} \hat{e}_{i(s+H)} \mathbf{1}(|t - s| \leq H), \quad (\text{SA-II.10.18})$$

and set

$$\hat{\Sigma}_{\text{LIN}} = \left(\hat{\Sigma}_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_l) \right)_{k, l \in [K_{\text{LIN}}]}.$$

The outer sum in (SA-II.10.18) uses the common index set $[n]$; the zero-weight and zero-residual conventions remove a unit whenever it is unavailable at either base date. The truncation at H is exact for the leading term because two forward residuals share a primitive idiosyncratic shock only when their base dates are at most H periods apart.

The next lemma shows that this feasible variance estimator is consistent.

Lemma SA-II.10.2 (Feasible Variance Consistency). *Suppose Assumptions SA-DGP, SA-FS, SA-PA, SA-LIN-M, SA-LIN-IV, SA-SEP, SA-FSB, and conditions (i)–(ii) of Lemma SA-II.10.1 hold. Also, assume*

$$\frac{H \log(nT)}{n} = o(1). \quad (\text{SA-II.10.19})$$

Then under the null hypothesis (SA-II.10.1),

$$\|\hat{\Sigma}_{\text{LIN}} - \Sigma_{\text{LIN}}\| = o_{\mathbb{P}} \left(\frac{J^d}{n(T - 2H)} \right). \quad (\text{SA-II.10.20})$$

The next theorem establishes the validity of the Wald test under the null.

Theorem SA-II.10.3 (Validity of the Forward-Split Wald Test). *Suppose the conditions of Lemma SA-II.10.1 hold and (SA-II.10.19) holds. Then $\widehat{\Sigma}_{\text{LIN}}$ is nonsingular with probability approaching one and the statistic in (SA-II.10.5) satisfies $W_{\text{LIN}} \rightsquigarrow \chi^2_{K_{\text{LIN}}}$ under the null hypothesis (SA-II.10.1). Consequently, the test that rejects when $W_{\text{LIN}} > \chi^2_{K_{\text{LIN}}, 1-\alpha}$ has asymptotic size α .*

Finally, the following lemma verifies Assumption SA-LIN-IV for the natural choice of instrument $\zeta_{it} = (1, \widehat{\beta}_{it}^\top)^\top$ under the primitive first-step conditions and an oracle relevance condition.

Lemma SA-II.10.4 (Old-Beta Instrument Verification). *Let*

$$x_{it} = (1, \beta_{it}^\top)^\top, \quad \mathbf{M}_{t,\beta}^0 = \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} x_{it} x_{i(t+H)}^\top.$$

Suppose Assumptions SA-DGP, SA-FS, SA-FSB, and SA-LIN-M hold,

$$\frac{T^{2/q}}{H} + \sqrt{\frac{\log T}{H}} = o(1), \quad R_{\text{FS}} + \frac{\log(nT)}{n} = o(1),$$

and the oracle old-future beta cross-moment is uniformly relevant:

$$\max_{H+1 \leq t \leq T-H} \{ \|\mathbf{M}_{t,\beta}^0\| + \|(\mathbf{M}_{t,\beta}^0)^{-1}\| \} \lesssim_{\mathbb{P}} 1. \quad (\text{SA-II.10.21})$$

Then the choice

$$\zeta_{it} = (1, \widehat{\beta}_{it}^\top)^\top$$

satisfies Assumption SA-LIN-IV.

Condition (SA-II.10.21) isolates the only substantive extra requirement for the old-beta choice: the predetermined beta must retain enough cross-sectional information about the future beta. Predetermination alone does not imply this cross-time relevance.

Remark SA-II.10.5 (Linear specification for excess returns). *When R_{it} denotes an excess return, the absence of arbitrage imposes the stronger zero-intercept restriction*

$$\mu_t(b) = c_t^\top b, \quad b \in \mathcal{B}, \quad t = 1, \dots, T.$$

The corresponding test is obtained by omitting the intercept from the period-by-period cross-sectional IV fit. Specifically, set $\widehat{x}_{i(t+H)} = \widehat{\beta}_{i(t+H)}$ and use a d -dimensional predetermined instrument ζ_{it} without a constant, with $\zeta_{it} = \widehat{\beta}_{it}$ as the natural choice. Define

$$\begin{aligned} \widehat{\mathbf{M}}_t &= \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \zeta_{it} \widehat{\beta}_{i(t+H)}^\top, \\ \widehat{\omega}_{t+H}^{\text{IV}} &= \widehat{\mathbf{M}}_t^{-1} \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \zeta_{it} R_{i(t+H)}, \end{aligned}$$

and construct the residuals as

$$\widehat{e}_{i(t+H)} = R_{i(t+H)} - \widehat{\beta}_{i(t+H)}^\top \widehat{\omega}_{t+H}^{\text{IV}}.$$

The fixed-grid moments, feasible covariance estimator, and Wald statistic are then constructed as above using these residuals. Under the corresponding d -dimensional version of Assumption [SA-LIN-IV](#), including nonsingularity of the $d \times d$ instrument–regressor matrix, and the remaining conditions of Theorem [SA-II.10.3](#), the same argument yields

$$W_{\text{LIN}} \rightsquigarrow \chi_{K_{\text{LIN}}}^2.$$

SA-II.10.5 Discussion

- (i) *Realized versus conditional slope.* The fitted slope estimates $\varpi_{t+H} = c_{t+H} + \tilde{f}_{t+H}$. Estimating only c_{t+H} would leave the realized factor component in the residual and would not give an idiosyncratic-scale test.
- (ii) *Why IV is necessary.* The forward split makes $v_{i(t+H)}$ independent of the old basis, but this fact alone does not make OLS valid. Specifically, for the OLS estimator of θ_{t+H} from regressing $R_{i(t+H)}$ on $\hat{x}_{i(t+H)}$, its population score at the true coefficient contains a component

$$\left\| \mathbb{E} \left[v_{i(t+H)} v_{i(t+H)}^\top \mid \mathcal{E}_\infty \right] \varpi_{t+H} \right\| = O_{\mathbb{P}}(H^{-1}).$$

Thus OLS would add an $O_{\mathbb{P}}(H^{-1})$ attenuation bias term to $\mathcal{R}_{\text{LIN}}(b)$ even though $v_{i(t+H)}$ is independent of the old partition. Any instrument satisfying Assumption [SA-LIN-IV](#) removes this term. The old beta estimate is one example because its first-step window does not overlap the future window.

- (iii) *A primitive verification of old–future relevance.* Condition [\(SA-II.10.21\)](#) can be verified under genuine sample-path smoothness of the beta process and conditionally random missingness. Suppose, for example, that there are possibly random paths $B_{iT} : [0, 1] \rightarrow \mathcal{B}$ such that $\beta_{it} = B_{iT}(t/T)$ and, for some $s_\beta > 0$, the paths are Hölder- s_β on $[0, 1]$ uniformly over $i \in [n]$, with their uniform bounds and Hölder constants bounded in probability. Then, since $H/T = o(1)$,

$$\delta_{\beta, H} := \max_{H+1 \leq t \leq T-H} \max_{i \in \mathcal{I}_t^{\text{LIN}}} \|\beta_{i(t+H)} - \beta_{it}\| \lesssim_{\mathbb{P}} \left(\frac{H}{T} \right)^{s_\beta \wedge 1} = o_{\mathbb{P}}(1).$$

Define the old-beta Gram matrix on the matched cross-section by

$$Q_{t, \beta}^0 = \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} x_{it} x_{it}^\top.$$

Compactness of $\mathcal{B}$ gives

$$\max_{H+1 \leq t \leq T-H} \|\mathbf{M}_{t, \beta}^0 - Q_{t, \beta}^0\| \lesssim_{\mathbb{P}} \delta_{\beta, H} = o_{\mathbb{P}}(1).$$

Under Assumption [SA-LIN-M](#), conditioning on the realized matched set preserves the original conditional beta-density bounds and cross-sectional independence. The density lower bound on the full-dimensional rectangle $\mathcal{B}$, $n_t^{\text{LIN}} \asymp n$, and uniform cross-sectional concentration under $\log(nT)/n = o(1)$ therefore imply

$$\min_{H+1 \leq t \leq T-H} \lambda_{\min}(Q_{t, \beta}^0) \gtrsim_{\mathbb{P}} 1, \quad \max_{H+1 \leq t \leq T-H} \|Q_{t, \beta}^0\| \lesssim_{\mathbb{P}} 1.$$

Singular-value perturbation therefore yields

$$\max_{H+1 \leq t \leq T-H} \{ \|\mathbf{M}_{t,\beta}^0\| + \|(\mathbf{M}_{t,\beta}^0)^{-1}\| \} \lesssim_{\mathbb{P}} 1,$$

which is (SA-II.10.21). We retain the high-level relevance condition in Lemma SA-II.10.4 because it also accommodates other persistence and missingness mechanisms.

- (iv) *Generated partition.* The proof conditions on the old generated partition. It does not compare that partition with a true-beta partition. The basis-switching restriction $J\mathbf{R}_{\text{FS}} = o(1)$ from the contemporaneous theory is therefore not needed for null size.
- (v) *Rate conditions.* Consider the proposed MSE-optimal tuning choices in Part III. For $(\mathbf{p}, \mathbf{q}) = (0, 0)$, they give

$$h \asymp T^{-1/3}, \quad H \asymp T^{2/3}, \quad J \asymp n^{1/(d+2)}.$$

The bias component of (SA-II.10.15) then requires

$$J\sqrt{T}\mathbf{B}_{\text{FS}} = o(1), \quad \text{or equivalently} \quad \mathbf{B}_{\text{FS}} = o\left(n^{-1/(d+2)}T^{-1/2}\right). \quad (\text{SA-II.10.22})$$

The implications depend on the order of the first-step local polynomial.

Local-constant first step. With a nonzero leading bias, if $\mathbf{B}_{\text{FS}} \asymp h \asymp T^{-1/3}$, then $J\sqrt{T}\mathbf{B}_{\text{FS}} \asymp JT^{1/6}$ does not vanish, so the proposed MSE-optimal bandwidth does not satisfy (SA-II.10.22).

Local-linear first step. If $s \geq 2$ and $\mathbf{B}_{\text{FS}} \lesssim h^2 \asymp T^{-2/3}$, write $T = n^\eta$, suppress logarithmic factors, and maintain the high-level instrument and covariance nondegeneracy conditions. The bias restriction gives the lower bound $\eta > 6/(d+2)$, while the other Gaussian approximation rates give the upper bound $\eta < 2(d+1)/(d+2)$. Thus

$$\frac{6}{d+2} < \eta < \frac{2(d+1)}{d+2}.$$

For the feasible Wald statistic, (SA-II.10.19) further requires $\eta < 3/2$. The resulting window is

$$\frac{6}{d+2} < \eta < \min \left\{ \frac{2(d+1)}{d+2}, \frac{3}{2} \right\}. \quad (\text{SA-II.10.23})$$

This window is nonempty when $d > 2$. It contains a subrange with $\eta < 1$, and hence permits $T \ll n$, only when $d > 4$.

Local-quadratic first step. If $s \geq 3$ and $\mathbf{B}_{\text{FS}} \lesssim h^3 \asymp T^{-1}$, the bias restriction now gives $\eta > 2/(d+2)$, so the feasible window becomes

$$\frac{2}{d+2} < \eta < \min \left\{ \frac{2(d+1)}{d+2}, \frac{3}{2} \right\}.$$

This window is nonempty for every $d \geq 1$. In particular, when $d = 1$, it contains the subrange $2/3 < \eta < 1$, which permits $T \ll n$. Thus, under the proposed MSE-optimal bandwidth, a higher-order first step is needed to accommodate $T \ll n$ in the lower-dimensional cases.

- (vi) *Variance stabilization.* Condition (SA-II.10.17) concerns the full rolling covariance functional, not pointwise convergence of the generated weights. One verification separates dates at which each grid point is away from reference-cell boundaries from boundary-near dates: the former have a stable cell label, while the latter are negligible under a time-averaged margin condition or are covered by local equivalence of the adjacent-cell rolling covariance kernels. Alternatively, the evaluation-point-anchored construction in Section SA-II.9, particularly Assumption SA-PA-G, fixes each grid point's cell label and basis coordinate; conditional quantile and Gram concentration can then control the remaining empirical-to-reference covariance difference. We retain the high-level condition here to accommodate either route.
- (vii) *Power against fixed persistent alternatives.* Let $\boldsymbol{\nu}_{0,T,\text{LIN}}$ denote the time-averaged population forward moment, after the population cross-time IV linear fit, evaluated on the fixed grid. If $\boldsymbol{\nu}_{0,T,\text{LIN}} \rightarrow_{\mathbb{P}} \boldsymbol{\nu}_{0,\text{LIN}} \neq 0$ and $\widehat{\boldsymbol{\nu}}_{\text{LIN}} - \boldsymbol{\nu}_{0,T,\text{LIN}} = o_{\mathbb{P}}(1)$, while $\widehat{\Sigma}_{\text{LIN}}$ is positive definite with probability approaching one and $\lambda_{\max}(\widehat{\Sigma}_{\text{LIN}}) = o_{\mathbb{P}}(1)$, then $W_{\text{LIN}} \rightarrow_{\mathbb{P}} \infty$ and the test is consistent. The nonzero-moment condition requires persistence and rules out cancellation over dates and grid points; without it, a nonlinear fixed alternative need not be detected.
- (viii) *Scope of the test.* Taking $K_{\text{LIN}} = 1$ gives a pointwise specification test; fixed K_{LIN} gives the Wald test above. A diverging-grid omnibus test requires a separate high-dimensional Gaussian approximation. The test uses an exactly identified $(d+1)$ -vector of instruments. An overidentified specification can be accommodated after replacing (SA-II.10.3) by the corresponding GMM fit and imposing the analogous Jacobian and weight-matrix conditions.
- (ix) *A fixed-period OLS alternative.* At one prespecified date $t > H$, one could instead compare the degree-one beta-sorted fit with the same-date full-cross-sectional OLS fit. Degree-one reproduction removes the within-cell discrepancy under the linear null. The nonparametric component has standard deviation $\sqrt{J^d/n}$, whereas the OLS sampling error is of smaller order $n^{-1/2}$. Unlike the forward-split test, this approach does not remove the generated-regressor and OLS attenuation distortions; it instead requires the sufficient centering condition $B_{\text{FS}} + H^{-1} = o(\sqrt{J^d/n})$. Use the paper's proposed local-constant point-estimation MSE tuning,

$$h \asymp T^{-1/3}, \quad H \asymp T^{2/3}, \quad J \asymp n^{1/(d+2)},$$

but construct both steps with degree-one fits. Then if $B_{\text{FS}} \lesssim T^{-2/3}$, the main restriction reduces, up to logarithmic factors from uniform remainder bounds, to $T \gg n^{3/[2(d+2)]}$. Thus $T = n^\eta$ with $3/[2(d+2)] < \eta < 1$ is admissible for every $d \geq 1$, permitting $T \ll n$.

SA-II.10.6 Proof of Lemma SA-II.10.1

Step 1: Remainders. Put

$$\begin{aligned} u_{it}^\circ &= \varepsilon_{i(t+H)} - \varpi_{t+H}^\top v_{i(t+H)}, & g_t &= \frac{1}{n_{\text{LIN}}^t} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \zeta_{it} u_{it}^\circ, \\ A_t(b) &= \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \widehat{w}_{it}(b) \widehat{x}_{i(t+H)}^\top, & A_t^\circ(b) &= \mathbb{E}[A_t(b) \mid \mathcal{K}_t]. \end{aligned}$$

For notational convenience, set $u_{it}^\circ = 0$ when $i \notin \mathcal{I}_t^{\text{LIN}}$. The IV normal equations give

$$\widehat{\theta}_{t+H}^{\text{IV}} - \theta_{t+H} = \widehat{\mathbf{M}}_t^{-1} g_t + \mathcal{B}_{t+H}^{\text{IV}}, \quad (\text{SA-II.10.24})$$

$$\mathcal{B}_{t+H}^{\text{IV}} := -\widehat{\mathbf{M}}_t^{-1} \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \zeta_{it} \varpi_{t+H}^\top (\check{\beta}_{i(t+H)} - \beta_{i(t+H)}). \quad (\text{SA-II.10.25})$$

Thus $\mathcal{B}_{t+H}^{\text{IV}}$ is the component of the IV estimation error induced by the smooth first-step bias. In particular,

$$\mathcal{B}_{\text{LIN}}(b) = -\frac{1}{T-2H} \sum_{t=H+1}^{T-H} A_t(b) \widehat{\mathbf{M}}_t^{-1} g_t - \frac{1}{T-2H} \sum_{t=H+1}^{T-H} A_t(b) \mathcal{B}_{t+H}^{\text{IV}}.$$

The partition, Gram, first-step moment, and instrument moment bounds imply

$$\max_{\substack{H+1 \leq t \leq T-H \\ k \in [K_{\text{LIN}}]}} \{ \|A_t(\mathbf{b}_k)\| + \|A_t^\circ(\mathbf{b}_k)\| \} \lesssim_{\mathbb{P}} 1.$$

Hence the bias terms in (SA-II.10.7) and (SA-II.10.24) satisfy

$$\max_{k \in [K_{\text{LIN}}]} \left\{ |\mathcal{B}_{\text{LIN}}(\mathbf{b}_k)| + \left| \frac{1}{T-2H} \sum_{t=H+1}^{T-H} A_t(\mathbf{b}_k) \mathcal{B}_{t+H}^{\text{IV}} \right| \right\} \lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}}. \quad (\text{SA-II.10.26})$$

Conditional on $\mathcal{K}_t$, g_t is centered by (SA-II.10.10). Cross-sectional independence, which is preserved after selection by Assumption SA-LIN-M, Assumption SA-LIN-IV(ii), the first-step moment bounds, and the weight bounds give

$$\begin{aligned} \mathbb{E}[\|g_t\|^2 \mid \mathcal{K}_t] &\lesssim_{\mathbb{P}} \frac{1 + \|\varpi_{t+H}\|^2/H}{n_t^{\text{LIN}}} \lesssim_{\mathbb{P}} \frac{1 + \|\varpi_{t+H}\|^2/H}{n}, \\ \max_{\substack{H+1 \leq t \leq T-H \\ k \in [K_{\text{LIN}}]}} \mathbb{E}[\|A_t(\mathbf{b}_k) - A_t^\circ(\mathbf{b}_k)\|^2 \mid \mathcal{K}_t] &\lesssim_{\mathbb{P}} \frac{J^d}{nH}. \end{aligned} \quad (\text{SA-II.10.27})$$

Boundedness of μ_t on the fixed rectangle $\mathcal{B}$ bounds c_t under the null. Assumption SA-DGP(iv) and Markov's inequality therefore give $(T-2H)^{-1} \sum_{t=H+1}^{T-H} (1 + \|\varpi_{t+H}\|^4) \lesssim_{\mathbb{P}} 1$. The multiplier $A_t^\circ(\mathbf{b}_k)(\mathbf{M}_t^\circ)^{-1} \zeta_{it}$ attached to each summand of g_t is fixed before the future-window shocks begin. Re-indexing the direct outcome shocks therefore gives a martingale sum. The rolling-window component is also centered; it has periodwise variance $O_{\mathbb{P}}((nH)^{-1})$ and overlaps for at most H adjacent base dates. Thus, for fixed K_{LIN} ,

$$\max_{k \in [K_{\text{LIN}}]} \left| \frac{1}{T-2H} \sum_{t=H+1}^{T-H} A_t^\circ(\mathbf{b}_k) (\mathbf{M}_t^\circ)^{-1} g_t \right| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{n(T-2H)}}. \quad (\text{SA-II.10.28})$$

It remains to replace the conditional multipliers by their sample counterparts. Assumption SA-LIN-IV(iii) and a matrix perturbation bound imply that $\widehat{\mathbf{M}}_t$ and $\widehat{\mathbf{M}}_t^{-1}$ are uniformly bounded in probability and

$$\begin{aligned} r_{\mathbf{M}} &:= \max_{H+1 \leq t \leq T-H} \|\widehat{\mathbf{M}}_t - \mathbf{M}_t^\circ\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{n}}, \\ \max_{H+1 \leq t \leq T-H} \|\widehat{\mathbf{M}}_t^{-1} - (\mathbf{M}_t^\circ)^{-1}\| &\lesssim_{\mathbb{P}} r_{\mathbf{M}}. \end{aligned}$$

Use the exact decomposition

$$A_t \widehat{\mathbf{M}}_t^{-1} - A_t^\circ (\mathbf{M}_t^\circ)^{-1} = (A_t - A_t^\circ) (\mathbf{M}_t^\circ)^{-1} + A_t \{\widehat{\mathbf{M}}_t^{-1} - (\mathbf{M}_t^\circ)^{-1}\}.$$

For the first term on the right-hand side, (SA-II.10.27), the uniform bound on $(\mathbf{M}_t^\circ)^{-1}$, the time-averaged factor-moment bound, and Markov's inequality give, uniformly over the fixed grid,

$$\frac{1}{T-2H} \sum_{t=H+1}^{T-H} \|A_t(\mathbf{b}_k) - A_t^\circ(\mathbf{b}_k)\|^2 \lesssim_{\mathbb{P}} \frac{J^d}{nH}, \quad \frac{1}{T-2H} \sum_{t=H+1}^{T-H} \|g_t\|^2 \lesssim_{\mathbb{P}} \frac{1}{n}.$$

Cauchy–Schwarz over the base dates therefore yields

$$\max_{k \in [K_{\text{LIN}}]} \left| \frac{1}{T-2H} \sum_{t=H+1}^{T-H} \{A_t(\mathbf{b}_k) - A_t^\circ(\mathbf{b}_k)\} (\mathbf{M}_t^\circ)^{-1} g_t \right| \lesssim_{\mathbb{P}} \frac{1}{n} \sqrt{\frac{J^d}{H}}. \quad (\text{SA-II.10.29})$$

Relative to the local standard-deviation scale, this bound satisfies

$$\sqrt{\frac{n(T-2H)}{J^d}} \frac{1}{n} \sqrt{\frac{J^d}{H}} = \sqrt{\frac{T-2H}{nH}} = o(1)$$

by (SA-II.10.14).

For the inverse-matrix term, (SA-II.10.27), the time-averaged factor-moment bound, and Markov's inequality give

$$\frac{1}{T-2H} \sum_{t=H+1}^{T-H} \|g_t\| \leq \left\{ \frac{1}{T-2H} \sum_{t=H+1}^{T-H} \|g_t\|^2 \right\}^{1/2} \lesssim_{\mathbb{P}} \frac{1}{\sqrt{n}}.$$

Together with the uniform bound on A_t , this gives

$$\max_{k \in [K_{\text{LIN}}]} \left| \frac{1}{T-2H} \sum_{t=H+1}^{T-H} A_t(\mathbf{b}_k) \{\widehat{\mathbf{M}}_t^{-1} - (\mathbf{M}_t^\circ)^{-1}\} g_t \right| \lesssim_{\mathbb{P}} \frac{r_{\mathfrak{M}}}{\sqrt{n}} \lesssim_{\mathbb{P}} \frac{\sqrt{\log(nT)}}{n}.$$

The last bound is absorbed by $O_{\mathbb{P}}\{\log(nT)/n\}$. Combining the two parts of the exact multiplier decomposition gives

$$\max_{k \in [K_{\text{LIN}}]} \left| \frac{1}{T-2H} \sum_{t=H+1}^{T-H} \left\{ A_t(\mathbf{b}_k) \widehat{\mathbf{M}}_t^{-1} - A_t^\circ(\mathbf{b}_k) (\mathbf{M}_t^\circ)^{-1} \right\} g_t \right| \lesssim_{\mathbb{P}} \frac{1}{n} \sqrt{\frac{J^d}{H}} + \frac{\log(nT)}{n}. \quad (\text{SA-II.10.30})$$

Combining (SA-II.10.26)–(SA-II.10.30) gives

$$\max_{k \in [K_{\text{LIN}}]} \{|\mathcal{R}_{\text{LIN}}(\mathbf{b}_k)| + |\mathcal{B}_{\text{LIN}}(\mathbf{b}_k)|\} \lesssim_{\mathbb{P}} \mathbf{B}_{\text{FS}} + \frac{1}{\sqrt{n(T-2H)}} + \frac{1}{n} \sqrt{\frac{J^d}{H}} + \frac{\log(nT)}{n}. \quad (\text{SA-II.10.31})$$

The restriction $J^{-d/2} = o(1)$ controls the normalized $\{n(T-2H)\}^{-1/2}$ term. Equations (SA-II.10.14) and (SA-II.10.15) control the remaining terms. Thus both non-leading terms in (SA-II.10.7), uniformly over the fixed grid, are $o_{\mathbb{P}}(\sqrt{J^d/(n(T-2H))})$.

Step 2: Leading term. The matched version of Lemma SA-II.3.3 described after Assumption SA-LIN-M,

Jensen's inequality over the H rolling lags, the first-step Gram bound, and $cn \leq n_t^{\text{LIN}} \leq n$ imply, under the absent-unit convention and uniformly over $k \in [K_{\text{LIN}}]$,

$$\sum_{r=1}^T \sum_{i=1}^n W_{ir}(\mathbf{b}_k)^2 \lesssim_{\mathbb{P}} \frac{(T-2H)J^d}{n}, \quad \sum_{r=1}^T \sum_{i=1}^n |W_{ir}(\mathbf{b}_k)|^3 \lesssim_{\mathbb{P}} \frac{(T-2H)J^{2d}}{n^2}. \quad (\text{SA-II.10.32})$$

Consequently, $\lambda_{\max}(\Sigma_{\text{LIN}}) \lesssim_{\mathbb{P}} J^d / \{n(T-2H)\}$. Condition (iv) implies that all eigenvalues of both $\bar{\Sigma}_{\text{LIN}}$ and Σ_{LIN} have this order.

Let $\mathbf{W}_{ir} = (W_{ir}(\mathbf{b}_1), \dots, W_{ir}(\mathbf{b}_{K_{\text{LIN}}}))^\top$. For fixed $a \in \mathbb{R}^{K_{\text{LIN}}}$, put

$$\widetilde{W}_{ir}(a) := a^\top \bar{\Sigma}_{\text{LIN}}^{-1/2} \mathbf{W}_{ir}.$$

The reference covariance is $\mathcal{E}_\infty$ -measurable. Hence, after ordering the primitive innovations first by date and then by unit, $\widetilde{W}_{ir}(a)$ is measurable before η_{ir} is observed. The sum of the conditional variances of the corresponding normalized martingale differences (terminal predictable quadratic variation of their sum) is

$$\frac{1}{(T-2H)^2} \sum_{r=1}^T \sum_{i=1}^n \varepsilon_{ir}^2 \widetilde{W}_{ir}(a)^2 = a^\top \bar{\Sigma}_{\text{LIN}}^{-1/2} \Sigma_{\text{LIN}} \bar{\Sigma}_{\text{LIN}}^{-1/2} a \rightarrow_{\mathbb{P}} \|a\|^2$$

by (SA-II.10.17). Let

$$\mathcal{J}_{ir-}^{\text{LIN}} := \mathcal{E}_\infty \vee \sigma(\eta_{js} : s < r, \text{ or } s = r \text{ and } j < i).$$

Since K_{LIN} is fixed, the third-moment bound in (SA-II.10.32) and (SA-II.10.16) also give the conditional Lyapunov bound

$$\frac{1}{(T-2H)^3} \sum_{r=1}^T \sum_{i=1}^n \mathbb{E} \left[|\widetilde{W}_{ir}(a) \varepsilon_{ir}|^3 \mid \mathcal{J}_{ir-}^{\text{LIN}} \right] \lesssim_{\mathbb{P}} \sqrt{\frac{J^d}{n(T-2H)}} = o_{\mathbb{P}}(1).$$

The martingale central limit theorem and the Cramér–Wold device therefore give

$$\bar{\Sigma}_{\text{LIN}}^{-1/2} \mathcal{S}_{\text{LIN}} \rightsquigarrow \mathcal{N}(0, I_{K_{\text{LIN}}}),$$

where $\mathcal{S}_{\text{LIN}}$ stacks the leading terms over the fixed grid. Condition (SA-II.10.16) and the Step 1 remainder bound then imply

$$\bar{\Sigma}_{\text{LIN}}^{-1/2} \widehat{\boldsymbol{\nu}}_{\text{LIN}} \rightsquigarrow \mathcal{N}(0, I_{K_{\text{LIN}}}). \quad (\text{SA-II.10.33})$$

Finally, (SA-II.10.16) and (SA-II.10.17), together with a matrix square-root perturbation bound, give

$$\left\| \Sigma_{\text{LIN}}^{-1/2} \bar{\Sigma}_{\text{LIN}}^{1/2} - I_{K_{\text{LIN}}} \right\| = o_{\mathbb{P}}(1).$$

Therefore

$$\Sigma_{\text{LIN}}^{-1/2} \widehat{\boldsymbol{\nu}}_{\text{LIN}} = \Sigma_{\text{LIN}}^{-1/2} \bar{\Sigma}_{\text{LIN}}^{1/2} \left(\bar{\Sigma}_{\text{LIN}}^{-1/2} \widehat{\boldsymbol{\nu}}_{\text{LIN}} \right) \rightsquigarrow \mathcal{N}(0, I_{K_{\text{LIN}}})$$

by (SA-II.10.33) and Slutsky's theorem. □

SA-II.10.7 Proof of Lemma SA-II.10.2

For this proof, put

$$\begin{aligned} \mathcal{T}_H &= \{H+1, \dots, T-H\}, & L_T &= |\mathcal{T}_H| = T-2H, & \ell_{nT} &= \frac{\log(nT)}{n}, \\ e_{nT}^2 &= \mathbf{B}_{\text{FS}}^2 + \ell_{nT}, & \rho_{nT}^2 &= \frac{1}{H} + \ell_{nT}, & u_{it}^\circ &= \varepsilon_{i(t+H)} - \varpi_{t+H}^\top v_{i(t+H)}, \\ g_t &= \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{T}_t^{\text{LIN}}} \zeta_{it} u_{it}^\circ. \end{aligned}$$

Unless stated otherwise, sums over t and s below are over $\mathcal{T}_H$, and shifted dates outside $\mathcal{T}_H$ are omitted. The generated weight $\hat{w}_{it}$ and the fitted residual-error components introduced below are set to zero outside the corresponding matched set. The latent block u_{it}° and its primitive shocks remain defined for every $i \in [n]$; zero weights remove unavailable units from all sample sums.

The assumptions used in this proof imply

$$\begin{aligned} \frac{H}{L_T} &= o(1), & \frac{J^d}{n} &= o(1), & H e_{nT}^2 &= o_{\mathbb{P}}(1), & H \rho_{nT} e_{nT} &= o_{\mathbb{P}}(1), \\ \frac{H}{n} &= o(1), & \sqrt{\frac{H J^d}{n L_T}} &= o(1), & H \sqrt{\frac{J^d}{n^2 L_T}} &= o(1). \end{aligned} \tag{SA-II.10.34}$$

For example, $H^2 \rho_{nT}^2 e_{nT}^2 = (1 + H \ell_{nT})(H \mathbf{B}_{\text{FS}}^2 + H \ell_{nT}) = o_{\mathbb{P}}(1)$. The matched-sample local-support and Gram bounds also give, uniformly over the fixed grid,

$$\begin{aligned} \max_{t,k} \sum_i \hat{w}_{it}(\mathbf{b}_k)^2 &\lesssim_{\mathbb{P}} \frac{J^d}{n}, & \max_{i,t,k} |\hat{w}_{it}(\mathbf{b}_k)| &\lesssim_{\mathbb{P}} \frac{J^d}{n}, \\ \max_{t,s,k,\ell} \sum_i \hat{w}_{it}(\mathbf{b}_k)^2 \hat{w}_{is}(\mathbf{b}_\ell)^2 &\lesssim_{\mathbb{P}} \frac{J^{3d}}{n^3}. \end{aligned} \tag{SA-II.10.35}$$

We use two primitive-shock calculations repeatedly. First, define

$$\kappa_{tr} := \mathbf{1}(r = t+H) - \frac{1}{H} \mathbf{1}(t \leq r \leq t+H-1) \varpi_{t+H}^\top w_{t+H-r}(t+H).$$

The uniform first-step Gram bound gives

$$\max_{s \in \mathcal{T}_H} \frac{1}{H} \sum_{a=1}^H \|w_a(s+H)\|^2 \lesssim_{\mathbb{P}} 1.$$

Therefore $u_{it}^\circ = \sum_{r=1}^T \kappa_{tr} \varepsilon_{ir}$ and

$$\sum_{r=1}^T \kappa_{tr}^2 \lesssim_{\mathbb{P}} 1 + \frac{\|\varpi_{t+H}\|^2}{H}. \tag{SA-II.10.36}$$

More generally, if Z_{is} is $\mathcal{K}_s$ -measurable, then

$$\begin{aligned} \sum_s Z_{is} u_{is}^\circ &= \sum_{r=1}^T \Gamma_{ir}(Z) \varepsilon_{ir}, \\ \Gamma_{ir}(Z) &:= Z_{i,r-H} - \frac{1}{H} \sum_{a=1}^H Z_{i,r+a-H} \varpi_{r+a}^\top w_a(r+a), \\ \sum_{i,r} |\Gamma_{ir}(Z)|^2 &\lesssim_{\mathbb{P}} \sum_{i,s} (1 + \|\varpi_{s+H}\|^2) |Z_{is}|^2, \end{aligned} \tag{SA-II.10.37}$$

where invalid subscripts are zero. Each $\Gamma_{ir}(Z)$ is measurable before ε_{ir} is observed.

Second, index primitive shocks by $a = (i, r)$ and $b = (j, r')$, write $\varepsilon_a = \varepsilon_{ir}$ and $\varepsilon_b = \varepsilon_{jr'}$, and order these compound indices lexicographically by date and unit. Let

$$\mathcal{J}_{ir-}^{\text{LIN}} := \mathcal{E}_\infty \vee \sigma(\eta_{js} : s < r, \text{ or } s = r \text{ and } j < i).$$

Suppose that $a < b$, A_{ab} is measurable before ε_a , and, for a deterministic sequence r_n , $\sum_{a < b} |A_{ab}|^2 = O_{\mathbb{P}}(r_n^2)$. Then bounded conditional fourth moments and successive conditioning give

$$\sum_{a < b} A_{ab} \varepsilon_a \varepsilon_b = O_{\mathbb{P}}(r_n). \tag{SA-II.10.38}$$

For a quadratic sum over $a \neq b$, we first combine the coefficients on (a, b) and (b, a) , which multiply the same shock product. We also use the analogous bound for a predictable linear shock sum: $\sum_{i,r} \Gamma_{ir} \varepsilon_{ir} = O_{\mathbb{P}}(r_n)$ whenever the coefficients are predictable and $\sum_{i,r} |\Gamma_{ir}|^2 = O_{\mathbb{P}}(r_n^2)$.

Step 1: Oracle covariance. For $k, \ell \in [K_{\text{LIN}}]$, let $\tilde{\Sigma}_{k\ell}$ denote (SA-II.10.18) with $(b, b') = (\mathbf{b}_k, \mathbf{b}_\ell)$ and $\hat{e}_{i(t+H)}$ replaced by u_{it}° . Because u_{it}° is a linear combination of primitive shocks dated in $[t, t+H]$, the truncation at H is exact. For example, if $t+H < s$, then $\hat{w}_{it}(\mathbf{b}_k) u_{it}^\circ \hat{w}_{is}(\mathbf{b}_\ell)$ is $\mathcal{K}_s$ -measurable, whereas $\mathbb{E}[u_{is}^\circ | \mathcal{K}_s] = 0$; the reverse ordering is symmetric. Re-indexing the retained products by primitive shock gives

$$\begin{aligned} \tilde{\Sigma}_{k\ell} &= \mathcal{D}_{k\ell} + \mathcal{O}_{k\ell}, \\ \mathcal{D}_{k\ell} &= \frac{1}{L_T^2} \sum_{r=1}^T \sum_{i=1}^n W_{ir}(\mathbf{b}_k) W_{ir}(\mathbf{b}_\ell) \varepsilon_{ir}^2, \\ \mathcal{O}_{k\ell} &= \frac{1}{L_T^2} \sum_{i=1}^n \sum_{r \neq r'}^n M_{i,rr'}^{k\ell} \varepsilon_{ir} \varepsilon_{ir'}. \end{aligned} \tag{SA-II.10.39}$$

Here $M_{i,rr'}^{k\ell}$ collects the coefficients on products of distinct primitive shocks. The diagonal coefficient is exactly the product of the primitive weights in (SA-II.10.12): every pair of residual terms containing the same primitive shock is retained because their base dates differ by at most H .

Diagonal component. The first two bounds in (SA-II.10.35), $|x+y|^4 \leq 8(|x|^4 + |y|^4)$, Jensen's inequality over the rolling average, and $L_T^{-1} \sum_t (1 + \|\varpi_{t+H}\|^4) \lesssim_{\mathbb{P}} 1$ give

$$\sum_{r=1}^T \sum_{i=1}^n |W_{ir}(\mathbf{b}_k)|^4 \lesssim_{\mathbb{P}} \frac{L_T J^{3d}}{n^3}. \tag{SA-II.10.40}$$

Since $W_{ir}(\mathbf{b}_k)W_{ir}(\mathbf{b}_\ell)$ is $\mathcal{J}_{ir-}^{\text{LIN}}$ -measurable, $W_{ir}(\mathbf{b}_k)W_{ir}(\mathbf{b}_\ell)(\varepsilon_{ir}^2 - \varsigma_{ir}^2)$ is a martingale difference and

$$\mathcal{D}_{k\ell} - \Sigma_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_\ell) = \frac{1}{L_T^2} \sum_{r=1}^T \sum_{i=1}^n W_{ir}(\mathbf{b}_k)W_{ir}(\mathbf{b}_\ell) (\varepsilon_{ir}^2 - \varsigma_{ir}^2).$$

Its conditional variance is bounded by (SA-II.10.40) and the conditional fourth-moment bound for ε_{ir} . Therefore

$$\mathcal{D}_{k\ell} - \Sigma_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_\ell) = O_{\mathbb{P}}\left(\sqrt{\frac{J^{3d}}{n^3 L_T^3}}\right) = o_{\mathbb{P}}\left(\frac{J^d}{n L_T}\right). \quad (\text{SA-II.10.41})$$

Off-diagonal component. For $t < s \leq t + H$, define the pre- s and s -onward components by

$$u_{it}^{<s} := -\frac{1}{H} \sum_{r=t}^{s-1} \varpi_{t+H}^\top w_{t+H-r}(t+H) \varepsilon_{ir}, \quad u_{it}^{\geq s} := u_{it}^\circ - u_{it}^{<s}.$$

Set $u_{it}^{<s} = 0$ and $u_{it}^{\geq s} = u_{it}^\circ$ when $s \leq t$; the reverse ordering is handled by interchanging t and s . Put $\Lambda_{t,h} = 1 + \|\varpi_{t+h+H}\|^2$. The first-step Gram and factor-moment bounds, $2|xy| \leq x^2 + y^2$, and $L_T^{-1} \sum_t (1 + \|\varpi_{t+H}\|^4) \lesssim_{\mathbb{P}} 1$ give

$$\begin{aligned} \max_{k \in [K_{\text{LIN}}]} \sum_{h=1}^H \frac{1}{L_T} \sum_t \sum_i \Lambda_{t,h} \widehat{w}_{it}(\mathbf{b}_k)^2 (u_{it}^{<t+h})^2 &\lesssim_{\mathbb{P}} \frac{J^d}{n}, \\ \sum_{h=1}^H \frac{1}{L_T} \sum_t \frac{\Lambda_{t,h}}{n} \sum_i (u_{it}^{<t+h})^2 &\lesssim_{\mathbb{P}} 1, \end{aligned} \quad (\text{SA-II.10.42})$$

where boundary terms are omitted.

For the products involving the s -onward component $u_{it}^{\geq s}$, all primitive shocks are dated at or after the later base date, so each coefficient is measurable before the earlier of the two shocks. Let $\mathcal{O}_{k\ell}^{\text{F}}$ denote their off-diagonal contribution and define $\{M_{i,rr'}^{k\ell,\text{F}}\}$ as the collected coefficients in

$$\mathcal{O}_{k\ell}^{\text{F}} := \frac{1}{L_T^2} \sum_{i=1}^n \sum_{r \neq r'}^n M_{i,rr'}^{k\ell,\text{F}} \varepsilon_{ir} \varepsilon_{ir'}.$$

The coefficients include the matched-set indicators and hence vanish when the relevant unit is unavailable. Equations (SA-II.10.35) and (SA-II.10.36), together with Jensen's inequality over the rolling averages, give

$$\sum_{r \neq r'} \sum_{i=1}^n |M_{i,rr'}^{k\ell,\text{F}}|^2 \lesssim_{\mathbb{P}} \frac{L_T H J^{3d}}{n^3}. \quad (\text{SA-II.10.43})$$

Combining the coefficients on (r, r') and (r', r) , which multiply the same product $\varepsilon_{ir} \varepsilon_{ir'}$, (SA-II.10.38) bounds this predictable quadratic shock sum by $O_{\mathbb{P}}\{\sqrt{L_T H J^{3d}/n^3}/L_T^2\}$.

For the pre- s component when s is the later base date, put

$$P_{is}^{k\ell} := \widehat{w}_{is}(\mathbf{b}_\ell) \sum_{h=1}^H \widehat{w}_{i,s-h}(\mathbf{b}_k) u_{i,s-h}^{<s}.$$

This coefficient is $\mathcal{K}_s$ -measurable. The maximum-weight bound in (SA-II.10.35), Cauchy–Schwarz over h , and (SA-II.10.42) imply

$$\max_{k,\ell} \sum_i \sum_s (1 + \|\varpi_{s+H}\|^2) |P_{is}^{k\ell}|^2 \lesssim_{\mathbb{P}} \frac{L_T H J^{3d}}{n^3}.$$

Equation (SA-II.10.37), applied with $Z_{is} = P_{is}^{k\ell}$, therefore rewrites $L_T^{-2} \sum_{i,s} P_{is}^{k\ell} u_{is}^\circ$ as a predictable linear shock sum whose coefficient-square sum is $O_{\mathbb{P}}(L_T H J^{3d}/n^3)$. The same argument applies after interchanging the base dates. Combining the two orderings of the base dates with the preceding off-diagonal contribution yields

$$\mathcal{O}_{k\ell} = O_{\mathbb{P}} \left(\sqrt{\frac{H J^{3d}}{n^3 L_T^3}} \right) = o_{\mathbb{P}} \left(\frac{J^d}{n L_T} \right). \quad (\text{SA-II.10.44})$$

Step 2: Fitted-residual error. We next replace u_{it}° by the fitted residual. From (SA-II.10.24), the IV matrix bounds, and (SA-II.10.27),

$$\frac{1}{L_T} \sum_{t=H+1}^{T-H} \|\hat{\theta}_{t+H}^{\text{IV}} - \theta_{t+H}\|^2 \lesssim_{\mathbb{P}} e_{nT}^2. \quad (\text{SA-II.10.45})$$

Moreover,

$$\hat{e}_{i(t+H)} - u_{it}^\circ = -\varpi_{t+H}^\top (\check{\beta}_{i(t+H)} - \beta_{i(t+H)}) - \hat{x}_{i(t+H)}^\top (\hat{\theta}_{t+H}^{\text{IV}} - \theta_{t+H}).$$

For $i \in \mathcal{I}_t^{\text{LIN}}$, let the left-hand side be δ_{it} , and set $\delta_{it} = 0$ otherwise. The first bound in (SA-II.10.35), the time-averaged factor moments, and (SA-II.10.45) imply

$$\max_{k \in [K_{\text{LIN}}]} \frac{1}{L_T} \sum_{t=H+1}^{T-H} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \hat{w}_{it}(\mathbf{b}_k)^2 \delta_{it}^2 \lesssim_{\mathbb{P}} \frac{J^d}{n} e_{nT}^2. \quad (\text{SA-II.10.46})$$

To make the residual replacement explicit, for arrays $a = \{a_{it}\}$ and $b = \{b_{it}\}$ define the banded bilinear form

$$\mathcal{H}_{k\ell}(a, b) := \frac{1}{L_T^2} \sum_{i=1}^n \sum_{t=H+1}^{T-H} \sum_{s=H+1}^{T-H} \hat{w}_{it}(\mathbf{b}_k) \hat{w}_{is}(\mathbf{b}_\ell) a_{it} b_{is} \mathbf{1}(|t-s| \leq H),$$

using the zero-weight convention for unavailable units. Since $\|a\|_k^2 := L_T^{-1} \sum_{t=H+1}^{T-H} \sum_{i=1}^n \hat{w}_{it}(\mathbf{b}_k)^2 a_{it}^2$, banded Cauchy–Schwarz gives

$$|\mathcal{H}_{k\ell}(a, b)| \leq \frac{2H+1}{L_T} \|a\|_k \|b\|_\ell. \quad (\text{SA-II.10.47})$$

Since $\hat{e}_{i(t+H)} = u_{it}^\circ + \delta_{it}$,

$$\hat{\Sigma}_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_\ell) - \tilde{\Sigma}_{k\ell} = \mathcal{H}_{k\ell}(u^\circ, \delta) + \mathcal{H}_{k\ell}(\delta, u^\circ) + \mathcal{H}_{k\ell}(\delta, \delta). \quad (\text{SA-II.10.48})$$

The inequality $2|xy| \leq x^2 + y^2$, the fact that each base date has at most $2H+1$ retained partners, and

(SA-II.10.46) give

$$\max_{k,\ell} |\mathcal{H}_{k\ell}(\delta, \delta)| \lesssim_{\mathbb{P}} \frac{J^d}{nL_T} H e_{nT}^2. \quad (\text{SA-II.10.49})$$

To decompose the two remaining cross terms, put

$$\begin{aligned} d_{it} &= \check{\beta}_{i(t+H)} - \beta_{i(t+H)}, & \check{x}_{i(t+H)} &= (1, \check{\beta}_{i(t+H)}^\top)^\top, & C_t^\circ &= (M_t^\circ)^{-1}, \\ h_t &:= \frac{1}{n_t^{\text{LIN}}} \sum_{j \in \mathcal{I}_t^{\text{LIN}}} \zeta_{jt} \varpi_{t+H}^\top d_{jt}, & \mathcal{B}_{t+H}^{\circ, \text{IV}} &:= -C_t^\circ h_t, \\ b_{it}^\circ &:= -\varpi_{t+H}^\top d_{it} - \check{x}_{i(t+H)}^\top \mathcal{B}_{t+H}^{\circ, \text{IV}}, & q_{it}^\circ &:= -\check{x}_{i(t+H)}^\top C_t^\circ g_t. \end{aligned}$$

For $i \in \mathcal{I}_t^{\text{LIN}}$, also define

$$r_{it} := -\left\{ \hat{x}_{i(t+H)}^\top \hat{M}_t^{-1} - \check{x}_{i(t+H)}^\top C_t^\circ \right\} (g_t - h_t).$$

Set $b_{it}^\circ = q_{it}^\circ = r_{it} = 0$ outside the matched set. Direct substitution gives the exact identity $\delta_{it} = b_{it}^\circ + q_{it}^\circ + r_{it}$. Consequently, the two cross terms in (SA-II.10.48) satisfy

$$\begin{aligned} &\mathcal{H}_{k\ell}(u^\circ, \delta) + \mathcal{H}_{k\ell}(\delta, u^\circ) \\ &= \mathcal{H}_{k\ell}(u^\circ, b^\circ) + \mathcal{H}_{k\ell}(b^\circ, u^\circ) + \mathcal{H}_{k\ell}(u^\circ, q^\circ) + \mathcal{H}_{k\ell}(q^\circ, u^\circ) \\ &\quad + \mathcal{H}_{k\ell}(u^\circ, r) + \mathcal{H}_{k\ell}(r, u^\circ). \end{aligned} \quad (\text{SA-II.10.50})$$

Step 3: Cross terms. Multiplier remainder. Assumption SA-LIN-IV(iii) gives

$$\max_{H+1 \leq t \leq T-H} \|\hat{M}_t^{-1} - (M_t^\circ)^{-1}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{n}}.$$

For $i \in \mathcal{I}_t^{\text{LIN}}$,

$$\hat{x}_{i(t+H)}^\top \hat{M}_t^{-1} - \check{x}_{i(t+H)}^\top C_t^\circ = (0, v_{i(t+H)}^\top) \hat{M}_t^{-1} + \check{x}_{i(t+H)}^\top (\hat{M}_t^{-1} - C_t^\circ).$$

The conditional score bound (SA-II.10.27), $\max_{i,t} \|d_{it}\| \leq B_{\text{FS}}$ w.p.a.1, and the time-averaged factor-moment bound imply

$$\frac{1}{L_T} \sum_t \|g_t - h_t\|^2 \lesssim_{\mathbb{P}} e_{nT}^2, \quad \frac{1}{L_T} \sum_t \mathbb{E}[\|g_t - h_t\|^4 \mid \mathcal{K}_t] \lesssim_{\mathbb{P}} e_{nT}^4.$$

Together with $\mathbb{E}[\|v_{i(t+H)}\|^2 \mid \mathcal{K}_t] \lesssim_{\mathbb{P}} H^{-1}$, the corresponding conditional fourth-moment bounds, and the maximum-instrument bound in Assumption SA-LIN-IV(ii), the multiplier identity above gives

$$\max_{k \in [K_{\text{LIN}}]} \frac{1}{L_T} \sum_{t=H+1}^{T-H} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \hat{w}_{it}(\mathbf{b}_k)^2 r_{it}^2 \lesssim_{\mathbb{P}} \frac{J^d}{n} \rho_{nT}^2 e_{nT}^2. \quad (\text{SA-II.10.51})$$

The corresponding weighted mean square for u° is $O_{\mathbb{P}}(J^d/n)$. Hence (SA-II.10.47) gives

$$\max_{k,\ell} \{|\mathcal{H}_{k\ell}(u^\circ, r)| + |\mathcal{H}_{k\ell}(r, u^\circ)|\} \lesssim_{\mathbb{P}} \frac{J^d}{nL_T} H \rho_{nT} e_{nT} = o_{\mathbb{P}}\left(\frac{J^d}{nL_T}\right), \quad (\text{SA-II.10.52})$$

by (SA-II.10.34).

First-step bias. Use the pre- s / s -onward split introduced in Step 1. By the definition of κ_{tr} above,

$$u_{it}^{\geq s} = \sum_{r=1}^T \kappa_{tr} \varepsilon_{ir} \mathbb{1}(r \geq \max(t, s)).$$

Define the s -onward and pre- s pieces explicitly by

$$\begin{aligned} \mathcal{F}_{k\ell}^{u,b} &:= \frac{1}{L_T^2} \sum_{i=1}^n \sum_t \sum_s \hat{w}_{it}(\mathbf{b}_k) \hat{w}_{is}(\mathbf{b}_\ell) u_{it}^{\geq s} b_{is}^\circ \mathbb{1}(|t-s| \leq H), \\ \mathcal{P}_{k\ell}^{u,b} &:= \frac{1}{L_T^2} \sum_{i=1}^n \sum_t \sum_s \hat{w}_{it}(\mathbf{b}_k) \hat{w}_{is}(\mathbf{b}_\ell) u_{it}^{\leq s} b_{is}^\circ \mathbb{1}(|t-s| \leq H), \\ \mathcal{H}_{k\ell}(u^\circ, b^\circ) &= \mathcal{F}_{k\ell}^{u,b} + \mathcal{P}_{k\ell}^{u,b}. \end{aligned}$$

The two weights and b_{is}° use only innovations dated before their base dates, and κ_{tr} is $\mathcal{E}_\infty$ -measurable. Define

$$F_{ir}^{k\ell} := \sum_t \sum_s \hat{w}_{it}(\mathbf{b}_k) \hat{w}_{is}(\mathbf{b}_\ell) b_{is}^\circ \kappa_{tr} \mathbb{1}(r \geq \max(t, s)) \mathbb{1}(|t-s| \leq H).$$

Then $F_{ir}^{k\ell}$ is measurable before ε_{ir} , and

$$\mathcal{F}_{k\ell}^{u,b} = \frac{1}{L_T^2} \sum_{i=1}^n \sum_{r=1}^T F_{ir}^{k\ell} \varepsilon_{ir}.$$

Equation (SA-II.10.35), the bound $\max_{i,t} \|d_{it}\| \leq \mathbf{B}_{\text{FS}}$ w.p.a.1, and the instrument moment, inverse-matrix, and time-averaged factor-moment bounds give

$$\max_{k,\ell} \sum_{i=1}^n \sum_{r=1}^T |F_{ir}^{k\ell}|^2 \lesssim_{\mathbb{P}} \frac{L_T H^2 J^{3d}}{n^3} \mathbf{B}_{\text{FS}}^2.$$

The predictable linear-shock-sum bound therefore yields

$$\max_{k,\ell} |\mathcal{F}_{k\ell}^{u,b}| \lesssim_{\mathbb{P}} \sqrt{\frac{H^2 J^{3d}}{n^3 L_T^3}} \mathbf{B}_{\text{FS}}.$$

For the pre- $(t+h)$ component, we use a direct bound rather than martingale cancellation. For $h = 1, \dots, H$, let

$$\begin{aligned} E_{k,h}^u &:= \frac{1}{L_T} \sum_t \sum_i \hat{w}_{it}(\mathbf{b}_k)^2 (u_{it}^{\leq t+h})^2, \\ E_{\ell,h}^b &:= \frac{1}{L_T} \sum_t \sum_i \hat{w}_{i,t+h}(\mathbf{b}_\ell)^2 (b_{i,t+h}^\circ)^2, \end{aligned}$$

where boundary terms are omitted. Equation (SA-II.10.42), the maximum-instrument bound, $n_t^{\text{LIN}} \asymp n$, and $\sup_t \|C_t^\circ\| \lesssim_{\mathbb{P}} 1$ give

$$\sum_{h=1}^H E_{k,h}^u \lesssim_{\mathbb{P}} \frac{J^d}{n}, \quad \sum_{h=1}^H E_{\ell,h}^b \lesssim_{\mathbb{P}} \frac{H J^d}{n} B_{\text{FS}}^2.$$

Cauchy–Schwarz first over (i, t) and then over h therefore bounds the pre- s piece explicitly as

$$\max_{k,\ell} |\mathcal{P}_{k\ell}^{u,b}| \leq \frac{1}{L_T} \sum_{h=1}^H \sqrt{E_{k,h}^u E_{\ell,h}^b} \lesssim_{\mathbb{P}} \frac{J^d}{n L_T} \sqrt{H} B_{\text{FS}}.$$

By interchanging t and s ,

$$\mathcal{H}_{k\ell}(b^\circ, u^\circ) = \mathcal{H}_{\ell k}(u^\circ, b^\circ) = \mathcal{F}_{\ell k}^{u,b} + \mathcal{P}_{\ell k}^{u,b}.$$

Consequently,

$$\max_{k,\ell} \{|\mathcal{H}_{k\ell}(u^\circ, b^\circ)| + |\mathcal{H}_{k\ell}(b^\circ, u^\circ)|\} \lesssim_{\mathbb{P}} \frac{J^d}{n L_T} \left\{ \sqrt{H} B_{\text{FS}} + H B_{\text{FS}} \sqrt{\frac{J^d}{n L_T}} \right\} = o_{\mathbb{P}} \left(\frac{J^d}{n L_T} \right). \quad (\text{SA-II.10.53})$$

The last equality follows from $\sqrt{H} B_{\text{FS}} = o(1)$ and (SA-II.10.34).

IV score. The maximum-instrument bound in Assumption SA-LIN-IV(ii) and (SA-II.10.35) imply

$$\max_{t,k} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \hat{w}_{it}(\mathbf{b}_k)^2 \|\zeta_{it}\|^2 \lesssim_{\mathbb{P}} \frac{J^d}{n}. \quad (\text{SA-II.10.54})$$

Substituting $g_s = (n_s^{\text{LIN}})^{-1} \sum_{j \in \mathcal{I}_s^{\text{LIN}}} \zeta_{js} u_{js}^\circ$ into q_{is}° , for $i \in \mathcal{I}_s^{\text{LIN}}$, gives

$$\begin{aligned} q_{is}^{\circ, \text{same}} &:= -\frac{\check{x}_{i(s+H)}^\top C_s^\circ \zeta_{is}}{n_s^{\text{LIN}}} u_{is}^\circ, \\ q_{is}^{\circ, \text{cross}} &:= -\frac{1}{n_s^{\text{LIN}}} \sum_{\substack{j \in \mathcal{I}_s^{\text{LIN}} \\ j \neq i}} \check{x}_{i(s+H)}^\top C_s^\circ \zeta_{js} u_{js}^\circ, \quad q_{is}^\circ = q_{is}^{\circ, \text{same}} + q_{is}^{\circ, \text{cross}}. \end{aligned}$$

Set both components to zero for $i \notin \mathcal{I}_s^{\text{LIN}}$. Thus the displayed decomposition is exact for every $i \in [n]$. Accordingly, define the same-unit and cross-unit pieces of the two score cross-products by

$$\begin{aligned} \mathcal{Q}_{k\ell}^{\text{same}} &:= \mathcal{H}_{k\ell}(u^\circ, q^{\circ, \text{same}}) + \mathcal{H}_{k\ell}(q^{\circ, \text{same}}, u^\circ), \\ \mathcal{Q}_{k\ell}^{\text{cross}} &:= \mathcal{H}_{k\ell}(u^\circ, q^{\circ, \text{cross}}) + \mathcal{H}_{k\ell}(q^{\circ, \text{cross}}, u^\circ), \\ \mathcal{H}_{k\ell}(u^\circ, q^\circ) + \mathcal{H}_{k\ell}(q^\circ, u^\circ) &= \mathcal{Q}_{k\ell}^{\text{same}} + \mathcal{Q}_{k\ell}^{\text{cross}}. \end{aligned}$$

The same-unit piece need not be centered. The multiplier $\check{x}_{i(s+H)}^\top C_s^\circ \zeta_{is}$ is uniformly bounded in probability and $(n_s^{\text{LIN}})^{-1} \lesssim n^{-1}$. Applying $2|xy| \leq x^2 + y^2$ to the banded sum, then using (SA-II.10.35) and the weighted

second-moment bound for u° used above, gives

$$\max_{k,\ell} |\mathcal{Q}_{k\ell}^{\text{same}}| \lesssim_{\mathbb{P}} \frac{J^d}{nL_T} \frac{H}{n}. \quad (\text{SA-II.10.55})$$

For $j \neq i$, Assumption [SA-LIN-M](#) preserves conditional independence of future primitive shocks across units after selection. For each pair t, s , split the residual associated with the earlier base date into its component before the later base date and its component from the later base date onward. In the latter component, all weights, instruments, and IV multipliers are fixed before both primitive shocks. Let $\mathcal{Q}_{k\ell}^{\text{cross},F}$ denote this contribution and define $\{N_{ij,rr'}^{k\ell,F}\}$ as the collected coefficients in

$$\mathcal{Q}_{k\ell}^{\text{cross},F} := \frac{1}{L_T^2} \sum_{\substack{i,j \in [n] \\ i \neq j}} \sum_{r=1}^T \sum_{r'=1}^T N_{ij,rr'}^{k\ell,F} \varepsilon_{ir} \varepsilon_{jr'}.$$

These coefficients include all matched-set indicators and are zero if a required unit is unavailable. Assumption [SA-LIN-IV](#)(ii), [\(SA-II.10.35\)](#), [\(SA-II.10.36\)](#), and Jensen's inequality give

$$\sum_{\substack{i,j \in [n] \\ i \neq j}} \sum_{r=1}^T \sum_{r'=1}^T |N_{ij,rr'}^{k\ell,F}|^2 \lesssim_{\mathbb{P}} \frac{L_T H^2 J^{3d}}{n^4}. \quad (\text{SA-II.10.56})$$

After ordering the compound shock indices, combine the coefficients on each pair and its reversal, which multiply the same shock product. The predictable quadratic-shock-sum bound in [\(SA-II.10.38\)](#) then gives

$$\max_{k,\ell} |\mathcal{Q}_{k\ell}^{\text{cross},F}| \lesssim_{\mathbb{P}} \sqrt{\frac{H^2 J^{3d}}{n^4 L_T^3}}.$$

Consider first $\mathcal{H}_{k\ell}(u^\circ, q^{\circ,\text{cross}})$ in the case $t < s$, so the IV-score term indexed by s has the later base date. Its pre- s contribution is linear in u_{js}° . For $j \in \mathcal{I}_s^{\text{LIN}}$, its $\mathcal{K}_s$ -measurable coefficient is

$$P_{js}^{k\ell,t < s} := -\frac{1}{n_s^{\text{LIN}}} \sum_{h=1}^H \sum_{\substack{i \in \mathcal{I}_s^{\text{LIN}} \\ i \neq j}} \hat{w}_{i,s-h}(\mathbf{b}_k) \hat{w}_{is}(\mathbf{b}_\ell) \{ \check{x}_{i(s+H)}^\top C_s^\circ \zeta_{js} \} u_{i,s-h}^{< s}.$$

As stipulated above, shifted base dates outside $\mathcal{T}_H$ are omitted. Set this coefficient to zero for $j \notin \mathcal{I}_s^{\text{LIN}}$. Since $n_s^{\text{LIN}} \asymp n$, Cauchy-Schwarz, [\(SA-II.10.42\)](#), [\(SA-II.10.35\)](#), and the uniform bounds on ζ_{js} , $\check{x}_{i(s+H)}$, and C_s° give

$$\max_{k,\ell} \sum_j \sum_s (1 + \|\varpi_{s+H}\|^2) |P_{js}^{k\ell,t < s}|^2 \lesssim_{\mathbb{P}} \frac{L_T H J^{3d}}{n^3}.$$

Equation [\(SA-II.10.37\)](#) therefore gives $O_{\mathbb{P}}\left(\sqrt{H J^{3d}/(n^3 L_T^3)}\right)$ for the corresponding predictable linear shock sum.

Next consider the same bilinear form in the case $s < t$, so the IV-score term indexed by s has the earlier

base date. For $s, t \in \mathcal{T}_H$ with $s < t$ and $i \in \mathcal{I}_s^{\text{LIN}}$, define

$$G_{i,s,t}^{<t} := \frac{1}{n_s^{\text{LIN}}} \sum_{\substack{j \in \mathcal{I}_s^{\text{LIN}} \\ j \neq i}} \check{x}_{i(s+H)}^\top C_s^\circ \zeta_{js} u_{js}^{<t}.$$

Set $G_{i,s,t}^{<t} = 0$ outside this domain. For $t \in \mathcal{T}_H$ and $i \in \mathcal{I}_t^{\text{LIN}}$, define

$$P_{it}^{k\ell, s < t} := -\widehat{w}_{it}(\mathbf{b}_k) \sum_{h=1}^H \widehat{w}_{i,t-h}(\mathbf{b}_\ell) G_{i,t-h,t}^{<t}.$$

Set $P_{it}^{k\ell, s < t} = 0$ outside this domain. Then $P_{it}^{k\ell, s < t}$ is $\mathcal{K}_t$ -measurable and is the coefficient multiplying the residual u_{it}° at the later base date t . Assumption [SA-LIN-IV\(ii\)](#) gives $|G_{i,s,t}^{<t}|^2 \lesssim_{\mathbb{P}} n^{-1} \sum_{j \in \mathcal{I}_s^{\text{LIN}}} (u_{js}^{<t})^2$. Cauchy–Schwarz over h , [\(SA-II.10.35\)](#), and the second bound in [\(SA-II.10.42\)](#) therefore give

$$\max_{k,\ell} \sum_i \sum_t (1 + \|\varpi_{t+H}\|^2) |P_{it}^{k\ell, s < t}|^2 \lesssim_{\mathbb{P}} \frac{L_T H J^{3d}}{n^3}.$$

Equation [\(SA-II.10.37\)](#) gives the same rate in the case $s < t$. The identity $\mathcal{H}_{k\ell}(q^{\circ, \text{cross}}, u^\circ) = \mathcal{H}_{\ell k}(u^\circ, q^{\circ, \text{cross}})$ covers the other cross term. Consequently,

$$\max_{k,\ell} |\mathcal{Q}_{k\ell}^{\text{cross}}| \lesssim_{\mathbb{P}} \sqrt{\frac{H^2 J^{3d}}{n^4 L_T^3}} + \sqrt{\frac{H J^{3d}}{n^3 L_T^3}} = o_{\mathbb{P}}\left(\frac{J^d}{n L_T}\right). \quad (\text{SA-II.10.57})$$

Relative to $J^d/(n L_T)$, the same-unit, cross-unit s -onward, and cross-unit pre- s bounds are

$$\frac{H}{n}, \quad H \sqrt{\frac{J^d}{n^2 L_T}}, \quad \text{and} \quad \sqrt{\frac{H J^d}{n L_T}}.$$

All three are $o(1)$ by [\(SA-II.10.34\)](#).

Conclusion. The quadratic and random-multiplier terms are controlled by [\(SA-II.10.49\)](#) and [\(SA-II.10.52\)](#). The bias and score terms are controlled by [\(SA-II.10.53\)](#), [\(SA-II.10.55\)](#), and [\(SA-II.10.57\)](#). Substitution into [\(SA-II.10.48\)](#) and [\(SA-II.10.50\)](#) gives, uniformly over the fixed grid,

$$\begin{aligned} \max_{k,\ell \in [K_{\text{LIN}}]} \left| \widehat{\Sigma}_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_\ell) - \widetilde{\Sigma}_{k\ell} \right| &\lesssim_{\mathbb{P}} \frac{J^d}{n L_T} H e_{nT}^2 + o_{\mathbb{P}}\left(\frac{J^d}{n L_T}\right) \\ &= o_{\mathbb{P}}\left(\frac{J^d}{n L_T}\right). \end{aligned} \quad (\text{SA-II.10.58})$$

Step 1, specifically [\(SA-II.10.41\)](#) and [\(SA-II.10.44\)](#), also gives

$$\max_{k,\ell \in [K_{\text{LIN}}]} \left| \widetilde{\Sigma}_{k\ell} - \Sigma_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_\ell) \right| = o_{\mathbb{P}}\left(\frac{J^d}{n L_T}\right).$$

The triangle inequality and fixed K_{LIN} therefore give

$$\left\| \widehat{\Sigma}_{\text{LIN}} - \Sigma_{\text{LIN}} \right\| \leq K_{\text{LIN}} \max_{k,\ell \in [K_{\text{LIN}}]} \left| \widehat{\Sigma}_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_\ell) - \Sigma_{\text{LIN}}(\mathbf{b}_k, \mathbf{b}_\ell) \right|$$

$$= o_{\mathbb{P}}\left(\frac{J^d}{nL_T}\right),$$

which is (SA-II.10.20). □

SA-II.10.8 Proof of Theorem SA-II.10.3

By Lemma SA-II.10.2, (SA-II.10.16), and (SA-II.10.17), $\lambda_{\min}(\Sigma_{\text{LIN}}) \gtrsim_{\mathbb{P}} J^d / \{n(T - 2H)\}$ and hence

$$\left\| \Sigma_{\text{LIN}}^{-1/2} \widehat{\Sigma}_{\text{LIN}} \Sigma_{\text{LIN}}^{-1/2} - I_{K_{\text{LIN}}} \right\| = o_{\mathbb{P}}(1).$$

It follows that $\widehat{\Sigma}_{\text{LIN}}$ is nonsingular with probability approaching one. Put $Z_{\text{LIN}} = \Sigma_{\text{LIN}}^{-1/2} \widehat{\nu}_{\text{LIN}}$ and $A_{\text{LIN}} = \Sigma_{\text{LIN}}^{-1/2} \widehat{\Sigma}_{\text{LIN}} \Sigma_{\text{LIN}}^{-1/2}$. Then

$$\begin{aligned} \widehat{\nu}_{\text{LIN}}^{\top} \widehat{\Sigma}_{\text{LIN}}^{-1} \widehat{\nu}_{\text{LIN}} &= Z_{\text{LIN}}^{\top} A_{\text{LIN}}^{-1} Z_{\text{LIN}}, \\ Z_{\text{LIN}}^{\top} A_{\text{LIN}}^{-1} Z_{\text{LIN}} &\rightsquigarrow \chi_{K_{\text{LIN}}}^2. \end{aligned}$$

The convergence follows from Lemma SA-II.10.1, Slutsky's theorem, and the continuous mapping theorem. Continuity of the chi-square distribution at its $1 - \alpha$ quantile gives the asserted asymptotic size. □

SA-II.10.9 Proof of Lemma SA-II.10.4

Write

$$\widehat{x}_{it}^{\text{old}} = (1, \widehat{\beta}_{it}^{\top})^{\top}, \quad \check{x}_{it} = (1, \check{\beta}_{it}^{\top})^{\top}.$$

Because the rolling window defining $\widehat{\beta}_{it}$ ends at $t - 1$, $\widehat{x}_{it}^{\text{old}}$ is $\mathcal{F}_{t-1}$ -measurable and hence $\mathcal{K}_t$ -measurable. This proves Assumption SA-LIN-IV(i).

Lemma SA-II.2.3, the boundedness of β_{it} in Assumption SA-DGP(iii), and $R_{\text{FS}} = o(1)$ give

$$\max_{H+1 \leq t \leq T-H} \max_{i \in \mathcal{I}_t^{\text{LIN}}} \|\widehat{\beta}_{it}\| \lesssim_{\mathbb{P}} 1.$$

Consequently,

$$\max_{H+1 \leq t \leq T-H} \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \|\widehat{x}_{it}^{\text{old}}\|^4 \lesssim_{\mathbb{P}} 1.$$

The last two displays verify Assumption SA-LIN-IV(ii).

The future-window decomposition and conditional mean-zero property imply

$$M_t^{\circ} = \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \widehat{x}_{it}^{\text{old}} \check{x}_{i(t+H)}^{\top}.$$

The first-step rate and boundedness of the true betas therefore give

$$\max_{H+1 \leq t \leq T-H} \|M_t^{\circ} - M_{t,\beta}^0\| \lesssim_{\mathbb{P}} R_{\text{FS}} = o_{\mathbb{P}}(1).$$

Equation (SA-II.10.21) and a matrix perturbation bound then yield

$$\max_{H+1 \leq t \leq T-H} \{ \|\mathbf{M}_t^\circ\| + \|(\mathbf{M}_t^\circ)^{-1}\| \} \lesssim_{\mathbb{P}} 1.$$

Finally, with $(0, v_{i(t+H)}^\top)$ interpreted as a row vector,

$$\widehat{\mathbf{M}}_t - \mathbf{M}_t^\circ = \frac{1}{n_t^{\text{LIN}}} \sum_{i \in \mathcal{I}_t^{\text{LIN}}} \widehat{x}_{it}^{\text{old}}(0, v_{i(t+H)}^\top).$$

Conditional on $\mathcal{K}_t$, the summands are centered and independent across units; Assumption SA-LIN-M preserves this property after restriction to the matched set. The first-step Gram bound and the conditional moment bounds in Assumption SA-DGP(ii) give

$$\mathbb{E}[\|v_{i(t+H)}\|^2 \mid \mathcal{K}_t] \lesssim_{\mathbb{P}} H^{-1}.$$

Applying the conditional Fuk–Nagaev inequality and a union bound over the base dates, using $T^{2/q}/H = o(1)$ and $cn \leq n_t^{\text{LIN}} \leq n$, yields

$$\max_{H+1 \leq t \leq T-H} \|\widehat{\mathbf{M}}_t - \mathbf{M}_t^\circ\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{nH}} + \frac{(nT)^{1/q} \log(nT)}{nH} \lesssim_{\mathbb{P}} \sqrt{\frac{\log(nT)}{n}}.$$

This proves the concentration part of Assumption SA-LIN-IV(iii) and completes the verification. □

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